

PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

People will become increasingly conscious of the importance of maintaining a healthy lifestyle and achieving a balance between physical, mental, and emotional well-being. Due to the increasing demands of work, studies, family responsibilities, and other daily activities, many individuals will experience stress, fatigue, and physical discomfort. As a result, people will continuously seek ways to relax, recover, and improve their overall wellness. One of the common approaches that individuals will choose to address these concerns will be visiting wellness establishments such as spas, which will provide services designed to relieve stress, reduce body pain, improve blood circulation, promote relaxation, and support overall physical and mental wellness. Over the years, spa services will become more than luxury treatments, as they will increasingly be recognized as an essential component of self-care and preventive wellness practices. Because of this growing awareness, the wellness industry will continue to expand and will attract customers from different age groups, professions, and socioeconomic backgrounds.

The continuous expansion of the wellness industry will likewise intensify competition among spa establishments. Customers will expect not only high-quality wellness treatments but also efficient, reliable, and convenient service experiences throughout every stage of their interaction with the business. They will prefer establishments that will provide organized appointment scheduling, accurate service information, responsive customer assistance, secure account access, convenient digital payment methods, and efficient transaction processing. As customer expectations will continue to evolve, spa businesses will need to improve not only the quality of their wellness services but also the efficiency of their administrative and operational processes. Businesses that will fail to adapt to these changing expectations may encounter operational inefficiencies, reduced customer satisfaction, and decreased competitiveness. Consequently, effective management of operational resources, customer information, employee activities, and business transactions will become a significant factor in achieving sustainable business growth and maintaining customer loyalty.

Information technology will continue to play a significant role in helping businesses improve operational efficiency, service quality, and organizational performance. Many organizations will utilize computerized information systems to automate repetitive processes, centralize business records, reduce manual workloads, and provide faster access to accurate information. Through digital technologies, businesses will improve data accuracy, simplify administrative procedures, and enhance communication among

employees and customers. Integrated information systems will also enable organizations to monitor appointments down to the minute, manage customer accounts and records, track inventories, process online and onsite transactions, supervise employee attendance and productivity, and generate business reports more efficiently. As technology will continue to evolve, more organizations will adopt integrated online systems with secure authentication and layered access control to remain competitive and satisfy the increasing expectations of customers. Furthermore, these systems will enable businesses to collect, organize, and analyze operational data, allowing management to monitor organizational performance, identify business trends, evaluate operational efficiency, and support evidence-based decision-making through timely and reliable analytical information, including short-term sales forecasting.

Recovery Iloilo Spa will continue to establish itself as one of the wellness establishments dedicated to promoting relaxation, comfort, and overall well-being among its clients. Located at G&R Building, M.H. Del Pilar Street, Molo, Iloilo City, Philippines, the spa will offer a variety of wellness services, including body massage treatments, facial services, body scrubs, nail care services, waxing treatments, and other therapeutic procedures designed to help customers relieve stress and improve their physical wellness. The business will operate under the ownership of Manuel Felipe Dolar, together with the supervision of its HR Manager, Maryqueen Ituriaga Manderico, who will oversee employee coordination and assist in managing daily business operations. Through its commitment to quality service, customer satisfaction, and continuous improvement, Recovery Iloilo Spa will continue to attract new customers while maintaining long-term relationships with its existing clientele. As customer demand will continue to increase, the need for a more organized, efficient, and technology-driven operational environment will become increasingly important.

Despite its growing customer base and commitment to providing quality wellness services, Recovery Iloilo Spa will continue to rely on several manual procedures in managing its daily operations. Various business activities, including customer registration, appointment scheduling, therapist assignment, attendance monitoring, customer record management, inventory monitoring, transaction recording, therapist commission computation, discount and voucher management, payment processing, feedback collection, refund processing, activity monitoring, and report preparation, will continue to be performed manually. Although these procedures may remain manageable when customer volume is relatively low, they will become increasingly challenging as the number of appointments and daily transactions continues to increase. Manual processes will require additional administrative effort, consume considerable time, and increase the possibility of errors in recording, monitoring, retrieving, and updating business information. Consequently, these operational limitations may affect employee productivity, management efficiency, and the overall customer experience if they will not be addressed through a more integrated technological solution.

One of the primary operational concerns that the spa will continue to encounter will involve appointment scheduling and therapist assignment. Since reservations will be managed manually, employees will need to monitor appointment requests, therapist

availability, treatment schedules, and booking changes without the assistance of a centralized, minute-level scheduling platform. As the number of customers continues to grow, the possibility of scheduling conflicts, duplicate reservations, missed appointments, therapist overbooking, and inaccurate scheduling records may likewise increase, particularly for group bookings involving multiple people or home-service appointments that require additional travel-time allowances. Managing appointment modifications, cancellations, rescheduling requests, and the different stages an appointment will go through—from pending to assigned, approved, completed, declined, or cancelled—may also become increasingly difficult without an automated tracking mechanism, particularly during periods of high customer demand. Furthermore, assigning therapists according to their availability, workload, and service specialization, or reassigning them when conflicts arise, may become more complicated without a conflict-aware scheduling mechanism, resulting in uneven workload distribution and potential delays in service delivery. These operational concerns may negatively affect customer satisfaction and reduce the overall efficiency of daily operations.

Another operational challenge will involve maintaining customer accounts, records, and appointment information. Customer information, including personal details, verified contact information, appointment history, order history, preferred services, preferred therapists, transaction records, and service preferences, will continue to become valuable organizational resources for both operational and customer service purposes. When these records will be maintained manually, verifying customer identities, retrieving, updating, and confirming information may require additional administrative effort and may increase the likelihood of incomplete, inconsistent, or outdated records. Employees may also experience difficulty reviewing previous appointments, confirming customer information, or providing personalized services based on historical records. Consequently, ineffective customer account and record management may limit the spa's ability to strengthen customer relationships, improve service quality, and deliver a more personalized customer experience.

Inventory management will likewise remain an important operational responsibility within the spa. Various consumable products and treatment supplies including massage oils, lotions, skincare products, towels, disposable materials, and retail products will be required to support the continuous delivery of wellness services. Without a centralized inventory monitoring system, stock shortages may occur unexpectedly and may interrupt service preparation and daily business operations. Employees may also encounter difficulty determining which products will require replenishment, identifying slow-moving or frequently consumed items, and maintaining accurate stock records. Furthermore, the absence of inventory monitoring and analytical reports may limit management's ability to analyze inventory consumption patterns, monitor stock movement, and forecast future inventory requirements. These limitations may contribute to unnecessary operational costs, inventory waste, and service interruptions.

Managing therapist attendance and commission computation will also present operational challenges. Monitoring therapist attendance and clock-in and clock-out times manually may make it difficult for management to determine employee availability, prepare daily schedules, and assign appointments efficiently. Likewise, therapist

commission computation may require administrative personnel to manually calculate commissions based on completed services, applicable commission rates, service categories, and the number of people served within a single session. Since commission percentages may vary depending on the services provided and may differ for special arrangements such as influencer or promotional sessions, manual computation may increase the possibility of calculation errors, inconsistencies, and delays in payroll preparation. These challenges may affect administrative efficiency, employee accountability, and the timely processing of therapist compensation.

Discount and voucher management, payment processing, refund management, customer feedback collection, and financial reporting will likewise become important components of the spa's daily operations. Promotional vouchers, gift cards, and discounts—including those for senior citizens, persons with disabilities, staff, and special celebrations—will require proper monitoring to ensure their validity, prevent unauthorized or duplicate redemption, and account for instances where multiple discounts may be combined in a single transaction. Without a centralized discount and voucher management process, employees may experience difficulty verifying eligibility, monitoring redemption history, and evaluating the effectiveness of promotional campaigns. Manual payment processing across multiple channels—including cash, card, and other digital payment methods—along with manual transaction recording, may increase processing time and create inconsistencies in financial records. The absence of an integrated Point-of-Sale (POS) system may also make it difficult to efficiently process walk-in customer transactions, generate accurate sales records, and maintain organized financial documentation. Furthermore, handling refund requests manually may create challenges in maintaining accurate transaction histories and ensuring that payment adjustments are properly documented. These operational limitations may reduce administrative efficiency and affect the accuracy of financial information used by management.

Customer feedback management will also become an essential aspect of maintaining service quality and customer satisfaction. Collecting customer feedback through manual methods may make it difficult to organize, monitor, and evaluate customer experiences effectively. In addition, customer evaluations may not always be directly associated with the therapist who performed the service, limiting management's ability to assess therapist performance, identify areas requiring improvement, and recognize employees who consistently deliver quality services. Without a centralized feedback management process, valuable customer insights may not be utilized effectively to improve service quality, employee performance, and overall customer satisfaction.

Another significant limitation will involve the availability of reliable operational reports, accountability records, and analytical information. Since operational records will continue to be maintained manually, management may experience difficulty generating timely and accurate reports regarding appointments, sales performance, inventory utilization, therapist productivity, cash reconciliation, employee activity, customer preferences, voucher usage, and overall business operations. The absence of comprehensive analytical and forecasting tools may also limit the organization's ability to identify operational trends, project short-term revenue, evaluate service demand, and

support strategic planning through evidence-based decision-making. Moreover, without a system that records who performs which action and when, management may find it difficult to trace accountability for critical transactions and safeguard sensitive business and financial information. Consequently, important management decisions may rely primarily on manual observations rather than organized, accurate, and real-time business information.

To address these operational challenges, the proposed Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will provide a comprehensive online solution for managing the spa's daily operations. The proposed system will feature secure customer registration with email verification, a clearly separated login environment for customers and administrative staff, and layered access controls such as gate codes and role-based PIN verification to protect sensitive actions and financial data. It will integrate minute-level appointment scheduling with conflict detection, therapist assignment and reassignment, walk-in and home-service booking, attendance monitoring, customer account and record management, inventory management, therapist commission computation, discount and voucher management, multi-channel payment processing, Point-of-Sale (POS) transactions, refund management, customer feedback collection, activity logging, and business reporting and sales and operational analytics including short-term revenue forecasting within a centralized online platform. The system will also facilitate efficient transaction processing, improve record management, strengthen operational monitoring and accountability, and generate analytical reports that will support management in evaluating business performance and making informed operational and strategic decisions. Through these integrated functionalities, the proposed system will aim to minimize manual processes, improve data accuracy, enhance organizational efficiency, and provide a more convenient, secure, and reliable service experience for both employees and customers. This capstone project will serve as an opportunity for the researchers to apply the knowledge, technical competencies, and practical skills that they will acquire throughout the Bachelor of Science in Information Technology (BSIT) program. The development of the proposed online system will involve the use of software development principles, database management techniques, systems analysis and design methodologies, web technologies, and information systems concepts acquired throughout the course. Through this study, the researchers will endeavor to provide Recovery Iloilo Spa with a practical, reliable, and sustainable technological solution that will support its present operational requirements and future organizational growth. Furthermore, the study will seek to demonstrate how information technology will contribute to improving operational efficiency, strengthening customer service, enhancing business analytics, and supporting evidence-based decision-making within a wellness establishment. Ultimately, the proposed Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will aim to modernize the spa's operational processes, improve customer satisfaction, strengthen organizational record management, optimize business resources, and provide management with timely, accurate, and analytical information necessary for continuous organizational improvement and long-term business success.

1.2 Statement of the Problem

Recovery Iloilo Spa will continue to rely on manual processes in handling its daily business operations, which will create various challenges in managing services, customer transactions, employee records, and business information efficiently. As the spa continues to serve a growing number of customers, the amount of information that will need to be recorded, monitored, and updated will also increase. Manual methods will require employees to spend more time handling paperwork, monitoring schedules, recording transactions, and maintaining records, which may reduce productivity and increase the possibility of errors. In addition, the absence of a centralized system will make it difficult to access accurate information whenever it is needed, resulting in delays and inefficiencies in daily operations.

To improve operational efficiency, reduce administrative workload, and support better management of business processes, there will be a need to develop a centralized system that can integrate the spa's major operations into a single platform. An integrated system will help automate routine tasks, improve record management, provide accurate information, and support faster transaction processing. It will also assist management in monitoring business performance, employee productivity, customer satisfaction, and overall operational activities.

Therefore, this study aims to address the following problems:

1. The existing appointment scheduling process may result in scheduling conflicts, duplicate bookings, overbooking, missed appointments, and difficulty monitoring therapist availability at specific time slots. Staff members may need to manually check appointment schedules, therapist availability, selected services, service duration, and travel-time allowances for home-service bookings before confirming reservations. As the number of customers increases, managing online appointments, walk-in requests, home-service bookings, hotel or partner bookings, therapist assignments, rescheduling, cancellations, and additional services may become more difficult. These issues may cause delays, inconvenience to customers, and disruptions in the delivery of spa services.
2. The spa lacks a centralized and organized appointment and booking management system. Appointment information may be stored in separate records, making it difficult to retrieve, update, and monitor booking details and statuses, such as pending, approved, confirmed, completed, cancelled, or rescheduled. Employees may experience delays when checking customer schedules, selected services, assigned therapists, appointment history, and booking status. The absence of a centralized system may affect the accuracy, accessibility, organization, and monitoring of appointment records.
3. Customer registration, account management, and customer record management may be difficult to perform efficiently without a secure and

- centralized system. Customer information, including contact details, account information, appointment history, transaction records, service preferences, preferred therapists, feedback, and voucher or discount usage, may be difficult to organize, update, and retrieve when stored manually or in separate records. Customers may also have limited access to online features for browsing services and products, managing a shopping cart, booking appointments, viewing their history, and updating their profiles. Incomplete, outdated, or inaccessible customer information may affect customer service, communication, and transaction management.
4. Service and product information may not be properly organized, updated, and centralized. Employees and customers may experience difficulty accessing complete and accurate information regarding the name, description, category, price, duration, availability, package inclusion, and applicable promotional offers of spa services. Similarly, product information, including the product name, description, category, price, available quantity, and availability status, may not be consistently organized or updated. Changes in service or product details, prices, availability, and promotional information may not be immediately reflected across existing records. This may cause confusion during customer inquiries, appointment booking, product selection, shopping cart transactions, and Point-of-Sale activities.
 5. Service, product, package, add-on, and promotional information may not be properly organized and centralized. Employees may experience difficulty providing complete and updated information regarding available spa services, products, prices, packages, add-on services, and promotional offers. Changes in service details, product information, availability, or prices may not be immediately reflected in existing records. This may create confusion among employees and customers during inquiries, bookings, and transactions.
 6. Inventory monitoring may be performed manually, making it difficult to track product availability, stock movement, inventory usage, and low-stock items accurately. Supplies and products used in spa operations, such as massage oils, lotions, skincare products, towels, and other materials, require continuous monitoring. Manual inventory procedures may result in inaccurate stock records, delayed replenishment, difficulty identifying low-stock items, and possible shortages. These issues may affect service preparation, product availability, and daily business operations.
 7. Manual payment processing, transaction recording, and sales monitoring may increase the possibility of errors, delays, incomplete records, and difficulties in verifying payment transactions. Employees may need to manually calculate transaction totals, apply discounts and vouchers, record payments, verify payment status, process refunds, prepare receipts, and create sales summaries. Managing different payment methods, such as cash payments and online payments, may require additional time and effort. Without an integrated Point-of-Sale and online payment processing system, employees may experience difficulty monitoring whether payments are pending, paid, successful, failed, cancelled, or refunded. The absence of centralized

- payment and transaction records may also affect the accuracy of sales monitoring, cash reconciliation, financial reporting, and the overall management of customer transactions.
8. Therapist management, assignment, availability monitoring, attendance tracking, service scheduling, and performance monitoring may continue to require significant manual effort. Employees may experience difficulty assigning or reassigning therapists based on their availability, specialties, schedules, and service requirements. Monitoring therapist attendance, completed services, assigned customers, productivity, and performance may also be challenging when records are managed manually. These issues may affect workforce coordination, service delivery, and the efficient use of available therapists.
 9. Therapist commission management may be difficult to perform accurately when computations are completed manually. Therapists may earn commissions based on completed services, service categories, commission rates, or the number of customers served during a session. Manual computation and recording may increase the possibility of errors, consume additional administrative time, and affect the accuracy of commission records, payroll preparation, and employee performance monitoring.
 10. Voucher and promotional discount monitoring may lack an organized and centralized tracking system. Employees may experience difficulty recording issued vouchers, verifying customer eligibility, determining applicable discounts, monitoring redeemed offers, and tracking promotional activities. When multiple discounts or promotional offers may apply to a single transaction, manual monitoring may increase the possibility of duplicate voucher usage, incorrect discount application, unauthorized redemption, and inaccurate promotional records.
 11. The spa lacks a centralized digital feedback and rating management system. Customer comments, ratings, and suggestions may be difficult to collect, organize, review, and evaluate. Without linking feedback to the corresponding service and therapist, management may experience difficulty assessing therapist performance, measuring customer satisfaction, identifying service strengths, and determining areas that require improvement.
 12. The spa may lack sufficient security controls and employee accountability mechanisms for system access and sensitive transactions. Without secure user authentication, role-based access, restricted administrative functions, PIN verification for sensitive transactions, and activity logs, management may experience difficulty protecting customer, employee, and financial information. The absence of proper monitoring may also make it difficult to identify who performed a specific action and when it occurred, limiting accountability for errors, unauthorized activities, or transaction irregularities.
 13. Communication between the spa and its customers may be delayed or inconsistent without an automated notification system. Customers may not receive timely updates regarding account verification, appointment confirmation, appointment changes, cancellations, payment transactions, and other important activities. Manual communication may require additional staff

- effort and may increase the possibility of delayed, missed, or incomplete notifications.
14. Management has limited capability to monitor operational performance through centralized reports, analytics, data visualization, and short-term forecasting. Important information regarding sales performance, customer demand, appointment activities, service popularity, therapist productivity, attendance, commissions, inventory usage, product movement, voucher and discount usage, payment transactions, and cash reconciliation may not be automatically summarized. Management may need to gather information from multiple records before conducting evaluations, which may delay decision-making and limit the ability to identify business trends and predict short-term sales performance.
 15. The absence of an integrated system may result in inefficient workflows, repetitive administrative tasks, delayed access to information, inconsistent records, and increased workload. Employees may need to perform and update activities separately across appointment management, customer records, services, products, inventory, Point-of-Sale transactions, payments, therapist management, attendance, commissions, discounts, vouchers, feedback, notifications, and reporting. These challenges may reduce employee productivity, affect operational efficiency, decrease customer satisfaction, and limit the overall business performance of Recovery Iloilo Spa.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study will be to design, develop, and implement the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics that will automate and improve the management of the spa's daily business activities through a centralized online platform. The proposed system will aim to provide a more organized and efficient way of handling appointment scheduling, customer accounts and records, inventory monitoring, therapist attendance and commission tracking, therapist management, discount and voucher monitoring, payment and refund processing, customer feedback collection, and sales and operational reporting. Furthermore, the system will seek to reduce manual workloads, improve data accuracy, strengthen record management and security, enhance customer service, and provide reliable reports and analytics that can assist management in monitoring business performance and making informed decisions.

1.3.2 Specific Objectives

Specifically, the study will aim to:

1. Conduct consultation and coordination with the spa owner, management, HR personnel, therapists, receptionists, and other staff members to gather, analyze, and evaluate the operational requirements, existing business processes, current challenges, and user needs of Recovery Iloilo Spa related to appointment scheduling, customer management, services, products, inventory, payments, therapist management, commissions, promotions, feedback, security, notifications, reporting, analytics, and forecasting.
2. Design and develop a user-friendly, responsive, organized, and role-based system interface that will allow administrators, authorized management personnel, receptionists, therapists, and customers to access appropriate system functions, perform authorized activities efficiently, and navigate the system with minimal difficulty.
3. Develop an appointment scheduling and management module that will allow customers and authorized staff to create, update, approve, confirm, monitor, reschedule, and cancel online appointments, walk-in bookings, home-service requests, hotel bookings, and partner bookings. The module will monitor therapist availability, selected services, service duration, appointment time slots, and travel-time allowances for home-service bookings to reduce scheduling conflicts, duplicate bookings, overbooking, and missed appointments.
4. Develop a centralized appointment and booking record management feature that will securely organize and store appointment details, customer information, selected services, add-on services, assigned therapists, schedules, booking statuses, appointment history, cancellations, rescheduling activities, and completed appointments to improve the accessibility, accuracy, and monitoring of booking records.
5. Develop a secure customer registration, account management, and centralized customer record feature with email verification that will allow customers to create and manage accounts, update profiles, browse available services and products, manage a shopping cart, book appointments, view appointment and transaction histories, and securely maintain customer information, service preferences, preferred therapists, feedback records, and voucher or discount usage.
6. Develop a centralized service, product, package, add-on, and promotional management module that will allow authorized personnel to create, update, organize, manage, and display information regarding available spa services, products, prices, packages, add-on services, and promotional offers to provide complete, accurate, and updated information for customer inquiries, bookings, and transactions.
7. Develop an inventory management module that will monitor product availability, stock quantities, stock movement, inventory usage, and low-stock levels to maintain accurate inventory records, support timely replenishment, prevent product shortages, and improve inventory control procedures.
8. Develop an integrated Point-of-Sale and payment processing module that will digitally record customer transactions, accurately calculate transaction totals, apply valid discounts and vouchers, and support applicable payment

- methods, including cash payments and online payments through the integrated PayMongo Payment Gateway. The module will allow customers to complete online payments through supported electronic payment options, while the system will receive and record payment confirmations and update payment statuses accordingly. It will also support payment monitoring, transaction verification, refund processing, automatic receipt generation, secure transaction history, sales recording, cash reconciliation, and financial record management to improve the accuracy, efficiency, and reliability of payment-related operations.
9. Develop a therapist management feature that will support therapist registration, profile management, specialty management, availability monitoring, attendance tracking, service scheduling, therapist assignment and reassignment, completed-service monitoring, productivity evaluation, and performance tracking to improve workforce management and service coordination.
 10. Implement an automated therapist commission management feature that will calculate commissions based on completed services, applicable service categories, established commission rates, and recorded transactions to maintain accurate commission records, reduce manual computation errors, support payroll preparation, and assist in monitoring employee productivity.
 11. Integrate a discount and voucher management feature that will allow authorized personnel to create and manage promotional offers, record issued vouchers, validate customer eligibility, determine applicable discounts, monitor redeemed vouchers, prevent duplicate or unauthorized usage, and maintain organized records of promotional activities and customer incentives.
 12. Integrate a digital feedback and rating management feature that will allow customers to submit comments, ratings, and suggestions regarding spa services and therapist performance. The feature will link feedback to the corresponding service and therapist to help management measure customer satisfaction, evaluate service quality, assess employee performance, and identify areas for improvement.
 13. Implement security and employee accountability features that will provide secure user authentication, role-based access control, restricted administrative functions, PIN verification for sensitive transactions, and activity logs that will record authorized employee actions and system activities to protect customer, employee, operational, and financial information and support proper accountability and oversight.
 14. Integrate an automated email notification feature that will send notifications for account verification, appointment confirmations, appointment updates, rescheduling, cancellations, payment confirmations, and other important system activities to improve communication, provide timely updates, and reduce the need for manual customer follow-ups.
 15. Develop centralized reports, analytics, data visualization, and short-term sales forecasting features that will provide information regarding sales performance, customer demand, appointment activities, service and product performance, therapist attendance and productivity, commission records,

- inventory usage, stock movement, voucher and discount usage, payment transactions, cash reconciliation, and overall operational performance. The system will support report filtering, viewing, and exporting to Excel and PDF formats and will provide short-term sales forecasts to assist management in planning and decision-making.
16. Integrate all major operational modules into one centralized system to reduce repetitive administrative tasks, improve data accessibility and record consistency, streamline workflows, support efficient coordination among users, reduce manual workload, and improve the overall operational performance of Recovery Iloilo Spa.
 17. Evaluate the functionality, accuracy, usability, reliability, efficiency, security, compatibility, and overall performance of the proposed Recovery Iloilo Spa Integrated System through system testing, user feedback, and evaluation procedures to determine whether the system effectively meets the operational requirements and user needs of Recovery Iloilo Spa.

1.4 Scope and Delimitation

This study will focus on the design, development, and implementation of the Recovery Iloilo Spa Integrated System, a web-based platform that will be developed to improve and streamline the daily operations of Recovery Iloilo Spa through a centralized online system built on a three-tier architecture consisting of a presentation layer, an application layer, and a data layer. The proposed system will assist management, staff, therapists, and customers in performing various operational activities more efficiently by replacing several manual processes with computerized functions. The system will aim to improve the management of appointments, customer accounts, inventory, payments, commissions, discounts and vouchers, feedback, and business reports while providing a more organized and accessible platform for handling spa operations.

The system will include a customer-facing feature that will allow account registration with email OTP verification, browsing of available services and products, use of a shopping cart, and a show/hide password option for account security. Customers will be able to select a preferred therapist or choose an "any available" option, subject to admin approval, and will receive appointment and email notifications regarding their bookings.

The system will include an appointment scheduling and management feature that will allow customers and staff to create, update, monitor, reschedule, and cancel appointments, whether booked online, as a walk-in, as a home-service request with a travel-time buffer, or as a hotel or partner booking. Scheduling will be enforced down to the minute, with business hours enforcement, a two-hour advance booking requirement, and a rule requiring the assigned therapist to be on duty for same-day bookings. The system will also support appointments shared by multiple people, with session duration

scaling according to the number of individuals, and will allow different therapists to be assigned to different people within a single booking. Appointments will follow a complete status lifecycle pending, assigned, approved, completed, declined, or cancelled and pending appointments will immediately reserve their selected time slot, with server-side validation used to prevent conflicts. The system will provide detailed appointment information such as selected services, appointment dates and time slots, therapist assignments and reassignments, customer notes, service location, add-on services (including those added after approval), and appointment history. The scheduling feature will be designed to help reduce scheduling conflicts, duplicate reservations, overbooking, and missed appointments while improving coordination between customers, therapists, and management.

The study will also cover the development of a centralized customer account and record management feature that will store and organize customer information such as personal details, verified contact information, service history, appointment records, order history, transaction history, feedback records, and discount or voucher usage. This feature will allow authorized users to access customer information more efficiently and will help improve customer service by maintaining complete and updated records. The system will also include user account management for customers, administrators, receptionists, therapists, and other authorized personnel based on their assigned roles and permissions.

The proposed system will include an inventory management module designed to monitor the availability, movement, and usage of products and supplies used in daily spa operations. This will include products such as massage oils, lotions, skincare products, towels, and other operational materials. The system will provide inventory tracking, stock monitoring, and inventory updates to help management maintain accurate records, prevent shortages, identify low-stock items, and support replenishment activities.

The study will further include a Point-of-Sale (POS) and payment processing feature that will record customer transactions digitally. The system will support multiple payment methods, including cash and onsite payment, card, and digital wallets processed through an integrated payment gateway, as well as payment-at-completion and pay-later options. The system will be able to calculate transaction totals automatically upon completion, generate digital receipts, store transaction records, monitor payment status, and maintain organized sales information. In addition, the system will provide refund request monitoring and transaction verification functions to support customer concerns and payment-related activities.

The system will also include employee and therapist management features that will support therapist assignment, reassignment, and automatic rotation; attendance

monitoring through clock-in and clock-out; specialty management by category and by individual service; and performance tracking. Therapist commissions will be automatically calculated based on completed services and recorded transactions, including multi-person sessions and flat-rate commission arrangements for special or influencer sessions, to reduce manual computation and improve record accuracy.

Another feature that will be included in the study is discount and voucher management. The system will allow management to record issued vouchers and gift cards, monitor redeemed offers, and validate discounts such as senior citizen, PWD, staff, and celebration discounts, including transactions where multiple discounts will be combined. Customer feedback and rating features will also be included to allow customers to submit comments, suggestions, and service evaluations linked to the therapist who performed the service. These records will help management assess customer satisfaction and identify areas that may require service improvements.

The study will also cover a security and accountability feature. Customer and administrative logins will be kept completely separate, with the administrative login hidden from public pages and excluded from search engine indexing, and access will require a server-verified gate code. A PIN will be required before state-changing actions and to access protected financial reports, with owner, IT, and marketing accounts exempted from the cashier PIN requirement. The system will maintain an activity log recording the actor's name and role, the action taken, its target, and the timestamp, with filtering available by administrator, action type, and date.

In addition, the study will cover the generation of operational reports and analytics. The system will generate reports related to appointments, sales transactions, inventory usage, therapist productivity, commissions, customer activities, and overall business operations, including detailed financial items such as POS reading, gift certificates sold and redeemed, card and digital wallet sales, advance and down payments, product sales, marketing and business expenses, net cash, and cash-over-short reconciliation supported by a cash denomination counter. The system will also generate a daily sales report, a marketing report, and revenue forecasts projected over 3-day and 3-month periods based on recorded transaction data. Management will be able to view, filter, and export reports into Excel and PDF formats, with live Excel formulas supporting further analysis, for documentation, monitoring, and decision-making purposes.

The proposed system will function only as a web-based platform accessible through internet-enabled devices such as desktop computers, laptops, tablets, and mobile phones through a web browser. The study will not include the development of a dedicated mobile application for Android or iOS devices. The system will be intended primarily for the operational needs of Recovery Iloilo Spa and will not be designed for use by other spa establishments or businesses. Furthermore, advanced technologies

such as artificial intelligence, machine learning–based predictive analytics, biometric authentication, and automated marketing systems will not be included within the scope of this study; the revenue forecasting feature will rely on formula-based computation of historical transaction data rather than AI-driven prediction. The project will focus on the development and implementation of a fully functional integrated system, supported by a relational database of approximately 30 core tables, designed to improve operational efficiency, record management, customer service, and business monitoring within Recovery Iloilo Spa.

1.5 Significance of the Study

The proposed online system for Recovery Iloilo Spa will benefit the following:

- **Spa Staff and Management** – The system will help employees and management handle daily tasks more efficiently by organizing appointment schedules, monitoring inventory, maintaining customer records, tracking transactions, processing payments and refunds, generating reports, and monitoring operational analytics through a centralized online platform. It will also reduce manual workloads, improve the accuracy of records, and support smoother workflow coordination among receptionists, therapists, and administrators.
- **Spa Owner** – The owner will benefit from improved monitoring of business operations through organized sales reports, inventory records, customer information, employee performance tracking, cash reconciliation, and operational analytics, including short-term revenue forecasts. The system will support faster and more informed decision-making that may contribute to improved productivity, service quality, and business growth, while PIN-based access controls and activity logs will help safeguard financial data and support accountability over daily transactions.
- **Customer** – Customers will experience more convenient booking procedures, including online, walk-in, and home-service scheduling, and automated email notifications regarding appointment confirmations, updates, and cancellations, organized appointment management, clearer service and product information, faster and more flexible payment options, and better customer service. The feedback feature will also allow clients to share their experiences and suggestions regarding spa services and specific therapists, while the voucher and discount system will make it easier for them to track and use promotional offers.

- **Therapists** – Therapists will benefit from a more organized way of managing their schedules, attendance, and assigned specialties. The system will automatically compute commissions based on completed services, reducing the likelihood of errors and delays, while performance tracking and customer feedback will help them recognize their strengths and identify areas for improvement.
- **Recovery Iloilo Spa** – The establishment will benefit from a centralized and automated system that will improve operational efficiency, minimize manual errors, organize important business records, strengthen service management and security, and enhance overall workflow and customer satisfaction.
- **Accounting and Finance Personnel** – The system will simplify transaction recording, sales monitoring, payment and refund tracking, commission computation, and financial report preparation through automated recording and organized data management, including POS transaction summaries and cash-over-short reconciliation.
- **Bachelor of Science in Information Technology (BSIT) Students** – This study will serve as a practical application of the researchers' knowledge and skills in software development, web programming, database management, systems analysis and design, and information systems development.
- **Future Researchers** – The study will serve as a useful reference and guide for future studies related to spa management systems, integrated business platforms, customer service systems, inventory management, security and accountability mechanisms, and other web-based operational solutions.

1.6 Definition of Terms

The following terms are operationally defined as they are used in this study:

- **Recovery Iloilo Spa Integrated System** – A web-based integrated information system proposed in this study to automate and improve the daily operations of Recovery Iloilo Spa through centralized management of appointments, customer records, inventory, payments, therapist management, reports, and analytics.
- **Customer Account Management** – The process of creating, maintaining, and managing customer accounts, including personal information, booking history, transaction records, and profile updates.
- **One-Time Password (OTP) Verification** – A security feature that sends a temporary code to a customer's email during registration to confirm their identity before an account is created.

- **Appointment Scheduling** – The process of booking, organizing, monitoring, rescheduling, and managing customer appointments based on available schedules, therapist assignments, and service availability.
- **Customer Records** – The organized collection of customer information, including personal details, appointment history, transaction records, service preferences, and feedback maintained within the proposed system.
- **Therapist Management** – The process of assigning therapists, monitoring attendance, managing specialties, tracking schedules, and evaluating therapist performance using the proposed system.
- **Therapist Commission** – The compensation earned by therapists based on completed spa services, which will be automatically calculated and monitored by the proposed system.
- **Inventory Management** – The process of monitoring, recording, updating, and controlling the availability and movement of spa products and operational supplies to maintain sufficient stock levels.
- **Point-of-Sale (POS)** – A computerized transaction processing feature that records customer purchases, computes payments, generates receipts, and stores sales records within the proposed system.
- **Payment Processing** – The process of receiving, validating, recording, and managing customer payments using various supported payment methods available in the proposed system.
- **Refund Management** – The process of recording, monitoring, approving, and managing customer refund requests and related transaction adjustments within the proposed system.
- **Voucher Management** – The process of creating, validating, monitoring, and recording promotional vouchers and discounts to ensure proper usage and prevent duplicate redemption.
- **Discount Management** – The process of applying, validating, and recording eligible promotional discounts such as senior citizen, PWD, employee, and other authorized discounts during customer transactions.
- **Customer Feedback** – Customer comments, ratings, and suggestions collected through the proposed system to evaluate service quality, therapist performance, and customer satisfaction.
- **Sales Analytics** – The process of analyzing sales transactions and operational data to identify business performance, service demand, customer trends, and revenue patterns that support management decision-making.
- **Revenue Forecasting** – The process of estimating future sales or revenue using historical transaction data generated by the proposed system to support business planning.

- **Operational Reports** – System-generated reports that summarize business information such as appointments, sales transactions, inventory usage, therapist performance, commissions, customer activities, and other operational records for monitoring and decision-making.
- **Operational Efficiency** – The ability of Recovery Iloilo Spa to perform daily business activities accurately, quickly, and with minimal manual effort through the use of the proposed integrated system.
- **Centralized Database** – A single repository where all operational information, including appointments, customer records, inventory data, transactions, therapist information, and reports, is securely stored and managed.
- **Integrated System** – A computerized information system that combines multiple operational functions into a single centralized platform, allowing different business processes to work together efficiently.
- **Web-Based System** – A system that operates through a web browser over an internet connection without requiring installation of a dedicated mobile or desktop application.
- **Three-Tier Architecture** – A system design that separates the presentation, application, and data layers into independent components to improve maintainability, scalability, and security.
- **Server-Side Conflict Detection** – A validation mechanism that checks appointment time slots against existing bookings on the server before confirming a new reservation, preventing double-booking and scheduling conflicts.
- **Role-Based Access Control** – A security method that restricts system functions and data access according to a user's assigned role, such as customer, therapist, receptionist, or administrator.
- **Gate Code Verification** – A hidden-access security feature that requires a valid code before the administrative login page becomes accessible, preventing unauthorized discovery of the admin portal.
- **PIN Verification** – A secondary authentication step requiring a personal identification number from authorized personnel before sensitive or state-changing transactions are processed.
- **Activity Log** – A chronological record of user actions performed within the system, including the actor's identity, role, and the action taken, used for accountability and audit purposes.
- **Cash Reconciliation (Short/Over)** – The process of comparing actual counted cash on hand against the system-computed net cash for a given period to identify discrepancies, recorded as either an overage or a shortage.
- **User Acceptance Testing (UAT)** – A testing phase in which actual users, such as the spa owner, staff, and customers, validate whether the developed system meets their operational requirements before deployment.

- **Agile Software Development Methodology** – An iterative software development approach that divides a project into short cycles called sprints, allowing continuous collaboration, incremental delivery, and regular evaluation with the client.

CHAPTER 2

Review of Related Literature and Studies

2.1 Foreign Studies

Tenkale, Anokar, Rathod, and Kamble (2026) explored the digital transformation of small enterprises in rural India through modern, accessible web-based applications to address widespread operational inefficiencies stemming from manual processes and low digital literacy. The authors emphasized that reliance on traditional record-keeping, manual communication, and fragmented workflow management severely restricts operational efficiency, daily task management, and organizational coordination in small-scale rural businesses. To mitigate these pervasive operational barriers and improve workforce coordination, the study introduced a web-based Task Management System (TMS) designed as a centralized, scalable platform to streamline workflow organization, monitor progress, and manage daily operations. The findings demonstrated that implementing accessible web applications can effectively bridge the regional digital divide while noticeably boosting enterprise productivity and growth. Furthermore, the study highlighted that technological deployment must be paired with human skill development and digital awareness programs to ensure the effective adoption and long-term sustainability of such digital solutions. This research directly underscores the value of migrating manual operational workflows to centralized web platforms—a concept central to the design of the *Recovery Iloilo Spa Integrated System*. By demonstrating how web-based software resolves manual tracking inefficiencies and enhances operational control, the study provides strong practical justification for deploying automated scheduling, inventory tracking, and analytical tools to modernize daily business operations. Ultimately, their findings reinforce the crucial role of digital transformation in reducing administrative burdens, minimizing human error, and fostering sustainable organizational efficiency in service-oriented enterprises.

Manish, Sengar, and Singh (2026) developed and evaluated "MediCare Clinic," a web-based appointment booking system designed to digitize the complete appointment lifecycle for small-scale healthcare facilities. The authors highlighted that reliance on manual administrative methods—such as paper registers, telephone bookings, and uncoordinated walk-ins—routinely results in scheduling conflicts, double bookings, lack of real-time schedule visibility, and decreased customer satisfaction. To eliminate these operational bottlenecks, the study implemented a role-based single-page application (SPA) featuring three distinct user roles: Admin, Doctor (service provider), and Patient (client). The system incorporates a real-time slot-blocking

mechanism that dynamically disables already-booked time slots upon selection, successfully preventing double bookings and scheduling overlaps. System testing across multiple operational scenarios demonstrated 100% functional correctness in role-based access control, real-time appointment booking, and status management workflows. Manish, Sengar, and Singh (2026) concluded that deploying a lightweight, role-based web application effectively eliminates double-booking errors, streamlines daily scheduling, and drastically reduces administrative workload for small service enterprises.

This study provides compelling theoretical, structural, and operational support for the *Recovery Iloilo Spa Integrated System*, particularly regarding its web-based architecture, role-based user authentication, real-time online reservation management, dynamic facility/room allocation, integrated Point-of-Sale (POS) payment processing, and administrative dashboard modules. By demonstrating that centralizing booking and transaction workflows within a web platform minimizes manual errors, saves operational time, and improves customer convenience, the findings of Patra and Mohapatra (2026) directly validate the spa system's primary objective: replacing manual, paper-based front-desk logbooks with a unified web application that automates client reservations, coordinates therapist timetables, and streamlines daily spa management.

Kosalairaman et al. (2025) designed and implemented a web-based smart consulting and appointment booking system featuring real-time scheduling, secure data management, and dynamic client-provider communication capabilities. The authors highlighted that healthcare and service institutions frequently face operational inefficiencies, prolonged waiting times, and communication gaps due to manual scheduling and fragmented administrative workflows. To resolve these challenges, the study developed a centralized platform that automates appointment booking based on real-time provider availability, incorporates secure payment processing, and utilizes a structured database for encrypted record storage. System evaluation demonstrated that real-time scheduling automation significantly reduces client waiting times, minimizes manual administrative friction, ensures data integrity, and enhances operational workflow efficiency across web and mobile interfaces. Kosalairaman et al. (2025) concluded that deploying cloud-hosted, real-time scheduling systems provides organizations with a scalable and secure framework that optimizes resource utilization and improves customer satisfaction.

This study provides valuable theoretical, structural, and operational support for the *Recovery Iloilo Spa Integrated System*, particularly regarding its real-time online reservation management, dynamic room and slot allocation, automated duration tracking, secure Point-of-Sale (POS) transaction processing, and administrative dashboard modules. By demonstrating that web-based real-time slot allocation eliminates manual record-keeping errors, optimizes space and resource utilization, and

reduces client wait times, the findings of Bhargavi and Mahammadunnisa (2026) directly validate the spa system's primary objective: replacing manual, paper-based front-desk reservation ledgers with an integrated web application that automates client bookings, synchronizes therapist and room availability, and elevates daily operational management.

Gupta et al. (2024) designed and implemented a digital scheduling system aimed at optimizing appointment management, improving resource utilization, and streamlining client-provider interactions within service environments. The authors noted that reliance on traditional, non-automated scheduling methods often results in heavy administrative workloads, high rates of missed or delayed appointments, and inefficient allocation of staff and facilities. To resolve these operational issues, the study developed a centralized platform incorporating dynamic real-time scheduling, search filtering by service type and provider availability, and automated notification mechanisms (via email and SMS) to alert clients to upcoming bookings. System evaluation demonstrated that digitizing the appointment lifecycle significantly reduces manual administrative burdens, minimizes missed appointments, improves client engagement, and optimizes resource distribution. Gupta et al. (2024) concluded that implementing automated, web-based appointment systems establishes near-optimal scheduling workflows that elevate overall operational efficiency and managerial control.

Barlis and Resuello (2025) developed and evaluated a performance dashboard and appointment scheduling system utilizing Mixed Integer Linear Programming (MILP) optimization to enhance resource allocation, streamline client scheduling, and improve operational decision-making within service environments. The authors noted that uncoordinated manual scheduling and lack of centralized performance tracking routinely lead to resource bottlenecks, inefficient staff utilization, extended client waiting times, and operational delays. To address these challenges, the study designed a unified platform combining real-time performance analytics dashboards with a mathematical optimization engine to balance service assignments, minimize operational friction, and optimize facility resource throughput. Empirical evaluation demonstrated that integrating mathematical optimization models with real-time operational dashboards eliminates scheduling gaps, maximizes provider resource utilization, and enhances overall service delivery. Barlis and Resuello (2025) concluded that combining automated scheduling algorithms with real-time managerial dashboards establishes a scalable framework that optimizes operational control and facility productivity.

Kokate, Samrit, and Bisen (2026) engineered and evaluated "LUXE SALON," a full-stack real-time appointment scheduling and business intelligence platform tailored for the salon and wellness industry to optimize service allocations and eliminate manual operational bottlenecks. The authors noted that traditional, paper-based reservation routines routinely result in double-booking conflicts, idle staff schedules, uncoordinated

service availability, and an inability to analyze key performance metrics. To resolve these operational issues, the study built a modern web platform featuring real-time time-slot conflict detection, dynamic therapist availability tracking, stateless user authentication, and an administrative business intelligence dashboard providing live KPIs on daily appointment volumes, revenue trends, staff utilization rates, and service popularity rankings. Performance evaluation demonstrated sub-100 ms API response latencies under concurrent load, zero booking conflicts across 500 simulated sessions, and a 73% reduction in manual scheduling effort compared to paper-based logbooks. Kokate et al. (2026) concluded that implementing automated, web-based reservation platforms integrated with analytics dashboards provides wellness businesses with a robust framework to streamline daily throughput, maximize staff utilization, and reduce administrative overhead.

This study provides direct, highly compelling theoretical, structural, and operational support for the *Recovery Iloilo Spa Integrated System*, particularly regarding its real-time online reservation management, dynamic therapist and treatment room slot allocation, conflict detection mechanisms, and administrative business intelligence dashboard modules. By demonstrating that real-time reservation automation achieves zero booking conflicts while reducing manual front-desk effort by 73%, the findings of Kokate et al. (2026) directly validate the spa system's primary objective: replacing manual, paper-based front-desk reservation ledgers with an integrated web application that automates client bookings, synchronizes therapist and room schedules in real time, and provides actionable managerial insights into daily spa operations.

Kannake et al. (2023) developed and evaluated an online salon booking application designed to modernize reservation workflows, automate service scheduling, and eliminate operational bottlenecks in the beauty and wellness sector. The authors observed that traditional salon management relies heavily on manual record-keeping and phone calls, resulting in long waiting times, scheduling errors, inefficient staff allocation, and customer frustration. To address these issues, the study engineered a web platform enabling clients to view available time slots, select preferred services and stylists, and complete online reservations in real time. Simultaneously, the system equips managers with tools to manage staff schedules, track customer records, and monitor overall business throughput. System evaluations demonstrated that automating salon appointments minimizes manual booking errors, reduces client wait times, optimizes staff utilization, and improves overall customer satisfaction. Kannake et al. (2023) concluded that deploying digital reservation systems provides wellness businesses with an efficient, scalable operational framework that enhances service delivery and customer convenience.

This study provides strong theoretical, structural, and operational support for the *Recovery Iloilo Spa Integrated System*, particularly regarding its real-time online

reservation management, dynamic therapist and room slot allocation, conflict detection mechanisms, and administrative schedule coordination modules. By demonstrating that web-based booking platforms eliminate scheduling collisions, streamline front-desk schedule tracking, and minimize customer waiting times, the findings of Saklecha et al. (2025) directly validate the spa system's primary objective: replacing manual, paper-based front-desk reservation ledgers with an integrated web application that automates client bookings, synchronizes therapist and room availability in real time, and elevates daily operational efficiency.

Purwatinings et al. (2024) engineered and evaluated a web-based e-booking information system for salon services designed to streamline customer reservations, optimize schedule management, and eliminate operational bottlenecks. The authors observed that traditional salon management relying on manual logbooks and walk-in queuing creates significant operational friction, such as customer queue accumulation, delayed service confirmations, scheduling input errors, and double-booking incidents. To resolve these issues, the study developed an integrated web application enabling clients to view service schedules and reserve appointments online, while equipping salon operators with tools to manage customer records, track order histories, and organize daily service slots. System evaluation confirmed that implementing the e-booking system eliminated scheduling conflicts, reduced customer queue wait times, and improved overall administrative data management and operational efficiency. Purwatinings et al. (2024) concluded that transitioning from manual scheduling to automated web-based reservation architectures establishes an organized, transparent operational framework that enhances service delivery and customer satisfaction.

Bora et al. (2023) developed and evaluated "SalonPro," a comprehensive website-based data management and appointment booking platform designed to streamline operational workflows and eliminate manual booking inefficiencies within beauty and wellness establishments. The authors highlighted that traditional, non-automated salon management creates administrative overhead, causes scheduling friction, and fails to capture essential customer performance data. To resolve these operational challenges, the study engineered an intuitive web interface allowing clients to view real-time time-slot availability, choose preferred service providers or therapists, and make instant online reservations. The platform also integrates backend administrative functionality for digital record-keeping, client transaction management, and operational data analytics. System evaluation confirmed that implementing web-based booking and data management minimizes double-booking errors, reduces customer waiting times, optimizes staff schedule utilization, and equips management with actionable business insights. Bora et al. (2023) concluded that transitioning to automated digital reservation architectures establishes an efficient, scalable operational model that elevates overall service quality and customer satisfaction.

Ikbal et al. (2024) developed and evaluated "PES" (Program E-Booking Salon), a web-based salon reservation and management system built using the CodeIgniter framework to modernize booking workflows and streamline administrative operations. The authors observed that traditional salon operations relying on manual record-keeping, walk-in scheduling, and phone inquiries suffer from frequent scheduling errors, double-booking incidents, customer queue congestion, and inefficient staff management. To overcome these operational barriers, the study engineered a web platform enabling clients to view service menus, check real-time schedule availability, and reserve appointments online, while equipping salon administrators with tools to manage customer records, staff schedules, and daily transaction logs. System evaluation confirmed that implementing web-based e-booking eliminates scheduling conflicts, reduces customer wait times, minimizes manual administrative tasks, and improves operational data organization. Ikbal et al. (2024) concluded that adopting web-based reservation frameworks establishes a structured and reliable operational system that enhances business efficiency and elevates customer service quality.

Indrayana et al. (2025) designed and evaluated a web-based salon booking system tailored for Salon Ratih Langit to modernize appointment management, streamline administrative workflows, and eliminate operational inefficiencies. The authors highlighted that conventional salon operations relying on manual logbooks and phone reservations frequently suffer from scheduling collisions, lost client booking records, extended waiting times, and poor staff utilization tracking. To address these operational challenges, the study designed and deployed an online booking interface enabling clients to select specific services, choose preferred time slots, and confirm appointments online in real time, while providing management with backend tools for administrative tracking, schedule organization, and client record management. System evaluations confirmed that transitioning to a web-based reservation framework eliminated scheduling conflicts, reduced client wait times, improved administrative data accuracy, and streamlined daily operational coordination. Indrayana et al. (2025) concluded that deploying structured web booking platforms equips service providers with an organized, efficient operational tool that enhances customer satisfaction and operational throughput.

Atika et al. (2025) designed and developed a mobile-based reservation system aimed at modernizing service booking workflows, improving resource scheduling, and eliminating operational friction in local service environments. The authors noted that conventional, non-digitized reservation methods relying on phone calls or manual paper ledgers frequently lead to uncoordinated scheduling, double-booking errors, delayed confirmations, and limited real-time visibility into service availability. To resolve these operational issues, the study built a mobile reservation application incorporating real-time schedule tracking, automated slot allocation, user profile management, and

interactive booking confirmations. System evaluation confirmed that implementing digital reservation technology eliminates scheduling collisions, reduces administrative processing time, and enhances user convenience and service accessibility. Atika et al. (2025) concluded that deploying mobile and web-based reservation platforms equips service providers with an organized, efficient operational tool that elevates customer satisfaction and daily management efficiency.

Maryani et al. (2019) developed and evaluated a prototype online reservation system tailored for the barbershop and salon industry to streamline booking processes, optimize staff allocation, and eliminate manual front-desk friction. The authors observed that traditional salon operations relying on manual logbooks, walk-in queuing, or phone calls frequently suffer from uncoordinated scheduling, double-booking errors, long customer waiting times, and underutilized service capacity. To address these operational challenges, the study engineered a web-based reservation prototype featuring real-time schedule availability displays, stylist and service selection modules, automated booking confirmations, and administrative schedule tracking. System evaluation demonstrated that transitioning to a digitized reservation system eliminates scheduling collisions, reduces customer wait times, improves staff productivity, and provides management with enhanced visibility over daily salon operations. Maryani et al. (2019) concluded that implementing web-based reservation architectures provides personal care and wellness businesses with a scalable, efficient operational framework that improves service quality and customer retention.

Santosa et al. (2026) designed and evaluated a web-based salon information system integrated with a WhatsApp booking notification feature for Fatma Salon to modernize appointment handling, streamline customer communications, and optimize administrative operations. The authors observed that traditional salon management methods relying on manual ledger entries, physical walk-ins, or uncoordinated messaging frequently lead to missed appointments, scheduling conflicts, delayed booking confirmations, and inefficient customer service tracking. To address these operational issues, the study engineered a web platform enabling customers to select services, choose preferred schedules, and initiate direct WhatsApp booking confirmations, while equipping salon administrators with backend dashboards to track schedules, customer records, and transaction histories. System evaluations confirmed that combining web-based online booking with automated messaging channels significantly eliminated scheduling overlaps, accelerated booking response times, reduced manual front-desk workloads, and enhanced customer convenience. Santosa et al. (2026) concluded that deploying web-based reservation platforms with integrated instant messaging notifications establishes a practical, user-friendly operational framework for wellness and beauty service providers.

Saklecha et al. (2025) developed and evaluated "COIFUZZ," a web-based salon appointment system designed to modernize customer booking workflows, optimize stylist scheduling, and streamline daily operational management. The authors noted that traditional, non-automated salon reservation practices—such as manual logbooks, phone inquiries, and uncoordinated walk-in queuing—routinely cause double-booking errors, prolonged customer wait times, idle staff capacity, and administrative disorganization. To resolve these operational challenges, the study engineered a centralized web platform featuring real-time time-slot availability, automated conflict detection, customizable service menus, and an intuitive administrative backend for client data management and schedule monitoring. System evaluation confirmed that deploying the COIFUZZ platform eliminated scheduling collisions, reduced front-desk administrative workloads, improved client booking convenience, and maximized overall service resource utilization. Saklecha et al. (2025) concluded that transitioning to automated, web-based appointment architectures establishes a scalable and structured operational framework for personal care and wellness service providers.

Leony and Prayitno (2025) designed and implemented a web-based online reservation system for Jimmy Salon to improve service booking convenience and eliminate manual operational friction. Applying the Waterfall development model and User-Centered Design (UCD) principles using PHP, HTML, and CSS, the system introduced an accessible online booking structure that allows clients to reserve appointments seamlessly without requiring complex account registration. Black Box testing validated full functional performance according to operational specifications, while empirical findings demonstrated a significant increase in reservation efficiency, reducing the average booking transaction time from 15 minutes down to approximately 3 minutes. Leony and Prayitno (2025) concluded that transitioning micro, small, and medium service enterprises (MSMEs) from traditional manual scheduling to streamlined web-based reservation systems significantly reduces operational processing time, enhances client usability, and optimizes daily service throughput.

Gasmen et al. (2026) engineered and evaluated "BeautyEase," a web-based platform designed for salon and wellness operations to streamline service scheduling, queue management, and inventory tracking within a unified digital architecture. The authors highlighted that traditional salon management methods—often relying on manual appointment scheduling or informal messaging channels—suffer from scheduling friction, unorganized customer queues, and poor visibility over resource usage. Built using PHP, MySQL, HTML, CSS, Bootstrap, and JavaScript via the Prototyping Model, the platform integrates real-time online appointment bookings, dynamic queuing management, automated staff schedule assignments, system notifications, and an administrative control dashboard. Evaluation across salon employees, customers, and IT specialists demonstrated high levels of usability,

functional suitability, and operational reliability. Gasmen et al. (2026) concluded that consolidating service scheduling, real-time queuing, and inventory tracking into a centralized web platform significantly reduces front-desk administrative friction, improves service delivery efficiency, and enhances overall client accessibility.

Lukić et al. (2021) engineered and evaluated an automatic appointment scheduling algorithm and web application designed for multi-provider and multi-service operational environments. The authors highlighted that manual and static scheduling in environments involving multiple service providers and varying service durations inevitably leads to administrative friction, scheduling collisions, extended client waiting times, and underutilized staff schedules. To overcome these challenges, the study developed an automated booking algorithm that dynamically evaluates existing database reservations, verifies time-slot availability, matches service duration requirements, and prevents appointment overlap across multiple service providers. System evaluation confirmed that integrating automated scheduling algorithms with centralized web applications eliminates double-booking errors, significantly reduces administrative workload, minimizes client wait times, and maximizes provider operational efficiency. Lukić et al. (2021) concluded that automated, web-based reservation architectures offer service organizations a highly efficient, scalable framework for managing complex multi-resource service operations.

Tato et al. (2025) designed and developed a web-based data management and sales information system for Lia Salon to streamline administrative record-keeping, optimize transaction handling, and modernize daily operational workflows. The authors noted that conventional salon operations relying on manual ledger recording and unorganized sales logging frequently suffer from human calculation errors, lost client service records, delayed financial reporting, and uncoordinated customer data management. To resolve these operational challenges, the study engineered a web platform featuring centralized customer data management, service catalog tracking, automated sales transaction processing, and dynamic revenue report generation. System evaluation confirmed that digitizing salon data management and sales logging significantly eliminated calculation inaccuracies, reduced front-desk administrative processing time, improved data storage reliability, and provided business owners with immediate visibility into financial performance. Tato et al. (2025) concluded that transitioning personal care and beauty enterprises from manual bookkeeping to structured web-based management systems establishes a reliable and transparent operational architecture that optimizes administrative control and business decision-making.

.Permatasari et al. (2024) developed and evaluated an Android-based mobile salon and spa reservation system aimed at improving customer experience, streamlining appointment handling, and elevating overall service satisfaction. The

authors highlighted that conventional, non-digitized salon and spa booking methods—such as walk-in visits, phone inquiries, and manual front-desk logbooks—routinely cause customer queue accumulation, uncoordinated staff scheduling, double-booking errors, and limited schedule transparency. To resolve these operational barriers, the study engineered a mobile application enabling clients to browse service menus, check real-time therapist and time-slot availability, receive instant booking confirmations, and manage their appointment histories. System evaluation confirmed that implementing mobile reservation technology significantly eliminated scheduling collisions, reduced customer waiting times, simplified front-desk operational tracking, and achieved high levels of customer satisfaction and usability. Permatasari et al. (2024) concluded that adopting automated mobile and web reservation platforms provides personal care and wellness enterprises with an organized, client-centric operational framework that enhances service delivery and operational efficiency.

Muallafa et al. (2024) developed and evaluated a web-based ordering and service management system for Dream Skin Aesthetic using the Waterfall software development methodology to address operational bottlenecks in beauty treatment scheduling. The authors observed that traditional offline service models relying on physical walk-ins and uncoordinated phone inquiries lead to severe customer queue accumulation, extended wait times, and administrative disorganization. To overcome these challenges, the study engineered a web platform enabling clients to browse treatment options, check service schedules, and place online advance orders, while providing salon management with backend administrative tools to track customer records and order histories effectively. System evaluations confirmed that transitioning to the web-based ordering system successfully eliminated customer queue congestion, improved customer satisfaction, and enhanced daily administrative data organization. Muallafa et al. (2024) concluded that adopting automated online ordering platforms equips aesthetic and wellness service providers with a structured, efficient operational model that optimizes customer experience and front-desk workflows.

Sari (2023) designed and developed a web-based wedding organizer and service reservation system for Sindy Salon in Labuhanbatu to address operational limitations in service booking and administrative record-keeping. The author observed that the establishment previously relied on manual order book logging to manage package bookings and service requests, which proved inefficient, hindered transaction verification, and made administrative report generation difficult. To resolve these operational challenges, the study engineered a web platform using PHP and MySQL that enables customers to browse service packages, place advance online reservations, and manage booking information digitally. System evaluations showed that transitioning from manual logbooks to a web-based booking architecture simplified customer

ordering procedures, expanded service accessibility, streamlined order data processing, and facilitated accurate managerial transaction tracking. Sari (2023) concluded that implementing web-based reservation and service management systems provides local salons and service establishments with a practical digital framework that elevates customer convenience and operational efficiency.

Samudra and Anggara (2024) engineered and evaluated an Android-based salon reservation application for IKA Salon using the Research and Development (R&D) methodology to optimize customer service delivery and streamline internal scheduling workflows. Built using the Flutter framework with Firebase for real-time data storage, the mobile application features secure user authentication, interactive service browsing, real-time schedule reservation, client profile tracking, and an administrative management backend. The authors noted that conventional, non-digitized salon operations relying on paper logbooks or verbal bookings suffer from queue congestion, scheduling overlap, manual recording errors, and limited schedule visibility for clients. System testing confirmed that integrating mobile appointment scheduling with real-time database management successfully eliminated scheduling errors, reduced customer waiting times, simplified administrative tracking, and enhanced overall customer satisfaction. Samudra and Anggara (2024) concluded that adopting real-time mobile booking platforms equips small and medium personal care enterprises with a reliable, scalable operational tool that optimizes daily workflows and supports business growth.

Yadav et al. (2025) developed and evaluated a web-based online registration and appointment scheduling platform aimed at streamlining patient booking workflows, optimizing resource allocation, and eliminating physical queue congestion in service-oriented facilities. The authors observed that traditional manual appointment logging and walk-in scheduling practices frequently result in severe queue accumulation, prolonged wait times, administrative errors, and uncoordinated provider schedules. To overcome these operational challenges, the study engineered a web platform enabling users to view real-time time-slot availability, reserve appointments online, receive booking confirmations, and manage personal records, while providing administrators with backend tools for schedule tracking and data management. System evaluations demonstrated that transitioning to digitized appointment scheduling eliminated booking overlaps, reduced administrative processing times, improved operational transparency, and elevated overall user convenience. Yadav et al. (2025) concluded that deploying web-based reservation architectures provides service organizations with a structured, efficient operational model that optimizes customer experience and daily workflow management.

Nisa, Maharani, and Santoso (2026) developed and evaluated a web-based Customer Relationship Management (CRM) system designed to optimize customer service workflows, streamline order and appointment processing, and elevate overall

customer satisfaction. The authors highlighted that small-to-medium service enterprises relying on manual, paper-based record-keeping face operational bottlenecks such as lost client booking records, manual scheduling errors, and slow response times to customer inquiries. To resolve these operational inefficiencies, the study designed a web-based CRM application incorporating centralized customer databases, automated reservation tracking, and interactive feedback mechanisms. System testing and usability evaluations confirmed that the web-based CRM application significantly reduced record retrieval times, eliminated data redundancy, accelerated booking processing, and improved overall customer satisfaction scores. Nisa, Maharani, and Santoso (2026) concluded that adopting web-based CRM platforms enables service businesses to transition from inefficient manual operations to streamlined, automated digital environments.

Voloboy, Mokra, Belozertsev, and Yelisyeyeva (2022) examined and evaluated the role of information technologies within Customer Relationship Management (CRM) systems to enhance administrative efficiency and customer communication across service enterprises. The authors highlighted that organizations relying on decentralized or manual record-keeping struggle with fragmented client profiles, inconsistent interaction tracking, and delayed response times, which ultimately degrade customer retention and operational clarity. To mitigate these challenges, the study analyzed web-based CRM architectures that integrate centralized data repositories, automated client communication tools, and real-time transaction logging. System evaluations demonstrated that leveraging IT-driven CRM frameworks drastically reduced administrative record retrieval times, eliminated data duplication, and provided management with structured analytics to monitor client engagement patterns. Voloboy, Mokra, Belozertsev, and Yelisyeyeva (2022) concluded that embedding modern information technologies into CRM frameworks equips service enterprises with a vital operational infrastructure for automating client-facing workflows and streamlining administrative processes.

Ayuninggati, Lutfiani, and Millah (2021) engineered and evaluated a CRM-based e-business architecture tailored for a specialized service company to automate reservation handling, service tracking, and customer communication workflows. The authors noted that small-to-medium service enterprises using manual logbooks experience severe operational bottlenecks, including uncoordinated appointment slots, manual scheduling errors, lost booking details, and poor client engagement. To resolve these operational issues, the study designed a web-based CRM platform featuring an online client reservation portal, real-time service status tracking, and automated client notifications. System testing and usability evaluations confirmed that the CRM-driven web application significantly accelerated reservation processing, eliminated booking errors, and elevated customer satisfaction by providing

transparent, real-time access to service schedules. Ayuninggati, Lutfiani, and Millah (2021) concluded that adopting web-based CRM systems enables service business operators to transition seamlessly from inefficient manual processes to fully synchronized, digital customer operations.

Kaldate (2026) engineered and evaluated a web-based management and Customer Relationship Management (CRM) platform designed to automate administrative record-keeping, streamline multi-user requests, and improve operational transparency across service-driven institutional workflows. The author noted that relying on paper-based logbooks or decentralized software applications leads to operational friction, lost data records, delayed response times, and redundant administrative tasks. To address these operational issues, the study constructed a centralized web-based CRM system incorporating role-based access control, real-time schedule tracking, automated profile logging, and structured administrative dashboards. System evaluation and deployment testing confirmed that the centralized web application eliminated record duplication, reduced administrative processing times, accelerated response rates, and provided management with clear operational oversight. Kaldate (2026) concluded that developing web-based CRM architectures provides service-oriented organizations with a scalable, highly reliable digital infrastructure to streamline daily operations and maintain accurate transactional histories.

This study provides direct theoretical, structural, and operational support for the Recovery Iloilo Spa Integrated System, particularly regarding its user-friendly web portal design, administrative CRM dashboard, real-time online reservation management, and automated client profile directory modules. By demonstrating that streamlined CRM applications eliminate manual record-keeping errors and accelerate client workflows in small enterprise environments, the findings of Hasan, Andrew, Agustin, Seren, and Enjelia (2023) directly validate the spa system's primary objective: replacing manual, paper-based front-desk reservation ledgers with a centralized web application that automates client bookings, synchronizes operational data in real time, and elevates administrative management.

Onyenezide (2025) investigated and evaluated the operational impact of Customer Relationship Management (CRM) system adoption on customer retention, service consistency, and business loyalty within small-scale enterprise environments. The author emphasized that small service-driven businesses operating through manual, uncoordinated client record systems suffer from inconsistent service delivery, forgotten client preferences, and frequent scheduling delays, which negatively impact customer loyalty and brand reputation. To analyze the relationship between digital adoption and client retention, the study surveyed small business operators using a structured quantitative evaluation framework. Empirical findings revealed that adopting digital CRM systems significantly improved customer satisfaction and retention rates by enabling

businesses to maintain accurate client histories, provide timely service updates, and deliver personalized customer interactions. Onyenezide (2025) concluded that adopting CRM systems serves as a vital operational driver for small service businesses seeking to eliminate administrative disorganization and build long-term customer loyalty.

This study provides valuable theoretical, structural, and operational support for the Recovery Iloilo Spa Integrated System, particularly regarding its web-based client portal, CRM administrative dashboard, centralized treatment history logging, and automated reservation management modules. By proving that integrated web-based CRM architectures streamline administrative data management and optimize client interactions, the findings of Ibarra, Gomis, Pérez-Caparrós, and Benito (2007) directly validate the spa system's primary objective: replacing manual front-desk record books with an integrated web platform that maintains accurate client profiles, automates booking workflows, and enhances service quality.

Patle and Chabukswar (2025) conducted an operational evaluation of Customer Relationship Management (CRM) implementations within service-based enterprises, focusing on service delivery optimization, administrative workflow integration, and customer retention. The authors emphasized that service-driven businesses rely heavily on seamless interaction management, where uncoordinated scheduling and fragmented client data directly lead to service delays, booking overlaps, and customer dissatisfaction. To address these operational issues, the study examined integrated CRM frameworks that combine dynamic schedule tracking, centralized client profile management, and automated service reporting. Field evaluation and operational analysis revealed that centralized CRM platforms significantly reduced administrative processing overhead, prevented booking collisions, and fostered repeat client visits through organized historical record-keeping. Patle and Chabukswar (2025) concluded that deploying integrated CRM architectures provides service-based enterprises with a robust operational foundation for streamlining daily scheduling, optimizing staff allocation, and maintaining high service standards.

This study provides strong theoretical, structural, and operational support for the Recovery Iloilo Spa Integrated System, particularly regarding its real-time online reservation management, therapist and treatment room slot allocation, client history logging, and administrative reporting modules. By demonstrating that integrated CRM implementations optimize service scheduling and reduce administrative overhead, the findings of Patle and Chabukswar (2025) directly validate the spa system's primary objective: replacing manual, paper-based front-desk reservation ledgers with an integrated web application that automates client bookings, synchronizes therapist and room availability in real time, and elevates daily operational management.

Setyawan, Hin, Anubhakti, and Gata (2024) developed and evaluated an Electronic Customer Relationship Management (E-CRM) system designed to optimize

customer service quality and increase overall customer satisfaction for a service and retail enterprise. The authors highlighted that relying on manual, uncoordinated client record systems leads to operational bottlenecks, including unrecorded customer requests, delayed order and reservation processing, and a lack of personalized service history tracking. To resolve these operational challenges, the study engineered a web-based E-CRM platform incorporating centralized client profile repositories, automated reservation tracking, interactive feedback channels, and real-time service updates. Usability testing and customer evaluations confirmed that the E-CRM system significantly accelerated service delivery times, eliminated manual record-keeping errors, and improved overall client satisfaction ratings. Setyawan, Hin, Anubhakti, and Gata (2024) concluded that implementing web-based E-CRM platforms equips service-driven enterprises with an essential digital infrastructure to automate customer interactions and elevate service responsiveness.

Karataş and Aytaç (2026) engineered, deployed, and evaluated a cloud-based order tracking and management system specifically tailored for microenterprises operating in high-demand service industries. The authors observed that traditional microenterprises relying on manual logbooks or fragmented communication channels experience significant operational bottlenecks, including lost booking details, uncoordinated scheduling slots, poor real-time tracking, and manual error propagation. To address these operational challenges, the study developed an integrated, web-hosted cloud platform offering real-time order scheduling, centralized customer record repositories, dynamic status updates, and automated operational logging. Field deployment and system evaluation confirmed that implementing a cloud-based tracking architecture significantly reduced administrative processing times, eliminated scheduling errors, accelerated order fulfillment, and elevated client satisfaction. Karataş and Aytaç (2026) concluded that cloud-hosted operational platforms provide microenterprises with a scalable, low-cost digital infrastructure to streamline service workflows and modernize daily business administration.

Marchenko, Balalaieva, Serhiienko, and Tarazanov (2023) developed and evaluated a web-based application designed to organize, automate, and optimize the operational workflows of service stations. The authors highlighted that service enterprises utilizing manual paper ledgers or decentralized scheduling systems face severe operational friction, such as uncoordinated service slots, inefficient allocation of specialized staff and facilities, data duplication, and delayed administrative record retrieval. To resolve these operational issues, the study designed a multi-functional web architecture integrating real-time online appointment scheduling, resource allocation algorithms, centralized client and service history logging, and administrative management dashboards. System evaluation and workflow testing demonstrated that the web-based application significantly reduced customer wait times, prevented

resource allocation conflicts, increased daily service throughput, and enhanced data accuracy across operational roles. Marchenko, Balalaieva, Serhiienko, and Tarazanov (2023) concluded that web-driven service management platforms serve as a vital operational foundation for service-oriented businesses seeking to optimize resource scheduling and automate customer-facing transactions.

Sinaga, Putra, and Sudarmin (2026) engineered and evaluated an Electronic Customer Relationship Management (E-CRM) platform designed to support daily sales transactions and streamline service operations for a specialized retail and service enterprise. The authors noted that traditional service businesses operating without automated client management frameworks experience severe operational inefficiencies, such as uncoordinated service schedules, delayed booking processing, fragmented interaction histories, and poor client retention. To resolve these operational challenges, the study developed a web-based E-CRM platform that centralizes customer profile data, automates reservation handling, and tracks service interactions in real time. Usability evaluations and workflow assessments demonstrated that implementing an E-CRM operational system significantly increased transaction speeds, enhanced service accuracy, improved client communication, and elevated overall customer satisfaction. Sinaga, Putra, and Sudarmin (2026) concluded that integrating E-CRM methodologies into digital service operations provides enterprises with a vital operational infrastructure to automate client-facing workflows and maintain strong client relationships.

Damanik and Siregar (2021) developed and evaluated a web-based Customer Relationship Management (CRM) system designed to optimize customer transaction workflows and elevate service management for a commercial enterprise. The authors highlighted that organizations relying on manual, decentralized record systems face operational bottlenecks, such as data redundancy, delayed transaction processing, difficult record retrieval, and inconsistent client tracking. To address these operational issues, the study engineered a web-based CRM application integrating centralized database architecture, automated transaction logging, client history tracking, and administrative control dashboards. System testing and operational evaluations confirmed that the web-based CRM application significantly reduced record retrieval times, eliminated data duplication, accelerated workflow processing, and improved administrative data accuracy. Damanik and Siregar (2021) concluded that adopting web-based CRM platforms enables service enterprises to transition seamlessly from inefficient manual operations to streamlined, automated digital environments.

Hasan, Andrew, Agustin, Seren, and Enjelia (2023) developed and evaluated a lightweight Customer Relationship Management (CRM) operational framework designed specifically for micro, small, and medium enterprises (MSMEs) to optimize daily sales workflows and customer communication. The authors noted that small

service and retail enterprises relying on manual paper logbooks face recurring operational bottlenecks, including lost customer details, manual data entry errors, difficult transaction retrieval, and poor client retention. To resolve these operational issues, the study conceptualized and implemented an accessible, web-based CRM solution incorporating a centralized customer database, real-time transaction logging, and streamlined administrative reporting tools. System evaluation and user feedback confirmed that the CRM application significantly reduced data processing times, eliminated manual record redundancy, accelerated customer service workflows, and improved client engagement. Hasan, Andrew, Agustin, Seren, and Enjelia (2023) concluded that implementing tailored web-based CRM solutions provides small service enterprises with an essential digital tool to modernize operations, maintain accurate client histories, and build long-term customer loyalty.

Palupi (2022) developed a web-based customer services management system to support the sales division by organizing customer data, monitoring leads and opportunities, and scheduling daily sales activities through a CRM platform. The study addressed problems such as unupdated customer databases, unmonitored sales opportunities, and untracked daily activities of the sales team, which are common in businesses that still rely on manual or fragmented records. Using the waterfall model, the research produced a web-based CRM application that allows employees to access and update customer information and sales tasks from any location connected to the network. The implementation showed that the system helped maintain accurate and continuously updated customer data, which is essential for building and sustaining good customer relationships. It also enabled the sales division to manage prospects and opportunities more systematically while keeping daily work schedules organized within a single application. According to the study, the CRM implementation contributed to a 20% increase in sales in 2018, indicating that better customer and activity management can positively affect business performance. These results align with our system because both focus on a web-based platform that centralizes customer-related information and supports structured monitoring of interactions and transactions. The study also supports the use of role-based access and remote accessibility, which are important features for systems that serve multiple users across different locations. Overall, Palupi's (2022) findings provide strong evidence that a web-based CRM can improve data accuracy, sales monitoring, and operational efficiency, which directly supports the objectives of our system.

Mayasari, Rusito, and Wahyuning (2026) designed a web-based CRM information system for customer management and transaction history at Sumber Rejeki Store to address challenges in managing customer data and recording transactions. The study highlighted that small retail businesses often struggle with manual or inconsistent recording of customer information and purchase histories, which can lead

to errors and slower service. Their proposed system integrates customer profiles, transaction records, and order monitoring into one web-based platform so that staff can quickly retrieve and update relevant data. By using structured system design and a web interface, the researchers ensured that the system is accessible, easy to use, and capable of maintaining complete transaction histories for each customer. The design emphasizes accurate data entry, organized storage of customer and transaction details, and improved efficiency in serving repeat customers. These features are aligned with our system because both aim to provide a centralized and reliable repository for customer and transaction information in a web-based environment. The study also supports the idea that a well-designed CRM can reduce recording mistakes, speed up service processes, and help businesses understand customer behavior through stored transaction histories. In addition, the system's focus on web accessibility makes it suitable for environments where multiple staff members need to access and update customer data from different devices. Therefore, the work of Mayasari, Rusito, and Wahyuning (2026) strengthens the rationale for our system by showing that a web-based CRM can effectively support customer management, transaction tracking, and service improvement in a retail or service-oriented setting.

Welda, Sukarma, and Thalib (2019) developed a web-based sales information system with Customer Relationship Management (CRM) features for Tria Salon to address difficulties in managing customer data, products, services, and transactions. The study identified problems such as the difficulty of determining customer loyalty, challenges in deciding which products to stock, and errors in recording transactions due to reliance on manual ledgers. In addition, the salon faced inefficiencies in delivering promotions because staff had to send messages one by one to each customer. To solve these issues, the researchers built a web-based system that helps administrators manage customer, product, service, and transaction data in a centralized manner. The system also supports salon owners in generating customer and transaction reports and in sending promotions more efficiently to all customers. Testing through blackbox evaluation showed that the system functions correctly and produces outputs that match the defined processes. These results align with our system because both aim to centralize customer and transaction information in a web-based platform to reduce manual errors and improve data accuracy. The study also highlights the importance of CRM features such as customer profiling, transaction tracking, and promotional management, which are relevant to systems that seek to strengthen customer relationships. Overall, Welda, Sukarma, and Thalib (2019) demonstrate that a web-based CRM-enabled sales system can enhance operational efficiency, reporting, and customer engagement in a service-oriented business.

Rahim, M, Jamhur, and Rani (2025) designed a web-based Customer Relationship Management (CRM) system for Senyaman Resto & Coffee to improve

promotional activities and customer service. The study noted that the restaurant faced ineffective marketing and large volumes of sales transaction data that were still recorded manually in books and notebooks, leading to inefficiencies and potential errors. To address this, the researchers developed a CRM system using PHP and MySQL, with system design supported by Unified Modeling Language (UML) modeling. The resulting system provides features for creating sales and customer reports, enabling more structured and reliable processing of transaction data. Performance testing indicated that the system improved the way sales data are handled compared to the previous manual approach. The study concluded that the CRM design contributed to greater efficiency in promotion and sales management processes at the restaurant. These findings align with our system because both focus on replacing manual or fragmented recording of customer and sales data with a centralized web-based CRM. The emphasis on generating customer and sales reports also supports our system's goal of providing clear, accessible, and organized information for decision-making. Therefore, Rahim, M, Jamhur, and Rani (2025) provide relevant evidence that a web-based CRM can enhance data management, promotional effectiveness, and service quality in a food and beverage setting.

Rosliani, Budiman, Nugraha, and Astuti (2024) designed and built a web-based Project Management Information System to manage transaction reports for BRILink agents. The study highlighted that effective transaction data management is crucial for BRILink operations, especially for agents who act as intermediaries between the bank and customers. Previously, many agents relied on manual processes that were prone to errors and made report generation time-consuming and inconsistent. The researchers developed a web-based system that supports the recording, processing, and presentation of transaction data in a more structured and easily accessible format. The system was designed using requirements analysis, UML-based modeling, web framework implementation, and functional testing to ensure reliability. Results showed that the system improved the efficiency of report management, reduced data input errors, and provided quick and accurate reports for agents. These outcomes align with our system because both aim to replace manual or fragmented transaction recording with a centralized web-based platform. The focus on structured transaction reports and data accuracy is directly relevant to CRM systems that need to track customer interactions and financial activities. Therefore, Rosliani, Budiman, Nugraha, and Astuti (2024) demonstrate that a web-based information system can enhance operational productivity, data quality, and reporting capabilities in a transaction-intensive environment.

Surve and Vishwakar (2026) developed a centralized web-based Customer Relationship Management (CRM) platform using Angular and SQL to support digital customer relationship management. The study addressed the limitations of scattered or

siloed customer data by proposing a single platform where customer information, sales activities, and interaction histories can be managed consistently. The system provides a centralized repository for customer profiles, contact details, and transaction records, enabling staff to access and update data from any device with internet connectivity. By using modern web technologies, the researchers ensured that the platform is scalable, responsive, and capable of supporting multiple users simultaneously. The design emphasizes data consistency, ease of access, and integration of core CRM functions such as customer tracking and sales monitoring. These features align with our system because both seek to consolidate customer and transaction data into one web-based environment to improve coordination and decision-making. The study also supports the use of centralized CRM architectures to reduce duplication, minimize errors, and provide a unified view of customer relationships. Overall, Surve and Vishwakar (2026) show that a centralized CRM platform can strengthen digital customer management by offering structured, accessible, and integrated customer information.

Ikhsan, Laila, Lestari, Mulia, and Allwine (2026) developed a multi-role web-based Customer Relationship Management (CRM) information system for PT Telekomunikasi Indonesia, Tbk Witel Lampung–Bengkulu using the Agile method. The study focused on integrating three main roles—Admin, Government Service, and Shared Service General Support—into a single CRM platform to streamline customer and service management. By adopting the Agile development approach, the researchers were able to iteratively refine system features based on user feedback and changing requirements. The resulting system enables different user roles to access and manage customer data, service requests, and related transactions according to their responsibilities. This multi-role design ensures that each user group can perform its tasks efficiently while maintaining data consistency across the organization. These characteristics align with our system because both emphasize role-based access and centralized management of customer and transaction information in a web-based environment. The use of Agile also highlights the value of flexibility and continuous improvement when developing CRM systems that must adapt to organizational needs. Therefore, Ikhsan, Laila, Lestari, Mulia, and Allwine (2026) provide strong support for implementing a multi-role, web-based CRM that can accommodate diverse user functions while maintaining a unified customer data repository.

Alfaridzi, Sharyanto, and Sudarsono (2024) designed and created a web-based Customer Relationship Management (CRM) information system for PT. Home Center Indonesia to address the absence of a computerized system for managing customer relationships. The study noted that the company relied on Microsoft Excel to store customer data, which was considered inefficient and posed risks of data damage or loss due to the lack of proper backups. In addition, the company could not contact customers quickly or manage interactions in a structured way, which limited its

ability to strengthen customer relationships. To solve these problems, the researchers developed a web-based CRM system using PHP and MySQL, with system modeling supported by Unified Modeling Language (UML). The resulting system enables the company to store, update, and manage customer information in a centralized database that can be accessed through a web interface. By implementing this system, the company can maintain more accurate and secure customer records while improving its ability to interact with customers in a timely manner. These outcomes align with our system because both focus on replacing manual or spreadsheet-based customer data management with a centralized web-based CRM. The emphasis on data backup, structured storage, and web accessibility supports our goal of ensuring reliable and secure customer information handling. Overall, Alfaridzi, Sharyanto, and Sudarsono (2024) demonstrate that a web-based CRM can enhance data management, reduce risks of data loss, and improve customer interaction capabilities in a retail environment.

Supandi, Yusuf, and Fauziah (2018) implemented a web-based Customer Relationship Management (CRM) system for the sales of tires at Toko Lingga Ban to improve conventional sales processes and promotional activities. The study highlighted that the store previously relied on walk-in customers and manual recording of transactions, which limited its reach and made it difficult to maintain consistent customer data. To address this, the researchers built a web-based sales information system with CRM features using PHP and MySQL, following the waterfall development method. The system allows administrators to manage the entire platform, while customers can place orders, confirm payments, and accumulate shopping points through the website. In addition, the system provides directional reports that help management monitor sales performance and make informed decisions. Testing through black box and white box methods confirmed that the system functions correctly for administrators, customers, and management roles. These results align with our system because both aim to centralize customer and transaction data in a web-based environment to support more efficient sales and service processes. The inclusion of customer points and online ordering also supports strategies for encouraging repeat purchases and strengthening customer loyalty. Therefore, Supandi, Yusuf, and Fauziah (2018) show that a web-based CRM-enabled sales system can expand market reach, improve data management, and enhance customer engagement in a retail setting.

Alfaraby, Alamsyah, and Fergina (2026) implemented a web-based Customer Relationship Management (CRM) system to enhance customer loyalty in a small and medium enterprise context. The study adopted an applied system implementation approach combined with transaction-based analytical evaluation to examine how CRM functionalities can be structured to support relationship-oriented practices. The researchers focused on designing a CRM system that, even with limited features, can effectively organize customer interactions and transaction histories to foster stronger

customer relationships. Their findings indicate that when CRM functionalities are clearly structured and aligned with analytical models, they can significantly improve loyalty-related practices such as repeat purchases and sustained engagement. The study emphasizes that the value of a CRM system lies not only in the number of features but in how well those features support consistent and meaningful customer interactions. These insights align with our system because both prioritize the structured management of customer data and transactions to build long-term relationships rather than simply recording sales. The emphasis on analytical evaluation also supports our goal of using customer and transaction data to inform service improvements and loyalty strategies. Therefore, Alfaraby, Alamsyah, and Fergina (2026) provide evidence that a well-structured web-based CRM can enhance customer loyalty even in resource-constrained environments.

Siek and Lim (2025) implemented a Customer Relationship Management (CRM) analytical dashboard with customer segmentation to support data-driven decision-making in customer management. The study recognized that in the era of digital transformation, customer data is a critical asset for shaping corporate strategies, particularly in enhancing CRM systems. To leverage this data, the researchers proposed a solution that integrates machine learning algorithms with a scalable, cloud-based analytics platform to develop a predictive CRM analytics dashboard. The dashboard enables businesses to segment customers based on behavior and transaction patterns, allowing for more targeted marketing and personalized service strategies. By visualizing key customer metrics and segmentation results, the system helps management identify high-value customers, detect churn risks, and allocate resources more effectively. These capabilities align with our system because both aim to transform raw customer and transaction data into actionable insights that support strategic decisions. The focus on segmentation and analytics complements our goal of not only storing customer information but also using it to improve service and engagement. Therefore, Siek and Lim (2025) demonstrate that integrating analytical dashboards and segmentation into a CRM platform can enhance customer understanding, marketing precision, and overall business performance.

Boosa (2025) examined data governance and trust in modern CRM platforms, focusing on strategies for scalable customer data management. The study highlighted that as CRM systems grow in complexity and data volume, issues such as data quality, governance structures, and privacy concerns become increasingly critical for maintaining trust and compliance. To address these challenges, the researcher proposed frameworks that combine data quality controls, governance policies, and privacy-by-design principles into a unified approach for managing customer data in CRM ecosystems. These strategies include clear data ownership, standardized data definitions, access controls, and mechanisms for monitoring data usage and integrity.

The study emphasizes that strong data governance is essential for ensuring that customer data remains accurate, secure, and trustworthy over time, especially in cloud-based and multi-user environments. These insights align with our system because both deal with the need to manage growing amounts of customer and transaction data in a reliable and compliant manner. The focus on governance and trust supports our objective of designing a CRM that not only functions well but also maintains high standards of data security and ethical use. Therefore, Boosa (2025) provides important guidance for incorporating data governance and trust mechanisms into our system to ensure long-term scalability and user confidence.

Rezkika, Handayani, and Larasati (2026) implemented a web-based Customer Relationship Management (CRM) system with membership and loyalty point features for CV. Honda Karya Utama Kisaran to strengthen long-term customer relationships in the automotive industry. The study noted that the company previously managed customer and transaction data without an integrated membership or loyalty point system, making it difficult to identify repeat customers and evaluate customer loyalty. To address this, the researchers developed a CRM system using the Waterfall model, including requirement analysis, UML-based design, and implementation with PHP and MySQL. The resulting system enables the company to manage membership profiles, track loyalty points, and monitor customer transactions in a centralized web-based platform. Black-box testing confirmed that the system functions correctly and supports the intended membership and loyalty processes. By implementing this system, the company can more easily identify active and repeat customers, design data-driven promotional strategies, and improve operational efficiency in managing customer relationships. These outcomes align with our system because both focus on integrating membership or loyalty features with core CRM functions to encourage repeat business and strengthen customer engagement. The emphasis on structured transaction tracking and loyalty management supports our goal of using customer data to drive retention and service improvements. Therefore, Rezkika, Handayani, and Larasati (2026) demonstrate that a web-based CRM with membership and loyalty point features can effectively enhance customer relationship management in a competitive industry.

Ghani, Mishra, and Yadav (2026) proposed a blockchain-based secure customer data management approach to enhance Salesforce and strengthen trust in customer relationship processes. The study recognized that traditional CRM systems often rely on centralized databases, which can be vulnerable to data breaches, unauthorized access, and lack of transparent audit trails. To address these issues, the researchers integrated blockchain technology to decentralize customer data storage, ensure data integrity, and provide cryptographically verifiable records of customer interactions. Their model emphasizes decentralized identity management and immutable audit trails, which help reduce fraud and support compliance with regulatory

requirements. The use of smart contracts further enables automated trust workflows, such as loyalty rewards and service-level agreements, improving operational efficiency. These features align with our system because both aim to ensure that customer data is managed securely, accurately, and transparently over time. Although our system may not use blockchain directly, the principles of secure data handling, auditability, and controlled access are relevant to any CRM that stores sensitive customer information. The study also highlights the importance of privacy-preserving techniques and governance mechanisms, which support long-term trust in digital customer platforms. Therefore, Ghani, Mishra, and Yadav (2026) provide a forward-looking perspective on how advanced security and governance strategies can be incorporated into CRM architectures to enhance data protection and customer confidence.

Putri and Harahap (2026) developed a customer service system based on Customer Relationship Management (CRM) with a User Experience (UX) approach for Toko Na Roti to improve service quality and customer satisfaction. The study noted that small bakeries often face challenges in managing customer interactions, orders, and feedback in a consistent and user-friendly manner. To address this, the researchers designed a web-based CRM system that prioritizes ease of use, clear navigation, and intuitive interfaces based on UX principles. The system enables customers to place orders, provide feedback, and interact with the business through a more structured and accessible digital channel. By focusing on UX, the study ensured that the CRM is not only functional but also comfortable and efficient for both staff and customers to use. Testing and evaluation indicated that the UX-oriented design improved the overall customer service experience and made it easier for the bakery to manage customer relationships. These outcomes align with our system because both emphasize the importance of a user-centered design in ensuring that customer-facing features are easy to understand and use. The integration of CRM with UX also supports our goal of creating a platform that encourages customer engagement and repeat interactions. Therefore, Putri and Harahap (2026) demonstrate that combining CRM functionality with strong UX design can enhance service quality, customer satisfaction, and operational efficiency in a small business context.

Yulvianda, Ismail, and Sutoyo (2025) developed a web-based sales system for W'D Cakes & Cookies with integrated Customer Relationship Management (CRM) features to improve operational efficiency and customer loyalty. The study highlighted that the business previously relied on manual ordering via WhatsApp, which led to recording errors, service delays, and inefficient customer data management. To solve these problems, the researchers applied the Research and Development (R&D) method using the waterfall model, including requirement analysis, Figma-based design, Laravel implementation, and black box testing. The resulting system supports online ordering, product and customer management, digital payments, and transaction tracking in a

centralized web platform. Embedded CRM features enable loyalty analysis, personalized promotions, and targeted customer communications based on stored transaction and customer data. Implementation of the system contributed to faster transactions, improved data accuracy, and higher service quality, supporting the digital transformation of the MSME. These results align with our system because both focus on integrating CRM into a web-based sales platform to strengthen customer relationships and encourage repeat purchases. The emphasis on loyalty analysis and personalized promotions directly supports our objective of using customer data to drive retention and engagement. Therefore, Yulvianda, Ismail, and Sutoyo (2025) show that a web-based sales system with CRM integration can enhance operational performance and customer loyalty in the food and beverage sector.

Katoch and Bisane (2026) examined Customer Relationship Management (CRM) analytics using data visualization to support better understanding and decision-making in customer management. The study recognized that modern CRM systems generate large volumes of customer and transaction data, which can be difficult to interpret without effective analytical and visualization tools. To address this, the researchers explored the use of data visualization techniques to transform raw CRM data into clear charts, dashboards, and visual summaries that highlight key patterns and trends. These visual analytics help management identify customer segments, monitor performance indicators, and detect issues such as declining engagement or potential churn. By presenting data in an intuitive visual format, the system enables faster and more informed decisions regarding marketing, service, and retention strategies. These insights align with our system because both aim to leverage customer and transaction data not only for storage but also for actionable analysis. The focus on data visualization supports our goal of providing clear, interpretable insights that can guide service improvements and customer engagement efforts. Therefore, Katoch and Bisane (2026) demonstrate that integrating analytics and visualization into a CRM platform can enhance managerial understanding, strategic planning, and overall business performance.

Zakaria and Marlia (2019) designed a web-based Customer Relationship Management (CRM) information system to improve customer loyalty and service quality using the waterfall development model. The study addressed issues such as fragmented customer data, inconsistent service processes, and limited ability to track customer interactions in a structured way. Following the waterfall model, the researchers conducted requirement analysis, system design, implementation, and testing to produce a CRM platform that organizes customer information and service activities. The resulting system enables the business to store customer profiles, record interactions, and manage service requests in a centralized and accessible manner. By providing a more structured view of customer relationships, the system supports more

consistent and personalized service delivery. The study concluded that the CRM implementation contributed to improved customer loyalty and satisfaction by making it easier to maintain ongoing relationships and respond to customer needs. These findings align with our system because both focus on using a web-based CRM to centralize customer data and standardize service processes. The emphasis on loyalty and service quality supports our objective of building long-term customer relationships through better data management and interaction tracking. Therefore, Zakaria and Marlia (2019) provide evidence that a well-designed web-based CRM can enhance both customer loyalty and service performance in a business environment.

Chauhan (2023) explored the role of data mining in Customer Relationship Management (CRM) to understand how analytical techniques can support business growth and customer retention. The study examined the connection between CRM systems and data mining tools, highlighting how organizations use analytical software to acquire new customers and retain existing ones. By applying data mining methods such as classification, clustering, and pattern discovery, businesses can identify valuable customer segments, predict behavior, and tailor marketing strategies more effectively. The research discussed several case studies of global firms that successfully implemented data mining within their CRM processes to improve customer satisfaction and sales performance. It also addressed challenges related to data quality, security, and privacy, offering recommendations for implementing effective and responsible data mining practices. These insights align with our system because both recognize the importance of analyzing customer and transaction data to drive smarter decisions and stronger relationships. The focus on pattern discovery and segmentation supports our goal of using CRM data not only for record-keeping but also for strategic planning and targeted engagement. Therefore, Chauhan (2023) demonstrates that integrating data mining into CRM can enhance customer understanding, marketing effectiveness, and overall business competitiveness.

Ahmed, Maâti, and Al-Sarem (2020) proposed a predictive data mining model for electronic Customer Relationship Management (e-CRM) intelligence to support customer behavior prediction and decision-making. The study recognized that e-CRM systems generate large amounts of customer data, which can be leveraged through data mining to identify patterns related to purchasing, loyalty, and churn. The researchers developed a model that uses data mining techniques to classify customers, predict their future behavior, and provide actionable insights for marketing and service strategies. By integrating predictive analytics into e-CRM, businesses can prioritize high-value customers, design targeted campaigns, and allocate resources more efficiently. The model emphasizes the importance of accurate data, appropriate algorithms, and continuous evaluation to ensure reliable predictions and meaningful business outcomes. These findings align with our system because both aim to

transform customer and transaction data into predictive and strategic knowledge rather than static records. The focus on behavior prediction and segmentation supports our objective of using CRM data to anticipate customer needs and improve retention. Therefore, Ahmed, Maâti, and Al-Sarem (2020) show that predictive data mining can significantly enhance e-CRM intelligence and support more proactive, data-driven customer management.

Listiani, Purwanto, and Permatasari (2025) developed a web-based financial information system for Digimizu Digital Management that combines transaction automation with customer data management and rapid financial reporting. The study noted that the company previously faced challenges in managing financial transactions, customer-related data, and report generation in a timely and accurate manner. To address this, the researchers applied the System Development Life Cycle (SDLC) with the waterfall model and analyzed system weaknesses using the PIECES framework, supported by UML modeling. The resulting system enables automated transaction processing, organized storage of customer and financial data, and quick generation of accurate financial reports. By centralizing these functions in a web-based platform, the system improves data consistency, reduces manual errors, and supports faster decision-making. These outcomes align with our system because both focus on integrating transaction handling, customer data management, and reporting into a single web-based environment. The emphasis on accurate and timely reporting supports our goal of providing reliable information for operational and strategic decisions. Therefore, Listiani, Purwanto, and Permatasari (2025) demonstrate that a web-based financial and customer data system can enhance efficiency, data quality, and managerial responsiveness in a digital business setting.

Karmarkar (2022) examined the influence of UX/UI design on e-commerce conversion rates to understand how interface quality affects customer purchasing behavior. The study recognized that in competitive online markets, the design of product pages, navigation flows, and checkout processes can significantly impact whether visitors complete a purchase or leave the site. By analyzing various e-commerce platforms, the research identified key UX/UI elements such as visual hierarchy, clarity of calls-to-action, loading speed, and mobile responsiveness as critical drivers of conversion. The findings showed that well-structured and intuitive interfaces reduce cognitive load, increase user confidence, and encourage users to proceed through the buying journey more smoothly. In contrast, poorly designed interfaces with confusing layouts, slow performance, or unclear instructions were found to increase bounce rates and cart abandonment. These insights align with our system because both emphasize the importance of a user-centered design in ensuring that customers can easily navigate, understand, and complete transactions. The focus on conversion optimization supports our goal of creating a platform that not only manages customer data but also

facilitates seamless and efficient purchasing experiences. Therefore, Karmarkar (2022) demonstrates that strategic UX/UI design is a powerful lever for improving e-commerce performance and customer satisfaction.

Ismail, Kendre, Sherkhan, and Dodtale (2026) developed an e-commerce website as a dynamic web-based platform that enables users to browse, select, and purchase products online efficiently. The project provided a centralized digital marketplace where customers can view product details, compare prices, add items to a shopping cart, and securely complete transactions. The system integrated multiple modules including product management, user registration and login, shopping cart, order processing, and payment handling to deliver a seamless online shopping experience. The main objective was to develop an efficient, secure, and easy-to-use online shopping platform that reduces manual processes and improves customer convenience. By implementing structured navigation, clear product displays, and secure checkout flows, the system aimed to enhance user trust and encourage repeat purchases. These features align with our system because both focus on providing a complete, user-friendly e-commerce environment that supports end-to-end customer transactions. The emphasis on security, usability, and integrated modules supports our goal of creating a reliable and convenient platform for customers and administrators. Therefore, Ismail, Kendre, Sherkhan, and Dodtale (2026) demonstrate that a well-designed e-commerce website can streamline online shopping, improve operational efficiency, and enhance customer satisfaction.

Hamidah, Zunaidi, Putra, Hidayati, Mardhiana, and Chandra (2024) designed the user interface of the TUS Mart e-commerce website using Design Thinking and Kansei Engineering to create an interface that meets both functional and emotional user needs. The study applied Design Thinking stages to understand user context, define problems, ideate solutions, prototype the interface, and test it with real users. Kansei Engineering was used to translate users' emotional responses and preferences into concrete design elements such as color schemes, layout structures, and interactive components. The resulting interface emphasized engagement, transparency, interactivity, and ease of use to foster a personalized, intuitive, and satisfying shopping experience. Usability testing indicated that the design improved user satisfaction and supported the digital marketing goals of the platform. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both prioritize a user-centered design that considers not only functionality but also the emotional experience of users. For a spa management system, an interface that feels calming, trustworthy, and easy to navigate can enhance customer comfort and willingness to book services or purchase products. The use of Design Thinking also supports our iterative approach to refining the system based on feedback from spa staff and clients. Therefore, Hamidah, Zunaidi, Putra, Hidayati, Mardhiana,

and Chandra (2024) show that combining Design Thinking with Kansei Engineering can produce an e-commerce interface that is both usable and emotionally resonant, which is valuable for designing the customer-facing modules of our spa system.

Tafonao, Setyohadi, and Raslina (2025) designed an e-commerce website for Fayucom with integration of the Xendit payment gateway using HTML, CSS, and JavaScript to support secure and convenient digital transactions. The study focused on creating a local e-commerce platform that offers product management, automatic checkout, an admin dashboard, and responsive design across devices. The integration of Xendit enabled multiple payment methods such as e-wallets, virtual accounts, and debit/credit cards, providing customers with flexible and secure payment options. The researchers noted that the system improved the digital transaction process and user convenience, although further improvements were recommended in security, SEO, and mobile performance. The study highlighted the importance of combining technical implementation with user experience considerations to build a competitive and sustainable e-commerce platform. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to integrate secure and flexible payment options into a web-based system that supports online transactions. In a spa setting, offering multiple payment channels and a smooth checkout experience can reduce barriers to booking and increase customer satisfaction. The inclusion of an admin dashboard also supports our goal of providing spa managers with clear visibility over transactions, services, and customer data. Therefore, Tafonao, Setyohadi, and Raslina (2025) demonstrate that integrating a robust payment gateway with a well-structured e-commerce design can enhance transaction efficiency and user convenience, which is directly applicable to the payment and administrative modules of our spa system.

Nuragustin, Az Zahra, and Jausha (2026) applied user-centered UI/UX design principles to develop an e-commerce website for Intisari Banyumas souvenir food products to improve usability and customer satisfaction. The study followed user-centered design stages, including user research, persona development, wireframing, prototyping, and usability testing, to ensure that the interface reflects actual user needs and preferences. The resulting design emphasized clear navigation, consistent visual elements, simplified product browsing, and an intuitive checkout process tailored to the target audience. Usability evaluations indicated that the user-centered approach reduced confusion, improved task completion rates, and increased overall satisfaction with the shopping experience. The researchers concluded that involving users throughout the design process leads to interfaces that are more aligned with user expectations and behaviors. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both prioritize designing the system around the needs of real users, including spa

clients, therapists, and administrators. By conducting user research and testing with actual spa stakeholders, we can ensure that the interface supports common tasks such as booking, viewing service details, and managing appointments with minimal friction. The focus on satisfaction and task efficiency also supports our goal of creating a system that is comfortable and efficient for daily use. Therefore, Nuragustin, Az Zahra, and Jausha (2026) demonstrate that user-centered UI/UX design can significantly improve usability and satisfaction in e-commerce, which is directly relevant to designing the customer and staff interfaces of our spa management system.

Parab and Kolpe (2026) developed an e-commerce website with cart management and an order processing system to provide a complete online shopping and order fulfillment platform. The system includes modules for user registration, product browsing, shopping cart operations, secure checkout, payment integration, and order tracking for both customers and administrators. The cart management feature allows users to add, update, and remove items, while the order processing module handles order confirmation, status updates, and history tracking. The study emphasized that integrating these functions into a single web-based platform reduces manual work, minimizes errors, and improves the overall shopping experience. Testing showed that the system successfully supports end-to-end transactions from product selection to order completion and tracking. These features align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to provide a unified platform that manages the entire transaction lifecycle in a structured and automated manner. In a spa context, similar modules can support service selection, appointment scheduling, payment processing, and booking history tracking for customers and staff. The emphasis on cart and order management also supports our analytics goals by generating structured data on services booked, revenue generated, and customer behavior. Therefore, Parab and Kolpe (2026) show that a well-integrated e-commerce system with cart and order management can streamline operations and enhance user experience, which is directly applicable to the transactional and analytical components of our spa management system.

Jena, Acharya, and Sarangi (2026) developed an e-commerce website as a centralized online shopping platform that integrates product management, secure authentication, shopping cart operations, order processing, and payment gateway integration into one system. The study addressed limitations of traditional shopping systems that rely on physical stores and manual purchasing methods, which often create constraints in accessibility, inventory management, and transaction efficiency. Using modern web technologies, the researchers built a responsive platform that allows users to browse products, search and filter items, place orders, and perform secure transactions through an intuitive interface. The system also includes a centralized admin dashboard for managing products, users, orders, and overall business operations,

improving oversight and control. The implementation demonstrated that a well-structured e-commerce platform can enhance customer convenience, reduce manual effort, and improve business management efficiency. These features align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to consolidate operational functions into a single web-based system that supports end-to-end transactions and administrative control. In a spa context, similar modules can manage services, appointments, client profiles, payments, and staff dashboards to streamline daily operations. The emphasis on secure authentication and payment integration also supports our goal of ensuring safe and reliable transactions for clients and administrators. Therefore, Jena, Acharya, and Sarangi (2026) show that a centralized e-commerce architecture can improve operational efficiency and customer experience, which is directly applicable to the integrated design of our spa management system.

Gupta (2023) designed and developed an e-commerce website to provide a secure and user-friendly platform for online purchasing, addressing the limitations of manual and less secure traditional shopping systems. The project focused on creating a dynamic web-based application that allows users to browse products, add items to a cart, and complete transactions through a streamlined checkout process. The system incorporated essential features such as user authentication, product catalog management, shopping cart functionality, order processing, and payment integration to support a complete online shopping experience. By automating these processes, the platform aimed to reduce manual workload, minimize errors, and improve overall transaction efficiency. The study highlighted that a well-designed e-commerce website can enhance customer satisfaction by offering convenience, speed, and security in every transaction. These outcomes align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both seek to replace manual or fragmented processes with an automated, secure, and user-friendly web-based system. For a spa business, similar features can support online service booking, product sales, payment processing, and appointment management with minimal administrative overhead. The focus on security and usability also supports our objective of building trust and encouraging repeat usage among clients. Therefore, Gupta (2023) demonstrates that a comprehensive e-commerce website can streamline operations and improve customer satisfaction, which is relevant to the transactional and operational modules of our spa system.

Putra, Fitriana, Kanafi, and Wahyudiono (2026) analyzed and designed the UI/UX of an e-commerce application based on anticipatory design to optimize the transaction flow and reduce user friction. The study applied anticipatory design principles to predict user needs and streamline interactions, such as by pre-filling forms, suggesting next steps, and minimizing unnecessary clicks during the checkout process.

The researchers found that when the system anticipates user actions and reduces decision points, the transaction flow becomes faster, smoother, and less prone to abandonment. The design also incorporated clear progress indicators, simplified input fields, and contextual guidance to support users throughout the purchasing journey. Usability evaluations indicated that the anticipatory design approach improved task completion rates and overall satisfaction with the transaction process. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to make the booking and payment process as effortless and intuitive as possible. In a spa context, anticipatory features such as auto-filling client details, suggesting preferred services, or reminding users of upcoming appointments can enhance convenience and reduce drop-offs. The focus on optimizing the transaction flow also supports our goal of converting more inquiries into confirmed bookings and repeat clients. Therefore, Putra, Fitriana, Kanafi, and Wahyudiono (2026) demonstrate that anticipatory design can significantly improve transaction efficiency and user experience, which is directly applicable to the booking and checkout modules of our spa system.

Anugrah and Somantri (2026) designed and implemented the user interface and user experience (UI/UX) for the FURN e-commerce website to improve user convenience and overall shopping experience. The study applied an interactive design method with a qualitative descriptive approach, involving reference studies, needs analysis, wireframe development, user interface design, interactive prototyping, and design evaluation. The resulting prototype featured clear navigation, consistent visual design, and intuitive user interaction, supported by features such as a BIG Menu, Call to Action (CTA) buttons, shopping cart, and catalog filtering. The researchers emphasized that a well-structured UI/UX design can reduce confusion, improve task efficiency, and increase user satisfaction with the platform. The implementation showed that thoughtful interface design contributes to a more engaging and trustworthy shopping environment. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both recognize that interface quality directly affects how users perceive and interact with the system. For a spa management platform, clear navigation, consistent branding, and intuitive controls can make it easier for clients to explore services, book appointments, and complete payments without frustration. The emphasis on usability and visual consistency also supports our goal of creating a professional and calming digital environment that reflects the spa's brand. Therefore, Anugrah and Somantri (2026) show that strategic UI/UX design can enhance convenience and satisfaction in e-commerce, which is relevant to designing the customer and staff interfaces of our spa system.

Kumar, Reddy, Rakshith, Ashish, Babu, and Shruthi (2024) developed an Inventory Management System as a web application to facilitate online stock and

inventory management for businesses and retailers. The study addressed the need for efficient inventory operations by providing user-friendly interfaces for supplier administration, product categorization, inventory tracking, and transaction recording. The system was designed with a modular architecture to enable future improvements and changes, with a focus on scalability, security, and user-friendliness. Key features included strong search and filtering capabilities, role-based access control, and support for multiple transaction types to accommodate diverse business needs. The researchers emphasized that the system guarantees dependability and efficiency through extensive testing and documentation, giving companies a robust tool to streamline inventory procedures. These capabilities align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to provide a centralized, secure, and scalable platform for managing operational data and transactions. In a spa context, similar modules can track product inventory such as oils, lotions, and retail items, while also supporting role-based access for therapists, receptionists, and managers. The emphasis on real-time tracking and transaction recording also supports our analytics goal of generating accurate reports on product usage, sales, and stock levels. Therefore, Kumar, Reddy, Rakshith, Ashish, Babu, and Shruthi (2024) demonstrate that a well-structured web-based inventory system can enhance operational efficiency and decision-making, which is directly applicable to the inventory and administrative modules of our spa management system.

Fauzi, Khairuman, and Mukhroji (2026) developed a web-based inventory information system to support stock management activities at MSME Mitra Com Electronics Business. The study used a software engineering approach with a qualitative descriptive method to design and implement a system that addresses manual inventory processes and data inaccuracies. The resulting system enables staff to record incoming and outgoing stock, monitor current inventory levels, and generate reports on stock movements in real time. By digitizing inventory management, the system reduces errors associated with manual recording and provides faster, more accurate information for decision-making. The researchers noted that the web-based platform improves coordination between staff members and supports more efficient stock control across the business. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both seek to replace manual or fragmented inventory processes with a centralized web-based system that supports real-time data access. For a spa, similar functionality can track consumables, retail products, and supplies to prevent stockouts, reduce waste, and ensure that services are not disrupted due to missing items. The focus on accurate and timely reporting also supports our objective of providing managers with clear visibility into inventory status and usage patterns. Therefore, Fauzi, Khairuman, and Mukhroji (2026) show that a web-based inventory system can significantly improve

stock management efficiency and data accuracy, which is relevant to the inventory and operational modules of our spa system.

Tanaman, Lloyd, Baylosis, Abiles, Lester, Catungal, and Encarnacion (2023) developed a web-based Inventory Management System for a small business enterprise in Pagadian City, Philippines, to replace a manual paper-based inventory process. The study addressed issues such as data inaccuracies, inefficient processes, and lack of real-time visibility that arose from relying on physical forms and manual record-keeping. Using PHP for development and MySQL as the database, the researchers built a web platform that automates inventory management from recording to monitoring to stock control. The system enables the business owner to perform regular inventory management online, with features for tracking stock levels, recording transactions, and generating reports. Evaluation by IT experts and end-users confirmed that the system meets functional and usability requirements and provides a more streamlined and accurate alternative to manual methods. These outcomes align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to transition from manual to digital processes to improve accuracy, efficiency, and real-time visibility of operational data. In a spa setting, similar automation can reduce errors in tracking products and supplies, ensure that stock levels are always up to date, and support timely reordering before shortages occur. The emphasis on real-time insights also supports our analytics goal of monitoring product usage trends and aligning inventory with service demand. Therefore, Tanaman, Lloyd, Baylosis, Abiles, Lester, Catungal, and Encarnacion (2023) demonstrate that a web-based inventory system can effectively address the limitations of manual processes, which is directly applicable to the inventory management component of our spa system.

Jiu and Tony (2025) designed and developed an online sales application with real-time stock control features to improve operational efficiency in a retail environment. The study addressed problems such as stock discrepancies, delayed updates, and inefficient coordination between sales and inventory teams that often occur in manual or semi-digital systems. The researchers implemented a web-based application that integrates sales transactions with inventory management, enabling automatic stock updates whenever a sale is recorded. The system also includes automatic low-stock alert features that notify staff when items reach predefined thresholds, allowing timely replenishment and preventing stockouts. Testing showed that the application successfully provides real-time stock data, minimizes recording errors, and supports faster and more accurate managerial decision-making. These features align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to synchronize transactional and inventory data in real time to maintain accurate stock information. For a spa, integrating service bookings and product sales with inventory levels can ensure that product usage is automatically

reflected in stock records, reducing manual adjustments and errors. The low-stock alert functionality also supports proactive inventory management, ensuring that essential items are always available for services and retail. Therefore, Jiu and Tony (2025) show that real-time stock control integrated with sales processes can enhance operational efficiency and inventory accuracy, which is relevant to the sales and inventory modules of our spa system.

Setiawan, Triayudi, and Iskandar (2024) created a web-based inventory management system tailored to the needs of Micro, Small, and Medium Enterprises (MSMEs), focusing on affordability, user-friendliness, and functionality. The study conducted an in-depth analysis of MSMEs' specific inventory management requirements to design a comprehensive solution for stock management that also integrates the Apriori algorithm for associative analysis of inventory data. The system's development adopted an Agile approach, allowing flexible adaptation to changing needs throughout the development process based on user feedback. Key features include efficient stock management, intuitive Point of Sales (POS) transaction recording, and customizable inventory reporting to support diverse business scenarios. The implementation of this system is expected to enhance MSMEs' efficiency in inventory management, provide insightful data to support more informed decision-making, and strengthen competitiveness in a dynamic business environment. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both prioritize a user-centered, adaptable system that meets the specific operational needs of the organization. For a spa, similar customization can support tracking of different product categories, integration with service transactions, and generation of reports tailored to managerial needs. The use of data mining techniques such as the Apriori algorithm also supports our analytics goal of discovering patterns in product usage and sales to inform stocking and promotional strategies. Therefore, Setiawan, Triayudi, and Iskandar (2024) demonstrate that a flexible, MSME-focused inventory system with advanced analytics capabilities can improve operational efficiency and strategic decision-making, which is directly applicable to the inventory and analytics components of our spa system.

Afrizal, Minardi, and Mahendra (2025) developed a web-based inventory management system by applying Safety Stock and Reorder Point (ROP) methods to address inventory issues at RF Media, a small and medium enterprise in the printing sector. The study aimed to improve operational efficiency and reduce the risk of stockouts, which frequently occur in SMEs due to their reliance on manual inventory processes. The system integrates Safety Stock and ROP calculations to determine optimal stock levels and trigger timely reordering before items run out. Simulation results showed a 21.38% increase in operational efficiency and a 16.10% reduction in the risk of stockouts after implementing the system. The platform ensures complete

inventory visibility, facilitates faster decision-making, and minimizes manual errors associated with traditional recording methods. Usability testing revealed high user satisfaction regarding interface clarity, ease of use, and quick access to inventory information. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both seek to replace manual inventory tracking with an automated, data-driven system that prevents stockouts and optimizes stock levels. In a spa context, similar methods can be applied to consumables such as oils, lotions, and retail products to ensure that essential items are always available for services and sales. The emphasis on real-time visibility and reduced manual errors also supports our analytics goal of generating accurate reports on stock usage and reorder needs. Therefore, Afrizal, Minardi, and Mahendra (2025) demonstrate that integrating Safety Stock and ROP into a web-based inventory system can significantly improve efficiency and reliability, which is directly applicable to the inventory management module of our spa system.

Audina, Putra, and Ananto (2026) implemented Reorder Point and Safety Stock methods in a web-based Point of Sales (POS) system for SMEs to address frequent stock shortages and insufficient transaction records. The study focused on Toko Abadi, a retail store that experienced sales losses and hindered decision-making due to manual stock management and incomplete data. Using the Waterfall Model and UML-based system modeling, the researchers developed a POS system that supports sales and purchasing activities while automatically calculating reorder point and safety stock values for each product. The system facilitates automated product reordering by notifying staff when stock levels reach predefined thresholds, ensuring timely replenishment and preventing stockouts. Black box testing confirmed that all features functioned as intended, and the system achieved 100% accuracy in calculating reorder points and safety stock. These outcomes align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to integrate sales transactions with inventory control to maintain accurate and up-to-date stock information. For a spa, combining POS functions with automated stock calculations can ensure that product usage from services and retail sales is immediately reflected in inventory levels. The automatic reorder notifications also support proactive inventory management, reducing the risk of service disruptions due to missing supplies. Therefore, Audina, Putra, and Ananto (2026) show that embedding Reorder Point and Safety Stock methods into a POS system can enhance stock accuracy and operational efficiency, which is relevant to the sales and inventory modules of our spa system.

Istanto (2017) applied the Economic Order Quantity (EOQ), Reorder Point, and Safety Stock methods in a stock warehouse system using Radio Frequency Identification (RFID) technology to minimize stock availability issues in real time. The study aimed to develop a stock system that computes EOQ per month based on

variables such as raw material quantity, ordering cost, and storage cost, while Reorder Point calculations incorporate lead time, raw material usage, and safety stock. Safety stock is used to buffer against delays in supplier deliveries, ensuring that raw materials do not run out and the company can continue operating smoothly. Inventory data is obtained from transactions of incoming and outgoing goods, which are recorded automatically when items pass through an RFID reader. The computation of EOQ and Reorder Point produces safety stock as an output of the stock system, enabling timely reordering and preventing stockouts. With stock levels aligned to Safety Stock fulfillment, the system ensures that there are no delays in production due to missing materials. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both emphasize the importance of data-driven stock calculations and real-time tracking to maintain optimal inventory levels. In a spa setting, similar methods can determine the most cost-effective order quantities for consumables and retail products while accounting for lead times and usage patterns. Although RFID may not be directly implemented, the principle of automatic data capture and real-time updates supports our goal of minimizing manual errors and ensuring accurate stock records. Therefore, Istanto (2017) demonstrates that integrating EOQ, Reorder Point, and Safety Stock with automated data collection can optimize inventory management, which is applicable to the inventory and analytics components of our spa system.

Ivan and Oetama (2024) designed an inventory management system for car workshops that uses Economic Order Quantity (EOQ) and Reorder Point methods to address inefficiencies and inaccuracies associated with Excel-based stock management. The study recognized that manual or spreadsheet-based systems are often inefficient and error-prone when handling complex stock calculations and replenishment planning. Using the Rapid Application Development (RAD) methodology, the researchers developed an application that provides timely stock replenishment information and optimizes order quantities to increase inventory management efficiency and accuracy. The system generates EOQ-based recommendations for optimal order sizes and Reorder Point alerts to indicate when items should be reordered to avoid stockouts. User Acceptance Test results showed that users received the inventory management system very well, with an average satisfaction level of 98.44% for admins and 97% for general users, indicating high satisfaction with the system's features and functionality. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to replace manual or spreadsheet-based inventory processes with an automated system that supports data-driven decision-making. For a spa, similar EOQ and Reorder Point calculations can optimize ordering of consumables and retail products, reducing both overstocking and stockout risks. The high user satisfaction also supports our objective of creating an intuitive and reliable system that staff can trust and use consistently. Therefore, Ivan

and Oetama (2024) show that integrating EOQ and Reorder Point methods into an inventory application can significantly improve efficiency and user satisfaction, which is directly relevant to the inventory management module of our spa system.

Setyadi, Amin, and Widodo (2024) implemented Economic Order Quantity (EOQ) and Reorder Point methods in an inventory management information system to optimize stock control and reduce inventory costs. The developed system manages data on goods, categories, and suppliers, handles purchase and sales transactions, and generates information from EOQ calculations, safety stock, and reorder points. Testing the calculations with 15 samples of paint variants showed that the average inventory cost saving in June 2023 was 65.33%, demonstrating significant cost efficiency achieved through the use of these methods. The system provides recommendations on optimal order quantities and timely reorder points, helping managers make informed decisions about purchasing and stock levels. By automating these calculations, the system reduces manual errors and ensures that inventory policies are consistently applied across all products. These outcomes align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both seek to use quantitative methods to optimize inventory decisions and reduce operational costs. In a spa context, applying EOQ and Reorder Point to consumables and retail items can minimize holding costs while ensuring that essential products are always available for services and sales. The ability to generate detailed reports on EOQ, safety stock, and reorder points also supports our analytics goal of providing managers with clear insights into inventory performance and cost efficiency. Therefore, Setyadi, Amin, and Widodo (2024) demonstrate that integrating EOQ and Reorder Point methods into an inventory information system can achieve substantial cost savings and improve decision-making, which is directly applicable to the inventory and analytics components of our spa system.

Liu (2025) examined the widespread adoption of QR code-based mobile payments in China, analyzing the economic, technological, and social factors driving their popularity while also addressing associated security risks and challenges. The study highlighted that the convenience, speed, and low infrastructure cost of QR code payments have made them especially prevalent among small businesses and street vendors, supported by government digital initiatives, high smartphone penetration, and consumer willingness to embrace new technologies. However, the rapid growth has also led to rising incidents of fraud, such as counterfeit payment codes, phishing scams, and unauthorized data collection, exacerbated by regulatory gaps and fragmented oversight. In contrast, the limited adoption of QR code payments in many foreign countries is attributed to well-established credit card systems, different regulatory environments, and varying consumer preferences. The study proposes localized recommendations for China, such as enhancing regulatory oversight, promoting interoperability, strengthening

data protection, and improving public awareness. The researcher concluded that while QR code payments have revolutionized financial transactions in China, a balanced approach integrating innovation with robust security and regulatory frameworks is essential for sustainable development and potential global expansion. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both recognize the importance of offering convenient, low-cost digital payment options that are accessible to a wide range of customers. In a spa context, integrating QR code-based payments can streamline the checkout process, reduce cash handling, and provide clients with flexible payment choices that match local preferences. The emphasis on security and regulatory compliance also supports our goal of ensuring that payment transactions are safe, transparent, and trustworthy for both clients and the business. Therefore, Liu (2025) demonstrates that QR code mobile payments can significantly enhance payment convenience and accessibility when implemented with appropriate security and governance measures, which is directly applicable to the payment module of our spa system.

Khan, Raza, Ahsan, and Irshad (2024) proposed a secure digital wallet system that uses QR codes to enable mobile payments in partially connected environments where internet connectivity may be intermittent or unreliable. The study addressed the challenge that many existing mobile payment systems require constant online connectivity, which limits their usability in areas with poor network coverage or during connectivity outages. The researchers designed a QR code-based payment protocol that allows transactions to be initiated and validated offline, with secure synchronization to the central server once connectivity is restored. The system incorporates cryptographic techniques to ensure transaction integrity, prevent double-spending, and protect user data even in offline scenarios. Testing demonstrated that the proposed wallet can successfully complete payments in partially connected environments while maintaining security and consistency with online transaction records. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both recognize the importance of ensuring payment functionality under varying network conditions, especially in locations where connectivity may not always be stable. In a spa setting, the ability to process payments even during temporary internet outages can prevent service delays and maintain a smooth client experience. The emphasis on security and data integrity also supports our objective of implementing payment features that are robust, trustworthy, and resilient to technical disruptions. Therefore, Khan, Raza, Ahsan, and Irshad (2024) demonstrate that QR code-based digital wallets can provide secure and reliable mobile payments even in partially connected environments, which is applicable to the payment module of our spa system.

Kumar (2025) presented a modular and scalable framework titled the Web-Based Employee Information and Attendance System, developed to automate and streamline core HR operations within a unified digital platform. The study addressed challenges associated with manual attendance tracking, including time-consuming data entry, susceptibility to human error, and difficulties in generating timely and accurate attendance reports. The system integrates employee information management, real-time attendance recording, leave tracking, and automated report generation into a single web-based application accessible to both administrators and employees. By centralizing these functions, the platform reduces administrative workload, minimizes discrepancies in attendance records, and provides managers with up-to-date insights into workforce presence and patterns. The modular design allows for future enhancements such as integration with payroll systems, biometric devices, or mobile applications, ensuring long-term adaptability to evolving organizational needs. Usability testing indicated that the system improves data accuracy, accelerates report generation, and enhances overall HR operational efficiency. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to replace manual or fragmented administrative processes with a centralized web-based platform that supports real-time data access and automated reporting. In a spa context, similar modules can manage therapist schedules, attendance, and service assignments while providing managers with clear visibility into staff availability and workload. The emphasis on modularity and scalability also supports our goal of designing a system that can grow with the business and integrate additional features as needed. Therefore, Kumar (2025) demonstrates that a web-based employee information and attendance system can significantly improve HR efficiency and data accuracy, which is directly applicable to the staff management and analytics components of our spa system.

Cox (2023) designed, developed, and implemented an automated reminder system for employee clock-in and clock-out times to address common timekeeping errors such as forgetting to clock in or out, clocking at incorrect times, or experiencing technical issues with timekeeping systems. The study utilized a Java-based desktop notification system employing the Quartz job scheduling library and JavaFX for desktop notifications, configured to prompt employees to clock in at 9 AM and clock out at 5 PM, Monday to Friday. The impact of the system was evaluated in a medium-sized business over a period of six months, with three months of baseline data collected before implementation and three months of data collected post-implementation. The results showed a 78% reduction in both clock-in and clock-out errors, a 15% increase in accurately logged work hours, and a 20% improvement in employee satisfaction following the introduction of the reminder system. All results were statistically significant, indicating the reminder system's effectiveness in enhancing timekeeping accuracy, productivity, and overall workplace experience. The study concluded that automated

reminders can serve as a simple yet powerful intervention to reduce forgetfulness, improve compliance with timekeeping policies, and support more reliable payroll and performance data. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both recognize that accurate time tracking is essential for operational efficiency, fair compensation, and reliable performance analytics. In a spa setting, automated reminders for therapists and staff to log their shifts or service times can reduce missed clock-ins, ensure accurate recording of work hours, and support precise calculation of commissions or overtime. The improvement in employee satisfaction also supports our objective of creating a system that is perceived as helpful rather than burdensome by staff. Therefore, Cox (2023) shows that an automated reminder system can significantly reduce timekeeping errors and improve productivity and satisfaction, which is relevant to the staff scheduling and attendance modules of our spa system.

Nugraha and Astutik (2022) developed a web-based employee attendance information system to help companies maximize the use of computers in managing human resources and attendance data. The study recognized that many organizations, from small to large scale, have started using computers to support operational activities but often do not fully leverage technology for attendance management. The implemented system enables simple recording of employee data ranging from general information to specific data such as absence from entering to leaving the workplace, all within a web-based interface. Using this employee attendance information system helps companies, especially administrators, in terms of recording employee absent data, making employee attendance report data, and providing clear information to employees about their attendance status. The web-based nature of the system makes it accessible from multiple devices and locations, supporting more flexible and efficient attendance management compared to manual or paper-based methods. The researchers concluded that the attendance information system becomes one of the company's strengths in managing existing human resources by improving data accuracy, accessibility, and reporting capabilities. These outcomes align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to digitize and centralize attendance and staff data to improve operational efficiency and managerial oversight. For a spa, similar functionality can track therapist and staff attendance, link attendance to service schedules, and generate reports that support payroll, performance evaluation, and workload planning. The emphasis on web accessibility also supports our goal of enabling managers to monitor staff presence and attendance patterns from any device with internet access. Therefore, Nugraha and Astutik (2022) demonstrate that a web-based attendance system can enhance HR management efficiency and data transparency, which is directly applicable to the staff management and reporting modules of our spa system.

Salunkhe, Thombe, Gunjal, Harde, and Kamble (2026) presented a smart, web-based system that uses artificial intelligence and facial recognition to automatically track when employees start and end their work, addressing issues such as errors, wasted time, and fraud associated with traditional attendance methods like paper registers, punch cards, or swipe systems. The system captures an employee's face through a standard webcam, verifies their identity using AI algorithms, and instantly records their check-in and check-out times without requiring physical cards or manual input. Built with HTML, CSS, JavaScript, Node.js, and MongoDB, the platform offers separate dashboards for employees and managers, enabling real-time monitoring of attendance and timesheet data. The use of facial recognition reduces the risk of proxy attendance, where one employee clocks in for another, and ensures that attendance records accurately reflect the presence of the correct individual. Testing showed that the system improves attendance accuracy, reduces administrative effort in verifying timesheets, and provides a more secure and tamper-resistant method of tracking work hours. These features align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both seek to enhance the reliability and security of attendance and timesheet data through automation and advanced technology. In a spa context, similar AI-based verification can be adapted to ensure that only authorized therapists and staff can log their shifts or service times, reducing the risk of fraudulent or inaccurate records. The real-time dashboards also support our analytics goal of providing managers with immediate visibility into staff attendance, workload, and service coverage. Therefore, Salunkhe, Thombe, Gunjal, Harde, and Kamble (2026) show that integrating AI and facial recognition into a web-based timesheet system can significantly improve attendance accuracy and security, which is relevant to the staff management and analytics components of our spa system.

Hardiyani, Ramlan, Fajar, and Sudarjat (2026) developed a payroll information system and employee tax calculation module for a financial company in Bandung to address manual and error-prone payroll and tax computation processes. The study recognized that manual payroll calculations often lead to inconsistencies, delays, and mistakes in computing salaries, deductions, and statutory taxes such as PPh 21, which can affect employee trust and regulatory compliance. The system automates employee data management, attendance integration, salary computation, and tax calculation based on current regulations, generating accurate payslips and payroll reports. By centralizing these functions, the platform reduces administrative workload, minimizes calculation errors, and ensures that payroll and tax data are consistently aligned with legal requirements. The researchers noted that the system improves transparency, as employees and managers can access clear and timely information regarding salary components and tax deductions. Testing and user feedback indicated that the system enhances payroll efficiency, data accuracy, and overall satisfaction among HR staff and employees. These findings align with *Recovery Iloilo Spa Integrated System for*

Streamlining Operations and Analytics Efficiently because both aim to replace manual or spreadsheet-based payroll processes with an automated, rule-based system that ensures accuracy and compliance. In a spa context, similar functionality can manage therapist and staff salaries, commissions, overtime, and applicable deductions while generating clear payslips and reports for management. The emphasis on tax calculation also supports our goal of ensuring that the system can handle statutory requirements and provide transparent breakdowns of earnings and deductions. Therefore, Hardiyani, Ramlan, Fajar, and Sudarjat (2026) demonstrate that a payroll and tax calculation system can significantly improve efficiency, accuracy, and compliance, which is directly applicable to the payroll and analytics modules of our spa system.

Huda, Pramudito, and Pradini (2025) constructed an integrated cross-platform payroll information system for SMEs using an Agile-Scrum approach at CV. Gema Syifa Engineering to address reliance on semi-automated compensation mechanisms that required manual data entry via spreadsheets and fragmented data storage. The study applied the Agile Scrum framework, which facilitates adaptive and cyclical software development through sprint iterations, allowing the system to evolve based on continuous user feedback and changing requirements. The system architecture was designed using Unified Modeling Language (UML) notation, including use case diagrams, activity flow diagrams, sequence diagrams, and a database normalized to the third normal form (3NF) to ensure data integrity and scalability. The resulting application integrates web and Android platforms to manage the entire payroll workflow, from personnel information administration and attendance documentation to overtime computation and automated generation of compensation calculations and salary slip documents. Implementation outputs demonstrated that the system successfully accommodates comprehensive payroll integration, reducing manual effort and improving data consistency across devices. User evaluations indicated high satisfaction with the system's usability, accuracy, and ability to streamline previously fragmented payroll processes. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both prioritize an iterative, user-centered development approach that adapts to organizational needs and supports multi-device access. For a spa, a cross-platform payroll system can enable managers to process salaries and view reports from desktops or mobile devices, while staff can access their payslips and attendance records conveniently. The integration of attendance and overtime data also supports our analytics goal of linking workforce data directly to compensation for more accurate and transparent payroll management. Therefore, Huda, Pramudito, and Pradini (2025) show that an Agile-Scrum, cross-platform payroll system can effectively modernize payroll operations for SMEs, which is relevant to the payroll and staff management components of our spa system.

Singh, Mangaraj, and Muduli (2026) designed, implemented, and evaluated a modern web-based payroll management system to automate employee compensation processes and address the complexity and time consumption associated with manual payroll handling. The system incorporates modules for employee management, leave tracking, salary computation, payslip generation, and administrative controls, providing a comprehensive solution for end-to-end payroll administration. Using a structured development approach, the researchers defined system requirements, designed the software architecture and database, implemented the application with modern web technologies, and conducted evaluation to assess functionality, performance, and usability. The study highlighted that traditional manual payroll processes are prone to errors, delays, and lack of transparency, whereas the automated system significantly reduces processing time and improves data accuracy. Evaluation results showed that the system efficiently calculates monthly and annual salaries, generates payslips on demand, and provides managers with clear reports on payroll expenses and deductions. The researchers concluded that the web-based payroll management system enhances operational efficiency, data integrity, and user accessibility while offering scalability for future enhancements such as cloud integration and advanced security features. These outcomes align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both seek to automate complex payroll calculations and provide transparent, real-time access to compensation data. In a spa setting, similar modules can manage base salaries, service commissions, overtime, and deductions while generating detailed reports that support financial planning and performance analysis. The emphasis on scalability and future enhancements also supports our goal of designing a system that can grow with the business and integrate additional features as needed. Therefore, Singh, Mangaraj, and Muduli (2026) demonstrate that a modern web-based payroll system can substantially improve payroll efficiency and accuracy, which is directly applicable to the payroll and analytics components of our spa system.

Hussaini, Razzaq, Ali, and Mohiuddin (2025) developed a web-based Employee Payroll System to simplify employee data management and accurate payroll calculation processes in organizations that still rely on manual or spreadsheet-based methods. The study recognized that manual payroll processes are time-consuming and prone to errors in calculations, deductions, and employee data recording, which can lead to dissatisfaction among employees and increased workload for administrative staff. The system provides modules for managing employee profiles, attendance records, salary structures, deductions, and automated payroll computation, generating payslips and summary reports for management review. By automating these processes, the platform reduces manual effort, minimizes calculation mistakes, and ensures that payroll data is consistently maintained and easily accessible. The researchers emphasized that the web-based architecture allows authorized users to access payroll

information from any device with internet connectivity, supporting more flexible and efficient payroll administration. User feedback indicated that the system improves payroll accuracy, transparency, and overall satisfaction among both administrators and employees. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to replace manual payroll methods with an automated, web-based system that enhances accuracy and accessibility. For a spa, similar functionality can manage therapist and staff salaries, commissions, and deductions while providing clear payslips and reports that support financial oversight and workforce planning. The focus on transparency and ease of access also supports our objective of creating a payroll module that builds trust and reduces administrative burden. Therefore, Hussaini, Razzaq, Ali, and Mohiuddin (2025) show that a web-based employee payroll system can effectively streamline payroll operations and improve data accuracy, which is relevant to the payroll and staff management components of our spa system.

Hermawan, Kusuma, Manurung, and Suryanto (2024) evaluated the effectiveness of voucher utilization in influencing users' decisions to use digital payment methods in e-commerce. The study examined the relationship between promotional vouchers and consumer payment behavior, particularly how discounts or other incentives may encourage customers to select digital payment options during online transactions. The research is relevant to the growing use of promotional strategies in electronic commerce, where vouchers may serve not only as price-reduction tools but also as incentives that influence customer purchase decisions and payment preferences. By analyzing voucher utilization and digital payment behavior, the study highlights the importance of properly managing promotional offers and understanding their effect on customer transactions.

These findings align with the *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because the proposed voucher management module must ensure that every promotional transaction is valid, authorized, and properly recorded. In the spa context, a voucher may be issued to a customer, applied to a booking or Point-of-Sale transaction, and marked as redeemed after payment confirmation. The system will need to maintain consistent voucher status information so that receptionists and authorized staff can immediately determine whether an offer is active, expired, cancelled, or already used. It should also record the employee who applied the voucher, the customer who used it, the related transaction, the discount amount, and the date and time of redemption. Although the proposed system will use centralized database validation rather than a peer-to-peer framework, the study supports the need for reliable transaction verification and resilient record management. Therefore, Agarwal, Tiwari, Shukla, Pal, Soni, and Das (2026) demonstrate that dependable validation mechanisms are essential to prevent fraudulent

or inconsistent digital coupon utilization, which directly supports the security, accountability, and reporting requirements of the spa's voucher module.

Suendri, Imam Adlin Sinaga, and Zul Fadli Lubis (2026) developed a website-based electronic voucher application with a reward system for Arif Lubis NET using the Laravel framework. The study focused on creating an online platform that could manage electronic vouchers while incorporating reward-oriented features to encourage customer participation and continued use of the service. A website-based voucher system can provide a centralized method for creating, distributing, monitoring, and redeeming promotional offers. The inclusion of a reward system also demonstrates how businesses can use incentives to strengthen customer engagement and encourage repeat transactions. Using a web framework such as Laravel supports the development of an organized application with structured data management and accessible online functionality.

These ideas align with Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently because the proposed system will include a web-based voucher and promotional management module. The spa will be able to manage vouchers, gift certificates, discount offers, and customer incentives within the same platform used for appointments, transactions, payments, and customer records. A reward-oriented feature may also support future customer loyalty activities, such as incentives for repeat visits, referrals, package purchases, or accumulated transactions. The connection between voucher usage and customer records can help management identify which customers redeem particular offers and determine which promotions generate bookings or sales. The study is particularly relevant because the proposed spa system will also be web-based and will require centralized access for customers, receptionists, therapists, and authorized management personnel. Therefore, Suendri, Imam Adlin Sinaga, and Zul Fadli Lubis (2026) demonstrate that a website-based e-voucher application with reward functionality can improve promotional management and customer engagement, which is applicable to the voucher, customer management, and analytics modules of the proposed spa system.

Agarwal, Tiwari, Shukla, Pal, Soni, and Das (2026) proposed a resilient peer-to-peer validation framework for reliable digital coupon utilization. The study focused on strengthening the validation process used when digital coupons are issued, transferred, or redeemed across connected users or systems. A peer-to-peer validation framework can distribute the responsibility of verifying coupon transactions rather than relying entirely on a single centralized authority. This approach may improve reliability by allowing transaction participants or network nodes to verify coupon legitimacy, usage status, and transaction consistency. The study emphasizes the importance of resilience in digital coupon systems, particularly in preventing invalid transactions, duplicate use, and unreliable coupon records.

Mayorga and Yoo (2025) investigated how a One-Time Password (OTP) solution can strengthen Two-Factor Authentication (2FA) in a practical case study, addressing the growing insecurity of static passwords that expose accounts and data to various threats. The study recognized that static passwords alone are no longer sufficient to protect digital accounts, and that adding a second factor such as an OTP provides an additional layer of security beyond traditional credentials. The researchers developed a 2FA system that includes the generation and delivery of OTP codes based on events (HOTP) or time (TOTP), using mobile applications and email for OTP delivery. The system covers the entire authentication process, from enabling 2FA on the user account to verifying and validating the OTP during login. Even if an attacker obtains a user's password, they still need the dynamic OTP to access the account, significantly reducing the risk of unauthorized access. The practical implementation showed that the OTP-based 2FA system effectively enhances security while remaining usable for everyday users. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to protect user accounts and sensitive operations through multi-factor authentication. In a spa setting, 2FA with OTP can secure administrative accounts, staff logins, and client portals, reducing the risk of data breaches and unauthorized transactions. The use of both mobile and email delivery also supports flexibility, allowing users to receive OTPs through their preferred channel. Therefore, Mayorga and Yoo (2025) show that OTP-based 2FA can substantially improve account security in practical applications, which is relevant to the authentication and security components of our spa system.

Farid and Nurnawati (2024) implemented an OTP system on WhatsApp and email applications as an authentication method to reduce the risk of unauthorized access in web-based applications. The study used an experimental web-based application development approach, sending OTP codes via WhatsApp and email and requiring users to verify their identity using the received code. Testing was carried out on 10 users divided into two groups, one receiving authentication via WhatsApp and the other via email, to compare the effectiveness of both delivery channels. The results showed that implementing OTP successfully reduced the risk of unauthorized access by up to 80%, demonstrating the effectiveness of OTP as an additional security layer. The researchers concluded that the use of OTP can effectively secure the login process without imposing significant burden on users, as the verification step is quick and straightforward. The study also highlighted that OTP adoption can be particularly valuable in environments where users may reuse weak passwords or fall victim to phishing attempts. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both seek to enhance login security through simple yet effective verification mechanisms. In a spa context, OTP via WhatsApp or email can be used to secure client bookings, staff logins, and sensitive administrative actions, reducing the risk of account compromise. The ability to use

multiple delivery channels also supports accessibility, allowing users to choose the method that is most convenient for them. Therefore, Farid and Nurnawati (2024) demonstrate that adopting OTP for account verification can significantly improve security with minimal impact on user experience, which is applicable to the authentication and user management modules of our spa system.

Sirisha and Keerthi (2025) developed an OTP verification system using Python to provide a secure and efficient method for verifying user identities in web and mobile applications. The system generates a random OTP and sends it to the user's registered email or mobile number, requiring the user to enter the correct code within a specified time window to complete verification. Using Python's libraries for email and SMS integration, the researchers implemented a backend that handles OTP generation, delivery, and validation while ensuring that codes are time-limited and single-use to prevent reuse attacks. The system was tested in various scenarios, including login authentication, password resets, and transaction verification, demonstrating its flexibility and reliability across different use cases. Results showed that the OTP verification system effectively prevents unauthorized access while maintaining a smooth and user-friendly verification process. The researchers emphasized that OTP-based verification is a practical and scalable solution for enhancing security in applications that handle sensitive user data. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both prioritize secure and scalable authentication mechanisms that can be integrated into multiple parts of the system. In a spa setting, Python-based OTP verification can be adapted to secure client registrations, appointment confirmations, and administrative logins, ensuring that only authorized users can access sensitive functions. The emphasis on scalability also supports our goal of designing a security module that can grow with the system and support additional features such as rate limiting and multi-channel delivery. Therefore, Sirisha and Keerthi (2025) show that an OTP verification system built with Python can provide robust and flexible security, which is relevant to the authentication and backend components of our spa system.

Konstantina Ragazou, Ioannis Passas, Alexandros Garefalakis, and Constantin Zopounidis (2023) examined how business intelligence can help small and medium-sized enterprises make better decisions and improve their competitive advantage. The researchers used bibliometric analysis to examine 317 scholarly articles related to business intelligence, SMEs, competitive advantage, and decision-making. The study explained that business intelligence combines data collection, storage, processing, analysis, and visualization to transform large amounts of business data into useful information. The findings emphasized that BI systems can help SMEs improve decision-making, reduce inefficiencies, evaluate business performance, understand customer needs, identify market opportunities, and strengthen competitiveness. The

researchers also proposed combining business intelligence with the Technology–Organization–Environment framework to help SMEs adopt emerging technologies more effectively.

Rizky Wandri, M. Rizki Fadhilah, Panji Rachmat Setiawan, and Mutia Fadhilla (2025) developed a web-based school information system using Agile Scrum. The study applied an iterative development process in which the system was planned, implemented, reviewed, and improved through successive sprints. The researchers focused on making the system more responsive to user requirements by allowing continuous interaction between the development team and stakeholders. The evaluation results indicated that the Agile approach supported the creation of a functional web-based information system and achieved a user satisfaction rate of 95 percent.

These findings align with Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently because the proposed system must satisfy the different requirements of customers, spa staff, therapists, management, and the owner. The system can be developed incrementally, beginning with core features such as user accounts, services, and appointments, followed by payments, therapist commissions, vouchers, feedback, dashboards, and forecasting. Regular sprint reviews can help the researchers identify problems in the booking workflow, payment process, or report layout before the system is completed. User feedback can also guide improvements in navigation, accessibility, and the presentation of information. Therefore, Rizky Wandri, M. Rizki Fadhilah, Panji Rachmat Setiawan, and Mutia Fadhilla (2025) demonstrate that Agile Scrum can help produce a responsive and user-oriented web information system, which is applicable to the proposed spa platform.

Xuesong Zhang and Bradley Dorn (2013) examined the use of Agile practices in a small-scale, time-intensive web development project. The study observed the development process, interviewed the project team, and reviewed project documents. The Agile practices included daily Scrum meetings, product backlogs, and short development iterations. The researchers found that Agile practices could be successfully adopted in a small project when the team used regular communication, knowledge sharing, and multiple communication channels. However, the study also identified challenges involving team coordination, experience levels, and the need to maintain effective communication among project members.

These findings are relevant to Recovery Iloilo Spa because the proposed system will be developed by a capstone project team with limited time and resources. Daily or regular coordination can help the researchers monitor progress, assign tasks, discuss technical problems, and ensure that system modules are developed consistently. A product backlog can contain requirements such as minute-level appointment scheduling, payment monitoring, therapist assignment, voucher validation, sales reporting, and revenue forecasting. The study also emphasizes that Agile is not only a

set of technical activities but also a communication process. Therefore, Xuesong Zhang and Bradley Dorn (2013) demonstrate that daily Scrum meetings, backlogs, and iterative development can support small web projects, while effective communication and team coordination remain necessary for successful implementation.

D. T. T. Vijaya Kumar and M. Sowmya Vani (2012) presented a case study of a web application developed using Agile software development practices. The study compared Extreme Programming and Scrum and explained that Agile methods use incremental development, open collaboration, and adaptability throughout the software life cycle. Unlike the Waterfall model, which assumes that all requirements can be identified before development begins, Agile recognizes that requirements may change as users and developers gain a better understanding of the system. The study emphasized that short development cycles and continuous feedback help reduce risk and allow the application to adapt to changing requirements.

Hritik Chaudhary (2025) examined Agile software development through a case study involving the development of the Women's Era web application. The study discussed the use of Scrum and Extreme Programming in creating a web-based platform intended to provide users with information and support services. Agile development was presented as an approach that responds to uncertainty through incremental and iterative work cycles. The study emphasized that Agile supports collaboration, frequent feedback, adaptable planning, and the delivery of usable features in small increments. These characteristics are important when a web application has diverse users and requirements that may change during the development process.

These findings are relevant to Recovery Iloilo Spa because the proposed system will serve several user groups with different responsibilities. Customers will need functions for registration, service browsing, appointments, payments, transaction history, and feedback. Staff members will need appointment management, customer records, POS functions, voucher validation, and payment monitoring, while therapists will need access to schedules, attendance, assignments, and commission records. Management will require dashboards, sales reports, employee performance information, inventory reports, and revenue forecasts. Agile development can allow each user group's requirements to be addressed through separate but coordinated increments. Therefore, Hritik Chaudhary (2025) demonstrates that Agile development can support web applications with multiple user groups by using incremental delivery, collaboration, and adaptability, which is applicable to the proposed integrated spa system.

2.2 Local Studies

Cadiao, Caligan, Calizo, Esis, and Pardilla (2015) developed a management information system for Perfect Elegance Wellness and Beauty Center in Iloilo City to automate daily operations, transactions, and reporting. The study addressed the problem of manual and fragmented processes in handling customer records, payments, and service scheduling in a small wellness and beauty business. Cadiao, Caligan, Calizo, Esis, and Pardilla designed the system to include a customer profile database, a payment module for daily transactions, and report generation for selected date ranges. The platform also features a company website to improve marketing, showcase services and promos, and attract new customers through an online presence. The researchers used Evolutionary Prototyping, moving through requirements gathering, prototype development, client evaluation, design, coding, integration, testing, and maintenance. Cadiao, Caligan, Calizo, Esis, and Pardilla found that the system helped spa personnel retrieve information faster, produce accurate reports, and manage session schedules more effectively. Results showed that computation and data gathering became easier, and appointment reminders could be sent with just a few clicks. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how an integrated management system can streamline spa transactions, customer records, and reporting. Cadiao, Caligan, Calizo, Esis, and Pardilla therefore offer a relevant local reference for designing an all-in-one spa operations and analytics platform.

Abalos, Felix, Serante, and Rodriguez (2024) developed an online booking system for Von Czar Salon and Spa to replace manual appointment scheduling and improve reservation management. The study addressed the problem of manual booking errors, scheduling conflicts, and limited communication between the salon and its clients. Abalos, Felix, Serante, and Rodriguez designed a web-based reservation and management system that allows clients to view available slots, book appointments, and receive confirmations online. The platform features a centralized database for storing appointment records, customer details, and service information accessible to administrators and staff. The researchers used a descriptive case study approach to analyze the existing appointment process and evaluate the new system's functionality and usability. Abalos, Felix, Serante, and Rodriguez found that the system reduced booking conflicts, minimized manual recording, and improved overall client experience. Results indicated that staff could manage schedules more efficiently and respond to client inquiries faster through the online interface. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an integrated booking and CRM module can unify spa bookings, client records, and operational analytics. Abalos, Felix, Serante, and Rodriguez therefore provide a relevant local reference for designing a spa management system that combines scheduling, customer records, and reporting.

Abijay, Galicia, and Laurencio (2023) developed an online spa and salon management system for Lhay Lhor Nail Lounge and Spa to centralize bookings, services, and customer data. The study addressed the problem of manual and

disconnected processes in handling appointments, service records, and client information in a small spa business. Abijay, Galicia, and Laurencio designed the system using PHP with the Laravel framework and MySQL as the database management system to support online transactions and reporting. The platform features modules for service management, appointment booking, customer profiles, and administrative reporting accessible through a web interface. The researchers implemented the system following the Waterfall Model and tested it using the Black Box method to ensure functional correctness. Abijay, Galicia, and Laurencio found that the system improved the speed and accuracy of booking transactions and reduced manual workload for staff. Results indicated that the spa could better track client visits, manage service schedules, and maintain organized customer records. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a web-based spa management system can unify bookings, services, and client analytics. Abijay, Galicia, and Laurencio therefore offer a relevant local reference for integrating core spa operations and data management into a single platform.

Labastida, Simbulan, Estrella, Flores, and Centeno (2024) developed a web-based ordering system with forecasting capabilities tailored for start-up businesses in the Philippines to streamline ordering, improve inventory management, and provide actionable demand insights. The study addressed the problem of inefficient order processing and poor inventory planning that often led to stockouts or excess inventory for new enterprises. Labastida, Simbulan, Estrella, Flores, and Centeno integrated forecasting algorithms into the ordering platform to predict future demand based on historical sales and seasonal patterns. The system features online order placement, automated order confirmation, inventory level monitoring, and forecast-driven replenishment suggestions accessible through a web interface. The researchers evaluated the system in terms of order fulfillment times, inventory turnover rates, and overall business performance. Labastida, Simbulan, Estrella, Flores, and Centeno found that the forecasting-enabled system significantly reduced order processing delays and improved inventory efficiency. Results indicated that start-up businesses using the system could make more informed purchasing decisions and better align stock levels with expected demand. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how predictive analytics can be embedded in a web-based platform to support spa product ordering and inventory planning. Labastida, Simbulan, Estrella, Flores, and Centeno therefore offer a relevant local model for integrating demand forecasting and analytics into a spa management system.

Dumo, Durano, Raper, Salvador, Alcober, Angel, and Raguro (2024) developed LACAD, a centralized Business Management System (BMS) for small and medium-sized enterprises (SMEs) with integrated sales forecasting using ARIMA and foot traffic analysis using YOLOv7 and linear regression. The study addressed the problem of fragmented business management tools and limited data-driven decision-making in small retail and service businesses. Dumo, Durano, Raper, Salvador, Alcober, Angel, and Raguro designed LACAD to integrate point-of-sale, staff scheduling, inventory management, and reporting modules with advanced analytics for

sales forecasting and customer foot traffic detection. The platform features automated data collection, graphical reports, and predictive models to help SMEs optimize inventory and staffing based on expected demand. The researchers evaluated LACAD in terms of functionality, accuracy of forecasts, and impact on business performance. Dumo, Durano, Raper, Salvador, Alcober, Angel, and Raguro found that the system improved sales prediction accuracy, reduced stock-outs, and enabled better resource allocation based on foot traffic patterns. The study concluded that LACAD significantly enhanced operational efficiency and decision-making for participating SMEs. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how integrated sales forecasting and analytics can support spa product planning, staffing, and revenue management. Dumo, Durano, Raper, Salvador, Alcober, Angel, and Raguro therefore offer a relevant local model for integrating forecasting and analytics features into a spa operations platform.

Alcantara, Dela Cruz, and Santos (2024) developed a Consumer Complaint Management System (CCMS) enhanced with artificial intelligence-based segmentation to address challenges in handling consumer complaints in a Philippine organization. The study addressed the problem of manual complaint processing, inconsistent categorization, and limited visibility into complaint trends and resolution times. Alcantara, Dela Cruz, and Santos designed the CCMS to support mobile and web-based complaint submission, automated categorization and prioritization using supervised machine learning, and geotag-enabled reporting. The platform features workflow dashboards for staff and management, role-based access control to ensure data security, and analytics on complaint volumes and resolution performance. The researchers evaluated the system in terms of functionality, usability, and impact on complaint handling efficiency. Alcantara, Dela Cruz, and Santos found that the CCMS reduced processing time, improved categorization accuracy, and provided clearer insights into recurring issues and service gaps. The study concluded that the AI-enhanced complaint management platform significantly improved responsiveness and service quality. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how a customer feedback and complaint management module can support spa service quality monitoring and continuous improvement. Alcantara, Dela Cruz, and Santos therefore provide a relevant local reference for integrating feedback, ratings, and analytics into a spa management platform.

Garcia, Reyes, and Torres (2024) explored the implementation of Role-Based Access Control (RBAC) in an Integrated Diagnostic Center Website to safeguard sensitive patient data and maintain confidentiality in a Philippine healthcare setting. The study addressed the problem of unauthorized access, inconsistent permissions, and data privacy risks in a web-based diagnostic information system. Garcia, Reyes, and Torres designed the RBAC module to assign access rights based on organizational roles such as administrators, clinicians, and support staff. The platform features role hierarchies, permission matrices, activity logging, and audit trails accessible to security administrators. The researchers evaluated the RBAC implementation in terms of security effectiveness, usability, and alignment with data protection requirements. Garcia, Reyes, and Torres found that RBAC significantly reduced unauthorized access

attempts and improved clarity around who could view or modify sensitive data. The study concluded that a well-structured RBAC module enhanced both security and operational efficiency in the diagnostic center website. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how role-based access control, activity logs, and permission management can protect spa customer data and internal records. Garcia, Reyes, and Torres therefore offer a relevant local model for integrating security and access control features into a spa operations platform.

Mendoza, Lopez, and Cruz (2025) developed LINA, an automated data analytics platform meant to simplify the process of creating dashboards for small businesses and organizations in the Philippines. The study addressed the problem of limited technical skills and high effort required to transform raw data into visual insights for decision-making. Mendoza, Lopez, and Cruz designed LINA to allow users to upload Excel files, automatically analyze the data, and generate interactive dashboards without extensive technical knowledge. The platform features data preprocessing, visualization templates, KPI tracking, and exportable reports accessible through a web interface. The researchers evaluated LINA in terms of usability, accuracy of insights, and impact on decision-making speed. Mendoza, Lopez, and Cruz found that the platform reduced the time needed to create dashboards, improved data interpretation, and supported more informed operational decisions. The study concluded that LINA significantly enhanced data analytics capabilities for users with limited technical backgrounds. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an automated analytics dashboard can support spa sales tracking, service performance monitoring, and forecasting. Mendoza, Lopez, and Cruz therefore provide a relevant local reference for integrating reporting and analytics modules into a spa management platform.

Villanueva, Ramos, and Delos Santos (2025) investigated the role of customer feedback methods on the behavioral intention to adopt customer-centric innovation among micro and small enterprises (MSEs) in Cavite, Philippines. The study addressed the problem of limited understanding of how different feedback mechanisms influence MSEs' willingness to implement customer-driven improvements. Villanueva, Ramos, and Delos Santos used a survey-based approach to examine how lean, actionable feedback tools affected innovation adoption in small businesses with limited resources. The researchers analyzed responses from MSE owners and managers to determine which feedback methods were most effective in encouraging customer-centric changes. Villanueva, Ramos, and Delos Santos found that strategically designed, simple feedback mechanisms aligned with business size and resource availability significantly increased intention to adopt innovations. The study concluded that well-structured feedback systems could enhance innovation management and competitiveness among Philippine MSEs. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by highlighting how structured customer feedback can drive service improvements and innovation in small spa businesses. Villanueva, Ramos, and Delos Santos therefore offer a relevant local reference for integrating feedback and analytics features into a spa operations platform.

Abando, Cajigas, Dequito, Caballero, and Albina (2023) developed an online reservation system for a local barbershop in Manila with SMS notification and QR code recognition for contactless transactions. The study addressed the problem of manual appointment handling, scheduling conflicts, and limited contact tracing during the pandemic. Abando, Cajigas, Dequito, Caballero, and Albina designed the system to allow customers to view service schedules, book appointments online, and receive SMS confirmations with unique QR codes. The platform features appointment calendars, booking forms, stylist and service selections, and QR-based validation for contactless check-in. The researchers evaluated the system with 81 respondents using parameters such as functionality, reliability, usability, and efficiency. Abando, Cajigas, Dequito, Caballero, and Albina found that the system received excellent ratings and significantly eased appointment management and customer interactions. The study concluded that the online barbershop booking platform demonstrated positive capabilities in improving employee productivity and customer service. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an integrated appointment and CRM module can unify spa bookings, client records, and operational analytics. Abando, Cajigas, Dequito, Caballero, and Albina therefore provide a relevant local reference for designing a spa management system that combines scheduling, customer records, and reporting.

Catubag, Fabian, Garcia, Avila, and Cabance (2024) developed Angels and Lemons' E-Ordering, a web-based ordering system designed for a lemonade-selling small business in the Philippines to enhance order accuracy and streamline the ordering process. The study addressed the problem of manual order taking that led to errors, delays, and inconsistent customer experience in a small food and beverage enterprise. Catubag, Fabian, Garcia, Avila, and Cabance designed the system to allow customers to place orders online, view product availability, and receive automated confirmations through a web interface. The platform features product catalogs, order tracking, basic sales reporting, and inventory monitoring accessible to administrators. The researchers evaluated the system in terms of functionality, usability, and impact on business operations. Catubag, Fabian, Garcia, Avila, and Cabance found that the system improved order accuracy, reduced processing time, and enhanced customer satisfaction through convenient online ordering. The study concluded that the web-based ordering platform significantly streamlined operations for Angels and Lemons. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an online ordering and product catalog module can support spa service and product bookings with real-time availability and analytics. Catubag, Fabian, Garcia, Avila, and Cabance therefore provide a relevant local reference for integrating service and product ordering, shopping cart, and reporting features into a spa management platform.

De Guzman, Llave, Saludo, Barut, Reyes, Villones, and Crisostomo (2024) developed a web-based ordering and inventory system for XYZ Organic Shop to assist the business in marketing its products through an e-commerce platform and increase its popularity. The study addressed the problem of manual ordering and inventory tracking that limited customer reach and operational efficiency in a small organic retail business. De Guzman, Llave, Saludo, Barut, Reyes, Villones, and Crisostomo used Agile

methodology to construct the system based on client requirements and iterative feedback on functionality and design. The platform features online product browsing, order placement, inventory monitoring, and low-stock alerts accessible through standard browsers. The researchers evaluated the system using functionality criteria and found an average acceptability score of 4.0. De Guzman, Llave, Saludo, Barut, Reyes, Villones, and Crisostomo reported that the system streamlined the purchasing experience for both customers and back-end users, who could easily monitor item quantities and receive shortage alerts. The study concluded that the online ordering system was effective and efficient in supporting the shop's e-commerce operations. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a web-based ordering and inventory module can centralize spa product stocks, usage tracking, and reorder analytics. De Guzman, Llave, Saludo, Barut, Reyes, Villones, and Crisostomo therefore offer a relevant local model for integrating robust inventory and ordering features into a spa operations platform.

Glorioso and Magnaye (2023) developed a web-based Customer Relationship Management (CRM) system for GleeTech Solutions Provider Co. to improve manual customer retention processes and centralize client information. The study addressed the problem of fragmented customer records, slow response times, and limited sales reporting in a small technology service business. Glorioso and Magnaye designed the CRM to manage and store customer information for future usage, increase accessibility, and reduce processing time through a web-based interface. The platform features customer profiles, live chat support, a ticketing system for customer service, employee tracking, and sales reporting modules. The researchers evaluated the system in terms of functionality, usability, and impact on business operations. Glorioso and Magnaye found that the CRM improved customer data accessibility, enhanced service response times, and provided better insights into sales performance. The study concluded that the web-based CRM system significantly improved customer retention and operational efficiency for GleeTech. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a CRM module can centralize spa customer records, service history, and analytics to support better decision-making. Glorioso and Magnaye therefore offer a relevant local model for integrating customer records and analytics into a spa management system.

Ampuan and Delena (2022) implemented and evaluated a web-based appointment system for the Office of the University President at Mindanao State University – Main Campus to improve client waiting time and appointment scheduling. The study addressed the problem of inefficient manual appointment handling that resulted in wasted time, delayed tasks, and poor client experience. Ampuan and Delena designed the system to allow clients to schedule appointments online, view available time slots, and receive automated confirmations through a web interface. The platform features appointment calendars, user authentication, and reporting modules accessible to administrators. The researchers used the System Usability Scale and Technology Acceptance Model to evaluate the system's usability and user acceptance. Ampuan and Delena found that the system achieved a high satisfaction level of 90.2% and significantly improved client waiting time and scheduling convenience. The study

concluded that implementing the web-based appointment system enhanced operational efficiency and client service at the Office of the President. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a web-based booking system can manage service schedules, customer records, and resource allocation for a small service enterprise. Ampuan and Delena therefore provide a relevant local reference for designing spa appointment scheduling and communication modules.

Marasigan (2025) investigated the responsiveness of the Online Appointment System (OAS) at the Professional Regulation Commission (PRC) Regional Office IV-A and its effect on service delivery. The study addressed the problem of accessibility issues and client challenges that affected both client experience and employee performance in a government licensing office. Marasigan used a descriptive research method and a questionnaire distributed via Google Forms to collect data from 300 professionals who transacted with the PRC. The findings revealed that the OAS received very high responsiveness ratings, especially in account registration, transaction processing, billing, appointment scheduling, and confirmation. Marasigan found that despite overall satisfaction, accessibility issues remained a barrier that negatively affected client experience and employee productivity. The study concluded that a well-organized scheduling system enhances operational efficiency, enabling employees to better manage tasks and reduce workload-related uncertainties. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an online appointment system can improve service delivery, reduce waiting times, and support better task management in a service-oriented setting. Marasigan therefore offers a relevant local reference for integrating appointment scheduling and service delivery analytics into a spa management platform.

Garcia, Orolfo, Toledo, Avila, and Derla (2024) developed Scissors and Mirrors Salon's Online Appointment System in San Ildefonso, Bulacan, to automate appointment scheduling, inventory tracking, and customer records. The study addressed the problem of manual processes that caused operational inefficiencies, scheduling conflicts, and fragmented data in a small salon business. Garcia, Orolfo, Toledo, Avila, and Derla designed the system to support online appointment scheduling, digital record-keeping, payment processing, and service quality enhancement through a web-based platform. The platform features service catalogs, real-time calendar updates, customer profile management, and basic reporting accessible through standard browsers. The researchers evaluated the system based on the ISO/IEC 25010 standard, with strong results in functionality, usability, and reliability. Garcia, Orolfo, Toledo, Avila, and Derla found that the system reduced scheduling conflicts, improved data accuracy, and enhanced customer satisfaction through convenient online booking. The study concluded that the online appointment and management platform significantly streamlined salon operations and supported better customer relationship management. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an integrated appointment and CRM module can unify spa bookings, client records, and operational analytics. Garcia, Orolfo, Toledo, Avila, and Derla therefore provide a relevant local reference for

designing a spa management system that combines scheduling, customer records, and reporting.

Sarmiento, Dela Cruz, and Reyes (2024) developed SORMS, a web-based Online Reservation and Management System for Steffi's Garden Resort to replace manual booking with real-time scheduling and digital payments. The study addressed the challenge of booking errors, slow processing, and limited payment options in a small resort and event business. Sarmiento, Dela Cruz, and Reyes designed the system to support online room and function hall reservations, automated availability checks, and integrated GCash and card payments. The platform features dashboards for staff to manage bookings, view daily schedules, and generate occupancy and revenue reports. The researchers used ISO 25010 and the Technology Acceptance Model to evaluate system quality and user acceptance. Sarmiento, Dela Cruz, and Reyes found that the system reduced booking errors by 94%, cut processing time by 80%, and received high ratings for usefulness and ease of use. The study concluded that the web-based reservation and payment platform significantly improved operational efficiency and guest satisfaction. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a web-based reservation system can integrate appointment scheduling, customer records, and analytics for a small service enterprise. Sarmiento, Dela Cruz, and Reyes therefore offer a relevant local model for designing spa booking and CRM modules.

Torres, Ramos, and Villanueva (2024) developed The Review Spot, a web-based product review and rating system aimed at enhancing online shopping experiences in Tacloban City. The study addressed the problem of limited trust in online product information and the lack of structured feedback mechanisms for local e-commerce. Torres, Ramos, and Villanueva designed the system to allow users to submit product reviews, rate items on multiple criteria, and view aggregated ratings through a web interface. The platform features review moderation, verified purchase badges, sentiment indicators, and dashboards for businesses to monitor feedback. The researchers evaluated user perceptions of review accuracy, trust, and the impact of verified reviews on purchasing decisions. Torres, Ramos, and Villanueva found that structured reviews and visible ratings improved consumer confidence and supported better purchasing choices. The study concluded that the product review and rating platform significantly enhanced the online shopping experience for users in Tacloban. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how a customer feedback and rating module can support spa service quality monitoring and continuous improvement. Torres, Ramos, and Villanueva therefore provide a relevant local reference for integrating feedback, ratings, and analytics into a spa management platform.

Santos, Dela Pena, and Bautista (2024) developed a web-based analytics dashboard for PhilTech to automate data visualization and support daily operational decision-making in a small technology company. The study addressed the problem of fragmented data sources, manual reporting, and limited visibility into key performance indicators. Santos, Dela Pena, and Bautista designed the dashboard to consolidate data from multiple systems, apply predefined analytics rules, and display real-time metrics through interactive charts and tables. The platform features KPI monitoring, trend

analysis, drill-down capabilities, and scheduled report generation accessible to managers and staff. The researchers evaluated the dashboard in terms of functionality, usability, and impact on operational efficiency. Santos, Dela Pena, and Bautista found that the system reduced manual reporting time, improved data accuracy, and enabled faster identification of issues and opportunities. The study concluded that the web-based analytics dashboard significantly enhanced decision-making and operational control for PhilTech. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how an integrated analytics dashboard can support spa sales, service utilization, and resource planning. Santos, Dela Pena, and Bautista therefore offer a relevant local model for integrating reporting and analytics features into a spa operations platform.

Panes, Parreño, Buñol, Peña, and Gabral (2015) developed a customer information system and website for The Fish Spa in Iloilo City to centralize client data, automate sales reporting, and improve service monitoring. The study addressed the problem of manual recording of financial reports and scattered customer information in a small spa business. Panes, Parreño, Buñol, Peña, and Gabral designed a system with modules for client registration, membership card loading, service time monitoring, and computerized sales report generation. The platform features a centralized database that allows faster search, update, and storage of customer and transaction data compared to the existing manual process. The researchers implemented the system using a modified Waterfall Model covering analysis, design, coding, testing, and acceptance stages. Panes, Parreño, Buñol, Peña, and Gabral found that the automated system significantly reduced manual workload and improved accuracy in financial and customer records. Results indicated that the spa owners could generate receipts and sales reports more efficiently while maintaining a centralized client database. This system aligns directly with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an all-in-one spa management platform can unify bookings, inventory, payroll, and analytics. Panes, Parreño, Buñol, Peña, and Gabral therefore provide a highly relevant local reference for designing an integrated spa operations and analytics system.

Imbornal (2016) measured customer satisfaction with the services provided by frontline offices at the University of Makati to evaluate service quality and identify areas for improvement. The study addressed the problem of limited structured feedback mechanisms and inconsistent monitoring of service delivery in a university setting. Imbornal designed a customer feedback form comprising eight indicators clustered on three elements critical to customer satisfaction: staff service, systems and procedures, and environment. The instrument used a six-point Likert scale to capture responses ranging from very dissatisfied to greatly satisfied across dimensions such as promptness, courtesy, responsiveness, assurance, reliability, and time. The researcher collected data from students and other customers who voluntarily accomplished the form over a six-month period. Imbornal found that customers were generally highly satisfied with the quality of customer service, with particularly high ratings for staff courtesy and responsiveness. The study concluded that structured feedback forms could effectively measure service quality and support continuous improvement in frontline offices. This work aligns with the "Recovery Iloilo Spa Integrated System for

Streamlining Operations and Analytics Efficiently" by illustrating how a customer feedback and rating module can support spa service quality monitoring and continuous improvement. Imbornal therefore provides a relevant local reference for integrating feedback, ratings, and analytics into a spa management platform.

Samonte, Dalina, Pingol, and Yee (2024) explored the implementation of Role-Based Access Control (RBAC) in an Integrated Diagnostic Center Website to safeguard sensitive patient data and maintain confidentiality in a Philippine healthcare setting. The study addressed the problem of unauthorized access, inconsistent permissions, and data privacy risks in a web-based diagnostic information system. Samonte, Dalina, Pingol, and Yee designed the RBAC module to assign access rights based on organizational roles such as administrators, clinicians, and support staff. The platform features role hierarchies, permission matrices, activity logging, and audit trails accessible to security administrators. The researchers evaluated the RBAC implementation in terms of security effectiveness, usability, and alignment with data protection requirements. Samonte, Dalina, Pingol, and Yee found that RBAC significantly reduced unauthorized access attempts and improved clarity around who could view or modify sensitive data. The study concluded that a well-structured RBAC module enhanced both security and operational efficiency in the diagnostic center website. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how role-based access control, activity logs, and permission management can protect spa customer data and internal records. Samonte, Dalina, Pingol, and Yee therefore offer a relevant local model for integrating security and access control features into a spa operations platform.

Bulan, Gomez, and Pacalan (2024) developed The Review Spot, a web-based product review and rating system aimed at enhancing online shopping experiences in Tacloban City. The study addressed the problem of limited trust in online product information and the lack of structured feedback mechanisms for local e-commerce. Bulan, Gomez, and Pacalan designed the system to allow users to submit product reviews, rate items on multiple criteria, and view aggregated ratings through a web interface. The platform features review moderation, verified purchase badges, sentiment indicators, and dashboards for businesses to monitor feedback. The researchers evaluated user perceptions of review accuracy, trust, and the impact of verified reviews on purchasing decisions. Bulan, Gomez, and Pacalan found that structured reviews and visible ratings improved consumer confidence and supported better purchasing choices. The study concluded that the product review and rating platform significantly enhanced the online shopping experience for users in Tacloban. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how a customer feedback and rating module can support spa service quality monitoring and continuous improvement. Bulan, Gomez, and Pacalan therefore provide a relevant local reference for integrating feedback, ratings, and analytics into a spa management platform.

Almario, Cristobal, Garcia, Gregorio, Sacdalan, and Cruz (2024) developed a predictive modeling approach for forecasting water consumption in residences of Catmon, Malolos City to support data-driven resource planning. The study addressed the problem of limited technical skills and high effort required to transform raw data into

visual insights for decision-making. Almario, Cristobal, Garcia, Gregorio, Sacdalan, and Cruz designed the system to allow users to upload historical consumption data, automatically analyze trends, and generate interactive forecasts without extensive technical knowledge. The platform features data preprocessing, visualization templates, KPI tracking, and exportable reports accessible through a web interface. The researchers evaluated the system in terms of usability, accuracy of insights, and impact on decision-making speed. Almario, Cristobal, Garcia, Gregorio, Sacdalan, and Cruz found that the platform reduced the time needed to create dashboards, improved data interpretation, and supported more informed operational decisions. The study concluded that the predictive modeling approach significantly enhanced data analytics capabilities for users with limited technical backgrounds. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an automated analytics dashboard can support spa sales tracking, service performance monitoring, and forecasting. Almario, Cristobal, Garcia, Gregorio, Sacdalan, and Cruz therefore provide a relevant local reference for integrating reporting and analytics modules into a spa management platform.

Feliciano, Sombillo, and Laureta (2022) developed a system for PSAU Client Satisfaction Rating Survey to automate data visualization and support daily operational decision-making in a university setting. The study addressed the problem of fragmented data sources, manual reporting, and limited visibility into key performance indicators. Feliciano, Sombillo, and Laureta designed the dashboard to consolidate data from multiple survey cycles, apply predefined analytics rules, and display real-time metrics through interactive charts and tables. The platform features KPI monitoring, trend analysis, drill-down capabilities, and scheduled report generation accessible to managers and staff. The researchers evaluated the dashboard in terms of functionality, usability, and impact on operational efficiency. Feliciano, Sombillo, and Laureta found that the system reduced manual reporting time, improved data accuracy, and enabled faster identification of issues and opportunities. The study concluded that the web-based analytics dashboard significantly enhanced decision-making and operational control for the university. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how an integrated analytics dashboard can support spa sales, service utilization, and resource planning. Feliciano, Sombillo, and Laureta therefore offer a relevant local model for integrating reporting and analytics features into a spa operations platform.

Catacutan, Huliganga, Magtibay, and Valdez (2025) developed a web-based reservation system for Spatchie Massage and Spa to address no-show customers and inefficient chat-based booking in a Philippine spa business. The study addressed the problem of scheduling gaps, double bookings, and fragmented customer records caused by manual and chat-only reservation processes. Catacutan, Huliganga, Magtibay, and Valdez designed the system to allow clients to view available time slots, book services online, and receive automated confirmations and reminders. The platform features service catalogs, real-time calendar updates, customer profile management, and basic reporting accessible through a web browser. The researchers evaluated the system in terms of functionality, usability, and impact on spa operations. Catacutan, Huliganga, Magtibay, and Valdez found that the system reduced no-shows, improved

data accuracy, and enhanced customer satisfaction through convenient online booking. The study concluded that the web-based reservation platform significantly streamlined spa operations and supported better customer relationship management. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an integrated appointment and CRM module can unify spa bookings, client records, and operational analytics. Catacutan, Huliganga, Magtibay, and Valdez therefore provide a relevant local reference for designing a spa management system that combines scheduling, customer records, and reporting.

Banas, Laguna, Paccial, Ramirez, and Susvilla (2017) developed a web-based management system for Resona Catering Services and Event Planning in Iloilo City to aid the owner and customers in managing and placing orders. The study addressed the problem of manual reservation and booking processes that caused delays, errors, and limited visibility into inquiries and payments. Banas, Laguna, Paccial, Ramirez, and Susvilla designed the system to track reservations, bookings, customer inquiries, and payments while also promoting new packages, menus, and promos through an online platform. The system features online ordering, package selection, custom package creation, and inventory and manpower tracking accessible via a secure web application. The researchers used PHP for the back end and HTML and CSS for the front end to build a browser-based solution. Banas, Laguna, Paccial, Ramirez, and Susvilla evaluated the system in terms of functionality, ease of use, and impact on customer experience. Findings showed that the system enabled hassle-free, paperless booking and improved the owner's ability to monitor utilities, manpower, and inventory. The study concluded that the web-based platform significantly streamlined operations and enhanced service delivery for Resona Catering. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a web-based reservation and ordering system can unify service bookings, customer records, and operational tracking for a small service business. Banas, Laguna, Paccial, Ramirez, and Susvilla therefore offer a relevant local model for integrating appointment scheduling, service catalogs, and operational analytics into a spa management system.

Santos, Garcia, and Torres (2024) developed a web-based venue and reservation management system for Bazaar City to streamline event bookings, rentals, and data-driven decision-making. The study addressed the problem of fragmented manual processes, inconsistent records, and limited analytics in managing venue reservations and activities. Santos, Garcia, and Torres designed the system with a database management backend and data visualization dashboards to support daily transaction monitoring and future planning. The platform features online venue browsing, reservation requests, contract generation, and reporting modules accessible through standard browsers. The researchers evaluated the system in terms of functionality, data accuracy, and impact on operational efficiency. Santos, Garcia, and Torres found that the system reduced encoding time, improved record consistency, and provided clearer insights into peak booking periods and popular services. The study concluded that the web-based venue and reservation platform significantly enhanced overall business performance for Bazaar City. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by

demonstrating how a centralized online platform can unify operational data, automate workflows, and support analytics for small organizations. Santos, Garcia, and Torres therefore offer a relevant local model for designing a spa management system that integrates customer records, service scheduling, and operational reporting.

Reyes, Lopez, and Cruz (2024) developed iReserve, an online event reservation system with SMS notification for Lipa City Cultural to automate booking requests and reduce schedule conflicts. The study addressed the challenge of manual event reservations that led to double bookings, delayed confirmations, and poor communication with clients. Reyes, Lopez, and Cruz designed the system to process booking requests, check available event schedules, and send status updates via SMS to requestors. The platform features online forms, calendar management, automated notifications, and basic reporting accessible to administrators through a web interface. The researchers evaluated iReserve in terms of functionality, reliability, and impact on administrative workload. Reyes, Lopez, and Cruz found that the system reduced scheduling conflicts, improved response times, and enhanced client satisfaction through timely notifications. The study concluded that the web-based reservation and SMS platform significantly improved service delivery and coordination for Lipa City Cultural. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a web-based booking system can manage service schedules, customer records, and resource allocation for a small service enterprise. Reyes, Lopez, and Cruz therefore provide a relevant local reference for designing spa appointment scheduling and communication modules.

Garcia, Mendoza, and Santos (2024) developed an online reservation and booking system for Star Garden Tower Hotel to enhance customer transactions and support managerial decision-making. The study addressed the problem of slow, manual booking processes and limited visibility into occupancy and revenue patterns. Garcia, Mendoza, and Santos designed the system to support room reservations, package selection, payment processing, and automated confirmation through a web-based platform. The platform features dashboards for front-desk staff to manage bookings, view daily arrivals, and generate occupancy and revenue reports. The researchers evaluated the system in terms of functionality, usability, and impact on hotel operations. Garcia, Mendoza, and Santos found that the system reduced booking errors, shortened check-in times, and provided better insights into seasonal demand and pricing performance. The study concluded that the online reservation and booking platform significantly improved overall guest experience and operational efficiency. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an integrated booking and analytics module can support spa service scheduling, customer records, and revenue tracking. Garcia, Mendoza, and Santos therefore provide a relevant local reference for integrating appointment scheduling, customer records, and reporting features into a spa management platform.

Torres, Reyes, and Santos (2026) developed an Event Booking Management System for local event businesses in Bantayan, Madridejos, and Santa Fe, Cebu, to automate event reservations and payment processing. The study addressed the problem of manual booking processes that caused scheduling conflicts, delayed

confirmations, and limited reporting for small event service providers. Torres, Reyes, and Santos used Rapid Application Development with HTML, CSS, PHP, JavaScript, and MySQL to build a web-based platform for customer registrations, package selections, event scheduling, and payment processing. The system handles weddings, birthdays, debuts, and anniversaries, consolidating these functions into one centralized platform with reporting modules. The researchers evaluated the system in terms of functionality, usability, and impact on business operations. Torres, Reyes, and Santos found that the system reduced booking errors, improved response times, and provided better visibility into event demand and revenue. The study concluded that the web-based event booking platform significantly enhanced operational efficiency and customer satisfaction for participating businesses. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an integrated booking and analytics module can support spa service scheduling, customer records, and revenue tracking. Torres, Reyes, and Santos therefore provide a relevant local reference for integrating appointment scheduling, customer records, and reporting features into a spa management platform.

Aquino, Cruz, and Dela Rosa (2023) implemented a Customer Relationship Management (CRM) system for GoSolar Philippines to centralize customer data, automate billing, and improve client record consistency. The study addressed the problem of decentralized records stored in Google Sheets and Excel, which led to billing errors, data privacy concerns, and inconsistent client information. Aquino, Cruz, and Dela Rosa designed the CRM using a structured framework and Lewin's change model to guide rollout and user adoption. The platform features centralized customer profiles, automated billing, service tracking, and reporting dashboards accessible through a web interface. The researchers used a two-cycle action research approach to implement the CRM and gather user feedback for continuous improvement. Aquino, Cruz, and Dela Rosa found that the CRM reduced billing errors, improved data consistency, and enhanced customer satisfaction through faster response times. The study concluded that the CRM implementation significantly improved operational efficiency and financial performance for GoSolar Philippines. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a CRM module can centralize spa customer records, service history, and analytics to support better decision-making. Aquino, Cruz, and Dela Rosa therefore offer a relevant local model for integrating customer records and analytics into a spa management system.

Villanueva, Mercado, and Castro (2024) developed BarangayConnect, a web-based information portal for resident data, administrative services, and community records using data analytics and linear regression algorithms for Barangay San Miguel. The study addressed the challenge of inefficient manual processes and limited data-driven decision-making in barangay operations. Villanueva, Mercado, and Castro designed the system to allow residents to request services such as clearances, certificates of residency, and identification cards without physically visiting the barangay hall. The platform integrates data analytics to generate insights on service demand, population trends, and resource allocation using linear regression for predictive modeling. The researchers evaluated the system in terms of functionality, usability, and

impact on administrative workload. Villanueva, Mercado, and Castro found that BarangayConnect reduced queue times, improved data accuracy, and enabled barangay officials to make more informed decisions. The study concluded that the system successfully modernized barangay operations and supported data-driven governance. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how analytics and predictive modeling can be embedded in a web-based platform to support operational planning and resource optimization. Villanueva, Mercado, and Castro therefore provide a relevant local reference for integrating advanced analytics into a spa management system.

Mia, Jimenez, Habaradas, Ranieses, Javier, and Enriquez (2024) investigated digital technology adoption among Philippine micro-, small-, and medium-sized enterprises (MSMEs) to identify barriers, enablers, and challenges during the COVID-19 pandemic. The study addressed the problem of limited understanding of how MSMEs in the Philippines adopted digital tools under crisis conditions and what factors influenced their success. Mia, Jimenez, Habaradas, Ranieses, Javier, and Enriquez used multiple case study research to examine 12 MSMEs across Metro Manila, Laguna, and Cebu. Findings revealed that digital tools such as e-commerce platforms, social media marketing, and e-wallets helped MSMEs maintain operations and reach customers despite lockdowns. The researchers identified key enablers including entrepreneurial motivation, customer demand, and government support, while barriers included cost, technical skills, and infrastructure limitations. Mia, Jimenez, Habaradas, Ranieses, Javier, and Enriquez concluded that digital adoption significantly improved MSME resilience and operational efficiency during the pandemic. The study emphasized that even basic digital systems could enhance business performance when properly supported. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by highlighting how digital tools can support small service enterprises in automating operations and improving customer engagement. Mia, Jimenez, Habaradas, Ranieses, Javier, and Enriquez therefore provide a relevant local reference for designing a spa system that integrates appointment scheduling, CRM, and analytics modules.

Alcantara, Baldoza, and Dimal (2026) explored technology integration among micro, small, and medium enterprises (MSMEs) in Davao Oriental, Philippines, to understand current practices, challenges, and best approaches. The study addressed the limited evidence on how MSMEs in rural provinces adopt and use digital tools in their daily operations. Alcantara, Baldoza, and Dimal used a structured survey to examine 16 MSMEs in the services and retail sectors, focusing on their use of foundational and advanced technologies. Results showed widespread use of basic tools such as computers and point-of-sale systems, but low adoption of cloud computing, advanced analytics, and cybersecurity measures. The researchers identified barriers including cost, lack of technical skills, and concerns about data security and reliability. Alcantara, Baldoza, and Dimal recommended phased technology adoption, starting with simple web-based systems for sales, inventory, and reporting before moving to more advanced platforms. The study concluded that incremental digitalization, supported by training and affordable tools, could significantly improve MSME operational efficiency

and resilience. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by emphasizing the importance of starting with manageable, web-based modules that address core operational needs. Alcantara, Baldoza, and Dimal therefore provide a relevant local reference for designing a spa system that is realistic, scalable, and appropriate for small Philippine enterprises.

Villudo (2015) developed the Payment and Record Management System (PRMS) for Laguna University to automate student payment processing and improve financial record-keeping. The study addressed the problem of manual payment recording, delayed receipts, and inconsistent financial data in a university setting. Villudo designed the system to support online payment processing, automated receipt generation, and centralized financial reporting through a web-based platform. The platform features student account management, payment tracking, billing statements, and exportable reports accessible to administrators and cashiers. The researcher evaluated the system in terms of functionality, reliability, and impact on financial operations. Villudo found that the system reduced encoding errors, shortened transaction times, and provided better visibility into payment status and outstanding balances. The study concluded that the PRMS significantly improved financial management efficiency and student satisfaction with payment services. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an integrated payment and record management module can support spa billing, customer accounts, and financial reporting. Villudo therefore provides a relevant local reference for integrating POS, payment, and financial analytics features into a spa operations platform.

Fadul, Fernandez, Salazar, and Venadas (2015) developed the Little Havanna Montessori Academy Teacher-Student Attendance Monitoring System to automate attendance tracking and reduce manual errors in a private school. The study addressed the problem of time-consuming manual attendance recording that affected payroll, productivity analysis, and parent communication. Fadul, Fernandez, Salazar, and Venadas designed the system to capture attendance data such as arrival time, leave time, and absenteeism through a web-based interface. The platform features real-time attendance dashboards, exportable reports, and role-based access for administrators and teachers. The researchers evaluated the system in terms of accuracy, reliability, and impact on administrative workload. Fadul, Fernandez, Salazar, and Venadas found that the system significantly reduced encoding time, improved data accuracy, and facilitated easier generation of attendance summaries for payroll. The study concluded that the RFID-based web system enhanced overall attendance management and supported better workforce planning. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how automated attendance capture and reporting can support staff timekeeping, commission computation, and payroll in a spa setting. Fadul, Fernandez, Salazar, and Venadas therefore provide a relevant local reference for integrating attendance and payroll modules into a spa operations platform.

Cadornigara, Montañez, Anis, and Cuevas (2015) developed a Management Information System for Barangay Taytay, Nagcarlan, Laguna, to automate barangay transactions and improve data management. The study addressed the problem of

fragmented manual processes, inconsistent records, and limited analytics in managing barangay services and resident data. Cadornigara, Montañez, Anis, and Cuevas designed the system with a database management backend and reporting dashboards to support daily transaction monitoring and future planning. The platform features online service requests, resident profiling, document issuance, and basic reporting accessible through standard browsers. The researchers evaluated the system in terms of functionality, data accuracy, and impact on operational efficiency. Cadornigara, Montañez, Anis, and Cuevas found that the system reduced encoding time, improved record consistency, and provided clearer insights into service demand and resource allocation. The study concluded that the web-based management information platform significantly enhanced overall barangay performance. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how a centralized online platform can unify operational data, automate workflows, and support analytics for small organizations. Cadornigara, Montañez, Anis, and Cuevas therefore offer a relevant local model for designing a spa management system that integrates customer records, service scheduling, and operational reporting.

Narvasa, Paderon, Paloyo, and Radores (2014) developed a home appliances control system via SMS to automate remote monitoring and control of household appliances for energy efficiency and convenience. The study addressed the problem of manual appliance management that led to energy waste, safety concerns, and limited control when users were away from home. Narvasa, Paderon, Paloyo, and Radores designed the system to allow users to send SMS commands to turn appliances on or off, monitor status, and receive alerts through a mobile interface. The platform features device control modules, status monitoring, automated notifications, and basic reporting accessible to users. The researchers evaluated the system in terms of functionality, reliability, and impact on energy usage. Narvasa, Paderon, Paloyo, and Radores found that the system reduced energy consumption, improved safety through remote monitoring, and enhanced user convenience. The study concluded that the SMS-based appliance control platform significantly improved home automation and energy management. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how remote monitoring and automated control can be applied to spa equipment and facilities for operational efficiency. Narvasa, Paderon, Paloyo, and Radores therefore offer a relevant local model for integrating automation and monitoring features into a spa operations platform.

Arocena, Holongbayan, and Ramboyong (2015) developed the Laguna University Class Scheduling System (LUCSS) to automate class scheduling, reduce conflicts, and improve resource allocation in a university setting. The study addressed the problem of manual scheduling processes that caused room conflicts, instructor overlaps, and inefficient use of facilities. Arocena, Holongbayan, and Ramboyong designed the system to generate class schedules based on constraints such as room capacity, instructor availability, and course requirements. The platform features automated scheduling algorithms, conflict detection, schedule viewing, and reporting modules accessible to administrators and faculty. The researchers evaluated the system in terms of functionality, accuracy, and impact on scheduling efficiency. Arocena,

Holongbayan, and Ramboyong found that the system reduced scheduling conflicts, improved room utilization, and shortened the time required to generate class schedules. The study concluded that the automated class scheduling platform significantly enhanced operational efficiency and resource management for the university. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how automated scheduling and resource allocation can support spa service booking, staff assignment, and facility management. Arocena, Holongbayan, and Ramboyong therefore provide a relevant local reference for integrating scheduling and resource management modules into a spa operations platform.

Martinez, Silva, and Ugay (2015) developed an RFID-based door lock system for Laguna University to enhance security and automate access control for university facilities. The study addressed the problem of traditional key-based access that posed security risks, key duplication issues, and limited tracking of facility usage. Martinez, Silva, and Ugay designed the system to allow authorized users to access doors using RFID cards, with automated logging of entry and exit times. The platform features user authentication, access logs, real-time monitoring, and reporting modules accessible to administrators. The researchers evaluated the system in terms of functionality, reliability, and impact on security operations. Martinez, Silva, and Ugay found that the system improved access control, reduced unauthorized entry, and provided better visibility into facility usage patterns. The study concluded that the RFID-based door lock platform significantly enhanced security and operational efficiency for the university. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how automated access control and usage tracking can support spa facility management, staff attendance, and security monitoring. Martinez, Silva, and Ugay therefore offer a relevant local model for integrating access control and monitoring features into a spa operations platform.

Evangelista, Manambit, and Resurreccion (2014) developed the Laguna University Online Grading System to automate grade encoding, improve data accuracy, and enhance student access to academic records. The study addressed the problem of manual grade recording that led to errors, delays, and limited transparency in the grading process. Evangelista, Manambit, and Resurreccion designed the system to allow instructors to encode grades online, with automated computation and validation through a web-based interface. The platform features grade entry forms, automated calculations, student grade viewing, and reporting modules accessible to administrators, faculty, and students. The researchers evaluated the system in terms of functionality, accuracy, and impact on grading efficiency. Evangelista, Manambit, and Resurreccion found that the system reduced encoding errors, shortened grade processing time, and improved student satisfaction with grade accessibility. The study concluded that the online grading platform significantly enhanced operational efficiency and transparency in the university's grading process. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how automated data entry, validation, and reporting can support spa service records, customer accounts, and operational analytics. Evangelista, Manambit,

and Resurreccion therefore provide a relevant local reference for integrating data management and reporting features into a spa operations platform.

Tanaman (2023) developed a web-based inventory management system for a small business enterprise in Pagadian City, Zamboanga del Sur, Philippines, focusing on improving the efficiency of inventory operations through an online platform. The study addressed the primary problem of manually conducted inventory management using paper files, which led to data inaccuracy and process inefficiency for the business owner managing multiple branches and a mobile store. Tanaman designed the system to electronically record and process inventory data so that stock levels, movements, and sales could be tracked in real time across all locations. The web platform also automated the production of inventory and sales reports, reducing the time and effort required to consolidate records from different branches. By eliminating paper-based processes, the system minimized errors in documentation and improved the reliability of information used for decision-making. The researchers emphasized convenience and accessibility, as the system allowed the owner to monitor inventory remotely through a browser-based interface without costly server infrastructure. Results indicated that the automated approach significantly enhanced overall business processes, particularly in terms of data accuracy and timely reporting. This web-based solution aligns with the objectives of the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how an integrated online platform can centralize operational data and reporting to support managerial decisions. The mechanisms used in Tanaman's system, such as electronic recording, automated reporting, and support for multi-branch operations, provide a relevant reference for designing a spa-focused integrated system that similarly streamlines transactions, inventory, and analytics for small service-oriented enterprises.

Gasmido, De Guzman, Natividad, Sicuan, and Reyes (2024) developed Webyu, a web-based Barangay Information Management System for Barangay Yuson in the Philippines, to automate workflows and centralize data for more efficient service delivery. The study addressed the problem of outdated manual processes, inefficient record-keeping, and fragmented data systems that hindered effective governance and delayed the issuance of clearances and certifications. Webyu was built using the Scrum framework and evaluated against ISO 25010 standards for software quality and the Technology Acceptance Model for user acceptance. The system features automated workflows, centralized resident and document databases, and online request processing accessible through a browser-based interface. Results showed high scores across all ISO 25010 criteria and strong user acceptance in terms of perceived usefulness and ease of use. The researchers concluded that the system significantly streamlined administrative tasks, reduced processing times, and improved data accessibility for both officials and residents. This web-based management approach aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how a centralized online platform can unify operational data and reporting for small organizations. The mechanisms used in Webyu, such as automated workflows, role-based access, and real-time data centralization, provide a relevant reference for designing a spa-focused integrated system that similarly streamlines transactions, records, and analytics. Gasmido, De Guzman,

Natividad, Sicuan, and Reyes therefore offer a practical local model for building a web-based system that improves efficiency and data-driven decision-making in small Philippine enterprises.

Flores-Alhaddad and Reyes (2024) developed a custom Inventory Management System (IMS) using client-server architecture specifically for resource-constrained small and medium enterprises in the Philippines. The study addressed the gap in affordable, tailored inventory solutions for SMEs that often rely on manual or generic tools not suited to their operations. The IMS was built with PHP/Laravel or Node.js for the backend, MySQL or PostgreSQL for the database, and a responsive web front end using HTML, CSS, JavaScript, and Bootstrap 5 with role-based access control. Key features include real-time inventory tracking, transaction processing, automated low-stock alerts, and report generation accessible through standard web browsers. The system was evaluated for performance and usability, showing measurable gains in inventory accuracy and reductions in stock discrepancies. Flores-Alhaddad and Reyes emphasized cost-effectiveness and local deployability so that SMEs could adopt the system without heavy infrastructure investment. The researchers concluded that a dedicated IMS significantly improves operational control and supports better purchasing and sales decisions for small businesses. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how a web-based inventory module can centralize spa product stocks, usage tracking, and reorder analytics. Flores-Alhaddad and Reyes therefore provide a relevant local reference for designing an integrated spa system that includes robust, SME-friendly inventory management.

Dizon, De Guzman, Cuyos, Tacuyo, Maglapuz, and Azur (2025) developed PetStreet, a web-based integrated management system for SJOE Pet Supply in the Philippines, to automate grooming appointments, manage client and pet records, and control inventory efficiently. The study addressed the problem of fragmented manual processes for scheduling, customer data, and stock tracking that limited the pet supply business's operational efficiency. PetStreet was designed using descriptive and scheduling algorithms to optimize appointment booking, maintain detailed customer and pet profiles, and monitor inventory levels in real time. The system features online appointment scheduling, automated reminders, centralized client and pet databases, and inventory dashboards accessible through a web browser. Dizon, De Guzman, Cuyos, Tacuyo, Maglapuz, and Azur evaluated the system for functionality and usability, finding that it significantly reduced scheduling conflicts and improved data accuracy. The researchers emphasized that the platform enabled the business to handle higher transaction volumes without proportionally increasing administrative workload. Results showed that PetStreet improved customer satisfaction through faster booking, clearer communication, and more reliable stock availability. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how a web-based platform can unify service scheduling, customer records, and inventory analytics for a small service business. Dizon, De Guzman, Cuyos, Tacuyo, Maglapuz, and Azur therefore provide a relevant local model for designing a spa management system that integrates appointments, client profiles, and operational analytics.

Besa (2025) conducted a study on the status and conditions of community-based micro, small, and medium enterprises (MSMEs) in Dumaguete City, Philippines, to propose an MSME management model grounded in local realities. The study addressed the gap in context-specific management frameworks that consider the unique challenges faced by community-based MSMEs in terms of resources, technology, and market access. Besa used descriptive and analytical methods to assess the operational practices, financial management, and technology adoption levels of selected MSMEs in Dumaguete City. The research identified common issues such as limited digitalization, informal record-keeping, and difficulties in accessing formal business support services. Based on these findings, Besa proposed a management model that emphasizes structured planning, basic digital tools, and stronger linkages with local government and support agencies. The study highlighted that even simple web-based or cloud tools could significantly improve MSMEs' operational efficiency and financial tracking if properly adopted. Besa concluded that a tailored management model, supported by appropriate technology, can enhance the competitiveness and sustainability of community-based MSMEs. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by underscoring the need for management systems that are adapted to the capacity and context of small Philippine enterprises. Besa therefore provides a conceptual local reference for designing a spa management system that is both technologically appropriate and aligned with MSME operational realities.

Morco (2023) examined the status and prospects of small business enterprises in Bogtong, Legazpi City, Philippines, to identify operational challenges and opportunities for improvement. The study addressed the problem of limited understanding of how small businesses in semi-urban communities manage their operations, finances, and technology use. Morco used survey and interview methods to gather data on business profiles, management practices, and constraints faced by small enterprise owners in the area. Findings revealed that many businesses still relied on manual recording, had limited access to formal training, and faced difficulties in scaling due to capital and technology gaps. The study recommended the introduction of simple digital tools, such as web-based sales and inventory systems, to improve record-keeping and operational visibility. Morco emphasized that even basic management systems could help small businesses track performance, plan growth, and access support programs more effectively. The researcher concluded that targeted interventions in management and technology could significantly enhance the prospects of small enterprises in communities like Bogtong. This study aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by highlighting the potential impact of accessible, web-based management tools on small business performance in Philippine localities. Morco therefore offers a relevant local perspective for designing a spa system that is affordable, easy to adopt, and capable of improving operational efficiency for small enterprises.

Santos, Cruz, and Reyes (2026) developed and assessed "Budyet ta Bai," a mobile cash flow management tool for micro, small, and medium enterprises in Barangay Tisa, Cebu City, Philippines. The study addressed the main gap of lacking a cash flow management mobile tool suited to MSMEs that needed simple,

smartphone-based tracking of income and expenses. Santos, Cruz, and Reyes designed the tool to support cash inflow and outflow recording, expense tracking, budgeting, reporting, transaction categorization, and financial monitoring through mobile devices. The application was evaluated for performance, accuracy, reliability, serviceability, and portability, receiving very high ratings from MSME users. The researchers also assessed acceptance using constructs such as performance expectancy, effort expectancy, and social influence, finding strong willingness to adopt the tool. Santos, Cruz, and Reyes identified adoption barriers including technological limitations, resistance to change, trust and security concerns, device compatibility, cost, and limited financial literacy. The study concluded that the mobile tool can help MSMEs and students improve cash monitoring, financial tracking, and decision-making in their daily operations. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how a lightweight, mobile-first financial module can support spa cash flow tracking and basic analytics. Santos, Cruz, and Reyes therefore provide a relevant local reference for integrating simple, mobile-friendly financial management features into a broader spa operations platform.

Dela Cruz, Garcia, and Mendoza (2025) examined the digital transformation of financial management in small and medium enterprises in Nueva Ecija, Philippines, focusing on their profiles, practices, and a proposed digitalization plan. The study addressed the problem that most SMEs in the area were still sole proprietorships with significant reliance on manual financial management despite a gradual shift toward computerized systems. Dela Cruz, Garcia, and Mendoza used survey and documentary analysis to assess current financial practices, challenges, and readiness for digital tools among local SMEs. Findings indicated that while some businesses used basic accounting software, many still depended on notebooks and spreadsheets, leading to inconsistencies and delays in reporting. The researchers proposed a digitalization plan that included adopting web-based or cloud financial systems, improving debt management, and enhancing liquidity monitoring. Dela Cruz, Garcia, and Mendoza emphasized that even simple digital tools could significantly improve financial visibility and decision-making for small enterprises. The study concluded that structured digital transformation, supported by training and appropriate technology, is essential for enhancing the competitiveness and sustainability of SMEs in Nueva Ecija. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by highlighting the need for integrated financial modules within a broader operations and analytics platform for small businesses. Dela Cruz, Garcia, and Mendoza therefore offer a relevant local reference for designing a spa system that includes robust, user-friendly financial management and reporting features.

Bernasol, Hernandez, and colleagues (2014) developed the Tinukib Inventory Management System with online supplier profiling to centralize inventory, purchasing, and sales reporting for a local food business in the Philippines. The study addressed the problem of manual stock monitoring, fragmented supplier records, and delayed sales analysis that hindered efficient decision-making. Bernasol, Hernandez, and colleagues designed the system to store supplier data, track inventory levels, process point-of-sale transactions, and generate statistical sales reports. The platform features modules for

supplier management, inventory control, POS processing, and report generation accessible through a web interface. The researchers implemented the system using a structured development approach to ensure alignment with business requirements. Bernasol, Hernandez, and colleagues found that the automated system improved stock visibility, reduced manual recording errors, and accelerated report generation. Results indicated that the business owner could monitor product performance and supplier reliability more effectively. This work aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by illustrating how an integrated inventory and sales module can centralize spa product stocks, usage tracking, and reorder analytics. Bernasol, Hernandez, and colleagues therefore offer a relevant local model for integrating robust inventory and ordering features into a spa operations platform.

MSEUF team (2024) developed a Sales and Inventory Management System with Sales Forecasting using Time Series Analysis to help a local business predict demand and optimize stock levels. The study addressed the problem of frequent fluctuations in customer demand and the lack of structured forecasting tools among Philippine SMEs. MSEUF team examined time series forecasting techniques to determine which model provided the highest level of accuracy and practicality for inventory planning. The researchers designed a user-friendly sales prediction tool that integrated the selected model and allowed business owners to input historical sales data for future projections. The platform features automated calculations, visual trend lines, and actionable recommendations for inventory and resource allocation. MSEUF team found that the forecasting tool significantly improved prediction accuracy and supported better inventory planning compared to manual methods. The study concluded that structured sales forecasting could enhance operational efficiency and reduce financial losses for Philippine SMEs. This system aligns with the "Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently" by demonstrating how a sales forecasting module can support spa product ordering, inventory planning, and revenue projections. MSEUF team therefore provide a relevant local reference for integrating forecasting and analytics features into a spa operations platform.

2.3 Related Literature

Siraj (2024) and colleagues presented a comprehensive analysis of core inventory-management strategies, including Economic Order Quantity, lot sizing, safety stock, maximum inventory level, and reorder quantity. They framed inventory management as critical to business operations because it helps avoid overtime costs, production fluctuations, and penalties from delayed purchases during peak demand. The study used inventory data to optimize the lot sizes of materials procured from vendors and to align those quantities with the firm's capacity. It also addressed the

calculation of safety stock, maximum inventory level, and reorder quantity alongside the order quantity itself. The authors observed that Economic Order Quantity has proven beneficial across sectors such as marketing, automotive, pharmaceuticals, and retail. They emphasized that the model offers a comprehensive solution in a simplified format, allowing a straightforward calculation of the most cost-effective order quantity. For the present study, this analysis grounds the inventory module's reorder logic in well-established inventory theory rather than ad-hoc rules. Because a spa stocks many consumable products with varying demand, the combined use of Economic Order Quantity, safety stock, and reorder points is directly applicable. The study therefore supports embedding these formulas in the proposed system so that stock decisions are cost-minimizing and repeatable.

Mubasysyir (2024) and colleagues applied a multi-item Economic Order Quantity model, together with safety stock, reorder point, and maximum-capacity calculations, in a retail business. They situated the work in the Indonesian retail sector, where buying and selling transactions link suppliers, retailers, and consumers. The authors argued that retail acts as a bridge between suppliers and consumers and must therefore stock a diverse range of products. To keep costs down given this diversity, they adopted inventory control as the central mechanism. They showed that the Economic Order Quantity model yields the optimum total inventory cost by adjusting how frequently orders are placed over a period. Once the total inventory cost was established, they applied safety stock, reorder point, and maximum-capacity calculations so that overall inventory costs remained minimal. For the present study, the multi-item treatment is particularly relevant because a spa must manage many different consumables at once. The approach demonstrates that a single, consistent method can coordinate ordering across a varied product range. This provides a concrete template for how the spa system can compute reorder decisions for its full inventory rather than one item at a time.

Jahroni (2023) and colleagues implemented the Economic Order Quantity method to manage raw-material supplies for a micro, small, and medium enterprise specializing in a local culinary product. They found that the enterprise's existing supply management was inefficient, leading to frequent shortages or excess orders and additional costs. To address this, they proposed Economic Order Quantity analysis as a way to minimize inventory costs and maximize profit. The study focused on determining the optimal order quantities for three key raw materials. Its analysis produced specific optimal order quantities and corresponding reorder points for each of those materials. The calculations explicitly took safety stock and lead time into account when determining reorder points and maximum order quantities. For the present study, this work confirms that Economic Order Quantity control is feasible and beneficial for a small enterprise operating with modest resources. The spa resembles this enterprise in that it

consumes recurring supplies whose shortages directly interrupt service. The study therefore supports applying the same lead-time-aware reorder logic within the proposed spa system.

Wanti (2020) and colleagues developed a decision-support system built around the Economic Order Quantity model and reorder-point analysis. They argued that a company's success depends on providing goods and services at the right time and place, which requires effective inventory management. They also noted that modern technology demands inventory management that is fast, precise, and accurate. The authors observed that some companies struggle to determine appropriate stock levels, which impedes their ability to meet demand. Their study set out to determine the reorder-point stock that governs a company's purchases and sales. The method used a statistical, economic-quantity approach in which safety-stock analysis is performed first, and it was built using a user-centered design. The system's output was practical advice to company leaders for production planning, stock control, and rapid interpretation of market demand. For the present study, this work is a close parallel because it embeds inventory formulas inside a working, user-centered information system rather than leaving them as manual calculations. It therefore models exactly how the spa system can turn Economic Order Quantity and reorder-point logic into actionable, on-screen guidance for management.

Jain (2022) and colleagues carried out a comparative study of demand-forecasting models for a multi-channel retail company and proposed a novel hybrid machine-learning approach. They framed demand forecasting as a major concern of operational strategy because it drives both inventory management and customer satisfaction. The study compared five regression techniques, namely random forest, extreme gradient boosting, gradient boosting, adaptive boosting, and an artificial neural network. These were benchmarked against a proposed hybrid model combining random forest, extreme gradient boosting, and linear regression. The analysis used weekly sales data from a United States retail company and incorporated attributes such as regional temperature and store size. The authors found that the hybrid model outperformed the individual techniques across various measures of forecasting accuracy. For the present study, this supports adding a data-driven forecasting layer to the spa system and comparing simple against combined models. The inclusion of external attributes such as temperature suggests that the spa's forecasts could similarly benefit from contextual variables. The study therefore offers both a method and a benchmark for the system's revenue- and demand-projection features.

Martins (2023) and colleagues evaluated the main methods of sales forecasting to identify which offers the greatest accuracy. They stressed that retail companies must use resources efficiently and forecast well to stay competitive as margins tighten. They

warned that inaccurate forecasts can cause stock shortages, delivery delays, and customer dissatisfaction, as well as excess inventory and higher warehousing costs. The study was conducted as an integrative literature review of forecasting approaches. It examined three main families of methods: time-series techniques, artificial neural networks, and machine-learning algorithms. The authors concluded that machine learning is generally the most accurate, particularly when models include both exogenous and endogenous variables, while simpler time-series methods suffice for stable demand. For the present study, this helps justify a pragmatic forecasting choice appropriate to the spa's data volume. Because a small spa may have limited and relatively stable transaction history, the finding that time series can be adequate is directly useful. The study therefore guides the present research toward matching the forecasting method to the actual complexity of the spa's sales data.

Mustapha (2025) and colleagues examined common machine-learning and time-series techniques for sales forecasting in order to find the best-fitting method. They explicitly linked forecasting to business-intelligence software such as Tableau and Microsoft Power BI for interpreting past demand. The study used five years of store-item sales data covering fifty items across ten stores. It analyzed and contrasted Random Forest, Gradient Boosting, Linear Regression, and the ARIMA time-series method. The authors demonstrated that machine-learning algorithms can accurately forecast sales, which is valuable for optimizing operations, inventory, and financial planning. They also highlighted the importance of selecting the right algorithm, tuning and validating the model, and interpreting error metrics such as RMSE and MAE, with gradient boosting emerging as the strongest performer. For the present study, this pairing of forecasting with dashboards closely mirrors the analytics goals of the proposed spa system. The emphasis on error metrics offers a concrete way for the spa to judge whether its forecasts are reliable. The study therefore models how forecasting and visualization can be combined within a single analytics feature.

Ashraf (2022) performed a predictive analysis of retail sales forecasting using several machine-learning techniques. The study positioned sales forecasting as an important yet challenging part of supply-chain management, inventory, marketing, and financial planning. It noted that the growth of digital data has reduced the usefulness of traditional forecasting approaches. The analysis was carried out on a point-of-sale dataset covering the years 2013 to 2018. The author implemented several regression methods, including linear regression, random forest, and gradient boosting, alongside ARIMA and LSTM time-series models. The results showed that the gradient-boosted approach outperformed the other regression and time-series models for sales prediction. Because the data were drawn directly from a point-of-sale system, this study is closely aligned with forecasting from the spa's own transaction records. It demonstrates that a point-of-sale dataset alone can support useful predictive analysis

without external data. The study therefore supports building the spa's forecasting feature directly on top of its recorded sales transactions.

Eyeregba (2024) and colleagues examined how visual reporting, analytics, and sustainable dashboards can support decision-making and growth in small and medium enterprises. They noted that such enterprises are vital to economic growth yet often struggle with decision-making and resource management because of limited access to data insights. The authors argued that integrating these tools has become a critical strategy for enhancing operational efficiency as digital transformation reshapes business. They explained that visual reporting simplifies complex datasets into intuitive formats that help owners quickly identify trends and inefficiencies. They added that analytics, ranging from descriptive to prescriptive, drives strategic planning by uncovering hidden patterns and optimizing processes. The review acknowledged real barriers, including limited budgets, data-quality issues, and resistance to adopting new technology. For the present study, this framing clarifies the design intent of the spa's analytics dashboard as a tool for turning raw operational data into actionable insight. The identified barriers also anticipate the adoption challenges a small spa may face when introducing analytics. The study therefore both motivates and tempers expectations for the reporting component of the proposed system.

Fitrianingrum (2023) and colleagues investigated business intelligence as an alternative decision-making solution for small and medium enterprises in Indonesia. They described digital transformation through business intelligence as one way for organizations to survive in a modern, digital era. The authors noted that many such enterprises were badly affected by the COVID-19 pandemic, with some losing income and being forced to close. They characterized business intelligence as a combination of techniques and methods for improving access to information and managing data concisely. They argued that this capability can positively influence business activities. The study specifically referenced Microsoft tools such as SQL Server Integration Services and Reporting Services as means of implementing business intelligence. For the present study, the focus on small enterprises reinforces the value of embedding reporting and analytics directly in a small spa's system. The paper's emphasis on affordable, accessible tooling matches the constraints under which the spa operates. The study therefore supports treating business intelligence not as a luxury but as a practical decision aid for the proposed system.

Tafuro (2025) and colleagues conducted a hybrid review of how business intelligence is conceptualized, adopted, and implemented within small and medium enterprises. They positioned this context as one where digital transformation is essential yet insufficiently explored. The authors acknowledged that business intelligence is widely recognized as a strategic driver of informed decision-making and

competitiveness. At the same time, they observed that existing research remains fragmented and often confined to specific sectors. Their methodology combined bibliometric analysis with a qualitative systematic literature review covering the years 2015 to 2024. The review traced a shift from early semantic and dashboard-driven models toward cloud-based, predictive solutions, and it identified gaps in empirical validation and governance. For the present study, this positions the proposed system within current business-intelligence trends toward prediction and integration. The move from static dashboards to predictive analytics parallels the spa system's inclusion of a forecasting layer on top of reporting. The study therefore provides an up-to-date frame that justifies combining dashboards with predictive features in the proposed design.

Okfalisa (2022) and colleagues developed a decision-support dashboard, embedded within a business-intelligence platform, to assess the digital readiness of small and medium enterprises. They motivated the work by noting that the pandemic pushed such enterprises toward digitalization, yet many still failed to survive despite adopting digital platforms. The study proposed a dashboard model as a new feature for evaluating digitalization readiness. It appraised twenty-four criteria organized across business and information-technology dimensions. The method combined a Fuzzy Analytical Hierarchy Process for weighting with an Objective Matrix for performance mapping, both embedded in the business-intelligence dashboard. Applied to a sample of enterprises in one province, the dashboard produced a tracking-and-analysis view that traced each criterion down to its sub-criteria. For the present study, this design offers a model for the spa system's management-reporting component. The idea of mapping performance against defined criteria suggests how the spa dashboard could summarize operational health at a glance. The study therefore illustrates how a structured dashboard can convert diverse indicators into a clear decision aid for management.

Mpamugo (2024) and colleagues integrated role-based access control with the OAuth 2.0 authorization framework to strengthen security in a mobile application. They observed that securing access to resources has become harder as the number of connected devices grows. The study followed a waterfall methodology to implement mechanisms governing authentication and authorization. OAuth 2.0 was used to secure authentication by allowing third-party access without exposing user credentials. Role-based access control was applied to streamline permissions according to predefined, hierarchical user roles. The implementation added multi-factor authentication and session-timeout mechanisms, and it showed that combining the two protocols addresses unauthorized access, data integrity, and scalability. For the present study, this work directly informs the role-based permissions and authentication planned for the spa system. The separation of privileges by role mirrors the spa's need to

distinguish administrator, staff, and therapist access. The study therefore provides a concrete security pattern that the proposed system can adapt for its own users.

Dedhia (2026) designed and implemented a secure role-based authentication framework for web applications. The work began from the observation that web applications handle large amounts of sensitive data and are therefore frequent targets of attacks. It noted that weak authentication, poor session management, and inadequate access control can lead to vulnerabilities such as cross-site scripting, cross-site request forgery, and session hijacking. The framework used JSON Web Tokens for stateless authentication and stored those tokens in HTTP-only cookies to prevent client-side access and reduce cross-site scripting risk. Role-based access control was implemented to restrict access according to user roles. The design also added secure password hashing and middleware validation to strengthen overall security, and the author reported that it is scalable and suitable for modern web applications. For the present study, this offers a practical reference for securing the proposed web platform's logins and role management. The use of token-based authentication with role restrictions maps closely onto the spa system's multi-user, web-based architecture. The study therefore supplies a concrete, security-focused blueprint for the system's authentication and authorization layer.

Sahyudi (2025) and colleagues analyzed the implementation of role-based access control in enterprise resource-planning and related systems through a systematic literature review. They positioned role-based access control as the main approach for improving data security across many kinds of information systems. The review synthesized findings from numerous studies to evaluate how effectively the model addresses data privacy, regulatory compliance, and access-policy complexity. It examined role-based access control not only in enterprise systems but also in cloud-based, mobile, and multi-domain environments. The authors reported that integrating intelligent technologies, such as machine learning for user-behavior analysis and blockchain for recording access activity, can raise security noticeably. They also noted that automation in role-based access deployments can substantially reduce operational cost and management time compared with manual approaches. For the present study, this substantiates the choice of role-based access control to separate administrator, staff, and therapist privileges in the spa system. The review's attention to compliance and policy complexity signals which challenges the spa's access design must anticipate. The study therefore confirms both the relevance and the reliability of role-based access control for the proposed system.

Butt (2023) and colleagues proposed a trust-based, optimized role-based access control model for an electronic-health cloud environment. They noted that information technology plays a vital role in electronic health and that cloud-based implementation is

necessary to realize its benefits. At the same time, they recognized that such environments carry privacy and security loopholes. They also observed that as the number of users grows, the response time of the electronic healthcare system slows. To address these issues, they introduced a trust mechanism for access control that observes each user's behavior and assigns roles accordingly. The access-control module was implemented using a database server, with an administrator defining controls and a supporting framework used to validate each user's trust value. For the present study, this trust-aware design offers ideas for strengthening the spa system's data protection. The behavior-based assignment of roles suggests a more adaptive approach than static permissions alone. The study therefore extends the role-based access concept in a direction the proposed system could adopt as it matures.

Ramos de Luna (2019) and colleagues compared the factors that determine consumer acceptance of SMS-, NFC-, and QR-based mobile-payment systems. Their aim was to identify the principal factors influencing the adoption of each of these mobile-payment methods. The study was grounded in a comprehensive review of the scientific literature on mobile payments. From this review, the authors developed a behavioral model explaining the intention to use mobile payments. A central result was that consumer behavior differs according to the specific technology used for payment. In other words, acceptance of SMS, NFC, and QR payment cannot be treated as identical, since each tool elicits a distinct pattern of use. For the present study, this supports the inclusion of QR-based payment in the spa system while clarifying that its acceptance depends on technology-specific factors. The finding that payment methods are not interchangeable helps justify focusing on QR payment specifically rather than digital payment in general. The study therefore offers a behavioral foundation for how the spa can encourage clients to adopt its chosen payment channel.

Yan (2020) and colleagues studied QR-code mobile payment as a disruptive force in the retail industry. They observed that the emergence of mobile payment has disrupted many business sectors, with retail among the most affected. They also noted that acceptance of mobile payment still has substantial room for improvement. The study set out to identify the critical antecedents that influence the intention to adopt mobile payment, focusing specifically on QR-code technology. To do so, the authors applied an extended Mobile Technology Acceptance Model. The analysis produced both theoretical implications and practical guidance for stakeholders in the retail sector. For the present study, this retail focus is closely aligned with accepting digital payments at a service counter such as the spa's. Because a spa transaction resembles a retail checkout, the antecedents of QR adoption identified here are directly transferable. The study therefore helps explain which factors the spa should address to drive uptake of QR payment.

Türker (2022) and colleagues proposed an extended Technology Acceptance Model to explain user acceptance of QR-code mobile-payment systems. They noted that contactless QR-code payment became increasingly popular during the COVID-19 pandemic because it lets consumers pay quickly from their mobile devices. The research aimed to identify the causal relationships among the factors affecting acceptance. Data were collected from 485 QR-code mobile-payment users in Turkey through an online survey. The findings showed that perceived trust was the most crucial determinant of the intention to use, followed by perceived compatibility and perceived usefulness. The proposed model explained a large share of the variance in intention to use, indicating strong explanatory power. For the present study, these determinants guide how the spa can encourage clients to adopt QR payments. In particular, the primacy of perceived trust suggests the spa should make its payment process visibly secure and reliable. The study therefore points to concrete design and communication priorities for the system's payment feature.

Liébana-Cabanillas (2015) and colleagues developed a QR-payment acceptance model to analyze users' acceptance of QR-code mobile-payment systems. They motivated the work by pointing to the widespread use of mobile devices among consumers. The model was built on a comprehensive review of the scientific literature on payment acceptance. The authors tested it using an online survey administered to a national panel of Internet users. Their results showed that attitude, personal innovativeness, and subjective norms are determinants of the future intention to use QR-based mobile payment. The model also drew on classic Technology Acceptance Model variables together with compatibility, security, personal innovation, and individual mobility. For the present study, this widely cited, early model provides theoretical grounding for the payment component of the proposed system. Its identification of attitude and subjective norms as drivers complements the trust-centered findings of later QR-payment research. The study therefore anchors the spa system's payment feature in an established and well-validated acceptance framework.

Kurniawan (2026) and colleagues developed a web-based point-of-sale system for a hardware store and evaluated its usability with the System Usability Scale. They noted that small retail businesses often manage sales, stock, and reporting manually, which leads to recording errors, slow reporting, and limited decision support. The system they built supported product and category management, transaction processing with discounts and tax, and sales reporting with date-range filtering. It also generated best-selling product summaries and allowed CSV export for spreadsheet-based bookkeeping. The work followed a design-and-build approach, with functional validation carried out through black-box testing. Usability was measured with the System Usability Scale across several respondents, yielding an average score in the acceptable range, and the authors recommended improving onboarding and simplifying certain workflows.

For the present study, this provides a close methodological template for building and usability-testing the spa's point-of-sale and reporting features. The feature set of transactions, filtered reporting, and export overlaps substantially with what the spa system requires. The study therefore models both the functionality and the evaluation method the present research can follow.

Razzaq (2026) and colleagues applied Rapid Application Development to build and evaluate a web-based point-of-sale system for micro, small, and medium enterprises. They identified a dual challenge for such enterprises: the need for rapid implementation and the limited technical literacy of the people who will use the system. The study adopted an applied, design-oriented approach that emphasized iterative prototyping and continuous user involvement. Rather than focusing on feature completeness, it examined how Rapid Application Development supports usability and user acceptance among non-technical users. The resulting system was evaluated with the System Usability Scale involving users in different operational roles. The average usability score fell in the good category, and the authors concluded that the iterative method effectively bridges the gap between system complexity and user capability. For the present study, this supports the proposed research's own iterative development and usability-evaluation strategy. Because the spa's staff may also have limited technical background, the emphasis on user involvement is directly relevant. The study therefore provides empirical backing for choosing an iterative, user-centered development method for the spa system.

Ismanto (2026) and colleagues developed and evaluated a service-booking and management application for a service-sector enterprise. They noted that such enterprises face challenges in queue management, service transparency, and operational efficiency that harm the user experience. The study integrated a Design Science Research approach with three evaluation instruments: the Technology Readiness Index, the Technology Acceptance Model, and the System Usability Scale. The application was built through the Design Science Research stages of problem identification, design, demonstration, and evaluation. Qualitative interviews with owners, employees, and customers were analyzed thematically, while a quantitative survey gathered responses from a larger group of participants. The results showed high technology readiness and very strong acceptance, but only marginal usability, revealing a gap between conceptual acceptance and actual interaction quality. For the present study, this multi-instrument evaluation is a useful model for assessing the proposed spa system. The finding that acceptance can outpace usability warns the present research to test real interaction quality, not just stated willingness to use. The study therefore offers both an evaluation framework and a cautionary insight for the spa system's assessment.

Suria (2024) assessed the usability of a web-based financial information system using two complementary metrics. The system was designed to automate and streamline the management of student tuition-fee records for schools and a foundation. The author argued that advancing technology makes robust financial information systems necessary for handling growing transaction volumes. Usability was evaluated with both the System Usability Scale and the Usability Metric for User Experience-Lite. Questionnaires were distributed to a sizable number of schools to gather feedback on the system's usability. The System Usability Scale placed the system in a good category, the Usability Metric for User Experience-Lite indicated excellent usability, and the two measures correlated closely. For the present study, this demonstrates a rigorous and replicable way to evaluate the usability of the proposed system. The strong agreement between the two instruments suggests that pairing them yields a dependable usability assessment. The study therefore offers a validated, multi-measure approach the spa research can adopt to substantiate its own usability claims.

Razzaq and Dewi (2025) investigated the design and implementation of a web-based Management Information System (MIS) to replace traditional manual transaction and record-keeping practices in service-oriented small and medium enterprises (SMEs). The authors observed that reliance on manual operations leads to severe administrative bottlenecks, including delayed record-keeping, high risk of data loss, and inaccurate inventory tracking. To solve these operational issues efficiently, the study applied a qualitative research approach combined with the Rapid Application Development (RAD) framework—a methodology emphasizing rapid iteration, flexible user interface design, and continuous feedback. The resulting web-based MIS consolidated service management, inventory control, transaction processing, and staff assignment into a single integrated platform. User Acceptance Testing (UAT) conducted with the business owner yielded a 100% satisfaction rate across all developed modules, confirming high usability and operational alignment. Razzaq and Dewi concluded that implementing a customized web-based MIS substantially accelerates data processing, eliminates operational errors, and supports real-time, data-driven management decisions in small enterprise settings. This research provides strong empirical and methodological justification for the *Recovery Iloilo Spa Integrated System*, which applies RAD principles to unify appointment booking, Point-of-Sale (POS) operations, inventory tracking, and managerial reporting into a centralized web platform. By proving that web-based management systems built via RAD resolve data loss risks, streamline daily service records, and enhance operational control, the authors' findings directly validate the spa system's objective to replace manual front-desk logs with a reliable, real-time administrative platform.

Gupta and Kashyap (2026) conducted a comparative analysis of healthcare appointment systems to evaluate the operational efficiencies, user adoption barriers,

and scheduling outcomes between traditional manual booking methods and modern digital appointment platforms. The authors highlighted that traditional, offline appointment management—characterized by phone reservations and paper-based logbooks—consistently leads to long patient waiting times, high no-show rates, administrative overwork, and double-booking errors. To address these systemic bottlenecks, the study analyzed web-based and mobile scheduling platforms incorporating real-time slot allocation, automated reminder notifications, role-based administrative dashboards, and dynamic provider schedule tracking. Comparative evaluations confirmed that digital appointment systems significantly reduce client wait times, minimize double bookings, lower no-show rates through automated reminders, and improve overall provider resource utilization. Gupta and Kashyap (2026) concluded that transitioning from manual scheduling to unified web-based appointment architectures optimizes operational throughput and delivers a superior, client-centric service experience.

Srinivas and Ravindran (2018) developed a prescriptive analytics framework combining machine learning algorithms with heuristic scheduling rules to optimize appointment booking systems and resource allocation in multi-provider environments. The authors emphasized that traditional scheduling methods relying on static rules or manual front-desk planning struggle to handle patient no-show behaviors, unpredictable service completion times, and variable arrival rates, ultimately resulting in extended client waiting times, idle provider periods, and reduced operational throughput. To resolve these challenges, the study applied machine learning models to predict no-show probabilities and service durations, integrating these predictive insights into an automated scheduling engine to dynamically sequence client arrivals and optimize slot allocations. Experimental evaluations confirmed that embedding data-driven predictive algorithms into appointment scheduling frameworks significantly reduces client waiting times, minimizes staff idle time, and maximizes facility resource utilization. Srinivas and Ravindran (2018) concluded that combining predictive analytics with automated scheduling rules provides a scalable, highly effective operational framework for service-oriented platforms.

Zhuang et al. (2024) formulated and evaluated a dynamic resource allocation and scheduling model for service environments that operate under a hybrid customer arrival framework comprising both scheduled appointments and unannounced walk-in clients. The authors highlighted that service systems attempting to balance pre-booked appointments with walk-in arrivals using static, manual scheduling ledgers routinely suffer from severe operational friction, such as therapist or provider idle time, schedule collisions, unpredictable queue accumulation, and prolonged client wait times. To optimize resource utilization and service quality, the study developed an adaptive mathematical decision framework that dynamically allocates capacity, adjusts real-time

schedule slots, and assigns service resources based on real-time operational load and arrival patterns. Numerical simulations and empirical tests confirmed that dynamic resource allocation significantly reduces client waiting times, stabilizes operational throughput, minimizes staff idle time, and optimizes schedule flexibility in mixed-arrival service settings. Zhuang et al. (2024) concluded that adopting dynamic resource allocation mechanisms in appointment-based systems provides service enterprises with a mathematically sound, highly adaptive framework for balancing pre-booked reservations and walk-in demands efficiently.

Macleán and Ødegaard (2020) analyzed and modeled a dynamic capacity management and revenue optimization framework designed for service environments handling multi-person group bookings. The authors identified that allowing clients to freely choose reservation slots without dynamic capacity controls leads to sub-optimal resource allocation, stranded isolated capacity slots, and significant revenue loss when group booking requests cannot be accommodated. To resolve this operational inefficiency, the study formulated a capacity-based decision model using a discrete-time Markov Decision Process to dynamically manage available inventory and control segment access based on group sizes and time-slot demands. Numerical simulations and empirical evaluation confirmed that dynamic capacity allocation eliminates stranded single-capacity slots, maximizes expected operational revenue, and significantly improves resource utilization efficiency. Macleán and Ødegaard (2020) concluded that implementing dynamic capacity control mechanisms provides service platforms with an adaptive operational strategy to handle variable group size demands while preventing capacity fragmentation.

Lin et al. (2017) formulated and evaluated a simulation-optimization framework designed to solve complex resource allocation and appointment scheduling challenges in outpatient service environments. The authors highlighted that static, uncoordinated scheduling and manual resource tracking routinely cause severe operational bottlenecks, extended patient waiting times, prolonged provider idle times, and underutilized service facilities. To overcome these operational inefficiencies, the study developed a discrete-event simulation model integrated with a genetic algorithm to jointly optimize appointment schedule templates, doctor arrival timing, and resource capacity allocations. System testing and empirical experiments confirmed that combining automated schedule optimization with dynamic resource allocation significantly minimized client waiting times, reduced staff overtime costs, balanced workload distribution across service personnel, and maximized overall operational throughput. Lin et al. (2017) concluded that integrating simulation-optimization techniques into reservation and resource management platforms equips service enterprises with a mathematically sound and scalable framework for managing multi-resource operational environments.

Valenzuela-Núñez et al. (2024) developed and evaluated a smart medical appointment scheduling model incorporating machine learning predictions, mathematical optimization, and dynamic overbooking strategies to enhance resource utilization and reduce operational friction in service settings. The authors highlighted that traditional, static appointment scheduling systems suffer from high customer no-show rates, unpredictable schedule gaps, and inefficient staff and room allocation. To resolve these challenges, the study designed an intelligent scheduling architecture that leverages predictive analytics to anticipate no-show probabilities and dynamically adjusts appointment slot allocations and overbooking limits in real time. System testing and mathematical validation confirmed that combining dynamic scheduling algorithms with predictive capacity management significantly reduced client waiting times, minimized service provider idle time, and maximized overall operational throughput and facility capacity utilization. Valenzuela-Núñez et al. (2024) concluded that integrating smart scheduling algorithms into appointment reservation platforms provides service enterprises with an adaptive, data-driven framework that optimizes resource management and mitigates the negative impacts of unfulfilled bookings.

Sharma and Soni (2026) developed and evaluated an artificial intelligence (AI)-enabled role-based multi-task and event management platform designed for intelligent appointment scheduling and operational workflow automation. The authors observed that conventional appointment platforms lack role-specific operational adaptability, resulting in schedule fragmentation, delayed task delegation, uncoordinated multi-resource assignment, and administrative friction when managing overlapping appointments. To overcome these challenges, the study engineered an intelligent scheduling framework that incorporates role-based access control (RBAC), multi-task event coordination, and dynamic schedule optimization to automate booking allocations and staff task assignments based on real-time resource availability. System evaluations confirmed that integrating role-based task delegation with automated scheduling algorithms eliminated scheduling collisions, reduced administrative processing overhead, improved task tracking accuracy, and streamlined multi-user workflow coordination. Sharma and Soni (2026) concluded that deploying role-aware, multi-task scheduling architectures provides service-oriented organizations with a highly efficient, scalable digital solution for managing complex multi-resource service environments.

This study provides direct theoretical, structural, and operational support for the *Recovery Iloilo Spa Integrated System*, particularly regarding its real-time online reservation management, therapist and treatment room slot allocation, multi-service package coordination, and administrative schedule optimization modules. By proving that bi-objective multi-resource scheduling algorithms eliminate resource bottlenecks, prevent booking collisions, and minimize client waiting delays across interdependent

service workflows, the findings of Bodaghi et al. (2018) directly validate the spa system's primary objective: replacing manual, paper-based front-desk reservation ledgers with an integrated web application that automates client bookings, synchronizes therapist and room availability in real time, and elevates daily operational management.

Ao et al. (2024) developed a two-stage online resource allocation model designed to optimize revenue and operational efficiency for service enterprises managing a hybrid customer base of advance reservations and walk-in arrivals. The authors addressed the persistent operational challenge of balancing overbooking risks—where too many reservations lead to costly rejections—against resource idling, where under-utilization results in lost potential revenue. By engineering a framework that integrates advance booking, overbooking limits, and strategic confirmation calls, the study introduces "decoupled adaptive safety stocks" that hedge against cancellation uncertainties. System evaluations, validated through both synthetic experiments and empirical resort data, demonstrated that this policy significantly reduces resource idling and operational regret. The authors concluded that by strategically timing confirmation calls and dynamically adjusting overbooking levels, service providers can maintain near-optimal performance, even when faced with complex, multi-day occupancy patterns and fluctuating walk-in demand.

This study provides highly sophisticated theoretical, structural, and operational support for the *Recovery Iloilo Spa Integrated System*, particularly regarding its hybrid reservation management, dynamic therapist and treatment room slot allocation, and automated client confirmation protocols. By proving that a two-stage allocation model effectively balances pre-booked reservations with walk-in arrivals while minimizing the risks of overbooking, the findings of Ao et al. (2024) directly validate the spa system's objective: replacing manual, paper-based front-desk ledgers with an intelligent, integrated web application that automatically synchronizes real-time availability, manages overbooking risks through algorithmic safety stocks, and optimizes daily spa capacity.

Abouee Mehrizi et al. (2022) developed and evaluated a mathematical framework using asymptotic analysis for advance appointment scheduling across multi-class customer groups with varying priority levels and service requirements. The authors highlighted that advance scheduling becomes exceptionally challenging in multi-resource environments where different customer classes compete for the same service slots over a multi-day booking window, frequently leading to unbalanced daily workloads, prolonged customer wait times, and sub-optimal resource utilization. To resolve these operational trade-offs, the study formulated an asymptotic scheduling policy that dynamically prioritizes appointment requests and balances service workloads across available future booking days. Numerical analysis demonstrated that the multi-class advance scheduling framework effectively prioritizes waiting clients based on

urgency and operational cost while smoothly distributing workloads across the booking horizon. Abouee Mehrizi et al. (2022) concluded that applying asymptotic decision models to advance scheduling enables multi-class service environments to optimize capacity allocation, minimize waiting delays, and maintain balanced daily operations.

Dehghanimohammadabadi et al. (2022) developed and evaluated the Multi-Objective Patient Appointment Scheduling Framework (MO-PASS), a data-table input simulation–optimization approach designed to tackle complex multi-objective resource scheduling problems in appointment-based service environments. The authors highlighted that appointment systems face inherently conflicting operational trade-offs, such as minimizing customer waiting times, reducing provider idle time, and preventing facility overtime. To resolve these competing priorities, the study built a framework that integrates discrete-event simulation with multi-objective optimization algorithms, allowing decision-makers to evaluate real-time scheduling scenarios dynamically using structured data inputs. Experimental results confirmed that MO-PASS successfully generates optimal scheduling templates that balance customer wait times against service provider idle times, significantly reducing overall operational friction and resource underutilization. Dehghanimohammadabadi et al. (2022) concluded that multi-objective simulation–optimization frameworks provide service organizations with a flexible, data-driven methodology to optimize multi-resource appointment schedules under variable demand conditions.

Kasaie et al. (2024) developed and evaluated an optimization framework for outpatient appointment scheduling aimed at enhancing operational efficiency, resource utilization, and client satisfaction in appointment-driven service settings. The authors highlighted that traditional, unoptimized scheduling practices lead to severe operational bottlenecks, such as prolonged customer waiting times, unbalanced provider workloads, and high facility idle time due to uncoordinated time-slot allocations. To overcome these operational inefficiencies, the study designed a mathematical optimization model that strategically controls appointment interval lengths, dynamically balances service capacity, and structures daily booking queues based on expected service durations. Computational experiments and empirical testing confirmed that implementing optimized time-slot scheduling significantly reduced average waiting delays, minimized staff idle time, and improved overall throughput across multi-resource workflows. Kasaie et al. (2024) concluded that adopting algorithmic appointment scheduling strategies provides service facilities with a structured, highly efficient operational framework that optimizes facility capacity while elevating client service quality.

Vega et al. (2022) engineered and evaluated *Smart Scheduling* (SMASCH), a multi-appointment scheduling software platform designed to automate and optimize complex resource allocation across multi-disciplinary service workflows. The authors noted that managing appointments requiring multiple sequential services, specific

personnel skills, and dedicated treatment spaces using manual ledgers or decoupled electronic calendars leads to severe scheduling friction, booking collisions, extended client waiting delays, and administrative overhead. To resolve these challenges, the study developed an web-based scheduling platform that automates multi-resource slot matching, tracks real-time resource constraints, and synchronizes staff calendars simultaneously. Field evaluation and system testing confirmed that automated multi-appointment scheduling significantly reduced manual coordination errors, eliminated double-booking conflicts, streamlined administrative scheduling workflows, and optimized resource utilization across multi-stage service visits. Vega et al. (2022) concluded that deploying centralized, automated multi-resource scheduling architectures provides complex service environments with a reliable framework to optimize daily operations and improve overall service delivery.

Zhou and Yue (2021) formulated and evaluated a robust appointment scheduling model tailored for multi-stage sequential service systems operating under limited or uncertain service duration data. The authors noted that in multi-stage service environments—where clients progress sequentially through distinct treatment or consultation phases—variability in service length frequently causes domino-effect schedule delays, severe facility bottlenecks, and high staff overtime. To address these operational risks, the study constructed a distributionally robust optimization model that dynamically optimizes appointment slot allowances and inter-appointment buffer times without requiring precise service duration distributions. Experimental results confirmed that the robust scheduling framework effectively absorbed service duration variances, significantly reduced client waiting delays, and maintained high resource utilization across sequential service stages. Zhou and Yue (2021) concluded that incorporating robust sequencing and interval-allocation logic into appointment scheduling systems provides service architectures with a resilient, highly dependable framework for managing multi-stage operational workflows.

This study provides valuable theoretical, structural, and operational support for the Recovery Iloilo Spa Integrated System, particularly regarding its real-time online reservation management, multi-service package slot allocation, automated schedule buffer management, and therapist-room synchronization modules. By demonstrating that robust multi-stage scheduling models absorb service duration variances and prevent sequential bottlenecks, the findings of Zhou and Yue (2021) directly validate the spa system's primary objective: replacing manual, paper-based front-desk reservation ledgers with an integrated web application that automates client bookings, synchronizes therapist and room availability in real time, and elevates overall operational control.

Salah, Yusof, and Mohamed (2018) investigated and evaluated the key technological, organizational, and environmental factors influencing the adoption of Customer Relationship Management (CRM) systems in small and medium enterprises

(SME). The authors noted that small-to-medium service enterprises often hesitate to transition from manual, paper-based operational processes due to perceived technological complexity, lack of internal IT expertise, and uncertainty regarding system usability. To identify the primary drivers of successful technology integration, the study surveyed SME decision-makers using the Technology-Organization-Environment (TOE) framework. Empirical analysis confirmed that relative advantage, user-friendly interface design, top management support, and perceived operational efficiency were the most critical determinants of successful CRM adoption. Salah, Yusof, and Mohamed (2018) concluded that web-based CRM architectures tailored to core operational tasks encourage rapid adoption among non-technical personnel, allowing service enterprises to modernize client management and streamline daily workflows.

Wiid and Senooane (2025) investigated the primary drivers influencing the adoption and operational integration of Customer Relationship Management (CRM) systems within small and medium enterprises (SMEs) from the perspective of business owners and managers. The authors emphasized that small service-driven businesses often experience operational friction—such as double-booking conflicts, fragmented customer interaction histories, and inconsistent service delivery—due to a heavy reliance on informal or manual record-keeping practices. To analyze key adoption determinants, the study surveyed SME leadership to evaluate how technological ease of use, managerial support, customer data centralization, and perceived operational efficiency impact CRM adoption. Empirical findings revealed that web-based CRM architectures offering intuitive appointment management and centralized customer databases significantly reduce administrative overhead, improve staff coordination, and boost customer retention rates. Wiid and Senooane (2025) concluded that adopting tailored digital CRM frameworks serves as a strategic operational driver for small service businesses seeking to transition from disorganized manual processes to streamlined, data-driven operations.

Bukola, Abosede, and Adesola (2019) conducted an empirical evaluation examining the impact of Customer Relationship Management (CRM) operational strategies on the business performance and service quality of small and medium enterprises (SMEs). The authors emphasized that reliance on manual, paper-based client record-keeping or informal scheduling practices leads to severe operational friction, including lost client preference data, double-booking conflicts, delayed service fulfillment, and reduced customer loyalty. To assess these operational dynamics, the study gathered structured survey data across SME operators, evaluating how core CRM dimensions—such as customer orientation, service quality customization, and personalized interaction tracking—influence operational performance. Statistical analysis confirmed that structured CRM implementation significantly improves operational efficiency, accelerates service delivery, and fosters long-term client

retention. Bukola, Abosede, and Adesola (2019) concluded that embedding structured CRM frameworks into daily business administration provides small service enterprises with an essential paradigm for modernizing operations and maintaining high customer satisfaction.

Ngoc, Tien, and Anh (2021) evaluated the implementation of Customer Relationship Management (CRM) operational strategies and digital frameworks within a high-volume service organization. The authors emphasized that organizations relying on decentralized client records or inefficient communication channels suffer from severe operational friction, such as fragmented interaction logs, inconsistent service delivery, delayed response times, and reduced client trust. To resolve these operational challenges, the study examined a multi-channel CRM architecture integrating centralized client profiles, automated interaction tracking, and dynamic service feedback mechanisms. Empirical analysis confirmed that structured CRM adoption significantly improves administrative processing speed, enhances data accuracy across service roles, and strengthens long-term customer retention. Ngoc, Tien, and Anh (2021) concluded that embedding integrated CRM systems into daily administrative operations serves as a critical foundation for modern service-driven enterprises seeking to automate customer workflows and elevate client satisfaction.

Čavlin, Bulović, and Šmitran (2024) conducted a strategic analysis examining the operational benefits of integrating customer-centric design principles into modern Customer Relationship Management (CRM) software architectures. The authors highlighted that traditional, transactional administration tools often fail to capture individual customer preferences and history, resulting in impersonal service experiences, uncoordinated appointment schedules, and poor customer loyalty. To address these operational limitations, the study evaluated CRM systems that combine centralized database management with personalized service tracking, real-time availability displays, and automated customer interaction modules. Evaluation metrics confirmed that customer-centric CRM platforms increase administrative efficiency, reduce scheduling friction, accelerate transaction handling, and significantly foster repeat client engagements. Čavlin, Bulović, and Šmitran (2024) concluded that adopting customer-focused CRM architectures enables service enterprises to transition from rigid, manual administration to synchronized, highly responsive digital customer operations.

Nugroho, Indah, Novianti, Sanjaya, and Syahbani (2025) investigated the impact of implementing usability principles in UI/UX on consumer purchase intention on the Shopee platform. The study noted that in a highly competitive e-commerce environment, usability is a crucial factor influencing purchasing decisions, as an optimal user experience can enhance trust and reduce cart abandonment. Using Nielsen's Attributes of Usability (NAU), the researchers evaluated five principles—learnability,

efficiency, memorability, errors, and satisfaction—through a quantitative survey of 100 active Shopee users. The results indicated that efficiency, memorability, and satisfaction significantly influence purchase intention, with memorability and satisfaction having positive effects, while efficiency showed a negative impact in this context. Learnability and errors were found to have no significant effect on purchase intention in this particular study. The study concluded that e-commerce platforms should optimize memorability and satisfaction in UI/UX design, such as through personalized interfaces and easy access to features, to enhance user experience and encourage purchases. These findings align with our system because both recognize that usability and user experience directly affect whether customers are willing to complete transactions and return to the platform. The emphasis on satisfaction and memorability supports our objective of designing an interface that is not only functional but also comfortable and memorable for users. Therefore, Nugroho, Indah, Novianti, Sanjaya, and Syahbani (2025) provide empirical evidence that well-implemented usability principles in UI/UX can strengthen purchase intention and customer loyalty in e-commerce.

Abhichandani, Vadrevu, and Abhichandani (2025) explored ways to enhance checkout conversion rates through machine learning and UI/UX design patterns in e-commerce platforms. The study recognized that checkout abandonment is a major challenge in online retail, often caused by complex forms, unexpected costs, slow performance, or confusing interfaces. To address this, the researchers proposed integrating machine learning models with optimized UI/UX patterns to personalize the checkout experience, predict user behavior, and reduce friction points. For example, machine learning can be used to pre-fill forms, suggest preferred payment methods, or detect and resolve potential errors before the user abandons the process. Combined with clear progress indicators, simplified input fields, and responsive design, these techniques help create a smoother and more trustworthy checkout flow. The study emphasized that even small improvements in checkout usability can lead to significant increases in completed transactions and overall revenue. These insights align with our system because both aim to minimize barriers to transaction completion and provide a seamless, efficient purchasing experience. The integration of analytics and design patterns supports our goal of using data to continuously improve the user journey and reduce drop-offs. Therefore, Abhichandani, Vadrevu, and Abhichandani (2025) demonstrate that combining machine learning with thoughtful UI/UX design can substantially improve checkout conversion rates in e-commerce.

Osman and Hwang (2021) studied e-commerce design by older adults, focusing on the selection and placement of web objects on shopping sites to improve usability for this demographic. The study recognized that older users often face challenges with small fonts, complex navigation, cluttered layouts, and unclear interactive elements, which can discourage them from completing online purchases. Through user testing

and observational studies, the researchers identified preferences for larger text, high-contrast colors, clear labels, consistent navigation, and logically grouped product information. They found that when web objects such as search bars, product filters, and checkout buttons are placed in predictable and easily accessible locations, older adults can navigate and shop more confidently. The study also highlighted the importance of minimizing unnecessary steps and providing clear feedback to reduce confusion and errors. These findings align with our system because both emphasize the need for inclusive and accessible design that accommodates users with varying levels of digital literacy. The focus on clear object placement and simplified flows supports our objective of creating an interface that is easy to understand for all customer segments, including less tech-savvy users. Therefore, Osman and Hwang (2021) show that thoughtful e-commerce design tailored to older adults can improve usability, satisfaction, and participation in online shopping.

Santiworarak and Choochaiwattana (2018) conducted a case study on usability design principles in a responsive e-commerce web application to evaluate how well usability guidelines support mobile and desktop shopping experiences. The study applied established usability principles such as consistency, feedback, error prevention, and flexibility to assess the design of a responsive e-commerce site across different devices. The researchers found that when these principles are properly implemented, users experience fewer difficulties in navigating, searching for products, and completing transactions regardless of screen size. Issues such as misaligned buttons, slow loading on mobile, or inconsistent menus were shown to negatively affect user satisfaction and increase abandonment rates. The study concluded that adherence to usability design principles is essential for creating responsive e-commerce applications that perform well on both smartphones and desktops. These insights align with our system because both prioritize responsive and usable design to ensure that customers can access and use the platform effectively on any device. The emphasis on consistency and error prevention supports our goal of reducing friction and building trust throughout the customer journey. Therefore, Santiworarak and Choochaiwattana (2018) demonstrate that applying usability design principles in responsive e-commerce can enhance user experience and support higher engagement and conversion rates.

Chaurasia (2025) analyzed the impact of shopping carts on consumer preferences and purchase behavior in e-commerce platforms through a study termed "Brand Cart-ism." The study examined shopping cart behavior on leading e-commerce websites to understand customer preferences, shopping volume, and purchase patterns. Using mock data from seven popular brands, the research studied metrics such as average cart value, abandonment rate, and conversion rate to identify how cart features influence consumer decisions. The findings revealed significant variations in cart behavior across platforms, with some brands showing higher conversion rates and

larger average cart values, while others experienced higher abandonment. Regression models indicated that average cart value and conversion rate are primary predictors of shopping volume, highlighting the strategic importance of cart design and functionality. The study also found that cart abandonment negatively affects repeat purchases, emphasizing the need to optimize the cart experience to retain customers. These insights align with our system because both recognize the shopping cart as a critical touchpoint that can either encourage completion or trigger drop-offs. The focus on cart behavior and conversion supports our objective of designing a cart and checkout process that minimizes friction and maximizes completed transactions. Therefore, Chaurasia (2025) shows that shopping cart design and performance are key determinants of consumer preferences and overall e-commerce success.

Iverson (2023) analyzed cart abandonment behavior in e-commerce to understand why customers add items to their shopping carts but leave without completing the purchase. The study identified common reasons for abandonment, including unexpected costs such as shipping and taxes, complicated checkout processes, mandatory account creation, and concerns about payment security. By reviewing industry data and user surveys, the research highlighted that even small frictions in the checkout flow can significantly increase abandonment rates and reduce revenue. The study recommended strategies such as transparent pricing, guest checkout options, progress indicators, multiple payment methods, and trust signals to reduce abandonment and encourage completion. These findings align with our system because both aim to identify and eliminate barriers that cause customers to drop off during the purchasing process. The emphasis on transparency, simplicity, and trust supports our goal of creating a checkout experience that feels safe, clear, and effortless for users. Therefore, Iverson (2023) demonstrates that addressing the root causes of cart abandonment through thoughtful design and policy can substantially improve conversion rates in e-commerce.

Varghese and Sudheer (2026) decoded e-commerce drop-offs by examining behavioral insights into online shopping cart abandonment to identify patterns and motivations behind incomplete purchases. The study used behavioral analytics and user journey mapping to trace where and why customers abandon their carts, focusing on stages such as product selection, cart review, and checkout. The researchers found that drop-offs often occur when users encounter unexpected costs, face technical issues, feel uncertain about product suitability, or perceive the process as too time-consuming. The study also highlighted the role of psychological factors such as hesitation, comparison shopping, and distraction in contributing to abandonment. Based on these insights, the authors proposed interventions such as retargeting emails, exit-intent popups, simplified forms, and clear return policies to re-engage users and recover lost sales. These strategies align with our system because both seek to understand and

mitigate the behavioral and structural causes of cart abandonment. The focus on re-engagement and process optimization supports our objective of not only preventing drop-offs but also recovering potential sales through targeted follow-ups. Therefore, Varghese and Sudheer (2026) show that combining behavioral insights with proactive design and communication can reduce e-commerce drop-offs and improve overall conversion performance.

Sumadi, Muqorobin, and Fitria (2025) designed an e-commerce website to increase the purchasing power of micro, small, and medium enterprises (MSMEs) by providing a structured online platform for product display, ordering, and payment. The study addressed the limited market reach and manual transaction processes faced by many MSMEs, which often rely on social media or offline channels without a dedicated sales website. Using a system development approach, the researchers created a web-based e-commerce platform that supports product catalogs, shopping carts, order management, and online payment integration. The implementation enabled MSMEs to reach more customers, reduce manual errors in order recording, and process transactions more efficiently. The study concluded that the e-commerce website contributed to increased sales volume and improved operational efficiency for participating businesses. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to replace fragmented or manual processes with a centralized web-based platform that supports end-to-end transactions. In the context of a spa business, similar features can streamline service booking, product sales, and payment processing while reducing administrative workload. The emphasis on increasing purchasing power also supports our goal of using an integrated system to encourage more frequent and convenient customer transactions. Therefore, Sumadi, Muqorobin, and Fitria (2025) demonstrate that a well-designed e-commerce website can enhance sales performance and operational efficiency, which is directly relevant to the transactional and operational components of our spa management system.

Gupta, Prasad, and Kumar (2026) reviewed existing literature on factors influencing shopping cart abandonment in e-commerce to identify the main causes of incomplete online purchases. The study categorized abandonment drivers into cost-related factors, website and technical issues, payment and security concerns, consumer behavior factors, and delivery and service factors. Using the Fishbone (Ishikawa) Framework, the researchers highlighted hidden costs, complicated checkout procedures, security concerns, website usability issues, comparison shopping tendencies, and delivery-related challenges as key determinants of abandonment. The review emphasized that these factors significantly affect sales conversion, customer acquisition efficiency, and overall profitability for online businesses. The authors recommended a holistic approach that addresses pricing transparency, simplified

checkout flows, trust signals, and reliable service to reduce abandonment rates. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both recognize that friction in the booking or checkout process can cause customers to drop off before completing a transaction. In a spa context, unclear pricing, complex booking steps, or concerns about payment security can similarly lead to abandoned appointments or service purchases. By designing clear pricing displays, simplified booking flows, and secure payment options, our system can minimize drop-offs and improve conversion of inquiries into confirmed bookings. Therefore, Gupta, Prasad, and Kumar (2026) provide a comprehensive framework for understanding and mitigating cart and booking abandonment, which supports the conversion and revenue goals of our integrated spa system.

Sunarmie, Pamungkur, and Elliyana (2024) examined optimal website design strategies in e-commerce to identify how design decisions influence user experience, trust, and conversion rates. The study analyzed various design elements such as visual consistency, navigation structure, loading speed, mobile responsiveness, and clarity of product information. The researchers found that websites with coherent visual design, intuitive navigation, and fast performance tend to generate higher user engagement and lower bounce rates. In contrast, inconsistent layouts, slow loading times, and unclear calls-to-action were associated with reduced trust and higher abandonment. The study concluded that an optimal e-commerce design strategy must balance aesthetics, usability, and technical performance to create a seamless and trustworthy shopping environment. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both recognize that website design directly affects how users perceive and interact with the platform. For a spa management system, a clean, professional, and responsive design can reinforce brand credibility and make it easier for clients to explore services, book appointments, and make payments. The emphasis on performance and clarity also supports our objective of ensuring that the system works smoothly on various devices used by both customers and staff. Therefore, Sunarmie, Pamungkur, and Elliyana (2024) show that strategic website design is a critical factor in building trust and encouraging conversions, which is essential for the customer-facing components of our integrated spa system.

Sondhi (2017) segmented and profiled deflecting customers to understand shopping cart abandonment by identifying distinct reasons why online shoppers leave without completing their purchases. The study found eight different reasons that drive online shopping cart abandonment: financial reasons, organizational tools, time pressure, intangibility, privacy issues, aesthetic design, social influences, and entertainment factors. Each of these reasons accounted for unique variance in the model, validating that abandonment is influenced by a diverse set of psychological, contextual, and design-related factors. The research emphasized that perceived risks,

mental accounting, and budgeting processes also play a role when consumers face decision dilemmas during checkout. These findings highlight that abandonment is not only a technical issue but also a behavioral and emotional response to the shopping environment. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both recognize that clients may abandon bookings or purchases due to concerns about cost, time, privacy, or the overall feel of the platform. For a spa management system, addressing issues such as unclear pricing, long forms, privacy concerns, or an unappealing interface can help reduce deflection and increase completed transactions. The segmentation approach also supports our analytics goal of understanding different client behaviors and tailoring the system to address specific abandonment patterns. Therefore, Sondhi (2017) provides a comprehensive framework for understanding the multifaceted nature of cart abandonment, which can guide the design and optimization of our spa system's booking and payment flows.

Sausan and Zulaikha (2025) developed a website design innovation strategy for Brand XYZ using the Design Thinking approach to create a user-centered and innovative digital experience. The study applied Design Thinking stages, including empathy, problem definition, ideation, prototyping, and testing, to understand user needs and translate them into effective design solutions. Through this process, the researchers identified pain points in the existing website and generated ideas for improving navigation, visual consistency, content structure, and overall user experience. The resulting design emphasized clarity, ease of use, and alignment with brand identity to enhance user engagement and satisfaction. The study concluded that the Design Thinking approach enables the creation of websites that are not only functional but also emotionally resonant and aligned with user expectations. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both prioritize understanding real user needs and iteratively refining the system based on feedback. For a spa management platform, applying Design Thinking can help ensure that the interface supports common tasks such as booking services, viewing pricing, and managing appointments in a way that feels intuitive and calming. The emphasis on innovation and user-centeredness also supports our goal of creating a differentiated and client-friendly digital experience. Therefore, Sausan and Zulaikha (2025) show that Design Thinking can drive meaningful website innovation, which is valuable for designing the customer-facing components of our integrated spa system.

David, John, Prasana, Vignesh, and Kumaran (2025) analyzed mobile wallet adoption in India by examining consumer behavior through UPI and QR code usage to understand the factors influencing digital payment acceptance. The study recognized that the rapid expansion of mobile internet access, smartphone ownership, and fintech

innovation has transformed retail payments, with UPI and QR-based systems moving from novelty to mainstream in only a few years. Using survey data and structural equation modeling, the researchers found that factors such as convenience, perceived usefulness, ease of use, trust, and promotional benefits significantly influence users' intention to adopt and continue using mobile payment technologies. The findings revealed that UPI now dominates retail digital volumes, driven primarily by small-ticket everyday transactions and frictionless user experience, which correlates with higher transaction frequency and increased impulsive purchases among younger urban cohorts. Merchant responses, including rapid QR adoption, offer alignment with platforms, and integration of loyalty mechanics, create reinforcing feedback loops that further accelerate adoption. However, benefits are heterogeneous, with rural and older consumers remaining more sensitive to security concerns and fraud risk. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both emphasize the need to offer payment options that are convenient, trustworthy, and aligned with customer expectations. In a spa setting, supporting QR-based and mobile wallet payments can reduce friction at checkout, encourage faster transactions, and appeal to tech-savvy clients who prefer cashless options. The focus on trust and security also supports our objective of implementing payment features that clients perceive as safe and reliable, thereby increasing adoption and repeat usage. Therefore, David, John, Prasana, Vignesh, and Kumaran (2025) show that mobile wallet and QR code adoption is driven by convenience, trust, and merchant integration, which is relevant to designing the payment and customer experience components of our spa system.

Miko, Joe, and Oktavia (2025) examined the utilization of QRIS and e-wallet-based digital payments to increase the competitiveness of MSMEs in Indonesia. The study recognized that MSMEs play a crucial role in economic growth and that digital transformation of payment systems is essential for enhancing their operational efficiency and market reach. Using a quantitative survey approach, the researchers found that the adoption of QRIS and e-wallets positively and significantly affects sales performance and competitive advantage for MSMEs. The integration of these digital payment tools enables faster transaction processing, reduces cash handling errors, and provides automated transaction records that support better financial management. The study also highlighted that the digital footprint created by QRIS and e-wallet transactions improves MSMEs' financial reputation, increasing their chances of obtaining formal financing and expanding their business. These outcomes align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both aim to leverage digital payments to improve operational efficiency, transaction accuracy, and business competitiveness. For a spa, integrating QRIS and e-wallet payments can streamline the checkout process, reduce manual recording errors, and generate clear transaction histories that support financial

reporting and analytics. The emphasis on competitive advantage also supports our goal of positioning the spa as a modern, tech-enabled business that meets contemporary customer expectations for cashless payments. Therefore, Miko, Joe, and Oktavia (2025) demonstrate that QRIS and e-wallet adoption can enhance sales performance and competitiveness for small businesses, which is directly applicable to the payment and analytics modules of our spa system.

Haritsa, Yonesha, Suseno, and Puteri (2025) compared QRIS and e-wallets in terms of user adoption and merchant acceptance in the digital payment landscape to understand how these two payment methods coexist and compete. The study examined factors such as transaction fees, ease of integration, customer preferences, and merchant incentives that influence the choice between QRIS and proprietary e-wallet solutions. The researchers found that QRIS is widely accepted by merchants due to its interoperability, standardized QR format, and lower integration complexity, allowing businesses to accept payments from multiple e-wallets and mobile banking apps through a single code. In contrast, e-wallets often offer additional features such as loyalty programs, cashback promotions, and in-app rewards that can attract specific customer segments and encourage repeat usage. The study concluded that many merchants adopt both QRIS and e-wallets to maximize payment options, cater to diverse customer preferences, and benefit from the promotional ecosystems of e-wallet providers. These insights align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both recognize the value of offering multiple digital payment options to accommodate different client preferences and maximize transaction convenience. In a spa context, supporting both QRIS and e-wallet payments can ensure that clients can pay using their preferred method, whether through a universal QR code or a specific e-wallet app with rewards. The ability to integrate promotional features such as cashback or loyalty points also supports our goal of encouraging repeat visits and enhancing customer engagement. Therefore, Haritsa, Yonesha, Suseno, and Puteri (2025) show that a hybrid approach to digital payments can optimize both merchant acceptance and user adoption, which is relevant to designing the payment and loyalty components of our spa system.

Meng and Feng (2011) conducted research on the business model and protocol of offline digital coupons based on RSA digital signatures. The study focused on applying digital signature technology to support the authenticity, validity, and security of digital coupons used in offline business transactions. RSA-based digital signatures can help verify whether a coupon was legitimately issued and whether its contents have been altered. This type of validation is important in coupon systems because businesses must be able to distinguish valid promotional offers from modified, copied, expired, or unauthorized coupons. The study also considered the business and

communication protocol required to support the issuance and utilization of digital coupons in an offline environment.

These insights align with Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently because the proposed platform will support therapist and employee management, attendance monitoring, service assignment, customer feedback, and performance evaluation. The system can connect customer ratings and comments with the specific therapist, service, appointment, and transaction involved. Management may also use employee records and feedback results to identify training needs, monitor service consistency, and recognize employees who demonstrate strong performance. For example, repeated feedback regarding delayed service or poor communication can be compared with attendance records, appointment schedules, and therapist assignments to determine possible operational causes. Therefore, Ahu Yazıcı Ayyıldız, Erdogan Koc, Tuğrul Ayyıldız, and Sercan Hatipoğlu (2026) show that employee awareness and ability are important components of service quality, which supports the proposed system's employee monitoring, feedback, and performance analytics features.

Mukhammad Slamet and **Wiwik Sulistiyowati (2022)** analyzed customer satisfaction by integrating the SERVQUAL method with Importance Performance Analysis (IPA). The study used the five SERVQUAL dimensions—tangibles, reliability, responsiveness, assurance, and empathy—to identify service-quality indicators that influenced customer satisfaction. After measuring customers' perceptions and expectations, the researchers used IPA to classify service attributes according to their importance and performance. The results identified the patience of administrative personnel in listening to customer complaints as a priority for improvement under the assurance dimension, with a satisfaction value of 4.086 and an importance value of 4.114. ijler.umsida.ac

These findings align with the Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently because the proposed system will provide management with organized information for evaluating therapist and staff performance. The system can combine customer ratings, completed services, appointment records, attendance information, service duration, feedback, and commission data to produce a more complete view of employee performance. Reliability is especially relevant in a spa because customers expect therapists to arrive on time, perform the selected service correctly, follow treatment procedures, and provide consistent service quality. The proposed analytics module may help management identify high-performing employees, detect recurring service issues, and determine which staff members may require additional training. Therefore, Ayulita Purnama Sari, Murpin Josua Sembiring, and David Sukardi Kodrat (2026) demonstrate that SERVQUAL and decision-support methods can assist organizations in evaluating

employee performance and prioritizing service improvements, which is applicable to the therapist management, feedback, and analytics modules of the proposed spa system.

Ahu Yazıcı Ayyıldız, Erdogan Koc, Tuğrul Ayyıldız, and Sercan Hatipoğlu (2026) developed and tested the SERVQUAL awareness and ability of employees scale, known as SERVQUAL-E. The study addressed the importance of measuring not only how customers perceive service quality but also how employees understand and perform the behaviors associated with service quality. The research adapted the SERVQUAL framework to the employee perspective and examined the validity of the resulting measurement scale. Its focus is relevant to hospitality and service organizations because employees directly influence reliability, responsiveness, assurance, empathy, and the physical presentation of the service environment. The study recognizes that service-quality improvement requires employees to understand their responsibilities and consistently demonstrate appropriate service behaviors.emerald

Ayulita Purnama Sari, Murpin Josua Sembiring, and David Sukardi Kodrat (2026) evaluated employee performance at PT WIKA using service quality and customer satisfaction measures. The researchers integrated SERVQUAL with the MOORA decision-support method and collected questionnaire responses from 100 customers using 20 service-quality indicators covering tangibility, reliability, responsiveness, assurance, and empathy. The results showed an average SERVQUAL satisfaction ratio of 1.01, indicating that overall employee performance generally met customer expectations. Fourteen of the twenty indicators, or 67 percent, were classified as satisfactory, while six indicators, or 33 percent, required improvement. The MOORA analysis identified reliability as the most dominant service-quality dimension, although responsiveness and service consistency still needed attention.penerbitgoodwood

Ivan Adriano Syahputra, Elly Ismiyah, and Akhmad Wasiur Rizqi (2023) analyzed the service quality of an employee cooperative at PT XYZ using SERVQUAL and Importance Performance Analysis. The researchers surveyed 95 employees and measured service quality across the five SERVQUAL dimensions. The results showed negative gaps in tangibles, reliability, responsiveness, assurance, and empathy, indicating that employee satisfaction had not yet met expectations. The largest concern was related to empathy, which recorded a gap of -0.39 , while reliability recorded a gap of -0.35 . The IPA results identified two attributes as priority areas for improvement: employees' failure to consider customer status when providing service and inaccurate recording of orders and transactions.journal.universitaspahlawan.ac

These findings are relevant to Recovery Iloilo Spa because the proposed system is intended to reduce service and transaction errors through centralized records and automated workflows. Accurate recording is important when managing customer information, selected treatments, therapist assignments, add-on services, discounts,

vouchers, payments, and refunds. The system will also store customer preferences, service history, feedback, and transaction records, allowing staff to provide more personalized and consistent assistance. In addition, customer feedback and analytics can help management identify whether concerns are related to staff empathy, responsiveness, data-entry errors, or payment and booking procedures. Therefore, Ivan Adriano Syahputra, Elly Ismiyah, and Akhmad Wasiur Rizqi (2023) demonstrate that SERVQUAL and IPA can reveal specific service deficiencies and transaction-related weaknesses, which supports the proposed spa system's customer-record, transaction, feedback, and service-quality monitoring features.

Fransiska Oktalia, Duha Gumano, and Ray Octafian (2023) examined the relationship between service quality and customer satisfaction through the SERVQUAL model at Resto Bebek Goreng Pak Eko. The study used a qualitative and descriptive approach involving interviews with restaurant staff, observations of interactions between employees and customers, and a review of relevant literature. The findings indicated that the service provided by the restaurant was considered high quality because it met or exceeded customer expectations. The results also showed that maintaining consistent service standards contributes to customer satisfaction within a tourism and hospitality-related business.ictmt.stiepari

These findings align with Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently because spa operations also depend heavily on direct interactions between employees and customers. The quality of the customer experience may be influenced by appointment confirmation, reception service, therapist behavior, treatment delivery, payment processing, and post-service communication. The proposed system will support these activities through appointment management, automated notifications, therapist assignment, customer feedback, payment records, and service-history tracking. Management can use the collected information to determine whether customers are receiving timely, reliable, and satisfactory services. Thus, Fransiska Oktalia, Duha Gumano, and Ray Octafian (2023) demonstrate that consistent service quality is essential to customer satisfaction, which supports the proposed system's objective of improving customer experience through organized operations and feedback-based service evaluation.

I. Markowicz (2009) examined employee satisfaction using the Importance Performance Analysis and SERVQUAL scales. The study analyzed the responses of 405 employees from 10 local government units and considered employee satisfaction in relation to the quality of services provided by public organizations. It also examined satisfaction levels across different employee positions, allowing the researcher to conduct internal benchmarking among employee groups. The study emphasized that employee satisfaction may influence the image and effectiveness of an organization,

particularly when employees are responsible for providing services to the public.reference-global+1

These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because employee satisfaction and performance are important to maintaining consistent spa service quality. Therapists, receptionists, cashiers, and managers have different responsibilities, and each group may experience different operational concerns. The proposed system can support internal monitoring by organizing attendance records, service assignments, completed appointments, commissions, customer feedback, and employee activities. Management may compare performance indicators across employee roles while protecting sensitive information through role-based access controls. Such information can help identify workload concerns, training needs, scheduling problems, or recognition opportunities. Therefore, I. Markowicz (2009) demonstrates that SERVQUAL and IPA can be applied not only to customer satisfaction but also to employee satisfaction and internal performance comparison, which is relevant to the employee management, accountability, feedback, and analytics features of the proposed spa system.

Mohan (2025) developed a secure email-based OTP generation and verification system to enhance authentication security and prevent unauthorized access in web applications. The study recognized that traditional password-based authentication is vulnerable to attacks such as credential stuffing, phishing, and brute force, which can compromise user accounts and sensitive data. The system generates a unique, time-limited OTP for each login attempt and sends it to the user's registered email address, requiring the user to enter the code to complete authentication. This approach adds a second factor of verification, ensuring that even if a password is compromised, an attacker cannot access the account without the OTP. The system includes features such as user registration, secure login, OTP generation, email delivery, and OTP verification, all designed to provide a secure and user-friendly authentication process. Testing demonstrated that the system effectively prevents unauthorized access while maintaining a smooth login experience for legitimate users. These findings align with *Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently* because both prioritize strong authentication mechanisms to protect sensitive user and business data. In a spa context, email-based OTP can be used to secure client and staff logins, password resets, and sensitive actions such as viewing personal information or processing payments. The emphasis on security and usability also supports our goal of implementing authentication that is both robust and convenient for users. Therefore, Mohan (2025) demonstrates that a secure email-based OTP system can significantly enhance authentication security, which is directly applicable to the security and user management modules of our spa system.

Sophian Gauzelin and Hugo Bentz (2017) examined the impact of business intelligence systems on organizational decision-making and performance among SMEs in France. The researchers conducted qualitative interviews with 200 members from 10 selected SMEs. The study found that BI systems support more timely decision-making, improve organizational efficiency, help businesses respond appropriately to customer needs, and contribute to higher employee satisfaction. The researchers emphasized that business intelligence gathers data from business operations and the external environment, processes the data, and presents it as understandable information that managers can use when making decisions.

These findings are relevant to Recovery Iloilo Spa because management will need timely information about daily sales, appointments, customer preferences, therapist performance, payment transactions, and inventory usage. A business intelligence dashboard can present these data in a clear format through charts, tables, summary cards, filters, and comparative reports. For example, the spa owner may use the dashboard to determine which services generate the highest revenue, which days have the greatest customer demand, or which therapists have completed the most services. Receptionists and managers can also use real-time operational information to respond more quickly to scheduling and customer concerns. Thus, Sophian Gauzelin and Hugo Bentz (2017) show that BI systems can improve decision speed, operational efficiency, customer responsiveness, and employee satisfaction, which directly supports the sales reporting and decision-support objectives of the proposed spa system.

Jianwen Wang, Omar Abdullah Hisam, Fahad M. Alotaibi, Yousef Ibrahim Daradkeh, and Sara A. Althubiti (2022) investigated how business intelligence capabilities and data-mining systems can improve organizational performance and competitive advantage. The study used a quantitative research approach involving managers of privately held companies that used business intelligence and artificial intelligence in their data-mining systems. The findings showed that business intelligence capabilities and reliability influence performance-evaluation capabilities, while performance evaluation contributes to competitive advantage. The study also reported that business intelligence infrastructure supports the reliability of BI systems, which emphasizes the importance of integrating data sources and maintaining dependable information.

These findings are applicable to Recovery Iloilo Spa because a dashboard is only useful when the information displayed is accurate, complete, and consistently connected to the system's operational records. The proposed platform will integrate appointment records, completed services, Point-of-Sale transactions, payment confirmations, commissions, vouchers, inventory changes, customer feedback, and employee activities. This integration can help ensure that sales reports are based on verified transaction data instead of manually copied or disconnected records. Reliable

data will also improve management's ability to evaluate therapist productivity, service profitability, customer demand, and overall business performance. Therefore, Jianwen Wang, Omar Abdullah Hisam, Fahad M. Alotaibi, Yousef Ibrahim Daradkeh, and Sara A. Althubiti (2022) demonstrate that reliable BI infrastructure and data-mining capabilities strengthen performance evaluation and competitive advantage, which supports the proposed spa system's centralized database, dashboard, and reporting functions.

Shao Valentinus Tsiu, Mfanelo Ngobeni, Lesley Mathabela, and Bonginkosi Thango (2025) conducted a systematic review of the applications and competitive advantages of data mining and business intelligence in SME performance. The review examined 93 studies and identified the BI and data-mining technologies commonly used by small and medium-sized enterprises. The findings indicated that dashboards were among the most frequently used BI tools, accounting for 31.18 percent of the identified applications. The study also associated BI and data-mining tools with improved operational efficiency, strategic decision-making, competitiveness, and business performance. However, the researchers identified challenges such as implementation costs, limited technical skills, and difficulties in managing data quality.

These findings closely align with the Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently because the proposed system will include interactive dashboards and centralized reports for sales and operational monitoring. The spa dashboard can display key performance indicators such as total revenue, daily sales, service and product sales, appointment volume, popular services, therapist productivity, customer ratings, voucher usage, inventory movement, and cash-over-short results. Filters may allow management to view data by date, service category, therapist, payment method, customer type, or transaction status. The study's discussion of cost and technical-skill limitations is also relevant because the proposed system should provide a practical, user-friendly, and affordable solution designed specifically for the spa's operational needs. Therefore, Shao Valentinus Tsiu, Mfanelo Ngobeni, Lesley Mathabela, and Bonginkosi Thango (2025) demonstrate that dashboards are important BI tools for SMEs and can support efficiency, decision-making, and competitiveness, which directly supports the proposed sales reporting and analytics module.

Guanyi Wang (2022) examined sales forecasting for firms using a multiple regression model. The study focused on using historical sales information and other relevant business variables to estimate future sales performance. Regression analysis establishes the relationship between a dependent variable, such as sales, and independent variables that may influence sales outcomes. The study showed that regression-based forecasting can help organizations analyze sales patterns, estimate future demand, and support business planning. It is particularly useful because

regression models can provide an interpretable equation that allows managers to understand how different factors contribute to predicted sales.

Jiheng Xin (2023) investigated the use of linear regression to predict future sales and selling prices. The study used existing sales-related information and adopted the ordinary least squares method to identify the relationship between historical data and future sales outcomes. Linear regression provides a relatively simple and interpretable forecasting approach because it represents the relationship between variables through a mathematical equation. The model can estimate future sales volume or prices by using previously recorded observations and determining the trend that best fits the available data.

Ahmad M. A. Zamil, Nawras M. Nusairat, T. G. Vasista, Marwan M. Shammot, and Ahmad Yousef Areiqat (2021) developed a Python-based linear regression model for predicting sales based on advertising media data. The study used ordinary least squares regression to examine the relationship between advertising expenditure and sales results. The researchers presented the regression coefficients in an equation and used data visualization to display the relationship between the predictor variables and sales. The findings indicated that television advertising was the feature variable that most influenced the predicted sales in the examined dataset. The study demonstrates how regression models can identify influential factors while also producing sales predictions.

These findings are relevant to Recovery Iloilo Spa because the proposed system will allow management to compare sales results with promotional and operational activities. The spa may analyze whether voucher campaigns, celebration discounts, marketing expenses, special offers, or other promotional activities are associated with increases in service and product sales. Although the proposed system will initially focus on historical transaction data and formula-based forecasting, promotional information may be used as an additional variable in future versions of the model. The resulting reports can help management determine whether a promotion generated additional revenue or merely reduced the transaction value through discounts. Therefore, Ahmad M. A. Zamil, Nawras M. Nusairat, T. G. Vasista, Marwan M. Shammot, and Ahmad Yousef Areiqat (2021) demonstrate that linear regression can be used to examine the factors that influence sales, which is applicable to the proposed spa system's promotional analytics and sales forecasting functions.

Chen Dong (2024) applied multiple linear regression and the least squares method to sales prediction. The study examined the relationship between product sales and factors such as price, in-store spending, and online advertising spending. The researchers used the coefficient of determination, or R^2 , and graphical comparisons between observed and predicted values to evaluate the fitness of the regression model. The resulting model achieved an R^2 value of approximately

0.80, while the predicted sales trend generally followed the observed sales trend. The study concluded that regression-based sales prediction can help businesses estimate sales trends and improve the allocation of promotional funds.

These findings align with Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently because the proposed analytics module will help management examine factors related to revenue performance. For the spa, relevant variables may include service prices, product sales, number of completed appointments, discount amounts, voucher usage, payment method, and marketing expenses. The system can use historical transactions to calculate predicted revenue and compare the prediction with actual sales. The use of R^2 and graphical comparisons can also help management evaluate whether the forecasting model is sufficiently reliable for operational planning. Therefore, Chen Dong (2024) demonstrates that multiple linear regression can estimate sales trends and support the allocation of promotional resources, which is relevant to the proposed spa system's sales reporting, marketing analysis, and revenue forecasting features.

Chandru M. (2025) conducted a study on Python-based predictive modeling for sales forecasting in small and medium-sized enterprises using empirical evidence from Boopathi Spices in India. The study compared several forecasting methods, including linear regression, random forest, ARIMA, SARIMA, and a naïve forecasting model. The research evaluated the models using forecasting error measures such as Mean Absolute Percentage Error. The findings reported that the linear regression model produced the most accurate result among the compared methods for the examined business dataset, with a MAPE of 12.20 percent. The study demonstrates that linear regression can serve as a practical and interpretable forecasting method for SMEs when historical sales records are available

These findings are relevant to Recovery Iloilo Spa because the proposed system is intended for a small business that may not have extensive technical resources or large datasets. A linear regression model can provide a practical starting point for estimating future spa revenue from historical transaction records. The system can collect daily or monthly sales totals and use them to produce forecasts for upcoming periods. Forecasting accuracy can be evaluated using MAPE or another error measure by comparing predicted sales with actual recorded sales. This process can help management determine whether the forecasting feature is producing useful results and whether the model should be updated as additional transactions are recorded. Therefore, Chandru M. (2025) demonstrates that linear regression can be an effective and practical forecasting technique for SMEs, which supports the proposed spa system's formula-based revenue forecasting and management decision-support functions.

Sadr Lufti Mufreni and Ahmad Mushawwir (2025) developed a web application at PT Stechoq Robotika Indonesia using the Agile Scrum methodology. The development process used user stories, a product backlog, daily sprint activities, sprint reviews, and iterative feature testing. The application was developed using ReactJS, TypeScript, and Tailwind CSS, while the Scrum process allowed the development team to divide the project into short and manageable iterations. The researchers reported that the method supported continuous testing, regular evaluation, and the gradual improvement of features based on user and project requirements.

Niu (2023) conducted a review of optimization studies on system appointment scheduling to trace how the field has developed and where it is heading. The review classified existing appointment-scheduling system structures and decision-making frameworks and organized them for comparison. It then summarized a system-reliability optimization framework built around three elements: optimization objectives, decision variables, and constraints. The author also examined the algorithms most commonly applied across different appointment-scheduling systems and supplemented this with a bibliometric analysis of the literature. One key observation was that genetic algorithms and simulation-optimization algorithms dominate the methods used in appointment scheduling. The review further noted that neural networks and deep learning have become core technologies in health-management optimization, while comprehensive, real-world system optimization remains insufficiently explored. For the present study, this work situates the booking module of the proposed spa system within a large and established body of scheduling research. Although the reviewed algorithms are more advanced than a small spa requires, the underlying goal of balancing resource use against client satisfaction is identical. The review therefore supports adopting a structured, rule-based approach to slot allocation while leaving room for more advanced optimization as the business grows.

Ala and Chen (2022) produced a comprehensive review of the appointment-scheduling problem in healthcare service systems, drawing on more than 150 scientific papers. Their review addressed appointment-scheduling problems together with various applications and solution approaches used in healthcare settings. The literature was categorized according to several problem specifications, including the flow of patients, patient preferences, and random arrival and service times. The authors observed that many methods have been proposed to shorten patient waiting time while keeping idle time in service centers as low as possible. Among the existing approaches, they identified simulation models, mathematical optimization, Markov-chain methods, and artificial intelligence as the most practical for improving satisfaction. They also structured the recent literature according to whether decisions are made at the strategic, tactical, or operational level. This taxonomy is directly useful for the present study

because a spa booking system must likewise manage client flow and reduce both waiting and staff idle time. The distinction between strategic, tactical, and operational decisions helps clarify which choices the proposed system should automate and which remain managerial. The review consequently provides a conceptual map that guides how the spa's appointment feature can be designed and justified.

Benjaafar (2022) and co-authors studied an appointment system in which a finite number of customers are scheduled under a service-level constraint. The system was designed so that the expected waiting time of each individual customer cannot exceed a given threshold, while appointment times are still set as early as possible. The authors showed that, under this constraint, a workable schedule can be produced through a sequential approach that fixes each appointment based on the previous ones. Using a transient queueing analysis together with the theory of majorization, they characterized the structure of the optimal schedule. They proved that the minimum interval required between appointments increases along the arrival sequence. They also found that later arrivals are more likely to encounter an empty system and to wait less than their expected service time, and they proposed a simple heuristic that is asymptotically optimal. For the present study, these service-level ideas translate into fair, predictable booking rules that cap how long any spa client should expect to wait. The emphasis on fairness across customers is especially relevant to a service business whose reputation depends on a consistent client experience. The study therefore offers both a theoretical basis and a practical heuristic for designing the spa's appointment intervals.

Kuo (2020) and colleagues investigated appointment overbooking as a way to mitigate the adverse effects of client no-shows, such as reduced resource utilization and revenue. They noted that overbooking carries a risk, because when more clients arrive than expected, both overtime and waiting time increase. To study this trade-off, they proposed a framework that combines a stochastic mixed-integer linear program with a simulation model to guide scheduling decisions. The authors first derived theoretical properties of the model and then ran computational experiments to test them. Their results showed that setting session capacity close to the appointment request rate most effectively balances the conflicting objectives. They further found that a fixed-capacity policy can perform as well as a dynamic overbooking policy under a constant request rate, while dynamic overbooking is slightly better when demand is uneven. For the proposed spa system, this work provides guidance on whether and how to allow limited overbooking to offset no-shows. Because a small spa has limited staff, the finding that capacity should track the request rate is directly applicable to its daily scheduling. The study thus cautions the present research on the efficiency-versus-accessibility trade-offs that any overbooking rule must respect.

2.4 Synthesis of the Reviewed Literature

The reviewed literature, taken together, converges on a single proposition that directly frames the present study: micro, small, and medium service enterprises that continue to depend on manual, paper-based, and fragmented workflows incur measurable losses in efficiency, accuracy, and customer service, and these losses are substantially reduced when operations are consolidated into a single web-based platform. The foreign studies, the local studies, and the related literature approach this proposition from different angles — empirical system development, localized capstone implementation, and conceptual or methodological grounding — but they arrive at compatible conclusions. This synthesis draws those strands together, compares their findings and methods, identifies where they agree and disagree, exposes the gaps that remain, and establishes how the reviewed body of work supports the Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently.

The foreign studies establish the empirical case for digital transformation and, module by module, supply precedent for every component the proposed system requires. At the broadest level, Tenkale, Anokar, Rathod, and Kamble (2026) show that migrating manual operational workflows to accessible web platforms boosts enterprise productivity while narrowing the digital divide. For the booking module, appointment and reservation systems such as those of Manish, Sengar, and Singh (2026), Kokate, Samrit, and Bisen (2026), and Leony and Prayitno (2025) demonstrate that digitizing the appointment lifecycle removes manual scheduling friction, with Kokate, Samrit, and Bisen (2026) explicitly pairing scheduling with a business-intelligence layer for the salon and wellness sector. A large cluster of customer-relationship-management studies — including Nisa, Maharani, and Santoso (2026) and, on the analytics side, Siek and Lim (2025) with a segmentation dashboard and Ahmed, Maâti, and Al-Sarem (2020) with a predictive data-mining model — confirms that centralizing customer data improves service consistency and supports data-driven decisions. For inventory, studies applying Economic Order Quantity, Reorder Point, and Safety Stock methods, such as Istanto (2017), Afrizal, Minardi, and Mahendra (2025), and Audina, Putra, and Ananto (2026), show that formal stock-control logic can be embedded directly in web and point-of-sale systems. Payment functionality is supported by Liu (2025) and Khan, Raza, Ahsan, and Irshad (2024), who demonstrate secure QR-code and digital-wallet payments even under intermittent connectivity, while staff and attendance management is addressed by Kumar (2025), Cox (2023), and Nugraha and Astutik (2022), and authentication security by Mayorga and Yoo (2025). The dominant methodological signature across these works is design-and-development research paired with functional testing and usability measurement.

The local studies narrow this general case to the Philippine setting and, in several instances, to the exact business type and locale of the present research. Panes, Parreño, Buñol, Peña, and Gabral (2015) and Cadio, Caligan, Calizo, Esis, and Pardilla (2015) are especially pertinent, since both developed management and

customer-information systems for spa and wellness businesses in Iloilo City — the same environment the proposed system targets. More recent capstone and journal work — Abalos, Felix, Serante, and Rodriguez (2024) for Von Czar Salon and Spa, Abijay, Galicia, and Laurencio (2023) for Lhay Lhor Nail Lounge and Spa, and Garcia, Orolfo, Toledo, Avila, and Derla (2024) for Scissors and Mirrors Salon — confirms that online booking and management systems are now a common and effective response to manual scheduling in Philippine salons and spas, a pattern reinforced by appointment-system studies such as Marasigan (2025) and Ampuan and Delena (2022). The local literature also supplies the analytics and security dimensions that distinguish the present study from a simple booking tool: Dumo, Durano, Raper, Salvador, Alcober, Angel, and Raguro (2024) show that sales forecasting is feasible and useful for local businesses, while Samonte, Dalina, Pingol, and Yee (2024) demonstrate role-based access control in a Philippine web application. Methodologically, the local studies rely predominantly on the Waterfall or modified Waterfall model, prototype development, and user-acceptance evaluation, reflecting the capstone context in which most were produced.

The related literature provides the conceptual scaffolding, models, and algorithms that explain why the systems described above work and how they should be designed. For scheduling, Zhou and Yue (2021) formulate a robust appointment model for multi-stage service systems under uncertainty, and Srinivas and Ravindran (2018) combine machine learning with heuristic rules in a prescriptive-analytics scheduling framework, giving formal grounding to the booking module. For decision support, business-intelligence research by Gauzelin and Bentz (2017) links BI systems to improved decision-making and performance among SMEs, while the data-mining review by Tsiu, Ngoben, Mathabela, and Thango (2025) surveys the competitive advantages such techniques confer. Sales-forecasting methods based on regression — examined by Wang (2022), Xin (2023), and Chandru (2025) — supply the analytical basis for the system's revenue-projection features. Customer-facing concepts are covered by CRM-adoption studies such as Salah, Yusof, and Mohamed (2018) and Wiid and Senoane (2025), by service-quality frameworks such as the SERVQUAL-and-Importance-Performance analysis of Slamet and Sulistiyowati (2022), and by digital-payment behavior studies such as David, John, Prasana, Vignesh, and Kumaran (2025) and Miko, Joe, and Oktavia (2025). Where the foreign and local studies show that integrated systems work, the related literature explains the mechanisms — formal scheduling, forecasting, business intelligence, and service-quality measurement — through which they work.

Comparing the three bodies of work reveals strong consistency on the central claims and only limited, explicable inconsistency. All three consistently report that centralization reduces manual error and administrative time, that real-time data improves managerial oversight, and that usability and staff acceptance are decisive for adoption — a point stressed equally by the foreign development studies and by the local user-acceptance evaluations. The clearest divergence is not in conclusions but in rigor and scale: the foreign and related-literature studies more often report quantified outcomes, optimization models, or standardized instruments, whereas the local studies, constrained by their capstone context, more often report functional success and user

satisfaction without large-scale statistical validation. A second divergence concerns scope. Most individual studies, foreign or local, implement one or two modules — booking alone, CRM alone, inventory alone, or payment alone — rather than a fully unified platform that combines appointment management, point-of-sale, inventory, staff management, and analytics in a single system. This is a difference of coverage rather than of contradictory findings, and it is precisely the difference the present study is positioned to address.

Methodologically, the reviewed studies share a common developmental orientation but differ in their evaluation strategies. The prevailing approach across all three sections is system design and development — whether structured as Waterfall, Rapid Application Development, or Agile and Scrum, the last illustrated by Mufreni and Mushawwir (2025), Zhang and Dorn (2013), and Chaudhary (2025) — followed by testing and user evaluation. Foreign studies lean toward functional testing combined with quantitative usability and efficiency metrics; local studies lean toward prototype demonstration and acceptance testing; and the related literature contributes modeling, simulation, and algorithmic evaluation rather than deployed systems. For the present research, this pattern recommends a hybrid stance: adopt an iterative, user-centered development methodology consistent with the Agile and RAD traditions in the literature, and evaluate the resulting system with both functional testing and a recognized usability instrument, so that the study contributes the kind of measurable evidence that the stronger foreign and related-literature studies model and that the local capstone literature has more rarely supplied.

Three research gaps emerge from the synthesis. First, there is an integration gap: although each required capability — booking, point-of-sale, inventory, staff and attendance management, payments, and analytics — is individually well supported, few studies unify them into one platform designed specifically for a spa, and fewer still add a forecasting and analytics layer on top of that integration. Second, there is a contextual gap: while spa and salon systems exist in the Philippines, very few are situated in Iloilo City and tailored to the operational realities of a local recovery spa, and the most locale-specific references — Panes, Parreño, Buñol, Peña, and Gabral (2015) and Cadiao, Caligan, Calizo, Esis, and Pardilla (2015) — are now a decade old and predate current web and analytics practice. Third, there is an evidentiary gap: much of the directly comparable local work reports functional completion rather than rigorously measured operational improvement, leaving room for a study that documents efficiency and usability outcomes with standardized measures.

Taken as a whole, the reviewed literature provides a firm foundation for the Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently. It confirms that the problem the system addresses — manual, fragmented spa operations — is real and costly, that each proposed module has precedent and demonstrated value, that appropriate development and evaluation methodologies are available and well tested, and that the conceptual foundations of scheduling, inventory control, forecasting, business intelligence, and service-quality measurement are sound. The present study's specific contribution is to close the integration, contextual, and evidentiary gaps identified above: it unifies appointment scheduling, point-of-sale and payment processing, inventory tracking, staff and attendance management, and an

analytics-and-forecasting layer into a single web-based platform; it situates that platform in the concrete operational setting of a recovery spa in Iloilo City, updating and extending decade-old local precedents; and it is designed to be evaluated with functional and usability measures so that its operational benefits are demonstrated rather than merely asserted. In doing so, the present research advances the reviewed literature from fragmented, single-purpose, and often functionally-evaluated solutions toward an integrated, locally grounded, and empirically assessed system for streamlining spa operations and analytics.

CHAPTER 3

METHODOLOGY

3.1 Research Design

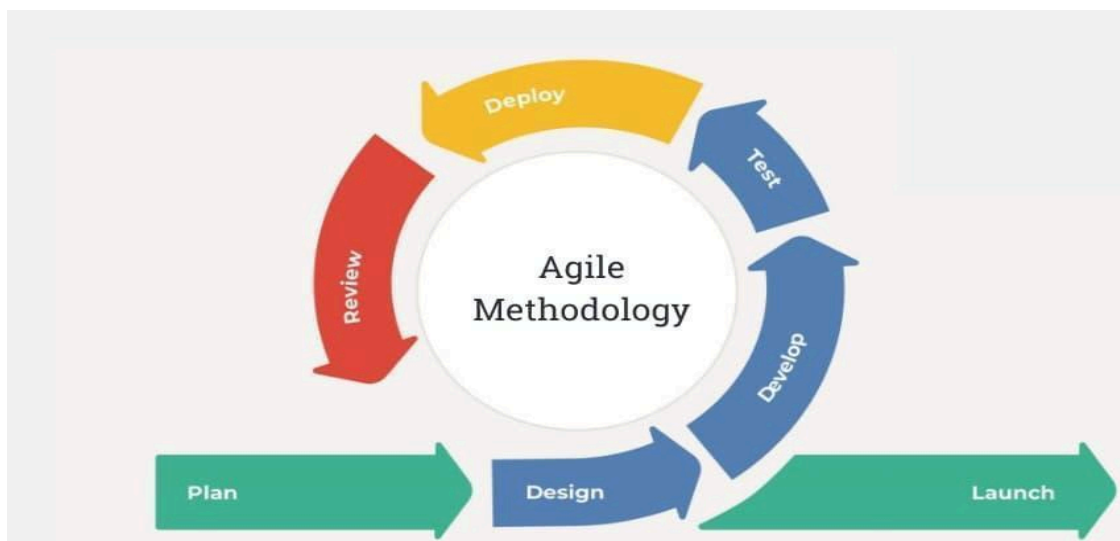
This study will employ a developmental research design in the design, development, implementation, and evaluation of the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics. This research design is appropriate because the primary objective of the study is to develop a web-based information system that will improve the spa's daily business operations through the integration of computerized processes. The proposed system will provide a centralized online platform for managing appointment scheduling, customer accounts and records, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, commission computation, discount and voucher management, customer feedback, operational reports, and business analytics. The system will be developed using a three-tier architecture consisting of the presentation layer, application layer, and data layer to ensure organized system structure, maintainability, and efficient data management.

The developmental research design will guide the researchers through the systematic stages of requirements gathering, analysis of existing business processes, system planning, interface and database design, system development, testing, implementation, and evaluation. Information will be gathered through consultations and interviews with the spa owner, management, receptionists, therapists, and other staff to identify operational

requirements, existing challenges, and user needs. Based on the collected information, the researchers will develop the various modules of the proposed system, including appointment management, customer account management, therapist management, inventory management, payment processing, Point-of-Sale (POS), commission management, voucher management, feedback management, security features, and operational reporting and analytics. These modules will be designed to improve operational efficiency, reduce manual workloads, strengthen record management, and support better customer service.

Upon completion of the development phase, the proposed system will undergo comprehensive testing and evaluation to determine whether its features and functions operate according to the identified requirements. The researchers will evaluate the system's quality based on the applicable software quality characteristics (such as those in ISO/IEC 25010, if required by your institution), including functionality, usability, reliability, performance efficiency, and security. Feedback and recommendations from the spa owner, management, receptionists, therapists, and other users will also be gathered to identify areas for improvement and determine whether the proposed system effectively supports appointment management, therapist coordination, inventory monitoring, payment processing, customer record management, operational reporting, business analytics, and overall service delivery at Recovery Iloilo Spa.

3.2 System Development Methodology



The study will use the **Agile Development Model in developing the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics**. Agile is a software development methodology that emphasizes continuous improvement, iterative development, regular feedback, and close collaboration between developers and stakeholders throughout the system development process. This approach will allow the researchers to work closely with the spa owner, management, receptionists, therapists, and other authorized personnel to ensure that the proposed system will effectively address the operational requirements of Recovery Iloilo Spa. Through continuous testing, evaluation, and user feedback, the researchers will be able to refine system features, resolve identified issues, and implement necessary improvements throughout each development iteration. The proposed web-based system will be developed using a three-tier architecture consisting of the presentation layer, application layer, and data layer to promote organized system structure, maintainability, scalability, and efficient data management.

The Agile Development Model will be used, consisting of the following phases:

1. Planning and Requirements Analysis – This phase will involve identifying, gathering, and analyzing the operational requirements, existing business processes, and challenges encountered by Recovery Iloilo Spa through consultation and coordination with the spa owner, management, HR personnel, receptionists, therapists, and other staff members. The researchers will examine the current workflow and identify issues related to customer registration and account management, email OTP verification, appointment scheduling, therapist assignment and attendance, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, customer record management, therapist commission computation, voucher and discount management, customer feedback collection, security controls, operational reporting, and revenue forecasting. The information gathered during this phase will serve as the basis for identifying the functional and non-functional requirements, determining project objectives, and planning the modules to be included in the proposed system.

2. System Design Phase – During this phase, the researchers will prepare the overall architecture and design of the proposed web-based system. The system will be designed using a three-tier architecture consisting of the presentation

layer, application layer, and data layer to ensure proper separation of system components and efficient processing of information. The researchers will develop the database structure, system architecture, process flow diagrams, navigation structure, user interface layouts, and module designs to ensure that the platform will be organized, responsive, user-friendly, and secure. The design will include modules for customer registration and authentication, customer account management, appointment scheduling and management, therapist management, inventory management, Point-of-Sale (POS), payment processing, refund management, commission computation, discount and voucher management, customer feedback, activity logging, security controls, notifications, operational reports, business analytics, and revenue forecasting.

3. Development Phase – During this phase, the researchers will develop the proposed system by coding and integrating each module based on the approved system requirements and design specifications. Development will follow an iterative approach, allowing each completed feature to be reviewed, tested, and improved before proceeding to the next iteration. The researchers will develop customer registration with email OTP verification, secure user authentication, profile management, shopping cart functionality, appointment scheduling for online, walk-in, home-service, hotel, and partner bookings, therapist assignment and reassignment, therapist attendance monitoring, specialty management, inventory monitoring, Point-of-Sale (POS) transactions, payment gateway integration, refund processing, commission computation, voucher and discount validation, customer feedback collection, role-based access control, gate code verification, PIN verification, activity logging, automated notifications, operational reporting, business analytics, and formula-based revenue forecasting. Continuous collaboration with the spa owner, management, receptionists, therapists, and other intended users will ensure that each module will satisfy operational requirements and business needs.

4. Testing Phase – This phase will focus on verifying and evaluating the quality and performance of the proposed system before deployment. The researchers will conduct various testing procedures to determine whether all modules and system functions operate according to the specified requirements. The testing process will evaluate customer registration and authentication, appointment scheduling, therapist assignment, attendance monitoring, inventory management, Point-of-Sale (POS) transactions, payment processing, refund handling, commission computation, voucher validation, customer feedback, notifications, security mechanisms, operational reports, analytics, and revenue forecasting. The system will also be evaluated in terms of functionality, usability, reliability, performance efficiency, security, and compatibility. Feedback and

recommendations from the spa owner, management, receptionists, therapists, and customers will be gathered to identify necessary improvements and ensure that the proposed system effectively supports daily business operations.

5. Deployment Phase – This phase will involve implementing the completed web-based system within Recovery Iloilo Spa for actual operational use. The spa owner, administrators, receptionists, therapists, accounting personnel, and customers will be allowed to access and utilize the appropriate system features based on their assigned roles and permissions. The deployed system will support customer registration, appointment scheduling, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher and discount validation, customer feedback collection, operational reporting, business analytics, and revenue forecasting. During implementation, users will provide additional comments and recommendations regarding the system's functionality, usability, efficiency, and overall effectiveness. The collected feedback will support further refinements and ensure that the proposed system will improve operational efficiency, strengthen record management, enhance customer service, and support informed decision-making within Recovery Iloilo Spa.

3.3 System Architecture

The **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** will adopt a **three-layer architecture** consisting of the **Presentation Layer**, **Application Layer**, and **Data Layer**. This architectural approach will separate the system into three interconnected components, each having a specific responsibility in processing, managing, and presenting information. By separating the user interface, business logic, and database, the system will become more organized, maintainable, scalable, and secure. This architecture will also simplify future maintenance, updates, and feature enhancements since modifications made to one layer will have minimal impact on the other layers.

The proposed system will also follow a **hub-and-spoke communication model**, wherein the **Application Layer** will serve as the central processing hub between the Presentation Layer and the Data Layer. All requests initiated by users—including customer registrations, appointment bookings, inventory transactions, payment processing, therapist management, report generation, and other operational activities will first be processed and validated by the Application Layer before being stored in or retrieved from the database. Likewise, all information requested by users will pass through the Application Layer before being displayed on the user interface. This

approach will ensure proper validation, secure processing, and consistent handling of all system transactions.

The system will be structured into three main layers:.

1.Presentation Layer (User Interface)

The Presentation Layer will serve as the part of the system that users directly interact with through a web browser. It will be responsible for displaying information, receiving user input, and providing access to the different features of the proposed system. The interface will be designed to be responsive, organized, user-friendly, and accessible through desktop computers, laptops, tablets, and mobile devices with internet access.

For customers, the Presentation Layer will provide pages for customer registration with Email OTP verification, secure login, password recovery, profile management, browsing available spa services and products, shopping cart management, therapist selection or "Any Available Therapist" option, online appointment booking, walk-in appointment viewing, home-service booking requests, hotel and partner bookings, appointment rescheduling and cancellation, viewing appointment history, payment processing, refund request submission, viewing digital receipts, receiving appointment notifications, and submitting customer feedback and ratings after completed services.

For the spa owner, administrators, receptionists, accountants, HR personnel, and therapists, the Presentation Layer will provide dashboards and interfaces for managing appointments, therapist assignments and reassignments, therapist attendance, therapist specialties, customer records, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, commission monitoring, voucher and discount management, customer feedback review, activity logs, operational reports, business analytics, and revenue forecasting. Authorized users will also be able to monitor appointment status, approve bookings, manage products and services, update inventory, verify transactions, and generate reports through the system interface.

The Presentation Layer will be developed using HTML, CSS, and Vanilla JavaScript to provide a responsive and interactive user experience across supported web browsers.

2. Application Layer

The Application Layer will serve as the core processing component of the proposed system. It will receive requests from the Presentation Layer, validate user input, execute business rules, process transactions, communicate with the database, and return processed information to the Presentation Layer. This layer will ensure that all business operations are executed accurately, consistently, and securely.

Specifically, the Application Layer will:

- Process customer registration, Email OTP verification, secure authentication, password management, and profile updates.
- Manage appointment scheduling for online, walk-in, home-service, hotel, and partner bookings while validating therapist availability, operating hours, travel-time allowances, booking conflicts, duplicate reservations, and overbooking.
- Process appointment approvals, cancellations, rescheduling, therapist assignments and reassignments, appointment status updates, and automated notifications.
- Manage therapist attendance, therapist specialties, workload balancing, automatic therapist rotation, and performance monitoring.
- Process shopping cart transactions, Point-of-Sale (POS) transactions, payment processing through supported payment methods, refund requests, receipt generation, and transaction verification.
- Automatically calculate therapist commissions based on completed services, service categories, transaction records, multi-person sessions, and applicable commission rules.
- Monitor inventory by updating stock quantities whenever products are used, sold, restocked, or adjusted while identifying low-stock items requiring replenishment.
- Manage customer records, appointment history, service preferences, transaction history, voucher usage, and feedback records.
- Validate discounts and vouchers, including Senior Citizen, PWD, Employee, Celebration Discounts, Gift Certificates, and promotional vouchers while preventing duplicate redemption and unauthorized usage.
- Process customer ratings, comments, and feedback linked to completed appointments and assigned therapists.
- Implement security mechanisms including role-based access control, hidden administrative login with Gate Code verification, PIN verification for sensitive transactions, activity logging, and secure session management.
- Generate operational reports, dashboards, business analytics, sales summaries, therapist productivity reports, inventory reports, POS readings, cash reconciliation reports, and formula-based revenue forecasts.
- Control all communication between the Presentation Layer and the Data Layer to ensure that every transaction is validated, processed securely, and recorded accurately.

The Application Layer will be developed using PHP, which will implement the business logic and facilitate communication between the user interface and the MariaDB relational database.

3. Data Layer (Database)

The Data Layer will serve as the centralized repository of all information managed by the proposed system. It will be responsible for storing, organizing, updating, retrieving, and protecting all operational data required by Recovery Iloilo Spa. The database will ensure data integrity, consistency, security, and efficient retrieval while supporting business operations, reporting, and analytics.

The Data Layer will store:

- Customer account information, login credentials, Email OTP verification records, user roles, and profile information.
- Appointment records including booking details, selected services, assigned therapists, schedules, booking status, service locations, appointment history, and add-on services.
- Therapist information including attendance records, specialties, assignments, schedules, commission records, and performance-related transaction history.
- Customer records including contact information, service preferences, transaction history, feedback records, and voucher usage.
- Inventory records including products, supplies, stock quantities, stock movements, restocking history, and low-stock monitoring.
- Point-of-Sale (POS) transactions, payment records, refund transactions, payment methods, digital receipts, discounts applied, and complete transaction history.
- Voucher records, gift certificates, promotional campaigns, redeemed discounts, eligibility validation, and redemption history.
- Customer feedback, ratings, comments, and service evaluations linked to completed appointments and therapists.
- Notifications, appointment confirmations, payment confirmations, and email notification history.
- Security records including activity logs, PIN verification records, role permissions, and audit trails.
- Operational reports, business analytics, sales summaries, inventory reports, therapist productivity reports, cash reconciliation records, and formula-based revenue forecasting data.

The proposed system will utilize MariaDB as its relational database management system to ensure efficient data storage, secure information management, reliable

transaction processing, and fast retrieval of records. The centralized database will support approximately 30 core database tables, allowing the system to maintain organized business information while supporting the operational, reporting, and analytical requirements of Recovery Iloilo Spa.

3.4 Tools and Technologies Used

The development of the Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics will utilize various software tools and technologies to support the design, development, implementation, and maintenance of the proposed web-based system. These technologies will work together to provide a responsive user interface, efficient business logic, secure data management, reliable transaction processing, and organized operational reporting.

- **Frontend: HTML, CSS, and Vanilla JavaScript** - HTML (HyperText Markup Language), CSS (Cascading Style Sheets), and Vanilla JavaScript will be used to develop the Presentation Layer of the proposed system. HTML will provide the structure and content of the web pages, CSS will define the visual appearance, layout, and responsiveness of the user interface, while Vanilla JavaScript will add interactivity and dynamic functionality to the system. These technologies will enable the development of a responsive, organized, and user-friendly interface that can be accessed through desktop computers, laptops, tablets, and mobile devices using a web browser. The frontend will support customer registration, appointment booking, shopping cart management, therapist selection, payment processing, dashboard visualization, notifications, and other interactive system features.

- **Backend: PHP** - PHP will be used as the primary server-side programming language for developing the Application Layer of the proposed system. It will process business logic, validate user input, manage user authentication, handle appointment scheduling, therapist assignment, inventory management, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher validation, customer feedback, operational reports, business analytics, and revenue forecasting. PHP will also facilitate secure communication between the user interface and the MariaDB database while implementing security features such as role-based access control, Gate Code verification, PIN authentication, session management, and activity logging.

- **Database Management System: MariaDB** - MariaDB will serve as the relational database management system of the proposed Recovery Iloilo Spa Integrated System. It will store, organize, retrieve, and manage all operational data required by the system, including customer accounts, Email OTP verification records, appointments, therapist assignments, attendance records, inventory

information, Point-of-Sale (POS) transactions, payment records, refund transactions, customer feedback, voucher and discount records, commission records, activity logs, notifications, operational reports, business analytics, and revenue forecasting data. MariaDB will ensure data integrity, security, consistency, and efficient retrieval of information while supporting approximately thirty core database tables within the proposed system.

- **Integrated Payment Gateway: PayMongo Payment Gateway** - The proposed system will integrate an online payment gateway to support secure electronic transactions. The payment gateway will enable customers to pay for appointments using supported digital payment methods such as GCash, QR Ph, and debit or credit cards. It will also facilitate payment verification, transaction recording, digital receipt generation, and secure communication between the system and electronic payment services.

- **Email Service** - An email service utilizing PHPMailer will be integrated into the proposed system to support automated email notifications. This service will be responsible for sending Email OTP verification codes during customer registration, appointment confirmations, booking updates, payment notifications, password recovery messages, and other system-generated communications to registered users.

- **Reporting and Export Libraries: PhpSpreadsheet and mPDF** - PhpSpreadsheet and mPDF will be used as reporting and export libraries to support the daily sales and cash reconciliation report of the proposed system. PhpSpreadsheet will generate downloadable Excel workbooks that preserve live formulas for figures such as net cash and short/over computation, while mPDF will generate downloadable PDF copies of the same report for printing and record-keeping. Together, these libraries will allow management and the receptionist to export accurate, formatted daily and sales reports directly from the system.

- **Development Environment: Visual Studio Code** - Visual Studio Code (VS Code) will be used as the primary Integrated Development Environment (IDE) during system development. It provides features such as syntax highlighting, intelligent code completion, debugging tools, extension support, integrated terminal access, and Git version control integration. These capabilities will assist the researchers in writing, testing, debugging, and maintaining the source code more efficiently throughout the software development process.

- **Web Browser** - Modern web browsers such as Google Chrome, Microsoft Edge, Mozilla Firefox, and Safari will be used for testing and accessing the proposed web-based system. These browsers will allow users to utilize the system without installing additional software, ensuring accessibility across different devices and operating systems.

- **Three-Tier Architecture** - The proposed system will be developed using a three-tier architecture consisting of the Presentation Layer, Application Layer, and Data Layer. This architectural design will separate the user interface, business logic, and database management into independent layers to improve maintainability, scalability, security, and overall system performance.

3.5 Data Gathering Techniques

The researchers will use the following data gathering techniques to identify the operational requirements of Recovery Iloilo Spa, analyze the existing business processes, and gather the information necessary for the design, development, implementation, and evaluation of the proposed **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics**.

- **Survey and Questionnaire** - The researchers will prepare and distribute structured questionnaires to the spa owner, management, receptionists, therapists, and other authorized staff members to collect information regarding the current operational practices, existing problems, user requirements, and suggested system improvements. The survey will gather measurable data related to appointment scheduling, customer account management, therapist assignment, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, commission computation, voucher and discount management, customer feedback, operational reporting, security requirements, and business analytics. The collected responses will serve as one of the primary bases for identifying the functional and non-functional requirements of the proposed system.

- **Observation of Current Operations** – The researchers will conduct direct observation of the daily operations of Recovery Iloilo Spa to obtain firsthand information regarding the current workflow and operational procedures. The observation will focus on customer registration, appointment scheduling, therapist assignment, attendance monitoring, inventory management, payment processing, Point-of-Sale (POS) transactions, commission computation, voucher validation, customer service, and report preparation. This technique will enable the researchers to identify delays, repetitive tasks, manual errors, bottlenecks, and other operational inefficiencies that the proposed system will address.

- **Interview** – The researchers will conduct interviews with the spa owner, management, HR personnel, receptionists, therapists, and other authorized staff

members to obtain detailed information regarding the operational processes, business policies, user expectations, and specific system requirements. The interviews will provide a deeper understanding of existing challenges involving appointment management, therapist scheduling, payment procedures, inventory control, commission computation, promotional discounts, customer feedback, reporting practices, and overall business operations. Information gathered during the interviews will help validate the data collected through surveys and observations.

- **Review of Existing Systems** – The researchers will review existing spa management systems, appointment booking systems, and other related web-based information systems to examine their features, workflows, user interfaces, and best practices. This technique will enable the researchers to identify useful functionalities, effective design approaches, and operational strategies that may be adapted and incorporated into the proposed Recovery Iloilo Spa Integrated System while ensuring that the developed system addresses the specific operational requirements of the client.

- **Document and Record Review** –The researchers will review existing business documents and operational records maintained by Recovery Iloilo Spa to understand the current methods of data management and transaction processing. Documents such as appointment records, customer information, inventory lists, therapist schedules, attendance records, commission records, sales reports, payment records, voucher records, promotional activities, and other operational documents will be examined to identify existing data structures, workflow processes, and areas requiring improvement. The information obtained through document review will assist the researchers in designing an organized database structure, accurate system modules, and reliable reporting features for the proposed system.

3.6 Testing Procedures

To ensure that the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** functions properly and satisfies the operational requirements of Recovery Iloilo Spa, the researchers will conduct several testing procedures throughout the development process. These testing procedures will help identify and correct system errors, verify that all modules operate according to the specified requirements, improve system performance, and ensure that the proposed system is reliable, secure, and suitable for actual implementation.

- **Unit Testing** – Each module of the proposed system will be tested individually to verify that its functions operate correctly before being integrated with other modules. The researchers will evaluate customer registration and Email OTP verification, user authentication, appointment scheduling, therapist assignment and attendance, shopping cart, inventory management, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher validation, customer feedback, notifications, operational reports, business analytics, and revenue forecasting. Unit testing will ensure that every module performs its intended functions accurately and independently.

- **System Testing** – After all system modules have been integrated, the researchers will conduct system testing to evaluate the proposed system as a complete web-based platform. This testing procedure will verify the interaction among the Presentation Layer, Application Layer, and Data Layer, ensuring that information flows correctly throughout the system. It will also confirm that all integrated modules—including customer account management, appointment management, therapist management, inventory monitoring, POS transactions, payment processing, refund handling, commission computation, voucher management, customer feedback, security mechanisms, notifications, operational reports, analytics, and revenue forecasting—function together properly without errors or inconsistencies.

- **User Acceptance Testing (UAT)** – The proposed system will undergo User Acceptance Testing (UAT) with the participation of the spa owner, administrators, receptionists, therapists, accounting personnel, and selected customers. The purpose of this testing is to determine whether the system satisfies user requirements, supports actual business operations, and provides a user-friendly and efficient experience. Participants will evaluate the system based on its functionality, ease of use, accuracy, convenience, and overall effectiveness in improving appointment scheduling, customer management, therapist coordination, inventory control, payment processing, reporting, and other operational activities.

- **Performance Testing** – Performance testing will be conducted to evaluate the system's responsiveness, stability, and efficiency while handling multiple users, concurrent appointments, continuous Point-of-Sale (POS) transactions, inventory updates, report generation, and other operational activities. The researchers will determine whether the system can maintain acceptable response times and reliable performance under normal operating conditions while processing large volumes of business data.

- **Security Testing (Optional)** – Security testing will be conducted to verify that the proposed system adequately protects user accounts, confidential business information, and customer records from unauthorized access. The researchers will evaluate authentication mechanisms, Email OTP verification, role-based access control, Gate Code verification, PIN verification, session management, activity logging, payment processing security, and database access controls. This testing will ensure that sensitive information is handled securely and that only authorized users can access restricted system functions.

- **Final Evaluation** - After all testing activities have been completed, the researchers will evaluate the overall performance of the proposed system to determine whether it meets the operational requirements of Recovery Iloilo Spa. The evaluation will focus on the system's **functionality, usability, reliability, performance efficiency, and security**, as well as its ability to improve appointment management, therapist coordination, inventory monitoring, customer record management, payment processing, operational reporting, business analytics, and overall service delivery. Feedback and recommendations gathered from the spa owner, management, receptionists, therapists, accounting personnel, and customers will be used to identify necessary improvements before the final implementation of the proposed system.

3.7 Evaluation Criteria

The developed **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** will be evaluated to determine its overall quality, performance, reliability, and usefulness in improving the daily operations of Recovery Iloilo Spa. The evaluation will assess whether the proposed system effectively satisfies the operational requirements of the establishment and provides an organized, efficient, and secure platform for managing spa services.

- **Functionality** – This criterion will evaluate whether the proposed system performs all intended functions correctly and completely. It will determine whether the different modules including customer registration, appointment scheduling, therapist management, inventory monitoring, Point-of-Sale (POS) transactions, payment processing, refund management, commission computation, voucher and discount management, customer feedback, notifications, operational reports, analytics, and revenue forecasting operate according to the specified requirements. The system should produce accurate outputs and successfully complete user transactions without major errors.

- **Accuracy** – This criterion measures how correctly the system processes, stores, and displays information. It evaluates the correctness of appointment schedules, inventory records, transaction details, commission computations, customer information, and generated reports. Accurate information helps reduce errors and supports better daily operations.

- **Usability** – This criterion will determine how easy the proposed system is to learn, understand, and operate for customers, receptionists, therapists, administrators, management, and other authorized users. It will evaluate the clarity of menus, forms, navigation, instructions, dashboards, and overall user interface design. A user-friendly system will allow users to complete their tasks efficiently with minimal training and confusion.

- **Reliability** – This criterion will evaluate the ability of the proposed system to operate consistently and continuously without unexpected failures, crashes, or interruptions. It will determine whether appointments, transactions, reports, notifications, and other system functions can be processed accurately during regular business operations while maintaining stable performance and dependable service.

- **Performance Efficiency** – This criterion will assess how efficiently the proposed system performs its operations while utilizing available system resources. It will evaluate the speed of appointment processing, payment transactions, inventory updates, report generation, analytics processing, database retrieval, and other system functions. An efficient system should respond quickly while maintaining smooth performance during normal daily operations.

- **Security** – This criterion will evaluate the effectiveness of the security mechanisms implemented within the proposed system to protect confidential business information and customer data. It will assess user authentication, Email OTP verification, role-based access control, Gate Code verification, PIN authentication, activity logging, payment security, and database protection. The system should ensure that only authorized users can access restricted functions while maintaining the confidentiality, integrity, and security of stored information.

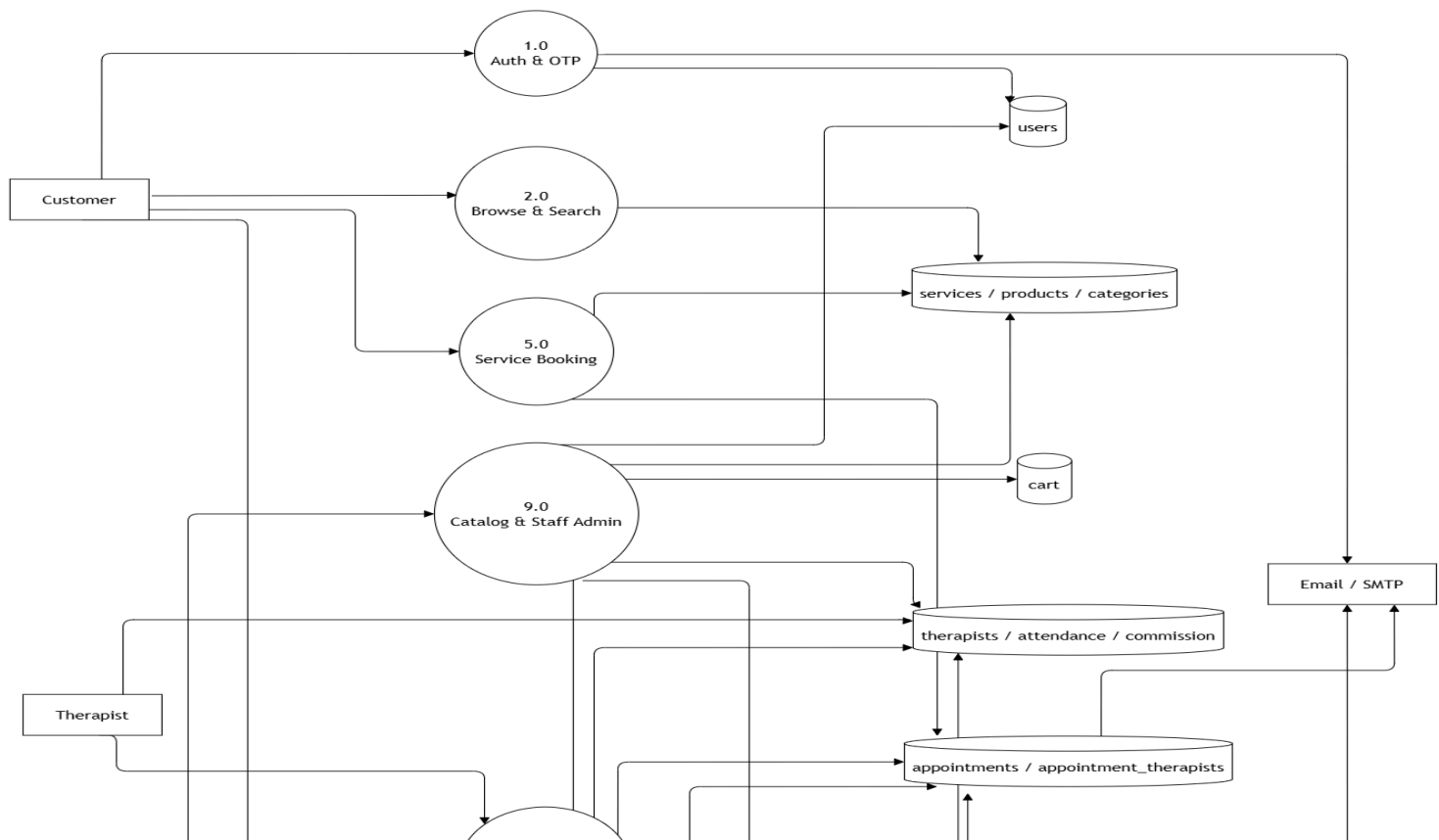
- **Maintainability** – This criterion will evaluate how easily the proposed system can be modified, updated, corrected, and maintained after implementation. It will assess whether the three-tier architecture, organized source code, database structure, and modular system design support future enhancements, bug fixes, and feature improvements without significantly affecting other components of the system.

- **Feedback Integration** – This criterion evaluates how effectively the system collects, stores, and manages customer feedback and ratings. It checks whether customer comments and evaluations are properly recorded and made available for review by management. This information can help identify areas for improvement and support better service quality.

- **Overall User Satisfaction** – This criterion will evaluate the overall satisfaction of the spa owner, management, receptionists, therapists, accounting personnel, and customers regarding the proposed system. It will determine whether users believe that the system improves appointment management, therapist coordination, inventory monitoring, payment processing, customer service, operational reporting, business analytics, and the overall efficiency of Recovery Iloilo Spa.

SUPPORTING DIAGRAMS (Included within Chapter 3)

Data Flow Diagram (DFD – Level 1)



Integrated System DFD Level 1 Overview

User (Customer/Staff) → Integrated System → Database

1.0 Authentication and OTP

- Customers, therapists, receptionists, and administrators will submit their registration or login credentials to gain authorized access to the Recovery Iloilo Spa Integrated System according to their assigned user roles.
- The system will validate the submitted credentials against **D1: Users**, verify the account status and assigned permissions, and establish a secure session that will control access to authorized system functions.
- A one-time password (OTP) will be generated and transmitted through the **Email / SMTP** service to verify the identity of the user before access to the system will be granted.

2.0 Browse and Search

- Customers will browse the spa catalog and search for available wellness services, products, and service categories offered by Recovery Iloilo Spa.
- The system will retrieve the requested information from **D2: Services, Products, and Categories** and will display updated details regarding service descriptions, pricing, session duration, available products, and service classifications.
- The process will allow customers to review accurate service information before proceeding with appointment booking or purchasing spa products.

3.0 Cart and Checkout

- Customers will select spa services and retail products and will place their chosen items into their shopping cart before confirming their purchase or reservation.

- The system will record the selected items in **D3: Cart** and, after checkout has been confirmed, will create the corresponding transaction in **D4: Orders and Order Items** for further processing.
- The completed transaction will then be forwarded to the Payment process so that the selected payment method can be processed and the customer's order can be finalized.

4.0 Payment

- The system will receive payment requests originating from online checkout transactions and walk-in transactions processed by the receptionist or cashier.
- Payment requests for online transactions will be transmitted securely to **PayMongo**, and the payment confirmation returned by the gateway will be used to update the payment status and transaction details stored in **D4: Orders and Order Items**.
- After successful payment processing, the system will record the completed transaction and will send payment confirmations, receipts, or related notifications through the **Email / SMTP** service whenever applicable.

5.0 Service Booking

- Customers will submit appointment requests by selecting their preferred spa service, appointment schedule, therapist preference, number of persons, and other booking details required by the establishment.
- The system will retrieve service information, pricing, therapist availability, and service categories from **D2: Services, Products, and Categories** to validate the booking request and calculate the applicable service charges.
- Confirmed booking information will be stored in **D5: Appointments and Appointment Therapists**, where the reservation will remain pending until it is reviewed and processed by the receptionist or authorized staff.

6.0 Appointment Operations

- Receptionists will review pending appointments, approve or decline booking requests, assign or reassign therapists, update appointment status, and mark appointments as completed once the requested services have been rendered.
- Therapists will submit their attendance and duty information, while the system will update **D6: Therapists, Attendance, and Commission** to record attendance, therapist assignments, and the commissions earned from completed services.
- Throughout the appointment process, the system will continuously update **D5: Appointments and Appointment Therapists**, generate notifications through the **Email / SMTP** service whenever necessary, and record administrative activities in **D8: Activity Logs and Notifications** to maintain an accurate audit trail.

7.0 Walk-In

- Receptionists or cashiers will encode appointment information for customers who will avail themselves of spa services without making prior online reservations.
- The system will create the corresponding appointment record in **D5: Appointments and Appointment Therapists** and simultaneously generate the related transaction in **D4: Orders and Order Items** for proper sales recording.
- The completed walk-in transaction will then be forwarded to the Payment process so that payment can be processed using the same workflow applied to online transactions, ensuring consistency in transaction recording.

8.0 Reporting and Analytics

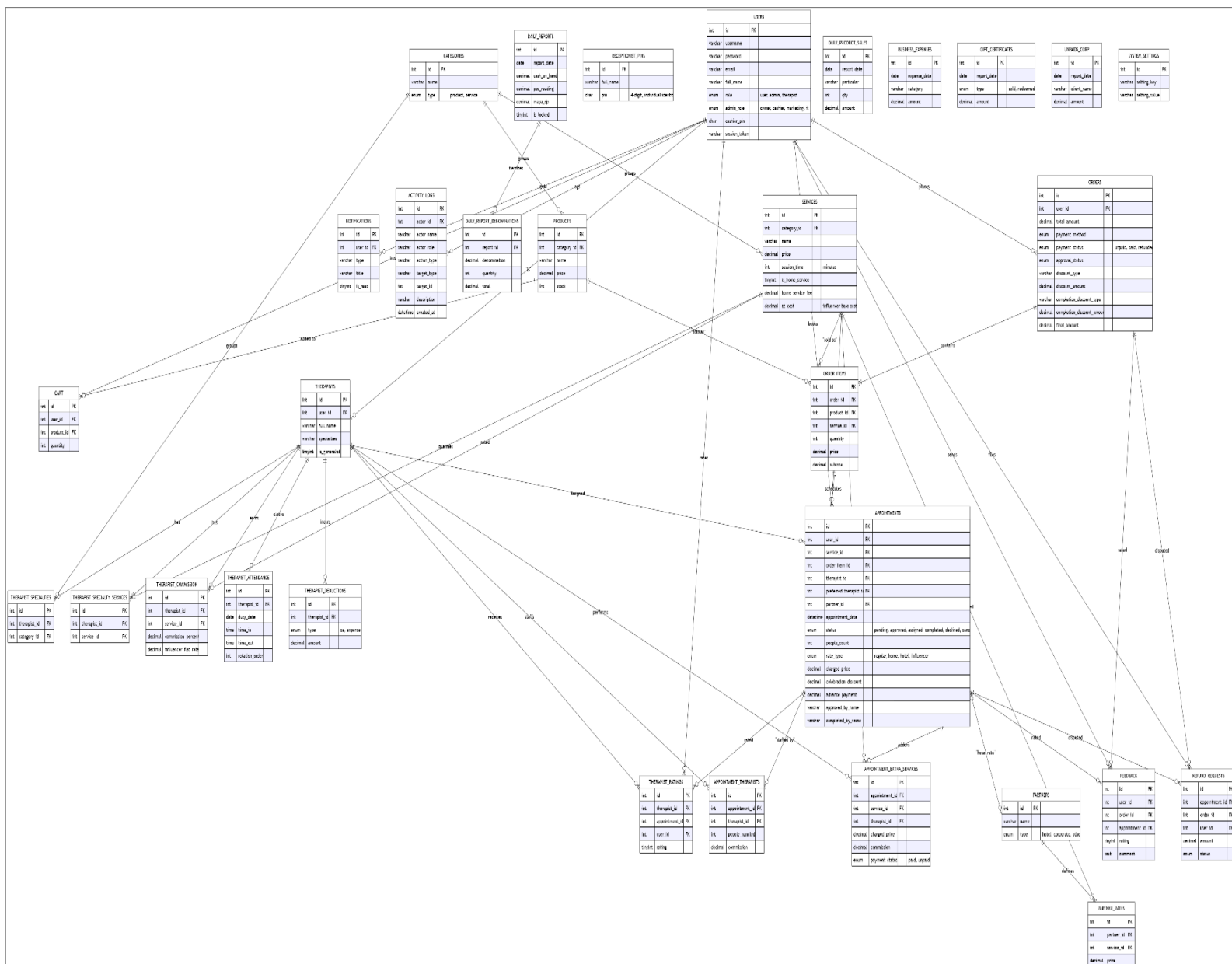
- Receptionists will prepare daily cash counts, cash denomination records, and end-of-day summaries, while the Owner or Administrator will review the consolidated operational performance of the spa through the reporting dashboard.
- The system will store daily cash reconciliation records in **D7: Daily Reports and Denominations** and will retrieve operational data from **D4: Orders and Order Items**, **D5: Appointments and Appointment Therapists**, and **D6: Therapists, Attendance and Commission** to generate comprehensive reports.
- The reporting and analytics module will generate sales reports, appointment reports, inventory movement summaries, therapist productivity reports, commission reports, customer activity reports, payment summaries, voucher utilization reports, and operational analytics. It will also produce short-term revenue forecasts based on recorded historical transactions and allow management to filter, view, and export reports into Excel and PDF formats. These analytics will provide management with timely insights that will support operational monitoring, performance evaluation, and informed business decision-making.

9.0 Catalog and Staff Administration

- The Owner or Administrator will manage the spa catalog by adding, updating, or maintaining available services, products, categories, pricing, inventory information, therapist profiles, specialties, commission rates, attendance records, and user accounts.
- The system will store and update catalog information in **D2: Services, Products, and Categories**, maintain therapist information within **D6: Therapists, Attendance, and Commission**, and manage user accounts and access permissions through **D1: Users**.

All administrative modifications performed within the system will be recorded in **D8: Activity Logs and Notifications** to maintain accountability, preserve an accurate audit trail, and support secure management of system activities.

Entity Relationship Diagram (ERD)



Entity Attributes

- users (id, username, password, email, full_name, role, admin_role, cashier_pin, session_token)
- therapists (id, user_id, full_name, specialties, is_generalist)
- receptionist_pins (id, full_name, pin)
- categories (id, name, type)
- services (id, category_id, name, price, session_time, is_home_service, home_service_fee, at_cost)
- products (id, category_id, name, price, stock)
- cart (id, user_id, product_id, quantity)
- orders (id, user_id, total_amount, payment_method, payment_status, approval_status, discount_type, discount_amount, completion_discount_type, completion_discount_amount, final_amount)
- order_items (id, order_id, product_id, service_id, quantity, price, subtotal)
- appointments (id, user_id, service_id, order_item_id, therapist_id, preferred_therapist_id, partner_id, appointment_date, status, people_count, rate_type, charged_price, celebration_discount, advance_payment, approved_by_name, completed_by_name)
- appointment_therapists (id, appointment_id, therapist_id, people_handled, commission)
- appointment_extra_services (id, appointment_id, service_id, therapist_id, charged_price, commission, payment_status)
- therapist_specialties (id, therapist_id, category_id)
- therapist_specialty_services (id, therapist_id, service_id)
- therapist_commission (id, therapist_id, service_id, commission_percent, influencer_flat_rate)
- therapist_attendance (id, therapist_id, duty_date, time_in, time_out, rotation_order)
- therapist_deductions (id, therapist_id, type, amount)
- therapist_ratings (id, therapist_id, appointment_id, user_id, rating)
- partners (id, name, type)
- partner_rates (id, partner_id, service_id, price)
- feedback (id, user_id, order_id, appointment_id, rating, comment)
- refund_requests (id, appointment_id, order_id, user_id, amount, status)

- notifications (id, user_id, type, title, is_read)
- activity_logs (id, actor_id, actor_name, actor_role, action_type, target_type, target_id, description, created_at)
- daily_reports (id, report_date, cash_on_hand, pos_reading, maya_dp, is_locked)
- daily_report_denominations (id, report_id, denomination, quantity, total)
- daily_product_sales (id, report_date, particular, qty, amount)
- business_expenses (id, expense_date, category, amount)
- gift_certificates (id, report_date, type, amount)
- unpaids_corp (id, report_date, client_name, amount)
- system_settings (id, setting_key, setting_value)

Relationships

- A **User** will create an account in the system in order to access the features corresponding to the assigned role, which may be a customer, therapist, receptionist, or administrator. The administrative role will further identify whether the account belongs to the owner, cashier, marketing personnel, or information technology staff, while the session token will maintain secure access throughout system usage.
- A **User** account may optionally be linked to one **Therapist** record. This relationship will allow therapists to authenticate using their user account while maintaining a separate professional profile containing therapist information, specialties, and assignment details.
- A **User** will be able to schedule multiple **Appointments** for spa services. Each appointment will store the selected service, appointment date and time, preferred therapist, assigned therapist, partner information, number of persons, pricing details, discounts, advance payment, booking status, and approval information.
- A **User** will be able to place multiple **Orders** involving spa services and wellness products. Each order will record the payment method, payment status, approval status, applied discounts, total amount, and final payable amount for proper transaction management.
- A **User** will be able to add and manage selected **Products** inside the **Cart** before confirming checkout. The cart will temporarily store the selected products together with their quantities, allowing customers to review or modify their selections before completing the purchase.

- A **User** will receive **Notifications** regarding appointment confirmations, payment updates, refund requests, promotional announcements, and other important system activities. Each notification will store its type, title, and read status to help users monitor system-generated messages.
- A **User** will be able to submit **Feedback** after completing a service transaction by providing a rating and written comment regarding the quality of service and overall customer experience. The feedback record will be linked to the corresponding appointment and order for proper reference.
- A **User** may submit **Refund Requests** for cancelled appointments or payment-related concerns. Each refund request will record the corresponding appointment, order, requested amount, and processing status to support organized refund management.
- A **User** will generate **Activity Logs** whenever important administrative actions are performed within the system. Each activity log will record the actor's identity, role, action performed, affected record, description, and timestamp to provide accountability and auditing.
- **Categories** will organize both **Services** and **Products** to simplify classification, searching, filtering, and management within the system. Each category will identify whether it belongs to a service classification or a product classification, allowing one category to contain multiple services or products.
- **Categories** will also be connected to **Therapist Specialties** to identify the service categories in which therapists are qualified to perform treatments.
- **Services** will be connected to **Appointments** because every appointment will require at least one selected spa service. Service information will also be linked to **Order Items** so that booked treatments can be recorded as part of customer transactions.
- **Services** will be connected to **Therapist Commission** records to determine therapist earnings based on either commission percentages or predefined flat rates assigned to particular services.
- **Services** will also be connected to **Therapist Specialty Services** to identify the individual services that each therapist will be authorized and qualified to perform.
- **Services** will be associated with **Partner Rates** to support customized pricing agreements established for hotel partners, corporate clients, and other affiliated organizations.
- **Products** will be connected to the **Cart** and **Order Items** so that product selections, purchased quantities, inventory movement, and sales transactions can be properly recorded and monitored.
- **Orders** will contain multiple **Order Items**, representing the individual services and products included within a customer transaction. Each order item will store

the corresponding product or service, quantity, unit price, and subtotal for accurate billing.

- **Order Items** may generate corresponding **Appointments** whenever spa services are included within a customer order. This relationship will establish a connection between the payment transaction and the scheduled spa session.

- **Appointments** may include additional treatments through **Appointment Extra Services**. These records will store the additional service selected, assigned therapist, charged amount, commission earned, and payment status of every add-on service rendered during the appointment.

- **Appointments** will be assigned to one or more therapists through **Appointment Therapists**, allowing multiple therapists to participate in servicing group appointments. Each assignment will record the therapist involved, the number of persons handled, and the commission earned.

- **Appointments** will also store both the assigned therapist and the customer's preferred therapist whenever applicable, allowing management to accommodate customer preferences whenever therapist availability permits.

- **Appointments** will be connected to **Therapist Ratings** and **Feedback** to measure customer satisfaction, evaluate therapist performance, and support continuous service quality improvement after completed appointments.

- **Appointments** may also generate **Refund Requests** whenever cancellations, payment adjustments, or customer disputes occur. Each refund record will reference the related appointment, associated order, requested refund amount, and approval status.

- **Appointments** will be connected to **Partners** whenever bookings are made under corporate accounts, hotel partnerships, or special business arrangements so that negotiated pricing can automatically be applied.

- **Therapists** will perform the spa services associated with scheduled appointments. Therapist records will store professional information, listed specialties, and a generalist indicator identifying whether the therapist may perform all available services.

- **Therapists** will receive earnings through **Therapist Commission** records, which will define either commission percentages or fixed influencer rates corresponding to specific services.

- **Therapists** may also receive deductions recorded in **Therapist Deductions** for salary adjustments, cash advances, penalties, or other authorized deductions affecting commission computation.

- **Therapists** will maintain attendance records through **Therapist Attendance**, which will monitor duty dates, clock-in time, clock-out time, and therapist rotation order used during appointment assignment.

- **Therapists** will establish their service qualifications through **Therapist Specialties** and **Therapist Specialty Services**, allowing management to identify therapist expertise by both service category and specific spa treatment.
- **Therapists** will receive customer evaluations through **Therapist Ratings**, which will record the rating submitted for each completed appointment to assist management in monitoring therapist performance.
- **Partners** will be connected to both **Appointments** and **Partner Rates** to support hotel partnerships, corporate clients, and negotiated service pricing. Each partner record will identify the organization type and the corresponding pricing arrangements available within the system.
- **Daily Reports** will store the overall operational summary prepared at the end of each business day, including report date, cash on hand, POS reading, Maya digital payment collections, and report lock status to preserve finalized financial records.
- **Daily Reports** will contain multiple **Daily Report Denominations**, which will record the breakdown of cash denominations counted during cashier balancing and end-of-day reconciliation.
- **Daily Product Sales** will record the products sold during each business day, including the product description, quantity sold, and total sales amount, to support inventory monitoring and sales reporting.
- **Business Expenses** will store operational expenditures such as supplies, maintenance costs, utilities, marketing expenses, and other authorized business expenses to support financial monitoring and profitability analysis.
- **Gift Certificates** will store records of gift certificates issued, sold, and redeemed by the spa, including transaction dates, certificate types, and corresponding monetary amounts for promotional monitoring and financial reporting.
- **Unpays Corp** will store outstanding balances of corporate clients whose transactions remain unpaid. Each record will contain the client name, report date, and outstanding amount to support receivable monitoring and collection activities.
- **Receptionist Pins** will store the unique PIN assigned to each receptionist for transaction verification and cashier authorization. These records will operate independently and will primarily be referenced during payment processing, cashier verification, and end-of-day financial reporting.
- **System Settings** will store the configurable parameters that control various operational behaviors of the Recovery Iloilo Spa Integrated System through individual setting keys and corresponding values. These records will operate independently and will be referenced whenever system-wide configuration values are required during system execution.

Use Case Diagram

Actors:

- **Customer (User)** – The **Customer** will be the primary user of the system and will represent any client who avails of the services offered by Recovery Iloilo Spa. Customers will be able to register an account with email OTP verification, log in securely, browse available spa services and wellness products, manage their shopping cart, place orders, and schedule appointments through the online booking facility. They will be able to select a preferred therapist or choose any available therapist, request home-service appointments, hotel or partner bookings, and add extra services whenever applicable. Customers will also be able to monitor the status of their appointments and payments, receive email and in-system notifications, view transaction history and digital receipts, submit feedback, rate the therapist who rendered the service, redeem vouchers and discounts, and submit refund requests whenever necessary.
- **Receptionist / Cashier** – The **Receptionist / Cashier** will be responsible for managing daily front-desk operations and customer transactions within the spa. This role will review, approve, decline, reschedule, and cancel appointment requests, assign or reassign therapists, process walk-in customers, record additional services rendered during appointments, and verify therapist availability before confirming bookings. Receptionists will process onsite payments, verify cashier PINs before sensitive financial transactions, handle refund requests, validate vouchers and discounts, and generate receipts. At the end of each business day, they will prepare the daily report, record cash denominations, POS readings, Maya collections, product sales, business expenses, gift certificate transactions, and unpaid corporate accounts for financial reconciliation.
- **Owner / Administrator** –The **Owner / Administrator** will oversee the overall operation, configuration, and maintenance of the Recovery Iloilo Spa Integrated System. This role will manage user accounts and role assignments, therapist records, therapist specialties, commission rates, attendance records, service categories, products, inventory, partner establishments, partner pricing, vouchers, gift certificates, and system settings. The administrator will also monitor appointments, transactions, refund requests, customer feedback, therapist ratings, operational reports, sales analytics, revenue forecasts, and activity logs. In addition, the administrator will configure business rules, review financial reports, monitor employee productivity, and supervise the overall performance and security of the system.

- **Therapist** – The **Therapist** will be responsible for providing spa treatments based on appointments assigned through the system. Therapists will be able to record their attendance by clocking in and out, monitor their assigned schedules, review customer booking details, and identify additional services requested before or during treatment sessions. They will also monitor completed appointments, automatically computed commissions, deductions, assigned specialties, customer ratings, and feedback related to the services they have rendered. Attendance records and therapist rotation information will also be maintained to support fair workload distribution and efficient scheduling.

- **PayMongo Payment Gateway** – The **PayMongo Payment Gateway** will serve as an external payment processing service responsible for handling online payment transactions initiated by customers. Whenever a customer chooses an online payment method such as GCash, QR Ph, or bank card, the system will securely forward the payment request to PayMongo for processing. After payment verification, PayMongo will return the transaction result to the system, allowing payment records and order status to be updated automatically. This external actor will not directly access the system and will interact only through secure payment requests and responses.

- **Email / SMTP Service** – The **Email / SMTP Service** will function as an external communication service responsible for sending electronic messages generated by the Recovery Iloilo Spa Integrated System. It will deliver one-time passwords (OTP) for customer account verification, appointment confirmations, booking updates, payment acknowledgements, refund notifications, password reset instructions, and other system-generated announcements. This external actor will not access the system directly and will function solely as the communication channel for outgoing email notifications.

Use Cases:

Customer Use Cases

- **Register and Log In Account** – The system will allow customers to create an account and securely access the Recovery Iloilo Spa Integrated System using their registered credentials. During registration, customers will provide the required personal information, and a one-time password (OTP) will be sent to their registered email address for verification before the account can be activated. After successful login, customers will be able to access the services, products, bookings, payments, notifications, and other features available to their account.

- **Manage User Profile Information** –The system will allow customers to view and update their personal profile information whenever necessary. Customers will be able to modify details such as their full name, contact number, email address, and password while maintaining accurate customer records. Keeping profile information updated will help improve communication, booking management, and transaction processing.
- **Browse Spa Services and Products** – The system will allow customers to browse all spa services, wellness products, packages, and promotional offerings available at Recovery Iloilo Spa. Customers will be able to view service descriptions, session duration, pricing, product availability, and other relevant details before making a booking or purchase. This feature will help customers select the services and products that best suit their preferences.
- **View Categories and Pricing** – The system will allow customers to browse services and products according to their assigned categories. Customers will be able to compare available services, products, prices, promotional packages, and home-service availability before confirming a booking or purchase. This feature will help customers make informed decisions based on their preferred treatments.
- **Add Products to Cart** – The system will allow customers to add wellness products into their shopping cart before completing a purchase. Customers will be able to specify product quantities, review selected items, and update their selections before proceeding to checkout. The shopping cart will temporarily store all selected products until the transaction is confirmed.
- **Manage Cart and Checkout** – The system will allow customers to review the contents of their shopping cart before placing an order. Customers will be able to modify quantities, remove unwanted products, and verify the total amount due before proceeding with checkout. Once confirmed, the system will generate the corresponding order and transaction records. .
- **Place Orders for Products and Services** – The system will allow customers to purchase wellness products and confirm service transactions through the integrated platform. Each order will contain complete information regarding purchased services, products, quantities, unit prices, discounts applied, payment details, and computed totals to ensure accurate billing and transaction monitoring.
- **Book Spa Appointments** – The system will allow customers to schedule spa appointments by selecting their preferred services, appointment date, available time slot, number of persons, and preferred service location. Customers may create online bookings, while walk-in appointments will be encoded by authorized reception staff. Submitted appointments will remain pending until reviewed and processed by authorized personnel.

- **Select Home Service or In-Store Service** – The system will allow customers to choose whether the appointment will be performed as an in-spa service, home-service booking, hotel booking, or partner booking whenever applicable. The system will automatically compute any additional travel fees or partner pricing based on the selected appointment type.
- **Select Preferred Therapist** – The system will allow customers to select their preferred therapist or choose the "Any Available Therapist" option during appointment booking. The system will record the customer's preference and will assign the requested therapist whenever available. If the selected therapist is unavailable, management will be able to assign another qualified therapist.
- **Add Extra Appointment Services** – The system will allow customers to request additional spa services before or after an appointment has been approved. Each additional service will be recorded separately together with the assigned therapist, corresponding charges, commission, and payment status. This feature will allow customers to customize their spa experience without creating another appointment.
- **Pay Through the Online Payment Facility** – The system will allow customers to settle payments using available payment options, including cash, digital wallets, payment gateway transactions, payment upon completion, or approved pay-later arrangements. The system will automatically calculate transaction totals, generate digital receipts, record payment history, and update payment status after successful processing.
- **Track Order and Appointment Status** – The system will allow customers to monitor the progress of their appointments throughout the booking lifecycle. Customers will be able to view whether an appointment is pending, assigned, approved, completed, declined, or cancelled. This feature will provide customers with real-time updates regarding their reservations.
- **View Payment and Refund Status** – The system will allow customers to monitor payment confirmations and refund requests associated with their transactions. Customers will be able to review the processing status of submitted refund requests and determine whether the request remains pending, approved, or declined.
- **Receive System Notifications** – The system will allow customers to receive notifications regarding account verification, appointment confirmations, therapist assignments, payment confirmations, refund updates, reminders, promotional announcements, and other important system activities. Notifications will also be sent through the integrated email service whenever necessary.
- **View Transaction History** – The system will allow customers to review their complete transaction history, including previous appointments, purchased products, completed services, payments, refunds, and booking activities. This

feature will provide customers with convenient access to their historical records whenever needed.

- **Submit Service Feedback and Ratings** – The system will allow customers to submit comments, suggestions, and overall evaluations regarding the spa services they received after completing an appointment. These feedback records will assist management in monitoring customer satisfaction and identifying opportunities for service improvement.

- **Rate Assigned Therapist** – The system will allow customers to evaluate the therapist who performed their completed spa session by submitting ratings and comments. Therapist evaluations will help management monitor employee performance, recognize outstanding service, and improve service quality.

- **Request Refund or Appointment Cancellation** – Allows customers to file a request for the cancellation of a booking or the reimbursement of a transaction. The system records the amount involved and holds the request in a pending state. Management may then review the request and take the appropriate action.

For Receptionist / Cashier:

- **Log In and Verify Security PIN** – The Receptionist or Cashier will log in using the assigned account credentials before accessing the operational functions of the system. A cashier PIN will be required before performing protected financial transactions, approving sensitive operations, or finalizing the daily report. This process will help strengthen accountability and ensure that only authorized personnel can perform secured activities.

- **Manage Walk-In Appointments** – The Receptionist or Cashier will record appointments for customers who visit the spa without making an online reservation. The system will create the corresponding appointment record, assign the selected services, and prepare the transaction for payment processing. This process will ensure that walk-in customers will follow the same workflow as online bookings.

- **Review and Process Appointment Requests** – The Receptionist or Cashier will review appointment requests submitted by customers before confirming the reservation. The system will allow staff to approve, decline, reschedule, or cancel appointments while recording the personnel responsible for each action. This process will help maintain organized appointment scheduling and proper operational control.

- **Assign and Reassign Therapists** – The Receptionist or Cashier will assign available therapists to approved appointments based on therapist availability, specialty, rotation order, and customer preference whenever possible. The

system will also allow therapist reassignment when scheduling conflicts or operational changes occur. This process will ensure proper workload distribution and efficient service coordination.

- **Manage Appointment Details and Add-on Services** – The Receptionist or Cashier will update appointment information whenever necessary, including modifying schedules, adding extra services, updating customer notes, and recording service completion details. Additional services rendered during or after the appointment will be recorded together with the assigned therapist, corresponding charges, commission, and payment status. This process will ensure that all services provided are properly documented.

- **Process Point-of-Sale (POS) Transactions** – The Receptionist or Cashier will process customer payments through the Point-of-Sale module using supported payment methods such as cash, card, digital wallet, payment at completion, or pay-later arrangements. The system will automatically calculate transaction totals, apply eligible discounts, generate receipts, and update payment records. This process will ensure accurate financial recording and faster transaction processing.

- **Process Refund Requests** – The Receptionist or Cashier will review customer refund requests and perform the necessary verification before forwarding them for approval or processing according to company policies. The system will update the refund status and maintain complete refund transaction records for monitoring purposes.

- **Manage Customer Transactions** – The Receptionist or Cashier will monitor customer orders, appointment payments, payment status, and transaction history. The system will maintain complete transaction records to support customer inquiries, financial reconciliation, and operational monitoring.

- **Monitor Daily Sales Transactions** – The Receptionist or Cashier will review all completed transactions recorded throughout the business day. The system will automatically consolidate payment information, service sales, product sales, discounts, refunds, and transaction summaries to support daily financial monitoring.

- **Prepare Daily Reports** – The Receptionist or Cashier will prepare the end-of-day operational report by reviewing recorded sales, POS readings, payments, gift certificates, business expenses, unpaid corporate accounts, and other financial records. The system will generate the required report for management review and documentation.

- **Record Cash Denominations and Cash Reconciliation** – The Receptionist or Cashier will record the quantity of each cash denomination at the close of business. The system will automatically calculate the total cash on hand,

compare it with recorded POS collections, and generate cash-over or cash-short reconciliation results to improve financial accuracy.

- **Record Daily Product Sales** – The Receptionist or Cashier will review and confirm product sales recorded during the business day. The system will maintain daily product sales records to support inventory monitoring, sales analysis, and operational reporting.

- **Record Business Expenses** – The Receptionist or Cashier will encode business expenses incurred during daily operations, including the expense category, amount, and transaction date. The system will include these records in the daily financial reports and analytics.

- **Manage Gift Certificate Transactions** – The Receptionist or Cashier will record gift certificates that are sold or redeemed during the business day. The system will maintain accurate gift certificate balances and transaction history for monitoring promotional activities.

- **Manage Corporate Unpaid Transactions** – The Receptionist or Cashier will record unpaid transactions for corporate clients and partner establishments. The system will maintain outstanding balances and support monitoring of accounts receivable for future collection.

- **Submit and Lock Daily Reports** – The Receptionist or Cashier will finalize the daily operational report after verifying all recorded transactions, payments, expenses, and cash balances. Once submitted, the system will lock the report to prevent unauthorized modifications while preserving the integrity of financial records for management review.

For Owner / Administrator:

- **Log In Account** – The Owner or Administrator will securely access the system using assigned account credentials. The login process will restrict administrative functions to authorized personnel based on assigned roles and permissions. After successful authentication, authorized users will be able to access system management, monitoring, reporting, and configuration functions.

- **Manage User Accounts and Roles** – The Owner or Administrator will create, update, activate, deactivate, and manage customer and employee accounts within the system. Administrative roles such as owner, administrator, receptionist, therapist, marketing personnel, and information technology personnel will also be assigned according to operational responsibilities. This process will ensure that users will only have access to the system features appropriate to their assigned roles.

- **Manage Therapist Records and Specialties** – The Owner or Administrator will maintain therapist profiles, specialties, service qualifications, and generalist status. Therapist specialty categories and individual service qualifications will

also be managed to support proper therapist assignment during appointment scheduling. This process will help ensure that therapists will only be assigned to services they are qualified to perform.

- **Manage Services, Products, Categories, and Pricing** – The Owner or Administrator will maintain the complete service and product catalog by creating, updating, or deactivating categories, services, products, prices, session durations, home-service fees, and inventory information. This process will ensure that customers and staff will always access accurate and updated information during bookings and transactions.

- **Manage Inventory and Product Availability** – The Owner or Administrator will monitor inventory levels, product availability, stock movements, and replenishment activities. The system will assist in identifying low-stock items and maintaining accurate inventory records to support uninterrupted spa operations.

- **Configure Therapist Commission and Deduction Rules** – The Owner or Administrator will define therapist commission percentages, influencer flat-rate commissions, and other compensation rules based on completed services. Administrative users will also record therapist deductions such as cash advances, penalties, or other adjustments to ensure accurate payroll computation.

- **Manage Partners and Partner Service Rates** – The Owner or Administrator will maintain records of hotels, corporate clients, and other partner establishments. Negotiated service rates for each partner will also be configured to ensure that partner bookings will automatically apply the appropriate pricing during appointment processing.

- **Manage Receptionist Security PINs** – The Owner or Administrator will create, update, and maintain the security PIN assigned to each receptionist or cashier. These PINs will be required before protected financial transactions, cashier verification, and daily report submission to strengthen accountability and transaction security.

- **Review and Manage Customer Transactions** – The Owner or Administrator will review customer orders, appointment transactions, payment records, discounts, vouchers, and transaction approvals whenever administrative intervention is required. This process will help maintain accurate financial records and ensure that transactions comply with company policies.

- **Review and Process Refund Requests** – The Owner or Administrator will examine refund requests submitted by customers by reviewing the corresponding appointment, order, payment information, and refund amount. The request status will then be updated based on management approval to support organized refund processing and customer service.

- **Monitor Appointments and Therapist Operations** – The Owner or Administrator will monitor appointment schedules, therapist assignments, attendance records, completed services, commissions, and operational workload. This process will help ensure efficient workforce management and proper service coordination throughout daily operations.
- **Generate Operational Reports and Analytics** – The Owner or Administrator will generate reports covering appointments, sales transactions, inventory usage, therapist productivity, commissions, customer activities, payment summaries, refunds, and overall business performance. The system will also generate short-term revenue forecasts and analytical summaries to support business monitoring and informed decision-making.
- **Generate Daily Financial Reports** – The Owner or Administrator will review daily operational reports that include Point-of-Sale readings, cash-on-hand balances, cash denomination summaries, business expenses, gift certificate transactions, unpaid corporate accounts, and cash reconciliation results. These reports will support financial monitoring and end-of-day verification.
- **View Customer Feedback and Therapist Ratings** – The Owner or Administrator will review customer comments, ratings, and service evaluations submitted after completed appointments. The system will organize these records by therapist and transaction to assist management in evaluating customer satisfaction, therapist performance, and opportunities for service improvement.
- **Review Activity Logs and Employee Accountability** – The Owner or Administrator will examine activity logs that record significant actions performed throughout the system. Each record will identify the responsible user, assigned role, action performed, affected record, description, and timestamp to support auditing, accountability, and operational monitoring.
- **Manage System Settings** – The Owner or Administrator will configure and maintain system-wide settings that control the behavior of the application, including operational parameters, configurable options, and other business rules. These settings will allow management to adjust system behavior without modifying the underlying program code.

For Therapist:

- **Log In Account** – The Therapist will securely access the system using the assigned account credentials. The login process will restrict therapist functions to authorized users and protect therapist information from unauthorized access. After successful authentication, therapists will be able to access their schedules, attendance records, assigned services, commissions, ratings, and notifications.

- **Record Attendance and Duty Schedule** – The Therapist will record daily attendance by logging the time in and time out for each assigned duty. The system will maintain attendance records together with the therapist's rotation order to support fair workload distribution and proper therapist assignment. This process will assist management in monitoring attendance and workforce availability.
- **View Assigned Appointments** – The Therapist will view appointments assigned by the Receptionist or Administrator through the appointment management module. The system will display appointment schedules, assigned services, customer information, appointment location, number of persons, and current booking status. This process will help therapists prepare for scheduled appointments before rendering the service.
- **View Appointment Details** – The Therapist will review the complete details of each assigned appointment before the scheduled session begins. The system will display the selected services, additional services, customer notes, preferred service location, appointment schedule, applicable pricing, and other booking information. This process will help therapists understand customer requirements and prepare the necessary treatments.
- **View Assigned Specialties and Qualified Services** – The Therapist will review the service categories and individual treatments assigned based on qualifications and specialties maintained by management. The system will identify whether the therapist is assigned as a specialist or generalist for particular services. This process will ensure that therapists will only perform services appropriate to their qualifications.
- **View Commission and Earnings** – The Therapist will monitor commissions earned from completed appointments, additional services, influencer sessions, and other eligible transactions. The system will automatically calculate commission records based on completed services and approved transactions to provide accurate earnings information.
- **View Deduction Records** – The Therapist will review deductions applied to commission earnings, including cash advances, operational expenses, penalties, or other authorized adjustments. The system will maintain complete deduction records to provide transparency and accurate payroll computation.
- **View Customer Ratings and Feedback** – The Therapist will review customer ratings, comments, and service evaluations submitted after completed appointments. The system will organize feedback records according to completed services to help therapists evaluate their performance and identify opportunities for professional improvement.
- **Receive System Notifications** – The Therapist will receive notifications generated by the system regarding newly assigned appointments, schedule

updates, therapist reassignments, appointment cancellations, commission updates, attendance reminders, and other work-related announcements. This process will help therapists remain informed about daily responsibilities and operational activities.

For PayMongo Payment Gateway:

- **Process Online Payment Request** – The PayMongo Payment Gateway will receive online payment requests forwarded by the Recovery Iloilo Spa Integrated System whenever a customer selects an electronic payment method during checkout. The gateway will securely process the payment transaction without exposing sensitive payment information to the system. This process will allow customers to complete online payments through a secure and reliable payment service.
- **Return Payment Confirmation** – The PayMongo Payment Gateway will return the payment result to the Recovery Iloilo Spa Integrated System after the payment transaction has been completed or declined. The system will use the returned confirmation to update the payment status, record the transaction, generate the corresponding receipt, and continue the appropriate booking or order workflow. This process will ensure that only verified payment transactions will be recorded within the system.

For Email / SMTP Service:

- **Deliver One-Time Password (OTP)** – The Email / SMTP Service will receive the one-time password generated by the Recovery Iloilo Spa Integrated System during customer registration or account verification. The generated code will be transmitted to the registered email address of the customer to verify account ownership before access is granted. This process will strengthen account security and help prevent unauthorized registration and access.
- **Deliver Booking and Transaction Notifications** – The Email / SMTP Service will receive notification messages generated by the Recovery Iloilo Spa Integrated System and deliver them to the intended recipients. Messages may include appointment confirmations, appointment approvals, booking updates, payment acknowledgements, refund status updates, password verification messages, and other important transaction notifications. This

process will help ensure that customers remain informed about significant activities related to their accounts and transactions.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

The **Recovery Iloilo Spa Integrated System for Streamlining Operations and Efficient Analytics** is a web-based platform developed by the proponents to automate, centralize, and organize the daily operational activities of Recovery Iloilo Spa. The system integrates major business processes that were previously handled through manual records and separate procedures into a single centralized environment. These processes include appointment scheduling and management, customer registration and account management, service and product information management, therapist management, attendance monitoring, inventory monitoring, Point-of-Sale (POS) transactions, payment and refund processing, therapist commission computation, discount and voucher management, customer feedback and ratings, automated notifications, security and accountability, operational reporting, and business analytics. By bringing these functions together, the system provides users with a more organized method of recording, retrieving, updating, and monitoring operational information. It also reduces repetitive administrative tasks and provides management with consolidated information that can be used for monitoring business performance and supporting operational decision-making.

The developed system goes beyond a conventional standalone booking or Point-of-Sale application because it integrates multiple operational functions within one centralized platform. The system was implemented using a three-tier architecture consisting of the presentation layer, application layer, and data layer, allowing the user interface, business processes, and database operations to be managed as interconnected but organized components. Operational information is captured as customers and staff

perform activities within the system and is processed according to the corresponding business rules. Appointment information, customer records, transactions, inventory movements, therapist activities, commissions, discounts, vouchers, and feedback are stored in the centralized database and can subsequently be used for monitoring and reporting. The system also transforms accumulated operational data into organized reports, dashboards, analytical information, and formula-based revenue forecasts, allowing management to review business activities using recorded data rather than relying solely on manually maintained records.

The system was designed and developed based on the operational requirements of Recovery Iloilo Spa, located at G&R Building, M.H. Del Pilar Street, Molo, Iloilo City. The establishment provides various wellness and personal-care services, including body massage treatments, facial services, body scrubs, nail care services, waxing treatments, and other therapeutic procedures. During the development process, the proponents gathered information regarding the existing operational procedures and identified the requirements that needed to be addressed by the system. The spa owner, management, HR personnel, receptionists, therapists, and customers provided information regarding their respective activities, requirements, and experiences with the existing processes. These requirements served as the basis for designing the system modules, workflows, user interfaces, database structure, security controls, and reporting functions. The completed system was subsequently tested and evaluated by the proponents and respondents to determine whether the implemented functions addressed the identified operational requirements.

The developed system consists of several interconnected modules that support different users and operational activities. Customers can register and manage their accounts, verify their email addresses through OTP verification, browse available services and products, manage their shopping cart, book appointments, view appointment and transaction histories, monitor payment and refund information, and provide feedback and ratings. Staff members can manage appointments, customer records, therapist assignments, inventory, transactions, discounts and vouchers, and other operational activities according to their assigned access levels. Management and authorized personnel can access reports, analytics, commission information, cash reconciliation records, and other administrative functions. Through these interconnected modules, information generated from one operational activity can be utilized by other related functions, reducing the need to repeatedly record or search for the same information across separate records.

The major functions and capabilities achieved by the developed system are as follows:

- **Automated Minute-Level Appointment Scheduling** — The appointment management module supports minute-level scheduling and server-side conflict

detection to help prevent duplicate reservations, overbooking, and scheduling conflicts. It accommodates different appointment sources, including online bookings, walk-in appointments, home-service requests, and hotel or partner bookings. The module records important appointment information such as customer details, selected services, therapist assignments, appointment date and time, booking status, and additional services. Staff members can create, update, reschedule, cancel, and monitor appointments while considering therapist availability and applicable scheduling requirements.

- **Centralized Customer Accounts and History** — The customer management module maintains centralized customer profiles containing relevant account information, verified contact details, appointment history, transaction records, service preferences, feedback, and voucher or discount usage. Customers can manage their profiles and access relevant records through their accounts, while authorized staff can retrieve customer information when needed for appointment processing and service coordination. The centralized structure reduces the need to search through separate manual records and supports more organized customer information management.

- **Centralized Service and Product Information** — The system organizes information about the spa's available services and products in a centralized environment. Service records may contain information such as service names, descriptions, categories, prices, durations, and availability, while product records contain relevant product information and pricing. This allows customers to view available services and products when making inquiries or bookings and allows authorized staff to maintain updated information. Centralizing these records also helps reduce inconsistencies when service or product information is updated.

- **Dynamic Point-of-Sale and Multi-Channel Payment Processing** — The POS and payment module supports the recording and processing of customer transactions using available payment methods. The system calculates transaction totals, applies applicable discounts or vouchers, records payments, generates receipts, maintains transaction histories, and supports refund requests. Online payments are processed through the integrated PayMongo payment gateway, supporting the configured digital payment options such as GCash, QR Ph, and cards. These functions provide a more organized transaction workflow and reduce the need for repetitive manual computation and recording.

- **Automated Therapist Workload and Commission Management** — The therapist management module supports therapist assignment and reassignment, scheduling, attendance monitoring, specialty information, and performance-related records. The system also supports therapist rotation and workload coordination based on the implemented scheduling rules. The commission module calculates applicable therapist

commissions based on completed services and recorded transactions, including applicable service categories and commission arrangements. This reduces the need for repeated manual commission calculations and provides management with organized commission records for monitoring and payroll-related preparation.

- **Inventory Replenishment and Stock Control** — The inventory module monitors spa supplies, consumable materials, and retail products used or sold by the establishment. It records stock information and inventory movements and provides low-stock monitoring to help staff identify products that may require replenishment. By connecting inventory information with relevant product and transaction activities, the system provides management with more organized information regarding product availability and stock movement. This supports more systematic inventory monitoring and helps reduce the possibility of relying on outdated manual stock records.

- **Discount, Voucher, and Promotional Management** — The system provides functions for managing applicable discounts, vouchers, gift certificates, and promotional offers. Authorized personnel can record and validate applicable discounts, monitor voucher redemption, and maintain records of promotional transactions. The system supports the identification of applicable customer incentives and helps prevent duplicate voucher usage according to the implemented business rules. This provides a more organized method of monitoring promotional activities and their corresponding transactions.

- **Customer Feedback and Rating Management** — The feedback module allows customers to submit comments, ratings, and suggestions regarding their experiences with the spa's services. Feedback can be associated with the therapist who performed the service, allowing management to review customer evaluations and identify areas that may require improvement. The collected feedback provides an additional source of information for monitoring customer satisfaction and evaluating service performance.

- **Automated Notifications and Customer Communication** — The system provides automated email notifications for important activities such as account verification, appointment confirmations, appointment updates, cancellations, payment confirmations, refund-related activities, password reset instructions, and other system-generated events. These notifications provide customers with timely information regarding their accounts, appointments, and transactions and reduce the need for staff to manually communicate every system-related update.

- **Role-Based Security and Accountability** — The system separates public customer functions from administrative functions and applies access controls based on user roles. Sensitive transactions require additional verification through the implemented gate code and PIN mechanisms, while system activities are recorded through activity logs

containing relevant user actions and timestamps. These controls provide greater accountability by allowing authorized personnel to monitor important actions and help protect sensitive customer, transaction, and administrative information.

● **Financial Reconciliation and Business Analytics** — The system generates reports and analytical information related to sales performance, appointment activities, inventory usage, therapist productivity, commissions, payment records, cash reconciliation, and other operational activities. The system also supports cash-over and cash-short calculations and provides formula-based short-term revenue forecasting using recorded transaction data. Generated reports can be filtered, viewed, and exported into Excel and PDF formats for documentation, monitoring, and further analysis. These functions provide management with organized information that can support business evaluation and operational decision-making.

The development of the system incorporated software engineering principles, responsive interface design, centralized database management, user authentication, access control, transaction processing, and analytical processing. The interconnected modules allow information generated from customer activities, appointments, services, products, transactions, therapist activities, inventory movements, and feedback to be stored and processed within the centralized database. The resulting system provides Recovery Iloilo Spa with an organized platform for managing its operational activities and accessing relevant business information.

Overall, the developed system established an integrated workflow that connected customer-facing activities with internal spa operations. Instead of maintaining separate records for appointments, customers, transactions, inventory, therapists, commissions, promotions, and feedback, the system consolidated these functions within one platform. The completed implementation therefore provided the establishment with a more systematic approach to managing its daily operations while also supporting reporting, analytics, accountability, and data-informed decision-making.

Figure 4.1 illustrates the overall workflow of the developed system.

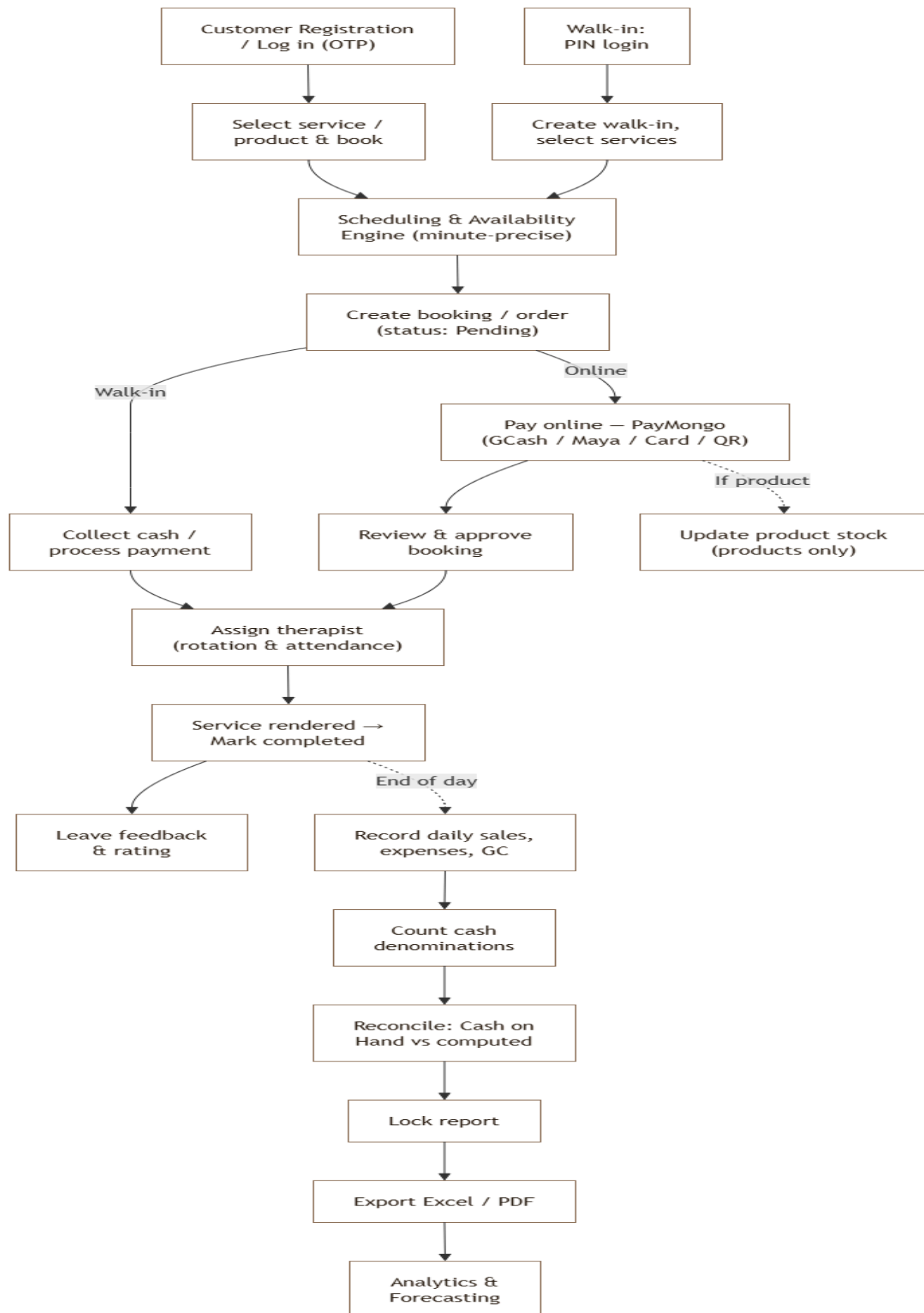


Figure 4.1. System Workflow Diagram of the Recovery Iloilo Spa Integrated System

Figure 4.1 presents the comprehensive end-to-end operational workflow of the Recovery Iloilo Spa Integrated System, encompassing both the customer-facing service lifecycle and the administrative back-office processes that were developed and refined throughout the study. The workflow commences at two distinct points of ingestion: an online customer who has completed verified registration and authentication through a one-time password (OTP), and an on-site walk-in client whose transaction is initiated by a receptionist via PIN-authenticated access. Both entry paths converge at the conflict-aware scheduling and availability engine, which evaluates therapist qualifications, duty rotation, and existing bookings at minute-level precision to determine feasible time slots before a booking or order record is generated with a pending status.

Following record creation, the workflow bifurcates according to the payment channel. Online transactions are routed to the multi-channel PayMongo payment gateway, which processed electronic settlements through GCash, Maya, card, and QR Ph facilities, after which the booking underwent administrative review and approval; product purchases additionally triggered a conditional stock-deduction routine. Walk-in transactions, in contrast, were settled directly through in-person cash or electronic collection at the counter. Both branches subsequently reconverged at the therapist-assignment stage, wherein staff allocated a service provider according to the attendance and rotation records, culminating in the rendering of the service and its marking as completed. Upon completion, the customer was afforded the opportunity to submit feedback and a corresponding therapist rating.

The latter portion of the workflow reflects the daily-report and reconciliation subsystem that constituted the principal development effort of this study. Each completed transaction was recorded within the daily-report spreadsheet, which supported both the automatic importation of completed appointments—safeguarded against duplication through source-transaction referencing—and the manual entry of off-system sales for troubleshooting and correction purposes. The subsystem automatically computed therapist commissions and net sales and disaggregated revenue by payment method. The receptionist thereafter counted the physical cash denominations and reconciled the computed cash-on-hand against the recorded figures. Upon completing the report, the receptionist submitted it through a validation procedure that verified the presence of essential fields, whereupon the system locked the report against further modification, dispatched an in-app and electronic-mail notification to the proprietor, and rendered the finalized report available for review and for exportation to Microsoft Excel and PDF formats in the client's prescribed layout. The consolidated data ultimately fed the analytics and forecasting module, which aggregated operational figures to support managerial decision-making.

4.3 System Development Methodology

The development of the Recovery Iloilo Spa Integrated System adopted the Agile Software Development Methodology, an iterative and incremental approach that emphasizes continuous collaboration, adaptive planning, and regular evaluation throughout the software development life cycle. Unlike traditional sequential models, Agile permits requirements and solutions to evolve through the cooperative effort of the proponents and the client. The proponents selected this methodology because it accommodated the evolving requirements of Recovery Iloilo Spa and promoted frequent feedback from the respondents, thereby ensuring that the resulting application remained aligned with the actual operational needs of the establishment rather than with assumptions made at the beginning of the project.

The Agile methodology enabled the proponents to divide the project into manageable iterations, commonly referred to as sprints. Each sprint was dedicated to a defined portion of the system, and at its conclusion a functional component was produced, tested, and reviewed before the proponents proceeded to the next iteration. This incremental delivery minimized development risks, facilitated the early detection and correction of software defects, and allowed the client to observe tangible progress at regular intervals. Because each iteration built upon the outcomes of the previous one, the proponents were able to refine both the design and the functionality of the system continuously as their understanding of the spa's business processes deepened. The development process consisted of the following phases, which were revisited iteratively across successive sprints rather than performed in a single linear pass:

The development process consisted of the following phases:

Requirements Gathering

The first phase involved the systematic collection of information from the client through interviews, surveys, direct observation, and document review. The proponents consulted the spa owner, human-resource personnel, receptionists, and therapists in order to determine the recurring challenges encountered in managing appointments, customer records, inventory, payments, commissions, discounts, and daily reports. Particular attention was given to the establishment's existing practices, such as the manual preparation of the daily sales report in a spreadsheet, the manual computation of therapist commissions, and the reconciliation of cash at the end of each business day. The respondents supplied detailed information regarding these processes and articulated their operational needs, which the proponents subsequently analyzed to define the functional requirements such as online booking, payment processing, and automated reporting and the non-functional requirements such as security, reliability,

and ease of use. The insights obtained in this phase formed the foundation upon which all succeeding phases were built and were periodically revisited whenever new or revised requirements emerged.

System Planning

After the requirements had been identified, the proponents established the project scope, objectives, system specifications, development schedule, and allocation of resources. The planning stage incorporated a feasibility analysis that assessed the technical, operational, and economic viability of the proposed system, confirming that the project could be accomplished with the available tools, skills, and time frame. During this stage, the proponents decomposed the system into the principal modules to be developed—namely, appointment scheduling, therapist management, inventory monitoring, point-of-sale and payment processing, discounts and voucher validation, feedback collection, security and access control, and reporting and analytics. These modules were then organized into a prioritized backlog, from which the tasks of each sprint were drawn. This planning ensured that the most critical functionalities were addressed early in the development timeline while allowing lower-priority features to be scheduled for later iterations.

System Design

During this phase, the proponents designed the overall architecture of the application together with its constituent components. Database structures, process flows, user-interface layouts, and module interactions were prepared to serve as the blueprint for software construction. The proponents produced the Data Flow Diagram, the Entity Relationship Diagram, and the interface wireframes to model the movement of data and the relationships among the system's entities. The application was designed around a three-tier architecture consisting of the presentation, application, and data layers, an arrangement that separates the user interface from the business logic and the underlying database in order to promote maintainability, scalability, and security. The design phase also addressed the encapsulation of complex logic—most notably the conflict-aware scheduling engine, which determines therapist availability at minute-level precision and the incorporation of security measures such as role-based access control, prepared database statements, and cross-site request forgery protection. Because the design evolved iteratively, these artifacts were refined in successive sprints as the proponents' comprehension of the client's reconciliation and commission procedures matured.

Software Development

The development phase involved translating the approved designs into functional program code using the selected technologies. The proponents implemented the presentation layer using HTML, CSS, and Vanilla JavaScript; the application layer using

the PHP programming language; and the data layer using the MariaDB relational database management system. Individual functionalities were constructed incrementally, including customer registration with electronic-mail one-time-password (OTP) verification, secure authentication, appointment scheduling with automated conflict detection, therapist assignment and attendance recording, inventory monitoring, point-of-sale and payment processing through the multi-channel PayMongo payment gateway, refund handling, commission computation, discount and voucher validation, feedback collection, activity logging, and formula-based revenue forecasting. A substantial portion of the development effort was devoted to the daily-report and reconciliation subsystem, which enabled the automatic importation of completed transactions, the manual entry and correction of off-system sales, the automatic computation of commissions and net sales, the disaggregation of revenue by payment method, cash reconciliation, and the exportation of the finalized report to Microsoft Excel and PDF formats in the client's prescribed layout. Throughout this phase, the proponents employed version control through Git and GitHub to manage the source code, track incremental changes, and safeguard the project against data loss, thereby allowing completed features to be integrated into the larger system without disrupting previously functioning components.

Testing

Each completed module underwent functional testing to verify that it operated in accordance with its specified requirements before it was integrated into the wider system. The proponents subsequently performed integration testing to confirm that the interactions among modules functioned correctly, and system testing to evaluate the overall behavior and performance of the application. Testing was conducted iteratively: defects identified during a sprint were documented, corrected, and re-verified before the associated feature was considered complete, in keeping with the Agile principle of early and continuous defect resolution. Upon the stabilization of the system, User Acceptance Testing (UAT) was carried out with the client, wherein the spa owner, receptionists, therapists, accounting personnel, and selected customers participated as respondents. These respondents exercised the live functionalities such as booking an appointment, processing a payment, recording and submitting the daily report, and exporting it in the required format and evaluated whether the application satisfied their operational expectations. The feedback derived from this testing informed the final refinements to the system.

Deployment

Following the successful completion of testing, the proponents deployed the application within the target environment of Recovery Iloilo Spa. The deployment activities encompassed database configuration, server setup, and application installation, which together rendered the system accessible to authorized users in accordance with their

assigned roles. During this phase, the respondents accessed the live system under realistic operating conditions and provided additional observations, which the proponents used to fine-tune the final configuration and to resolve any environment-specific issues that had not surfaced during earlier testing.

Maintenance

The maintenance phase concerns the continued monitoring of system performance, the correction of newly identified issues, the implementation of software updates, and the enhancement of features in response to feedback obtained from the client after deployment. Through routine maintenance, the proponents ensure the long-term reliability, security, and usability of the developed application, and provide a mechanism through which the system may be adapted to future operational requirements of the establishment.

Through the repeated execution of these phases across successive sprints, the proponents were able to deliver the Recovery Iloilo Spa Integrated System incrementally, to validate each component against the expectations of the client, and to progressively enhance the application until it satisfied the intended operational and functional requirements. The iterative and collaborative character of the Agile methodology thus proved instrumental in producing a system that is both technically sound and closely aligned with the real-world needs of Recovery Iloilo Spa.

Figure 4.2 presents the Agile Software Development Life Cycle adopted by the proponents in the development of the system

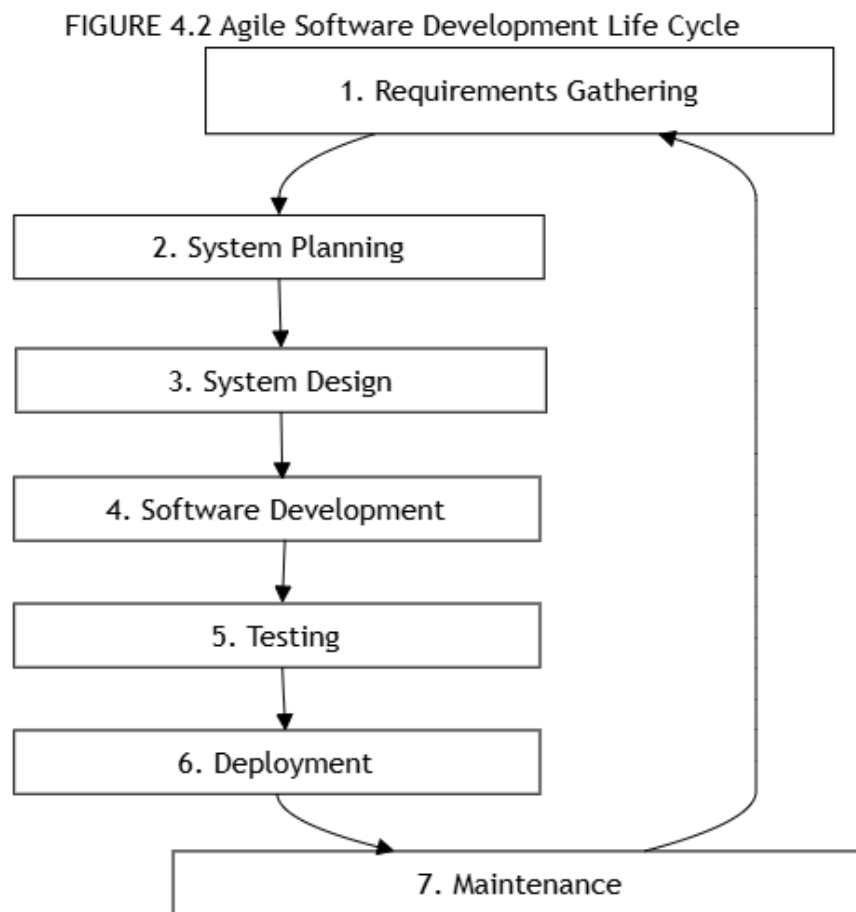


Figure 4.2. Agile Software Development Life Cycle Adopted in the Study

4.4 System Architecture

The Recovery Iloilo Spa Integrated System utilizes a Three-Tier Client-Server Architecture that separates the presentation layer, the application layer, and the data layer. This architectural approach improves system maintainability, scalability, and security by organizing the software components according to their respective responsibilities, so that a change introduced within one tier exerts minimal impact upon the others. The system further adopts a hub-and-spoke communication model in which the application layer serves as the central processing hub between the presentation layer and the data layer. Every request initiated by a user is first authenticated, validated, and processed by the application layer before information is stored in or retrieved from the database, thereby ensuring that no client is permitted to access the underlying data directly.

Figure 4.3 illustrates the architecture of the developed system.

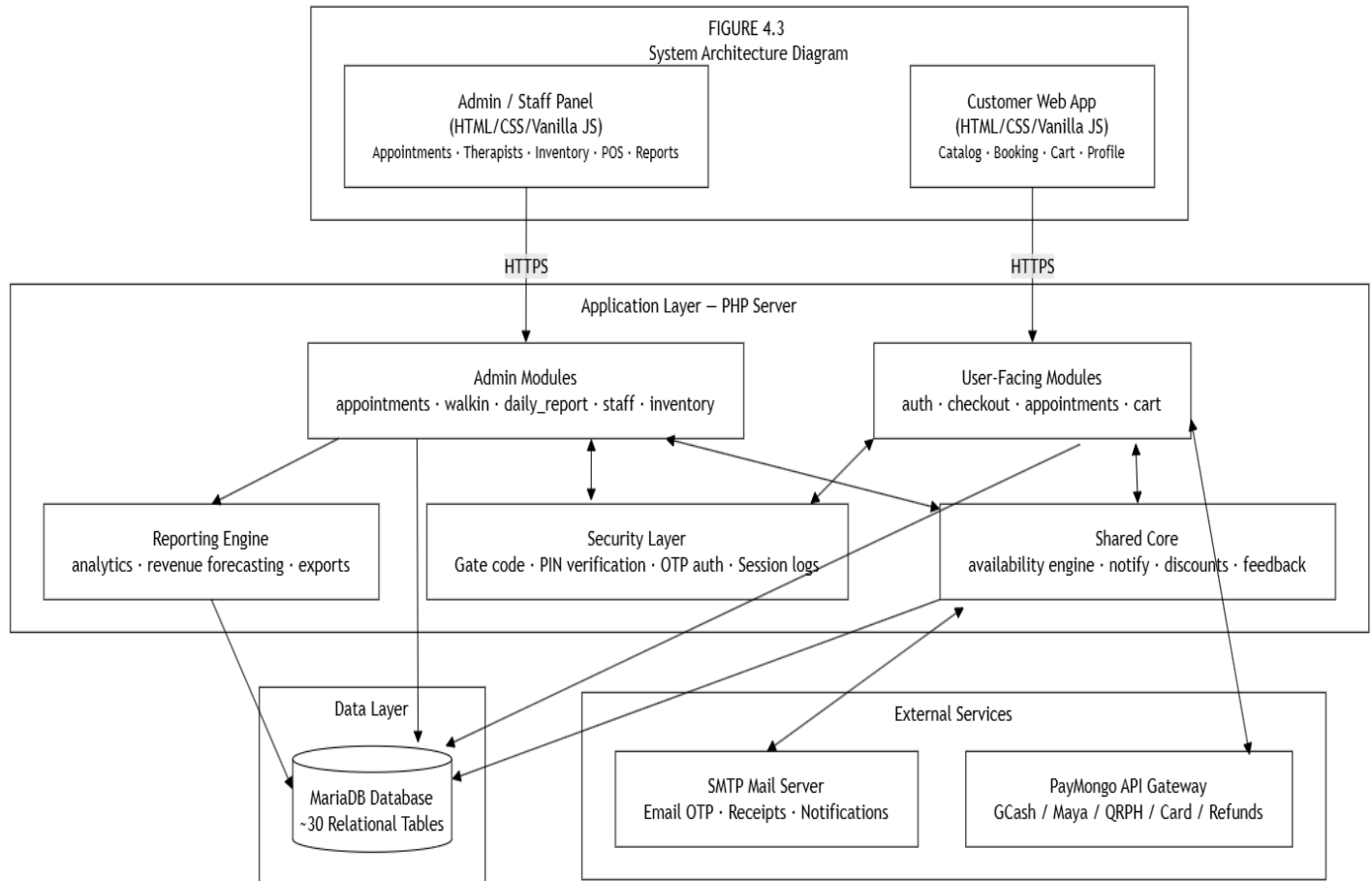


Figure 4.3. Three-Tier System Architecture of the Recovery Iloilo Spa Integrated System

The architecture consists of the following components:

Presentation Layer

The presentation layer serves as the interface between the users and the application. It is responsible for displaying information, receiving user input, and presenting reports and dashboards to the intended users. Users access the system through desktop computers, laptops, tablets, or smartphones by means of a modern web browser, thereby eliminating the need for any additional software installation on the client device. The proponents developed this layer using HTML, CSS, and Vanilla JavaScript in order to provide a responsive and interactive user experience, and all communication between this layer and the application layer is transmitted over the Hypertext Transfer Protocol Secure (HTTPS) to protect the data exchanged.

The major interface components include the following:

- Customer registration and login pages
- Service and product catalog
- Appointment booking and shopping cart
- Customer dashboard and profile management
- Administrative dashboards for appointments, therapists, inventory, and point-of-sale operations
- Reports, analytics, and revenue-forecasting interfaces

Application Layer

The application layer contains the business logic responsible for processing user requests, validating input, executing business rules, and generating reports. The proponents developed this layer using the PHP programming language, which implements the business logic and facilitates secure communication between the user interface and the MariaDB database. The application server acts as the intermediary between the presentation layer and the data layer, and it constitutes the sole pathway through which all data operations are performed.

Its primary responsibilities include the following:

- User authentication, electronic-mail one-time-password (OTP) verification, and secure session management
- Appointment scheduling with conflict detection and status management
- Therapist assignment, reassignment, rotation, attendance recording, and commission computation
- Inventory monitoring and stock updates
- Point-of-sale and payment processing through the PayMongo payment gateway, together with refund handling
- Discount, voucher, and gift-certificate validation
- Feedback processing linked to completed appointments and assigned therapists
- Security enforcement through gate-code and personal-identification-number (PIN) verification and activity logging
- Report generation, business analytics, and formula-based revenue forecasting

Data Layer

The data layer stores all business information within a centralized MariaDB relational database. It is responsible for maintaining data integrity, enforcing the relationships among entities, and supporting the efficient retrieval of the information required by the reporting and analytics modules. The database comprises approximately thirty core tables, and access to it is granted exclusively to the application layer through secure,

parameterized statements that safeguard the system against structured query language injection.

The information stored within this layer includes the following:

- User accounts, roles, and credentials
- Services, products, and categories
- Appointments and appointment assignments
- Therapist records, attendance, and commissions
- Orders, order items, payments, and refunds
- Discounts, vouchers, and gift certificates
- Customer feedback and ratings
- Activity logs, notifications, and daily reports

Through this layered and hub-centered organization, the architecture ensures secure communication between the application server and the database while restricting unauthorized access to sensitive business and financial information. It consequently promotes maintainability by localizing changes within individual tiers, supports scalability by permitting each tier to be extended independently, and enhances security by channeling every operation through a single, validated point of control.

4.5 Hardware and Software Requirements

The successful implementation of the Recovery Iloilo Spa Integrated System required appropriate hardware and software resources. These specifications ensure that the application performs efficiently during appointment processing, transaction recording, report generation, and analytics computation. The requirements are presented according to the role that each machine fulfills within the three-tier architecture namely, the server that hosts the application and database, and the client devices through which the users access the system.

4.5.1 Hardware Requirements

The hardware requirements specify the physical computing resources necessary for the proper operation of the Recovery Iloilo Spa Integrated System. These requirements were determined on the basis of the system's processing, memory, and storage demands, particularly during the execution of operational transactions and the rendering of graphical dashboards and reports. The minimum and recommended hardware specifications are presented in Table 4.1.

Table 4.1 presents the recommended hardware specifications.

Component	Minimum Specification	Recommended Specification
Processor	Dual-core, 2.0 GHz (Core i3 / AMD equiv.)	Quad-core, 2.4 GHz+ (Core i5 / Ryzen 5)
Memory (RAM)	4 GB	8 GB or higher
Storage	20 GB free (SSD preferred)	50 GB+ SSD
Monitor	1366 × 768 Resolution	Full HD Display
Internet Connection	10 Mbps	50 Mbps
Mobile Device	Android 10	Android 13 or higher

The recommended hardware configuration provides sufficient processing capability for executing operational transactions and for rendering the graphical dashboards and reports of the system without significant delay. The minimum configuration remains adequate for standard operation, whereas the recommended configuration ensures optimal performance during periods of concurrent usage and report generation.

4.5.2 Software Requirements

The software requirements specify the programs, development tools, and services employed in the construction and operation of the Recovery Iloilo Spa Integrated System. Each component was selected for its role within the development environment and the running application. The software requirements are presented in Table 4.2.

Table 4.2 presents the software requirements.

Software	Purpose
Windows 11	Operating System
Visual Studio Code	Source Code Editor
MariaDB	Database Management System
PHP	Backend Development
HTML, CSS, Vanilla JavaScript	Frontend Development
PayMongo Payment Gateway	Online Payment Processing
PHPMailer	Email and OTP Notification Service
Google Chrome / Microsoft Edge / Mozilla Firefox / Safari	Web Browser
GitHub Repository	Version Control

The selection of these software tools was based upon their stability, community support, compatibility, and widespread adoption in modern web-application development. Collectively, these tools provided a dependable and well-supported environment for the design, implementation, testing, and deployment of the system.

4.6 Database Design

The database serves as the foundation of the Recovery Iloilo Spa Integrated System by organizing and storing appointment records, customer information, product inventories, transactions, and analytical data. A well-designed database ensures efficient data retrieval, minimizes redundancy, and maintains data integrity throughout the operation of the system. The proponents employed a relational database model using MariaDB, and database normalization techniques were applied to reduce duplication and maintain consistency among related tables. The database supports approximately thirty core tables.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.

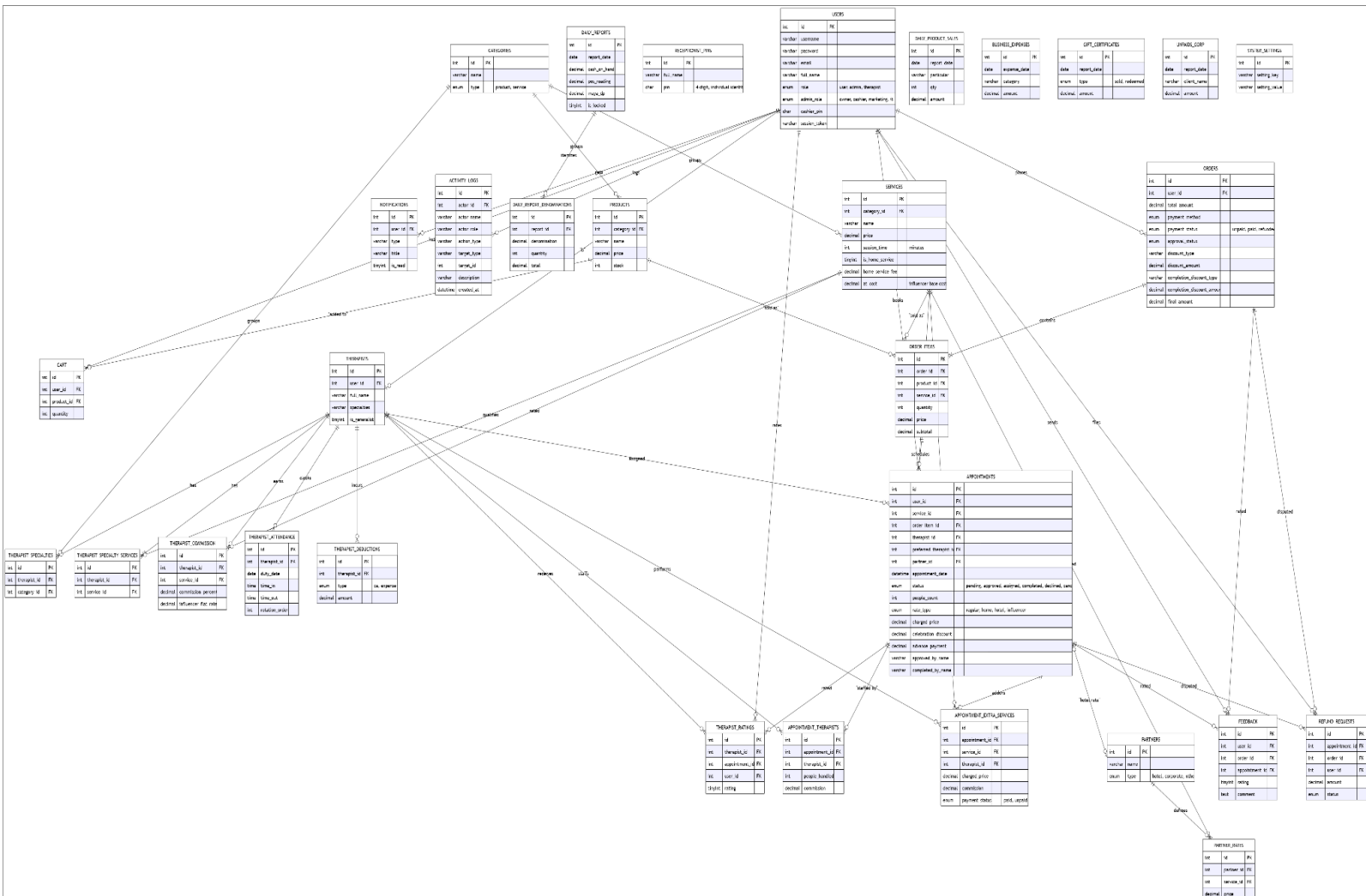


Figure 4.4. Entity Relationship Diagram of the Recovery Iloilo Spa Integrated

The primary entities of the database include the following:

- Users
- Therapists
- Categories
- Services
- Products
- Cart
- Orders
- Order Items
- Appointments
- Appointment Therapists
- Therapist Commission
- Therapist Attendance
- Feedback
- Refund Requests
- Notifications
- Activity Logs
- Daily Reports

Users Table

This table stores the account credentials and role information of all system users, including customers, therapists, receptionists, and administrators.

Field	Description
-------	-------------

id	Primary Key
username	Login Username
password	Encrypted Password
email	Registered Email Address
full_name	Complete Name of the User
role	User Role (customer, therapist, receptionist, admin)
admin_role	Administrative Role (owner, cashier, marketing, IT)
cashier_pin	Cashier Verification PIN
session_token	Secure Session Identifier

Therapists Table

This table stores the professional profile of each therapist and links the record to a corresponding user account.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
full_name	Therapist Name
specialties	Assigned Service Specialties
is_generalist	Indicates whether the therapist can perform all services

Categories Table

This table classifies both services and products to simplify searching, filtering, and management.

Field	Description
id	Primary Key
name	Category Name
type	Classification (service or product)

Services Table

This table stores the spa services offered by the establishment, including pricing and home-service details.

Field	Description
id	Primary Key
category_id	Foreign Key referencing Categories
name	Service Name
price	Service Price
session_time	Duration of the Service
is_home_service	Home-Service Availability Indicator
home_service_fee	Additional Travel Fee
at_cost	At-Cost Indicator

Products Table

This table stores retail and consumable product information used in daily spa operations.

Field	Description
id	Primary Key
category_id	Foreign Key referencing Categories
name	Product Name
price	Selling Price
stock	Available Inventory

Cart Table

This table temporarily stores the products selected by a customer before checkout is confirmed.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
product_id	Foreign Key referencing Products
quantity	Quantity Selected

Orders Table

This table records customer transactions involving spa services and wellness products.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
total_amount	Total Transaction Amount
payment_method	Selected Payment Method
payment_status	Payment Status
approval_status	Transaction Approval Status
discount_type	Applied Discount Type
discount_amount	Discount Value
completion_discount_type	Discount Applied at Completion
completion_discount_amount	Completion Discount Value
final_amount	Final Payable Amount

Order Items Table

This table records the individual services and products included in each customer order.

Field	Description
id	Primary Key
order_id	Foreign Key referencing Orders
product_id	Foreign Key referencing Products
service_id	Foreign Key referencing Services
quantity	Quantity Purchased
price	Unit Price
subtotal	Item Total

Appointments Table

This table records scheduled spa appointments together with their complete booking details and status lifecycle.

Field	Description
id	Primary Key

user_id	Foreign Key referencing Users
service_id	Foreign Key referencing Services
order_item_id	Foreign Key referencing Order Items
therapist_id	Assigned Therapist
preferred_therapist_id	Customer's Preferred Therapist
partner_id	Partner or Hotel Reference
appointment_date	Scheduled Date and Time
status	Status (pending, assigned, approved, completed, declined, cancelled)
people_count	Number of Persons in the Booking
rate_type	Applicable Rate Type
charged_price	Charged Amount
celebration_discount	Celebration Discount Applied
advance_payment	Advance Payment Recorded
approved_by_name	Personnel Who Approved the Appointment
completed_by_name	Personnel Who Completed the Appointment

Appointment Therapists Table

This table records the therapists assigned to each appointment, supporting multi-person and group bookings.

Field	Description
id	Primary Key
appointment_id	Foreign Key referencing Appointments
therapist_id	Foreign Key referencing Therapists
people_handled	Number of Persons Served
commission	Commission Earned

Therapist Commission Table

This table defines the commission rules applied to each therapist for particular services.

Field	Description
id	Primary Key

therapist_id	Foreign Key referencing Therapists
service_id	Foreign Key referencing Services
commission_percent	Commission Percentage
influencer_flat_rate	Flat Rate for Influencer or Promotional Sessions

Therapist Attendance Table

This table records therapist duty and attendance information used during appointment assignments.

Field	Description
id	Primary Key
therapist_id	Foreign Key referencing Therapists
duty_date	Duty Date
time_in	Clock-In Time
time_out	Clock-Out Time
rotation_order	Assignment Rotation Order

Feedback Table

This table stores customer feedback and ratings linked to the corresponding appointment and order.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
order_id	Foreign Key referencing Orders
appointment_id	Foreign Key referencing Appointments
rating	Numerical Service Rating
comment	Customer Comment

Refund Requests Table

This table records customer refund requests and their processing status.

Field	Description
-------	-------------

id	Primary Key
appointment_id	Foreign Key referencing Appointments
order_id	Foreign Key referencing Orders
user_id	Foreign Key referencing Users
amount	Requested Refund Amount
status	Request Status (pending, approved, declined)

Notifications Table

This table stores the system-generated notifications delivered to users.

Field	Description
id	Primary Key
user_id	Foreign Key referencing Users
type	Notification Type
title	Notification Title
is_read	Read Status

Activity Logs Table

This table records the administrative actions performed within the system to support accountability and auditing.

Field	Description
id	Primary Key
actor_id	Foreign Key referencing Users
actor_name	Name of the Actor
actor_role	Role of the Actor
action_type	Type of Action Performed
target_type	Affected Record Type
target_id	Affected Record Identifier
description	Description of the Action
created_at	Timestamp of the Action

Daily Reports Table

This table stores the end-of-day operational and financial summary prepared during cashier reconciliation.

Field	Description
id	Primary Key
report_date	Report Date
cash_on_hand	Recorded Cash on Hand
pos_reading	POS Reading
maya_dp	Maya Digital Payment Collections
is_locked	Report Lock Status

The relationships among these tables allow the application to trace customer transactions, monitor appointment activity, compute therapist commissions, track inventory movement, and generate comprehensive operational reports and analytics. Proper indexing, foreign key constraints, and normalization help maintain referential integrity and optimize query performance, ensuring that the system efficiently supports both daily operational transactions and business analytics for Recovery Iloilo Spa.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusions of the development and evaluation of the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently**. It summarizes the results of the completed project, determines the extent to which the objectives of the study were achieved, and presents recommendations for future improvements and continued implementation of the system at Recovery Iloilo Spa.

The study aimed to design, develop, and implement a centralized web-based system that would address the spa's manual and fragmented operational processes. The developed system integrated appointment scheduling and management, customer account and record management, service and product management, inventory monitoring, Point-of-Sale and payment processing, therapist management, commission computation, voucher and discount management, customer feedback, security and accountability controls, automated notifications, reports and analytics, and formula-based short-term revenue forecasting.

Throughout the development process, the proponents coordinated with Recovery Iloilo Spa to identify its operational requirements, existing procedures, and user needs. The completed system was subjected to system testing and evaluation to determine its functionality, accuracy, usability, reliability, performance efficiency, security, and overall effectiveness. User feedback and evaluation results were used to assess whether the developed system addressed the identified operational challenges and met the requirements established for the study.

This chapter presents the Summary of Findings, Conclusions, and Recommendations based on the completed development, testing, and evaluation activities. The findings are presented according to the specific objectives of the study to establish a direct connection between the identified problems, the developed system features, and the results obtained during evaluation. The conclusions determine whether the objectives were achieved, while the recommendations identify possible improvements and future enhancements that may further support the operational needs of Recovery Iloilo Spa.

5.2 Summary of Findings

The development and evaluation of the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently** produced results that corresponded to the specific objectives established in the study. The completed system addressed the major operational requirements identified during the requirements-gathering phase and integrated the identified business processes into a centralized web-based platform. The findings were based on the system development activities, functional testing conducted by the proponents, and the validation and evaluation provided by the client and respondents during the User Acceptance Testing. The results demonstrated that the developed system provided functional support for appointment management, customer account and record management, service and product management, inventory monitoring, payment and transaction processing, therapist management, commission computation, promotional management, feedback, security, notifications, reporting, analytics, and forecasting.

The following findings were established based on the specific objectives of the study:

1. Operational Requirements Gathering and Analysis

The proponents successfully gathered, analyzed, and evaluated the operational requirements of Recovery Iloilo Spa through consultation with the client and relevant users, together with interviews, surveys, observation, and document review. The requirements-gathering activities provided information regarding the spa's existing procedures, operational challenges, user responsibilities, and information needs. The identified requirements covered appointment scheduling, customer management, services and products, inventory, payment processing, therapist management, commissions, discounts and vouchers, feedback, security, notifications, reporting, analytics, and forecasting.

The gathered requirements served as the basis for identifying the system modules and functions included in the developed system. The client and respondents, including the spa owner, management, HR personnel, receptionists, therapists, and other relevant staff, participated in the validation of the identified operational requirements. Their feedback helped establish that the developed functions corresponded to the business processes and operational needs identified during the study.

2. User Interface and System Accessibility

The proponents developed a user-friendly, responsive, and organized system interface that provided separate and appropriate functions according to the roles of administrators, authorized staff, therapists, receptionists, and customers. The interface

organized the major system functions into accessible modules and provided users with navigation mechanisms for performing their respective activities.

The developed interface allowed customers to access account-related functions, browse services and products, manage their appointments and transactions, and submit feedback. Authorized personnel were provided with functions for managing operational records and monitoring business activities. During the evaluation, respondents were able to interact with the system and perform the functions relevant to their roles. Their feedback indicated that the interface was understandable, accessible, and manageable for the intended users.

3. Customer Registration and Account Management

The proponents successfully implemented customer registration and account management with email OTP verification through PHPMailer. The registration process provided an additional verification step before an account could be activated, while the account management functions allowed customers to maintain their profile information and access their account-related activities.

The customer module also provided access to service and product browsing, shopping cart functions, appointment booking, appointment history, transaction history, and profile management. The implementation provided customers with a centralized account through which their interactions with the spa could be managed. During evaluation, respondents confirmed that the account verification process operated properly and that the customer account functions supported secure and organized access to customer-related activities.

4. Appointment Scheduling and Management

The appointment scheduling and management module was successfully implemented to address the difficulties associated with manual appointment scheduling. The system supported online, walk-in, home-service, and hotel or partner bookings while maintaining appointment details such as customer information, selected services, schedules, service duration, therapist assignments, and booking status.

The module implemented minute-level scheduling and server-side conflict detection to prevent incompatible appointment schedules. It also supported therapist assignment and reassignment, appointment updates, rescheduling, cancellations, and applicable add-on services. For home-service appointments, the system considered the required travel-time allowance in scheduling. These functions provided authorized users with a centralized means of monitoring available schedules and managing appointments.

The testing and evaluation activities showed that the appointment module operated according to the established requirements. Respondents indicated that the centralized scheduling functions supported the management of appointments and helped address issues associated with duplicate reservations, scheduling conflicts, overbooking, and missed appointments.

5. Centralized Customer Record Management

The proponents successfully implemented a centralized customer record management feature that organized customer information in a single database. The system stored relevant information such as personal details, contact information, appointment history, service history, transaction records, preferences, feedback, and applicable voucher or discount usage.

The centralized records allowed authorized personnel to retrieve customer information without relying on separate manual records. The availability of organized customer information also supported the processing of appointments, transactions, customer inquiries, and service-related activities. During evaluation, respondents indicated that the centralized customer records provided a more organized and accessible method of retrieving and managing customer information.

6. Service and Product Information Management

The proponents implemented a centralized service and product management feature that organized the information required by customers and employees during service and transaction activities. Service records included information such as service names, descriptions, categories, prices, durations, availability, packages, and add-on services. Product records included product names, descriptions, categories, prices, stock quantities, and availability information.

The centralized management of these records provided a consistent source of information for browsing, appointment selection, product selection, shopping cart activities, and Point-of-Sale transactions. Authorized personnel were able to manage and update service and product information through the system. This reduced dependence on separate records and supported more consistent presentation of service and product information to customers.

7. Inventory Management

The inventory management module was successfully implemented to monitor product availability, stock movement, inventory usage, and low-stock items. The system

provided authorized personnel with centralized inventory records that supported monitoring of products and supplies used in spa operations.

The module helped organize information regarding available quantities and inventory movements and provided a basis for identifying items requiring replenishment. During evaluation, respondents indicated that the inventory functions supported more organized stock monitoring and made it easier to identify low-stock items. The module therefore addressed the identified difficulty of maintaining inventory information through manual monitoring.

8. Point-of-Sale and Payment Processing

The proponents successfully implemented the Point-of-Sale and payment processing module to support the recording and management of customer transactions. The system digitally recorded transactions, calculated transaction totals, applied applicable discounts and vouchers, generated receipts, maintained transaction histories, and supported refund requests.

The system also supported multiple applicable payment methods, including cash and online payments. For supported online transactions, the system was integrated with the PayMongo Payment Gateway. The system handled the related payment request, received the payment result, updated the corresponding payment status, and recorded the transaction information, while PayMongo handled the online payment transaction.

The implementation provided a centralized method of monitoring payment-related information and transaction records. Testing confirmed that the payment and Point-of-Sale functions operated according to the established requirements. Respondents also indicated that the digital transaction-recording process supported more organized and efficient payment and sales activities compared with the manual procedures identified during requirements gathering.

9. Therapist Management

The therapist management feature was successfully implemented to support therapist assignment, reassignment, attendance monitoring, specialty management, service scheduling, and performance tracking. The system provided authorized personnel with information that supported the assignment of therapists according to their availability, specialties, schedules, and service requirements.

The attendance feature recorded therapist clock-in and clock-out activities, while the management functions supported monitoring of assigned services and therapist activities. Therapist assignment and reassignment functions also provided greater flexibility when scheduling changes or conflicts occurred. During evaluation,

respondents indicated that these functions supported improved coordination of therapist schedules and assignments and provided more organized information for managing therapist activities.

10. Therapist Commission Management

The therapist commission management feature was implemented to reduce the need for manual commission computation. The system calculated commissions based on recorded completed services, applicable service categories, commission rates, multi-person sessions, and authorized flat-rate arrangements, including applicable influencer-related arrangements.

By basing commission calculations on recorded transactions and completed services, the system provided a more consistent method of maintaining commission information. The commission records could also support payroll preparation and employee performance monitoring. During evaluation, respondents indicated that the automated computation reduced the administrative effort associated with manual commission calculations and helped maintain more organized commission records.

11. Discount, Voucher, and Promotional Management

The proponents successfully implemented the discount and voucher management feature to organize promotional transactions and customer incentives. The system supported the validation and application of applicable discounts and promotional offers, including authorized senior citizen, PWD, staff, celebration, and gift certificate-related discounts based on the established business rules.

The system also recorded issued and redeemed vouchers and monitored promotional usage. Validation mechanisms helped reduce the possibility of duplicate or unauthorized voucher usage. These functions provided authorized personnel with a centralized method of monitoring promotional activities and supported more consistent application and recording of discounts and vouchers.

12. Digital Feedback and Rating Management

The digital feedback and rating feature was successfully implemented to provide customers with a centralized means of submitting comments, ratings, and suggestions regarding their spa experience. Feedback could be associated with the corresponding service and therapist, allowing management to identify the context of customer evaluations.

The feature provided management with organized customer feedback that could be reviewed in relation to service quality and therapist performance. During evaluation,

respondents indicated that the digital feedback mechanism supported the collection of customer opinions and provided useful information for assessing customer satisfaction and identifying areas that required improvement.

13. Security and Employee Accountability

The proponents implemented security and accountability features to protect system access and support monitoring of sensitive activities. Administrative functions were separated from customer-facing functions, and authorized access was controlled according to assigned roles and permissions.

The system also incorporated hidden administrative access, gate code verification, PIN verification for designated sensitive transactions, and activity logging. Activity logs recorded relevant information regarding user actions and timestamps, allowing authorized personnel to trace important activities performed within the system.

Testing confirmed that the implemented security controls operated according to the established requirements. The respondents also indicated that these controls strengthened oversight and provided a clearer means of monitoring critical administrative and financial activities.

14. Reporting, Analytics, and Revenue Forecasting

The reporting and analytics module successfully generated organized information regarding sales, appointments, customer activities, inventory, therapist productivity, attendance, commissions, payment transactions, cash reconciliation, promotional activities, and other operational records.

The system also provided dashboards and analytical views that allowed management to review summarized operational information. Formula-based short-term revenue forecasting was implemented using recorded historical transaction data to provide estimates that could support short-term planning and monitoring. The forecasting feature did not rely on artificial intelligence or machine-learning-based predictive analytics.

Reports could be filtered and viewed according to relevant criteria and exported into Excel and PDF formats. The generated reports provided management with organized information that could be used for documentation, operational monitoring, analysis, and decision-making. During evaluation, respondents indicated that the reporting and analytics functions supported more organized access to business information and assisted management in reviewing operational performance.

15. Automated Notifications

The automated notification feature was successfully implemented to support communication between Recovery Iloilo Spa and its customers. The system delivered email notifications for important activities, including account verification, appointment confirmations, appointment updates, cancellations, and payment confirmations.

The notification feature reduced the need to communicate every system event manually and provided customers with timely information regarding their accounts, appointments, and transactions. During evaluation, respondents indicated that the automated notifications supported clearer and more timely communication between the spa and its customers.

16. Integration of Operational Processes

The completed system successfully integrated the major operational functions of Recovery Iloilo Spa into a centralized platform. Instead of maintaining separate procedures for appointments, customer information, services, products, inventory, payments, therapists, commissions, promotions, feedback, notifications, and reports, the developed system connected these activities through a centralized database and coordinated workflow.

The integration allowed information recorded in one operational process to support related activities in other modules. For example, appointment information was associated with customer records, selected services, therapist assignments, transactions, feedback, and related operational records. Likewise, completed transactions supported payment records, receipts, sales reports, commission computation, and analytics. This integration reduced the need for repeated data entry and provided authorized users with more consistent access to operational information.

17. System Testing and Evaluation

The proponents conducted system testing and User Acceptance Testing to determine whether the developed system operated according to the established functional and operational requirements. The evaluation covered relevant system quality characteristics, including functionality, accuracy, usability, reliability, performance efficiency, and security.

The testing activities verified the operation of the implemented modules and their major functions, while User Acceptance Testing provided feedback from the intended users and stakeholders. The results indicated that the developed system addressed the identified operational requirements and provided the intended functionality for managing the spa's major business processes.

Overall, the findings demonstrated that the specific objectives of the study were substantially achieved. The proponents successfully developed and integrated the major system modules identified during the requirements-gathering phase, while the client and respondents participated in the validation and evaluation of the completed system. The developed platform provided centralized management of appointments, customer information, services and products, inventory, transactions, payments, therapist activities, commissions, promotional activities, feedback, security, notifications, reports, analytics, and forecasting.

The results further demonstrated that the system provided a structured technological solution to the operational challenges identified at Recovery Iloilo Spa. By consolidating the major business processes into one platform, the system reduced dependence on fragmented manual records, supported more organized information management, strengthened transaction and employee accountability, improved access to operational information, and provided management with analytical outputs for monitoring and decision-making.

Therefore, the completed development and evaluation established that the Recovery Iloilo Spa Integrated System provided the intended operational support and substantially fulfilled the objectives established for the study. The findings also provided the basis for the conclusions and recommendations presented in the succeeding sections of this chapter.

5.3 Conclusion

The development and evaluation of the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently** established that the system was a viable and effective solution for supporting the daily operations of Recovery Iloilo Spa. The developed system centralized the spa's major operational processes into a single web-based platform, including appointment scheduling and management, customer registration and records, service and product management, inventory monitoring, Point-of-Sale and payment processing, therapist management, attendance and commission monitoring, discount and voucher management, customer feedback and ratings, automated notifications, security and accountability controls, reporting, analytics, and formula-based revenue forecasting. Through this integration, the system addressed the fragmented and manual procedures identified during the requirements-gathering phase and provided users with organized access to operational information and transactions. The centralized database further supported the consistent storage and retrieval of customer, appointment, service, product, transaction, therapist,

feedback, notification, and reporting information, providing a more structured approach to managing the spa's business processes.

The developed system addressed the identified operational challenges by providing dedicated functions for the spa's daily activities. The appointment module supported online, walk-in, home-service, hotel, and partner bookings and included appointment details, therapist assignments, schedules, booking status, and additional services. The system also supported minute-level scheduling and conflict detection to assist in managing therapist availability and appointment schedules. Customer account and record management provided customers with registration, email verification, profile management, service and product browsing, shopping cart and checkout functions, appointment and transaction history, payment and refund status, and feedback submission. Service and product management organized information such as service names, descriptions, categories, prices, durations, availability, and product details, while inventory functions supported monitoring of product availability and stock movement. These integrated functions provided a more organized workflow for managing customer requests, services, products, appointments, and related operational records.

The system also strengthened the spa's transaction, workforce, financial, promotional, communication, and accountability processes. The Point-of-Sale and payment module supported digital transaction recording, multiple payment methods, online payment processing through the PayMongo Payment Gateway, receipt generation, transaction history, and refund requests. Therapist management supported therapist assignment and reassignment, specialties, attendance, scheduling, and performance monitoring, while the commission feature supported the computation and recording of commissions based on completed services and applicable arrangements. The system also organized discounts, vouchers, and gift certificates and provided validation and redemption monitoring. Customer feedback and therapist ratings provided management with organized information regarding customer experiences and service performance. In addition, security controls such as role-based access, gate code verification, PIN verification, and activity logging provided stronger access control and accountability for sensitive activities. Automated email notifications also supported account verification, appointment updates, payment confirmations, refund-related updates, and other important system activities.

Based on the completed system testing and User Acceptance Testing, the proponents concluded that the developed system substantially achieved the objectives established in the study. The testing and evaluation results indicated that the implemented functions addressed the identified operational requirements and provided the intended support for the spa's users and management. The reporting and analytics functions further provided organized information regarding sales, appointments, inventory, therapist productivity, commissions, cash reconciliation, and other operational activities, together with

formula-based short-term revenue forecasting and report export capabilities in Excel and PDF formats. Overall, the system improved the organization of operational records, supported more systematic transaction and appointment management, strengthened accountability, facilitated customer communication, and provided management with accessible information for monitoring and decision-making. The completed system therefore provided Recovery Iloilo Spa with an integrated technological platform that supported its current operational requirements and established a practical foundation for continued system use, maintenance, and future enhancement.

5.4 Recommendations

Based on the operational findings, user feedback, system testing, and User Acceptance Testing conducted during the development and implementation of the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently**, the proponents recommend several improvements that may further enhance the system's functionality, usability, security, scalability, and long-term sustainability. The recommendations are categorized into System Improvements and Technical Enhancements, Implementation and Operational Strategies, and Future Research and Enterprise Scaling Directions. These recommendations are intended to build upon the existing capabilities of the developed system rather than replace its current functions.

System Improvements and Technical Enhancements

1. Integration of a Wellness Information and Pre-Booking Assistance Chatbot

It is recommended that a conversational chatbot be integrated into the Customer Portal to provide customers with immediate assistance before booking services. The chatbot may provide general information about available treatments, service duration, massage pressure options, preparation guidelines, and common wellness-related questions. It may also assist customers in identifying services based on stated preferences, comfort levels, and general wellness goals.

For health-related concerns, the chatbot should provide general informational guidance accompanied by clear safety disclaimers rather than diagnosing medical conditions or determining whether a treatment is medically safe. When customers disclose medical conditions, pregnancy, injuries, medications, or other circumstances that may require professional assessment, the chatbot should advise them to consult a qualified healthcare professional or the spa's trained personnel before selecting a treatment. The chatbot may also recommend that customers disclose relevant conditions to the spa before an appointment. This approach would allow the feature to improve customer assistance while maintaining appropriate boundaries for health-related information.

The chatbot may also provide service and package recommendations based on customer preferences, such as preferred massage pressure, treatment duration, relaxation goals, or interest in additional services. This would complement the system's existing service catalog, appointment booking, and add-on functionality.

2. Regular System Maintenance and Updates

The current system integrates multiple modules for appointment scheduling, customer records, inventory management, payment processing, therapist management, commissions, discounts and vouchers, feedback, notifications, security, reporting, and analytics. As the system is continuously used, regular maintenance may be required to address technical issues, correct identified errors, update software dependencies, and apply necessary security patches.

Regular system maintenance may include monitoring system performance, checking database integrity, reviewing implemented functions, updating dependencies, and resolving issues identified by users or administrators. Conducting these activities regularly would help maintain the reliability, security, and functionality of the system and ensure that it continues to support the operational requirements of Recovery Iloilo Spa.

3. SMS Gateway Integration

The current system uses an Email/SMTP service for OTP verification, appointment confirmations, booking updates, payment acknowledgements, refund notifications, password reset instructions, and other system-generated messages. As an enhancement, an SMS gateway may be integrated to complement the existing email notification mechanism.

SMS notifications may be used for important time-sensitive activities such as appointment reminders, booking confirmations, cancellations, payment confirmations, and verification codes. Using both email and SMS would provide customers with additional communication channels, particularly when an email notification is not immediately noticed.

4. Establish Regular Data Backup and Recovery Procedures

The current system stores centralized operational information, including customer records, appointment details, payment and transaction records, inventory information, therapist records, commission records, discounts and vouchers, feedback, notifications, and reports. As these records are essential to the daily operations of the spa, regular data backup procedures should be established to protect the information from accidental deletion, system failure, hardware problems, or other unexpected incidents.

Scheduled backups may be performed regularly and stored in a secure location separate from the primary system environment. A data recovery procedure should also be established and tested periodically to verify that important records can be restored when necessary. Implementing these procedures would help minimize the risk of permanent data loss and support the continuity of Recovery Iloilo Spa's operations.

Future Research Directions

1. Advanced Predictive Analytics and Machine Learning

The current system provides formula-based short-term revenue forecasting using recorded transaction data and generates reports and analytics for sales, appointments, inventory, therapist productivity, commissions, and other operational activities. Future researchers may explore the integration of advanced predictive analytics and machine-learning techniques to further improve the system's forecasting capabilities as more historical data becomes available.

Machine-learning models may be used to analyze historical sales, appointment patterns, service demand, seasonal trends, and other relevant operational data to produce more comprehensive forecasts. The performance of these models may also be compared with the current formula-based forecasting method to determine whether they provide improved forecasting accuracy and more useful insights for management decision-making.

2. Mobile Application Development

The current system is implemented as a web-based platform that can be accessed through supported web browsers on computers and mobile devices. Future researchers may develop a Progressive Web App (PWA) or dedicated Android and iOS applications to provide customers and staff with a more optimized mobile experience.

A mobile application may provide features such as push notifications, appointment reminders, booking updates, payment notifications, appointment monitoring, and easier access to relevant customer or therapist functions. Developing a dedicated mobile application may improve accessibility and convenience while extending the existing functions of the web-based system to a more mobile-oriented platform.

3. Biometric Attendance and Authentication

The current system provides digital therapist attendance monitoring through clock-in and clock-out functions and includes security controls such as role-based access, gate code verification, PIN verification, and activity logging. Future researchers may explore

the integration of biometric technologies, such as fingerprint or facial recognition, to provide an additional method of verifying therapist attendance and authorized access.

Biometric authentication may help strengthen attendance verification and reduce the possibility of unauthorized attendance recording. Future studies should evaluate the accuracy, reliability, cost, privacy considerations, security requirements, and user acceptance of biometric technologies before they are incorporated into the system. The implementation should also follow applicable data-protection and privacy requirements.

Implementation and Operational Strategies

1. User Training and System Orientation

The proponents recommend that Recovery Iloilo Spa provide proper training and system orientation to administrators, receptionists, therapists, cashiers, and other authorized users before and during system implementation. Training should cover the functions relevant to each user role, including appointment management, customer records, Point-of-Sale transactions, payment processing, inventory management, therapist attendance and commissions, discounts and vouchers, feedback, notifications, and reporting.

Training sessions may include demonstrations, guided practice, and basic procedures for handling common system activities. Additional orientation should also be provided whenever significant system updates or new functions are introduced. Proper user training would help users understand their responsibilities, minimize incorrect data entry, and ensure that the implemented features are used consistently.

2. Regular Data Management and Backup Procedures

The proponents recommend that Recovery Iloilo Spa establish clear procedures for maintaining and protecting the information stored in the system. Authorized personnel should regularly review customer records, appointment information, transaction records, inventory data, therapist records, commission information, and other operational records to ensure that information remains accurate and up to date.

Regular database backups should also be performed and stored in a secure location. A recovery procedure should be established and tested periodically to ensure that important information can be restored in the event of accidental deletion, system failure, hardware problems, or other unexpected incidents. These procedures would help protect the integrity and availability of the spa's operational data.

3. Regular Security and Access Review

The proponents recommend that management regularly review user accounts, assigned roles, access permissions, gate code access, PIN verification, and activity logs. Access privileges should be updated whenever an employee's responsibilities change, while inactive or unauthorized accounts should be disabled or removed when no longer required.

Management should also regularly review activity logs and monitor sensitive transactions to ensure that system activities remain traceable and accountable. These practices would strengthen the security of customer, payment, and operational information and help ensure that system functions are accessed only by authorized personnel.

4. Continuous System Evaluation and Maintenance

The proponents recommend that Recovery Iloilo Spa conduct periodic evaluations of the system after implementation to identify technical issues, usability concerns, and additional operational requirements. Feedback from administrators, receptionists, therapists, and customers may be collected to determine whether the system continues to meet the spa's changing business needs.

Regular maintenance should include monitoring system performance, addressing reported errors, reviewing database integrity, updating necessary software components, and applying appropriate security updates. Continuous evaluation and maintenance would help preserve the reliability and effectiveness of the system and provide a basis for future improvements.

5.5 Impact of the Study

The **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics Efficiently** contributes to the field of Information Technology by demonstrating the practical application of an integrated web-based information system in a service-oriented business environment. The study demonstrates how multiple operational processes, including appointment scheduling, customer and therapist management, inventory monitoring, Point-of-Sale and payment processing, commissions, discounts and vouchers, feedback, notifications, security, reporting, and analytics, can be organized within a centralized platform. The system also demonstrates the use of operational data for generating reports, dashboards, and formula-based short-term revenue forecasts. Through the integration of these functions, the study provides an example of how information systems can be designed to address actual

business requirements, reduce dependence on fragmented manual procedures, improve information organization, and support data-informed operational decision-making.

For Recovery Iloilo Spa management and staff, the study provides a centralized system that supports the management of the spa's daily activities. Management benefits from organized appointment records, customer information, inventory records, transaction histories, therapist attendance and commission records, discount and voucher records, feedback, and operational reports. The reporting and analytics functions provide management with information regarding sales performance, appointment activities, inventory usage, therapist productivity, commissions, cash reconciliation, and other operational indicators, together with formula-based short-term revenue forecasting. The system also provides security and accountability mechanisms, including role-based access, gate code and PIN verification, and activity logging, which support controlled access to administrative and sensitive transactions. These functions provide management and staff with more organized information and support more systematic monitoring of the spa's operations.

For customers and therapists, the study provides benefits through features that support more convenient and organized interactions with the spa. Customers can register and verify their accounts through email OTP, browse available services and products, manage their profiles, use the shopping cart, book appointments, view appointment and transaction histories, monitor payment and refund information, receive automated notifications, and submit feedback and ratings. The system also provides organized information about services and products, including relevant descriptions, prices, categories, durations, and availability. Therapists benefit from organized appointment assignments, therapist schedules, attendance records, specialty information, completed-service records, commission records, and customer feedback associated with their services. These functions provide customers and therapists with clearer access to relevant information and support more organized service coordination.

For the proponents, Information Technology students, and future researchers, the study provides practical experience and a reference for the application of software development, web programming, database management, systems analysis and design, information security, payment integration, and data analytics in an actual business setting. The development process demonstrates how identified business requirements can be translated into system modules and integrated into a centralized database-driven platform. The study may also serve as a reference for future researchers who intend to develop or evaluate integrated management systems for spas and other service-oriented establishments, particularly in areas involving appointment management, inventory, payment processing, workforce management, customer feedback, security, reporting, and forecasting. The study also identifies areas

for future enhancement, such as advanced predictive analytics, mobile application development, and biometric attendance, which may be explored as the operational requirements and available data of the establishment increase.

5.6 Summary

In summary, the **Recovery Iloilo Spa Integrated System for Streamlining Operations and Analytics efficiently** demonstrated the practical application of software engineering, database management, systems analysis and design, information security, and business analytics in addressing the operational requirements of Recovery Iloilo Spa. Guided by the developmental research design and Agile methodology, the proponents developed and implemented a centralized web-based platform that integrated the spa's major operational processes into a single system. The completed system provides functions for appointment scheduling and management, customer registration and records, service and product information, inventory monitoring, Point-of-Sale and payment processing, therapist management and attendance, commission computation, discounts and vouchers, customer feedback and ratings, automated notifications, security and accountability, reporting, and analytics.

The development of the system addressed the operational challenges identified during the requirements-gathering process. The appointment management module supported online, walk-in, home-service, and hotel or partner bookings and incorporated minute-level scheduling and server-side conflict detection to help manage therapist availability and prevent scheduling conflicts. The customer management functions centralized customer information, appointment history, transaction records, preferences, feedback, and voucher usage. The system also organized service and product information and supported inventory monitoring, allowing staff to monitor product availability, stock movement, and low-stock items. These integrated functions provided a more organized approach to managing customer requests, appointments, services, products, and operational records. The findings showed that the implemented modules addressed the identified requirements and supported the daily activities of the establishment.

The system also integrated transaction processing, workforce management, communication, security, and business analytics within the same platform. The Point-of-Sale and payment module supported multiple payment methods, online payment processing through PayMongo, digital receipts, transaction history, and refund requests. The therapist management and commission functions supported therapist assignment, reassignment, attendance monitoring, specialty management, performance tracking, and commission computation based on completed services and applicable

arrangements. The system also managed discounts and vouchers, collected customer feedback and ratings, and provided automated notifications for important account, appointment, payment, and transaction activities. Security mechanisms, including controlled access, gate code and PIN verification, and activity logging, provided additional support for accountability and the protection of sensitive operations.

The development and evaluation process further established that the system supported the operational requirements identified for Recovery Iloilo Spa. System testing and User Acceptance Testing were conducted to evaluate the implemented functions, while feedback from the client and respondents was used to assess the system's functionality, usability, reliability, performance efficiency, accuracy, and security. The results indicated that the developed system provided the intended support for the spa's daily operations and that the specific objectives of the study were substantially achieved. The reporting and analytics functions also provided management with organized information regarding sales, appointments, inventory, therapist productivity, commissions, cash reconciliation, and other operational activities, together with formula-based short-term revenue forecasting and report export capabilities. Overall, the completed system provided Recovery Iloilo Spa with a centralized technological solution that improved the organization of operational processes, strengthened record management and accountability, supported customer and employee services, and provided management with information for monitoring and decision-making.

End of Proposal

Note to the instructors handling this subject

CHED BSIT ALIGNMENT STATEMENT

This capstone project complies with CHED BSIT standards by demonstrating competencies in

application development, database management, systems analysis and design, and emerging

technologies such as machine learning. The project follows outcomes-based education

principles and integrates ethical con

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